

Chemistry 2e

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CHAPTER 21

Nuclear Chemistry

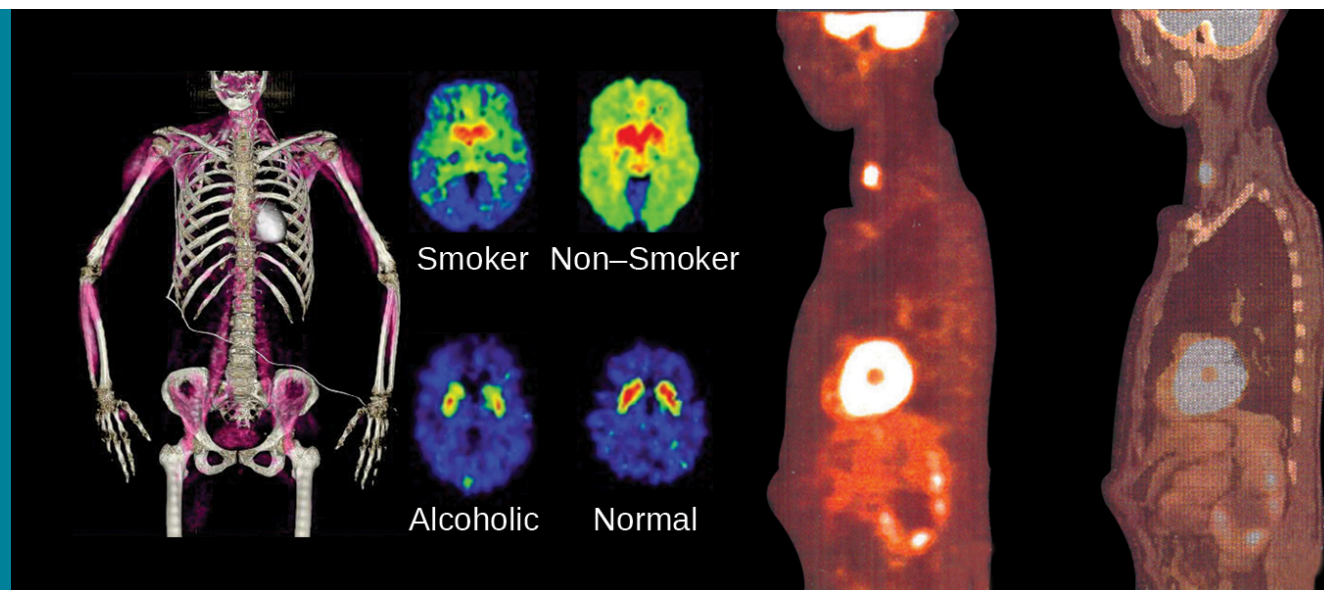


FIGURE 21.1 Nuclear chemistry provides the basis for many useful diagnostic and therapeutic methods in medicine, such as these positron emission tomography (PET) scans. The PET/computed tomography scan on the left shows muscle activity. The brain scans in the center show chemical differences in dopamine signaling in the brains of addicts and nonaddicts. The images on the right show an oncological application of PET scans to identify lymph node metastasis.

CHAPTER OUTLINE

21.1 Nuclear Structure and Stability

21.2 Nuclear Equations

21.3 Radioactive Decay

21.4 Transmutation and Nuclear Energy

21.5 Uses of Radioisotopes

21.6 Biological Effects of Radiation

INTRODUCTION The chemical reactions that we have considered in previous chapters involve changes in the *electronic* structure of the species involved, that is, the arrangement of the electrons around atoms, ions, or molecules. *Nuclear* structure, the numbers of protons and neutrons within the nuclei of the atoms involved, remains unchanged during chemical reactions.

This chapter will introduce the topic of nuclear chemistry, which began with the discovery of radioactivity in 1896 by French physicist Antoine Becquerel and has become increasingly important during the twentieth and twenty-first centuries, providing the basis for various technologies related to energy, medicine, geology, and many other areas.

21.1 Nuclear Structure and Stability

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe nuclear structure in terms of protons, neutrons, and electrons
- Calculate mass defect and binding energy for nuclei
- Explain trends in the relative stability of nuclei

Nuclear chemistry is the study of reactions that involve changes in nuclear structure. The chapter on atoms, molecules, and ions introduced the basic idea of nuclear structure, that the nucleus of an atom is composed of protons and, with the exception of ^1_1H , neutrons. Recall that the number of protons in the nucleus is called the

atomic number (Z) of the element, and the sum of the number of protons and the number of neutrons is the mass number (A). Atoms with the same atomic number but different mass numbers are isotopes of the same element. When referring to a single type of nucleus, we often use the term **nuclide** and identify it by the notation ${}^A_Z\text{X}$, where X is the symbol for the element, A is the mass number, and Z is the atomic number (for example, ${}^{14}_6\text{C}$). Often a nuclide is referenced by the name of the element followed by a hyphen and the mass number. For example, ${}^{14}_6\text{C}$ is called “carbon-14.”

Protons and neutrons, collectively called **nucleons**, are packed together tightly in a nucleus. With a radius of about 10^{-15} meters, a nucleus is quite small compared to the radius of the entire atom, which is about 10^{-10} meters. Nuclei are extremely dense compared to bulk matter, averaging 1.8×10^{14} grams per cubic centimeter. For example, water has a density of 1 gram per cubic centimeter, and iridium, one of the densest elements known, has a density of 22.6 g/cm^3 . If the earth’s density were equal to the average nuclear density, the earth’s radius would be only about 200 meters (earth’s actual radius is approximately 6.4×10^6 meters, 30,000 times larger). [Example 21.1](#) demonstrates just how great nuclear densities can be in the natural world.



EXAMPLE 21.1

Density of a Neutron Star

Neutron stars form when the core of a very massive star undergoes gravitational collapse, causing the star’s outer layers to explode in a supernova. Composed almost completely of neutrons, they are the densest-known stars in the universe, with densities comparable to the average density of an atomic nucleus. A neutron star in a faraway galaxy has a mass equal to 2.4 solar masses (1 solar mass = $M_\odot$ = mass of the sun = $1.99 \times 10^{30} \text{ kg}$) and a diameter of 26 km.

(a) What is the density of this neutron star?

(b) How does this neutron star’s density compare to the density of a uranium nucleus, which has a diameter of about 15 fm (1 fm = 10^{-15} m)?

Solution

We can treat both the neutron star and the U-235 nucleus as spheres. Then the density for both is given by:

$$d = \frac{m}{V} \quad \text{with} \quad V = \frac{4}{3}\pi r^3$$

(a) The radius of the neutron star is $\frac{1}{2} \times 26 \text{ km} = \frac{1}{2} \times 2.6 \times 10^4 \text{ m} = 1.3 \times 10^4 \text{ m}$, so the density of the neutron star is:

$$d = \frac{m}{V} = \frac{m}{\frac{4}{3}\pi r^3} = \frac{2.4 (1.99 \times 10^{30} \text{ kg})}{\frac{4}{3}\pi (1.3 \times 10^4 \text{ m})^3} = 5.2 \times 10^{17} \text{ kg/m}^3$$

(b) The radius of the U-235 nucleus is $\frac{1}{2} \times 15 \times 10^{-15} \text{ m} = 7.5 \times 10^{-15} \text{ m}$, so the density of the U-235 nucleus is:

$$d = \frac{m}{V} = \frac{m}{\frac{4}{3}\pi r^3} = \frac{235 \text{ amu} \left(\frac{1.66 \times 10^{-27} \text{ kg}}{1 \text{ amu}} \right)}{\frac{4}{3}\pi (7.5 \times 10^{-15} \text{ m})^3} = 2.2 \times 10^{17} \text{ kg/m}^3$$

These values are fairly similar (same order of magnitude), but the neutron star is more than twice as dense as the U-235 nucleus.

Check Your Learning

Find the density of a neutron star with a mass of 1.97 solar masses and a diameter of 13 km, and compare it to the density of a hydrogen nucleus, which has a diameter of 1.75 fm (1 fm = $1 \times 10^{-15} \text{ m}$).

Answer:

The density of the neutron star is $3.4 \times 10^{18} \text{ kg/m}^3$. The density of a hydrogen nucleus is $6.0 \times 10^{17} \text{ kg/m}^3$. The neutron star is 5.7 times denser than the hydrogen nucleus.

To hold positively charged protons together in the very small volume of a nucleus requires very strong attractive forces because the positively charged protons repel one another strongly at such short distances. The force of attraction that holds the nucleus together is the **strong nuclear force**. (The strong force is one of the four fundamental forces that are known to exist. The others are the electromagnetic force, the gravitational force, and the nuclear weak force.) This force acts between protons, between neutrons, and between protons and neutrons. It is very different from the electrostatic force that holds negatively charged electrons around a positively charged nucleus (the attraction between opposite charges). Over distances less than 10^{-15} meters and within the nucleus, the strong nuclear force is much stronger than electrostatic repulsions between protons; over larger distances and outside the nucleus, it is essentially nonexistent.

 LINK TO LEARNING

Visit this [website \(http://openstax.org/l/16fourfund\)](http://openstax.org/l/16fourfund) for more information about the four fundamental forces.

Nuclear Binding Energy

As a simple example of the energy associated with the strong nuclear force, consider the helium atom composed of two protons, two neutrons, and two electrons. The total mass of these six subatomic particles may be calculated as:

$$\begin{array}{ccccc} (2 \times 1.0073 \text{ amu}) + & (2 \times 1.0087 \text{ amu}) + & (2 \times 0.00055 \text{ amu}) = & 4.0331 \text{ amu} \\ \text{protons} & \text{neutrons} & \text{electrons} & \end{array}$$

However, mass spectrometric measurements reveal that the mass of an ${}^4_2\text{He}$ atom is 4.0026 amu, less than the combined masses of its six constituent subatomic particles. This difference between the calculated and experimentally measured masses is known as the **mass defect** of the atom. In the case of helium, the mass defect indicates a “loss” in mass of $4.0331 \text{ amu} - 4.0026 \text{ amu} = 0.0305 \text{ amu}$. The loss in mass accompanying the formation of an atom from protons, neutrons, and electrons is due to the conversion of that mass into energy that is evolved as the atom forms. The **nuclear binding energy** is the energy produced when the atoms’ nucleons are bound together; this is also the energy needed to break a nucleus into its constituent protons and neutrons. In comparison to chemical bond energies, nuclear binding energies are *vastly* greater, as we will learn in this section. Consequently, the energy changes associated with nuclear reactions are vastly greater than are those for chemical reactions.

The conversion between mass and energy is most identifiably represented by the **mass-energy equivalence equation** as stated by Albert Einstein:

$$E = mc^2$$

where E is energy, m is mass of the matter being converted, and c is the speed of light in a vacuum. This equation can be used to find the amount of energy that results when matter is converted into energy. Using this mass-energy equivalence equation, the nuclear binding energy of a nucleus may be calculated from its mass defect, as demonstrated in [Example 21.2](#). A variety of units are commonly used for nuclear binding energies, including **electron volts (eV)**, with 1 eV equaling the amount of energy necessary to move the charge of an electron across an electric potential difference of 1 volt, making $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$.



EXAMPLE 21.2

Calculation of Nuclear Binding Energy

Determine the binding energy for the nuclide ${}^4_2\text{He}$ in:

- (a) joules per mole of nuclei
- (b) joules per nucleus
- (c) MeV per nucleus

Solution

The mass defect for a ${}^4_2\text{He}$ nucleus is 0.0305 amu, as shown previously. Determine the binding energy in joules per nuclide using the mass-energy equivalence equation. To accommodate the requested energy units, the mass defect must be expressed in kilograms (recall that $1 \text{ J} = 1 \text{ kg m}^2/\text{s}^2$).

(a) First, express the mass defect in g/mol. This is easily done considering the *numerical equivalence* of atomic mass (amu) and molar mass (g/mol) that results from the definitions of the amu and mole units (refer to the previous discussion in the chapter on atoms, molecules, and ions if needed). The mass defect is therefore 0.0305 g/mol. To accommodate the units of the other terms in the mass-energy equation, the mass must be expressed in kg, since $1 \text{ J} = 1 \text{ kg m}^2/\text{s}^2$. Converting grams into kilograms yields a mass defect of $3.05 \times 10^{-5} \text{ kg/mol}$. Substituting this quantity into the mass-energy equivalence equation yields:

$$E = mc^2 = \frac{3.05 \times 10^{-5} \text{ kg}}{\text{mol}} \times \left(\frac{2.998 \times 10^8 \text{ m}}{\text{s}} \right)^2 = 2.74 \times 10^{12} \text{ kg m}^2 \text{ s}^{-2} \text{ mol}^{-1}$$

$$= 2.74 \times 10^{12} \text{ J mol}^{-1} = 2.74 \text{ TJ mol}^{-1}$$

Note that this tremendous amount of energy is associated with the conversion of a very small amount of matter (about 30 mg, roughly the mass of typical drop of water).

(b) The binding energy for a single nucleus is computed from the molar binding energy using Avogadro's number:

$$E = 2.74 \times 10^{12} \text{ J mol}^{-1} \times \frac{1 \text{ mol}}{6.022 \times 10^{23} \text{ nuclei}} = 4.55 \times 10^{-12} \text{ J} = 4.55 \text{ pJ}$$

(c) Recall that $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$. Using the binding energy computed in part (b):

$$E = 4.55 \times 10^{-12} \text{ J} \times \frac{1 \text{ eV}}{1.602 \times 10^{-19} \text{ J}} = 2.84 \times 10^7 \text{ eV} = 28.4 \text{ MeV}$$

Check Your Learning

What is the binding energy for the nuclide ${}^{19}_9\text{F}$ (atomic mass: 18.9984 amu) in MeV per nucleus?

Answer:

148.4 MeV

Because the energy changes for breaking and forming bonds are so small compared to the energy changes for breaking or forming nuclei, the changes in mass during all ordinary chemical reactions are virtually undetectable. As described in the chapter on thermochemistry, the most energetic chemical reactions exhibit enthalpies on the order of *thousands* of kJ/mol, which is equivalent to mass differences in the nanogram range (10^{-9} g). On the other hand, nuclear binding energies are typically on the order of *billions* of kJ/mol, corresponding to mass differences in the milligram range (10^{-3} g).

Nuclear Stability

A nucleus is stable if it cannot be transformed into another configuration without adding energy from the outside. Of the thousands of nuclides that exist, about 250 are stable. A plot of the number of neutrons versus the number of protons for stable nuclei reveals that the stable isotopes fall into a narrow band. This region is known as the **band of stability** (also called the belt, zone, or valley of stability). The straight line in [Figure 21.2](#) represents nuclei that have a 1:1 ratio of protons to neutrons (n:p ratio). Note that the lighter stable nuclei, in general, have equal numbers of protons and neutrons. For example, nitrogen-14 has seven protons and seven neutrons. Heavier stable nuclei,

however, have increasingly more neutrons than protons. For example: iron-56 has 30 neutrons and 26 protons, an n:p ratio of 1.15, whereas the stable nuclide lead-207 has 125 neutrons and 82 protons, an n:p ratio equal to 1.52. This is because larger nuclei have more proton-proton repulsions, and require larger numbers of neutrons to provide compensating strong forces to overcome these electrostatic repulsions and hold the nucleus together.

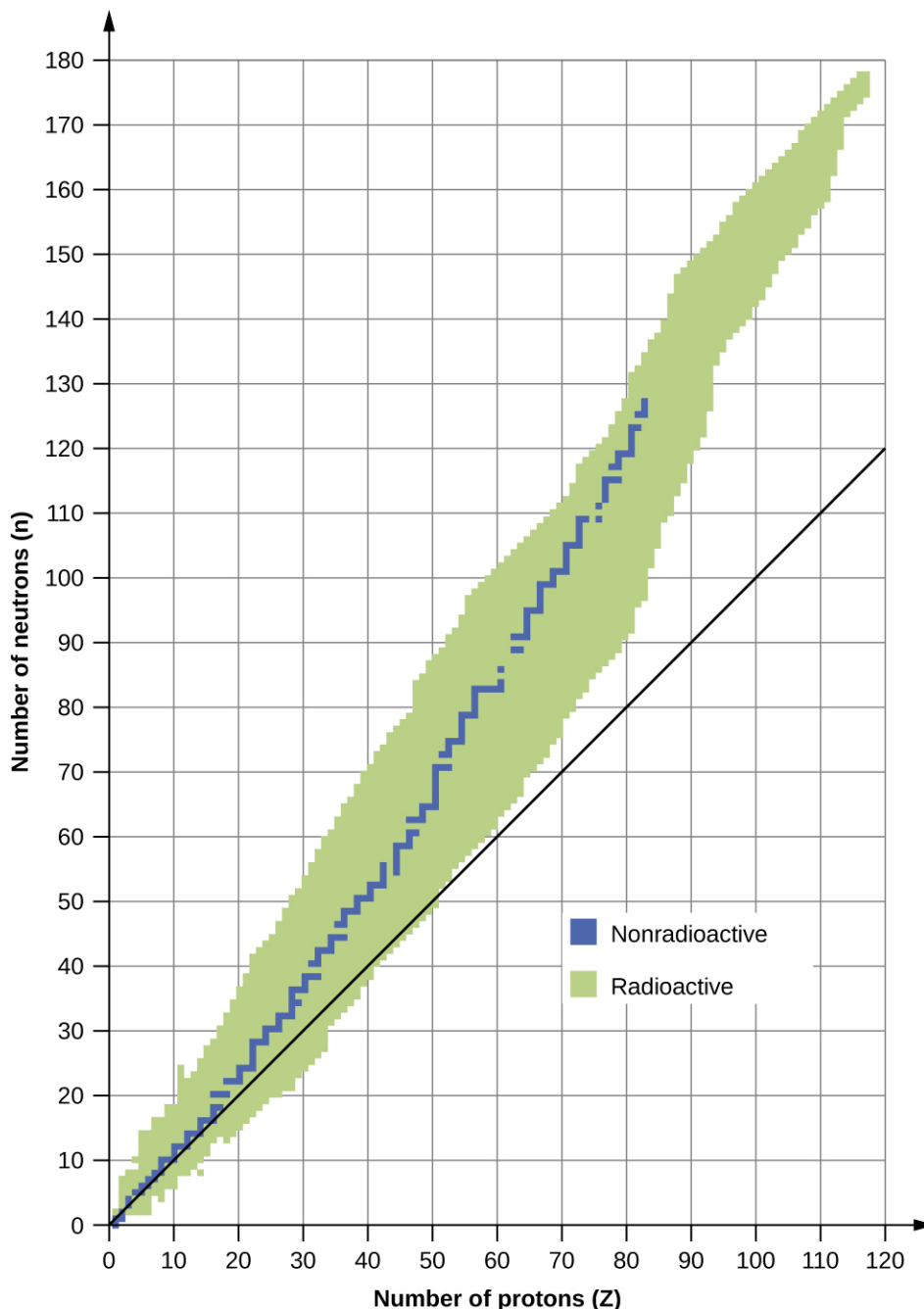


FIGURE 21.2 This plot shows the nuclides that are known to exist and those that are stable. The stable nuclides are indicated in blue, and the unstable nuclides are indicated in green. Note that all isotopes of elements with atomic numbers greater than 83 are unstable. The solid line is the line where $n = Z$.

The nuclei that are to the left or to the right of the band of stability are unstable and exhibit **radioactivity**. They change spontaneously (decay) into other nuclei that are either in, or closer to, the band of stability. These nuclear decay reactions convert one unstable isotope (or **radioisotope**) into another, more stable, isotope. We will discuss the nature and products of this radioactive decay in subsequent sections of this chapter.

Several observations may be made regarding the relationship between the stability of a nucleus and its structure.

Nuclei with even numbers of protons, neutrons, or both are more likely to be stable (see [Table 21.1](#)). Nuclei with certain numbers of nucleons, known as **magic numbers**, are stable against nuclear decay. These numbers of protons or neutrons (2, 8, 20, 28, 50, 82, and 126) make complete shells in the nucleus. These are similar in concept to the stable electron shells observed for the noble gases. Nuclei that have magic numbers of both protons and neutrons, such as ${}^4_2\text{He}$, ${}^{16}_8\text{O}$, ${}^{40}_{20}\text{Ca}$, and ${}^{208}_{82}\text{Pb}$, are called “double magic” and are particularly stable. These trends in nuclear stability may be rationalized by considering a quantum mechanical model of nuclear energy states analogous to that used to describe electronic states earlier in this textbook. The details of this model are beyond the scope of this chapter.

Stable Nuclear Isotopes

Number of Stable Isotopes	Proton Number	Neutron Number
157	even	even
53	even	odd
50	odd	even
5	odd	odd

TABLE 21.1

The relative stability of a nucleus is correlated with its **binding energy per nucleon**, the total binding energy for the nucleus divided by the number of nucleons in the nucleus. For instance, we saw in [Example 21.2](#) that the binding energy for a ${}^4_2\text{He}$ nucleus is 28.4 MeV. The binding energy *per nucleon* for a ${}^4_2\text{He}$ nucleus is therefore:

$$\frac{28.4 \text{ MeV}}{4 \text{ nucleons}} = 7.10 \text{ MeV/nucleon}$$

In [Example 21.3](#), we learn how to calculate the binding energy per nucleon of a nuclide on the curve shown in [Figure 21.3](#).

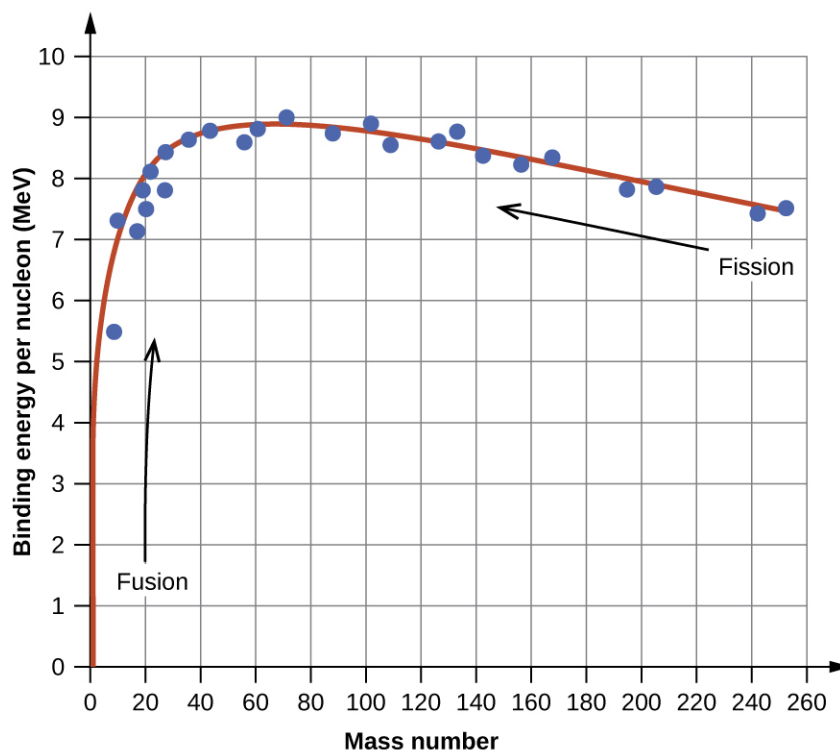


FIGURE 21.3 The binding energy per nucleon is largest for nuclides with mass number of approximately 56.



EXAMPLE 21.3

Calculation of Binding Energy per Nucleon

The iron nuclide $^{56}_{26}\text{Fe}$ lies near the top of the binding energy curve (Figure 21.3) and is one of the most stable nuclides. What is the binding energy per nucleon (in MeV) for the nuclide $^{56}_{26}\text{Fe}$ (atomic mass of 55.9349 amu)?

Solution

As in Example 21.2, we first determine the mass defect of the nuclide, which is the difference between the mass of 26 protons, 30 neutrons, and 26 electrons, and the observed mass of an $^{56}_{26}\text{Fe}$ atom:

$$\begin{aligned}\text{Mass defect} &= [(26 \times 1.0073 \text{ amu}) + (30 \times 1.0087 \text{ amu}) + (26 \times 0.00055 \text{ amu})] - 55.9349 \text{ amu} \\ &= 56.4651 \text{ amu} - 55.9349 \text{ amu} \\ &= 0.5302 \text{ amu}\end{aligned}$$

We next calculate the binding energy for one nucleus from the mass defect using the mass-energy equivalence equation:

$$\begin{aligned}E &= mc^2 = 0.5302 \text{ amu} \times \frac{1.6605 \times 10^{-27} \text{ kg}}{1 \text{ amu}} \times (2.998 \times 10^8 \text{ m/s})^2 \\ &= 7.913 \times 10^{-11} \text{ kg}\cdot\text{m}^2/\text{s}^2 \\ &= 7.913 \times 10^{-11} \text{ J}\end{aligned}$$

We then convert the binding energy in joules per nucleus into units of MeV per nuclide:

$$7.913 \times 10^{-11} \text{ J} \times \frac{1 \text{ MeV}}{1.602 \times 10^{-13} \text{ J}} = 493.9 \text{ MeV}$$

Finally, we determine the binding energy per nucleon by dividing the total nuclear binding energy by the number of

nucleons in the atom:

$$\text{Binding energy per nucleon} = \frac{493.9 \text{ MeV}}{56} = 8.820 \text{ MeV/nucleon}$$

Note that this is almost 25% larger than the binding energy per nucleon for ${}^4_2\text{He}$.

(Note also that this is the same process as in [Example 21.1](#), but with the additional step of dividing the total nuclear binding energy by the number of nucleons.)

Check Your Learning

What is the binding energy per nucleon in ${}^{19}_9\text{F}$ (atomic mass, 18.9984 amu)?

Answer:

7.810 MeV/nucleon

21.2 Nuclear Equations

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Identify common particles and energies involved in nuclear reactions
- Write and balance nuclear equations

Changes of nuclei that result in changes in their atomic numbers, mass numbers, or energy states are **nuclear reactions**. To describe a nuclear reaction, we use an equation that identifies the nuclides involved in the reaction, their mass numbers and atomic numbers, and the other particles involved in the reaction.

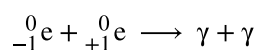
Types of Particles in Nuclear Reactions

Many entities can be involved in nuclear reactions. The most common are protons, neutrons, alpha particles, beta particles, positrons, and gamma rays, as shown in [Figure 21.4](#). Protons (${}_1^1\text{p}$, also represented by the symbol ${}_1^1\text{H}$) and neutrons (${}_0^1\text{n}$) are the constituents of atomic nuclei, and have been described previously. **Alpha particles** (${}_2^4\text{He}$, also represented by the symbol ${}_2^4\alpha$) are high-energy helium nuclei. **Beta particles** (${}_{-1}^0\beta$, also represented by the symbol ${}_{-1}^0\text{e}$) are high-energy electrons, and gamma rays are photons of very high-energy electromagnetic radiation. **Positrons** (${}_{+1}^0\text{e}$, also represented by the symbol ${}_{+1}^0\beta$) are positively charged electrons (“anti-electrons”). The subscripts and superscripts are necessary for balancing nuclear equations, but are usually optional in other circumstances. For example, an alpha particle is a helium nucleus (He) with a charge of +2 and a mass number of 4, so it is symbolized ${}_2^4\text{He}$. This works because, in general, the ion charge is not important in the balancing of nuclear equations.

Name	Symbol(s)	Representation	Description
Alpha particle	${}^4_2\text{He}$ or ${}^4_2\alpha$		(High-energy) helium nuclei consisting of two protons and two neutrons
Beta particle	${}^0_{-1}\text{e}$ or ${}^0_{-1}\beta$		(High-energy) electrons
Positron	${}^0_{+1}\text{e}$ or ${}^0_{+1}\beta$		Particles with the same mass as an electron but with 1 unit of positive charge
Proton	${}^1_1\text{H}$ or ${}^1_1\text{p}$		Nuclei of hydrogen atoms
Neutron	${}^1_0\text{n}$		Particles with a mass approximately equal to that of a proton but with no charge
Gamma ray	γ		Very high-energy electromagnetic radiation

FIGURE 21.4 Although many species are encountered in nuclear reactions, this table summarizes the names, symbols, representations, and descriptions of the most common of these.

Note that positrons are exactly like electrons, except they have the opposite charge. They are the most common example of **antimatter**, particles with the same mass but the opposite state of another property (for example, charge) than ordinary matter. When antimatter encounters ordinary matter, both are annihilated and their mass is converted into energy in the form of **gamma rays (γ)**—and other much smaller subnuclear particles, which are beyond the scope of this chapter—according to the mass-energy equivalence equation $E = mc^2$, seen in the preceding section. For example, when a positron and an electron collide, both are annihilated and two gamma ray photons are created:



As seen in the chapter discussing light and electromagnetic radiation, gamma rays compose short wavelength, high-energy electromagnetic radiation and are (much) more energetic than better-known X-rays that can behave as particles in the wave-particle duality sense. Gamma rays are a type of high energy electromagnetic radiation produced when a nucleus undergoes a transition from a higher to a lower energy state, similar to how a photon is produced by an electronic transition from a higher to a lower energy level. Due to the much larger energy differences between nuclear energy shells, gamma rays emanating from a nucleus have energies that are typically millions of times larger than electromagnetic radiation emanating from electronic transitions.

Balancing Nuclear Reactions

A balanced chemical reaction equation reflects the fact that during a chemical reaction, bonds break and form, and atoms are rearranged, but the total numbers of atoms of each element are conserved and do not change. A balanced nuclear reaction equation indicates that there is a rearrangement during a nuclear reaction, but of nucleons (subatomic particles within the atoms' nuclei) rather than atoms. Nuclear reactions also follow conservation laws, and they are balanced in two ways:

1. The sum of the mass numbers of the reactants equals the sum of the mass numbers of the products.
2. The sum of the charges of the reactants equals the sum of the charges of the products.

If the atomic number and the mass number of all but one of the particles in a nuclear reaction are known, we can identify the particle by balancing the reaction. For instance, we could determine that ${}^{17}_8\text{O}$ is a product of the nuclear reaction of ${}^{14}_7\text{N}$ and ${}^4_2\text{He}$ if we knew that a proton, ${}^1_1\text{H}$, was one of the two products. [Example 21.4](#) shows how we can identify a nuclide by balancing the nuclear reaction.



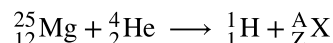
EXAMPLE 21.4

Balancing Equations for Nuclear Reactions

The reaction of an α particle with magnesium-25 ($^{25}_{12}\text{Mg}$) produces a proton and a nuclide of another element. Identify the new nuclide produced.

Solution

The nuclear reaction can be written as:



where A is the mass number and Z is the atomic number of the new nuclide, X. Because the sum of the mass numbers of the reactants must equal the sum of the mass numbers of the products:

$$25 + 4 = A + 1, \text{ or } A = 28$$

Similarly, the charges must balance, so:

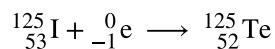
$$12 + 2 = Z + 1, \text{ and } Z = 13$$

Check the periodic table: The element with nuclear charge = +13 is aluminum. Thus, the product is $^{28}_{13}\text{Al}$.

Check Your Learning

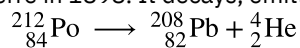
The nuclide $^{125}_{53}\text{I}$ combines with an electron and produces a new nucleus and no other massive particles. What is the equation for this reaction?

Answer:

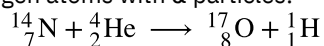


Following are the equations of several nuclear reactions that have important roles in the history of nuclear chemistry:

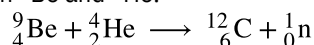
- The first naturally occurring unstable element that was isolated, polonium, was discovered by the Polish scientist Marie Curie and her husband Pierre in 1898. It decays, emitting α particles:



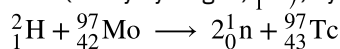
- The first nuclide to be prepared by artificial means was an isotope of oxygen, ^{17}O . It was made by Ernest Rutherford in 1919 by bombarding nitrogen atoms with α particles:



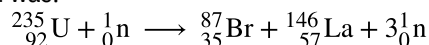
- James Chadwick discovered the neutron in 1932, as a previously unknown neutral particle produced along with ^{12}C by the nuclear reaction between ^9Be and ^4He :



- The first element to be prepared that does not occur naturally on the earth, technetium, was created by bombardment of molybdenum by deuterons (heavy hydrogen, ^2H), by Emilio Segre and Carlo Perrier in 1937:



- The first controlled nuclear chain reaction was carried out in a reactor at the University of Chicago in 1942. One of the many reactions involved was:



21.3 Radioactive Decay

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Recognize common modes of radioactive decay
- Identify common particles and energies involved in nuclear decay reactions
- Write and balance nuclear decay equations
- Calculate kinetic parameters for decay processes, including half-life
- Describe common radiometric dating techniques

Following the somewhat serendipitous discovery of radioactivity by Becquerel, many prominent scientists began to investigate this new, intriguing phenomenon. Among them were Marie Curie (the first woman to win a Nobel Prize, and the only person to win two Nobel Prizes in different sciences—chemistry and physics), who was the first to coin the term “radioactivity,” and Ernest Rutherford (of gold foil experiment fame), who investigated and named three of the most common types of radiation. During the beginning of the twentieth century, many radioactive substances were discovered, the properties of radiation were investigated and quantified, and a solid understanding of radiation and nuclear decay was developed.

The spontaneous change of an unstable nuclide into another is **radioactive decay**. The unstable nuclide is called the **parent nuclide**; the nuclide that results from the decay is known as the **daughter nuclide**. The daughter nuclide may be stable, or it may decay itself. The radiation produced during radioactive decay is such that the daughter nuclide lies closer to the band of stability than the parent nuclide, so the location of a nuclide relative to the band of stability can serve as a guide to the kind of decay it will undergo ([Figure 21.5](#)).

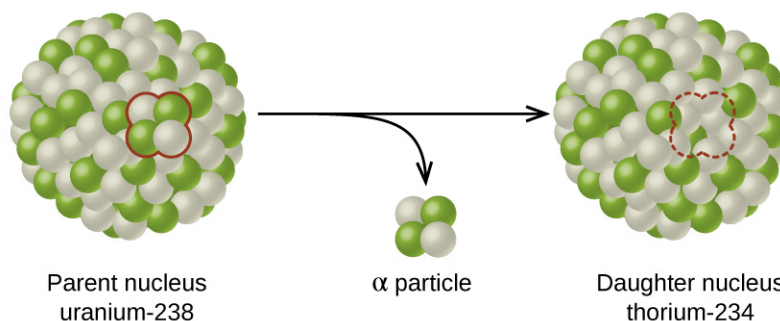


FIGURE 21.5 A nucleus of uranium-238 (the parent nuclide) undergoes α decay to form thorium-234 (the daughter nuclide). The alpha particle removes two protons (green) and two neutrons (gray) from the uranium-238 nucleus.

LINK TO LEARNING

Although the radioactive decay of a nucleus is too small to see with the naked eye, we can indirectly view radioactive decay in an environment called a cloud chamber. Click [here \(http://openstax.org/l/16cloudchamb\)](http://openstax.org/l/16cloudchamb) to learn about cloud chambers and to view an interesting Cloud Chamber Demonstration from the Jefferson Lab.

Types of Radioactive Decay

Ernest Rutherford’s experiments involving the interaction of radiation with a magnetic or electric field ([Figure 21.6](#)) helped him determine that one type of radiation consisted of positively charged and relatively massive α particles; a second type was made up of negatively charged and much less massive β particles; and a third was uncharged electromagnetic waves, γ rays. We now know that α particles are high-energy helium nuclei, β particles are high-energy electrons, and γ radiation compose high-energy electromagnetic radiation. We classify different types of radioactive decay by the radiation produced.

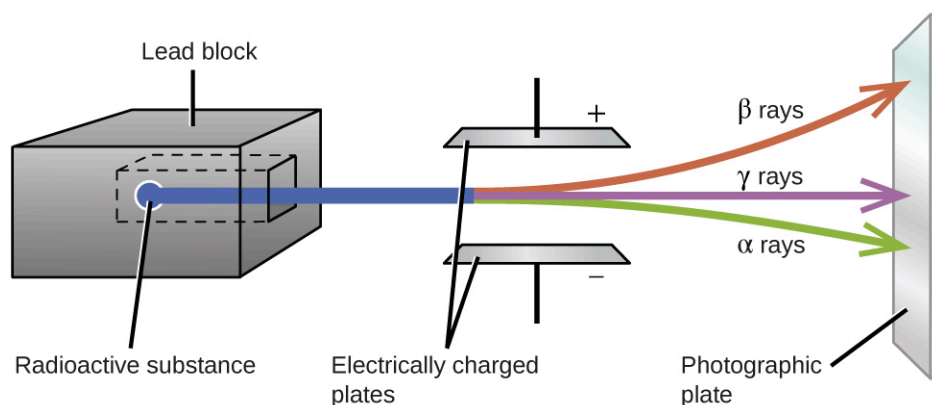
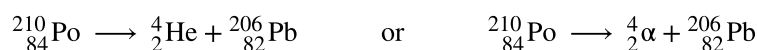


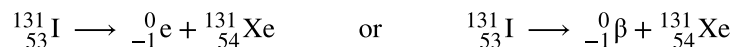
FIGURE 21.6 Alpha particles, which are attracted to the negative plate and deflected by a relatively small amount, must be positively charged and relatively massive. Beta particles, which are attracted to the positive plate and deflected a relatively large amount, must be negatively charged and relatively light. Gamma rays, which are unaffected by the electric field, must be uncharged.

Alpha (α) decay is the emission of an α particle from the nucleus. For example, polonium-210 undergoes α decay:



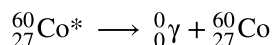
Alpha decay occurs primarily in heavy nuclei ($A > 200$, $Z > 83$). Because the loss of an α particle gives a daughter nuclide with a mass number four units smaller and an atomic number two units smaller than those of the parent nuclide, the daughter nuclide has a larger $n:p$ ratio than the parent nuclide. If the parent nuclide undergoing α decay lies below the band of stability (refer to [Figure 21.2](#)), the daughter nuclide will lie closer to the band.

Beta (β) decay is the emission of an electron from a nucleus. Iodine-131 is an example of a nuclide that undergoes β decay:



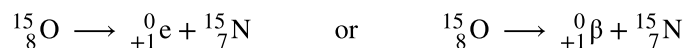
Beta decay, which can be thought of as the conversion of a neutron into a proton and a β particle, is observed in nuclides with a large $n:p$ ratio. The beta particle (electron) emitted is from the atomic nucleus and is not one of the electrons surrounding the nucleus. Such nuclei lie above the band of stability. Emission of an electron does not change the mass number of the nuclide but does increase the number of its protons and decrease the number of its neutrons. Consequently, the $n:p$ ratio is decreased, and the daughter nuclide lies closer to the band of stability than did the parent nuclide.

Gamma emission (γ emission) is observed when a nuclide is formed in an excited state and then decays to its ground state with the emission of a γ ray, a quantum of high-energy electromagnetic radiation. The presence of a nucleus in an excited state is often indicated by an asterisk (*). Cobalt-60 emits γ radiation and is used in many applications including cancer treatment:



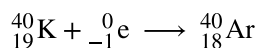
There is no change in mass number or atomic number during the emission of a γ ray unless the γ emission accompanies one of the other modes of decay.

Positron emission (β^+ decay) is the emission of a positron from the nucleus. Oxygen-15 is an example of a nuclide that undergoes positron emission:



Positron emission is observed for nuclides in which the $n:p$ ratio is low. These nuclides lie below the band of stability. Positron decay is the conversion of a proton into a neutron with the emission of a positron. The $n:p$ ratio increases, and the daughter nuclide lies closer to the band of stability than did the parent nuclide.

Electron capture occurs when one of the inner electrons in an atom is captured by the atom's nucleus. For example, potassium-40 undergoes electron capture:



Electron capture occurs when an inner shell electron combines with a proton and is converted into a neutron. The loss of an inner shell electron leaves a vacancy that will be filled by one of the outer electrons. As the outer electron drops into the vacancy, it will emit energy. In most cases, the energy emitted will be in the form of an X-ray. Like positron emission, electron capture occurs for “proton-rich” nuclei that lie below the band of stability. Electron capture has the same effect on the nucleus as does positron emission: The atomic number is decreased by one and the mass number does not change. This increases the n:p ratio, and the daughter nuclide lies closer to the band of stability than did the parent nuclide. Whether electron capture or positron emission occurs is difficult to predict. The choice is primarily due to kinetic factors, with the one requiring the smaller activation energy being the one more likely to occur.

Figure 21.7 summarizes these types of decay, along with their equations and changes in atomic and mass numbers.

Type	Nuclear equation	Representation	Change in mass/atomic numbers
Alpha decay	${}^A_Z\text{X} \Rightarrow {}^4_2\text{He} + {}^{A-4}_{Z-2}\text{Y}$		A: decrease by 4 Z: decrease by 2
Beta decay	${}^A_Z\text{X} \Rightarrow {}^0_{-1}\text{e} + {}^A_{Z+1}\text{Y}$		A: unchanged Z: increase by 1
Gamma decay	${}^A_Z\text{X} \Rightarrow {}^0_0\gamma + {}^A_Z\text{Y}$		A: unchanged Z: unchanged
Positron emission	${}^A_Z\text{X} \Rightarrow {}^0_{+1}\text{e} + {}^A_{Z-1}\text{Y}$		A: unchanged Z: decrease by 1
Electron capture	${}^A_Z\text{X} + {}^0_{-1}\text{e} \Rightarrow {}^A_{Z-1}\text{Y}$		A: unchanged Z: decrease by 1

FIGURE 21.7 This table summarizes the type, nuclear equation, representation, and any changes in the mass or atomic numbers for various types of decay.

CHEMISTRY IN EVERYDAY LIFE

PET Scan

Positron emission tomography (PET) scans use radiation to diagnose and track health conditions and monitor medical treatments by revealing how parts of a patient’s body function (Figure 21.8). To perform a PET scan, a positron-emitting radioisotope is produced in a cyclotron and then attached to a substance that is used by the part of the body being investigated. This “tagged” compound, or radiotracer, is then put into the patient (injected via IV or breathed in as a gas), and how it is used by the tissue reveals how that organ or other area of the body functions.

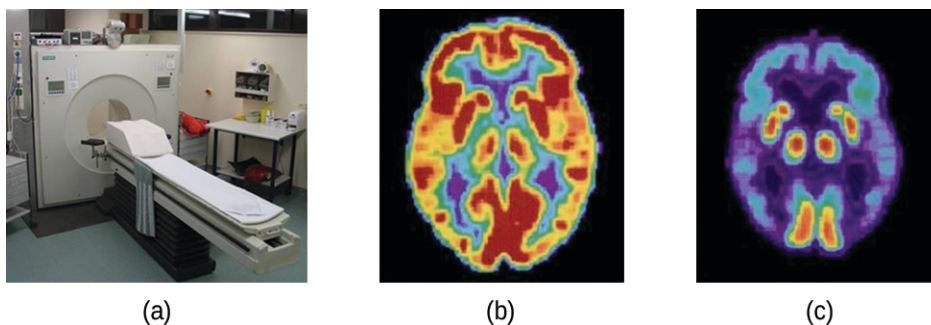


FIGURE 21.8 A PET scanner (a) uses radiation to provide an image of how part of a patient's body functions. The scans it produces can be used to image a healthy brain (b) or can be used for diagnosing medical conditions such as Alzheimer's disease (c). (credit a: modification of work by Jens Maus)

For example, F-18 is produced by proton bombardment of ^{18}O ($^{18}_8\text{O} + {}^1_1\text{p} \rightarrow {}^{18}_9\text{F} + {}^1_0\text{n}$) and incorporated into a glucose analog called fludeoxyglucose (FDG). How FDG is used by the body provides critical diagnostic information; for example, since cancers use glucose differently than normal tissues, FDG can reveal cancers. The ^{18}F emits positrons that interact with nearby electrons, producing a burst of gamma radiation. This energy is detected by the scanner and converted into a detailed, three-dimensional, color image that shows how that part of the patient's body functions. Different levels of gamma radiation produce different amounts of brightness and colors in the image, which can then be interpreted by a radiologist to reveal what is going on. PET scans can detect heart damage and heart disease, help diagnose Alzheimer's disease, indicate the part of a brain that is affected by epilepsy, reveal cancer, show what stage it is, and how much it has spread, and whether treatments are effective. Unlike magnetic resonance imaging and X-rays, which only show how something looks, the big advantage of PET scans is that they show how something functions. PET scans are now usually performed in conjunction with a computed tomography scan.

Radioactive Decay Series

The naturally occurring radioactive isotopes of the heaviest elements fall into chains of successive disintegrations, or decays, and all the species in one chain constitute a radioactive family, or **radioactive decay series**. Three of these series include most of the naturally radioactive elements of the periodic table. They are the uranium series, the actinide series, and the thorium series. The neptunium series is a fourth series, which is no longer significant on the earth because of the short half-lives of the species involved. Each series is characterized by a parent (first member) that has a long half-life and a series of daughter nuclides that ultimately lead to a stable end-product—that is, a nuclide on the band of stability ([Figure 21.9](#)). In all three series, the end-product is a stable isotope of lead. The neptunium series, previously thought to terminate with bismuth-209, terminates with thallium-205.

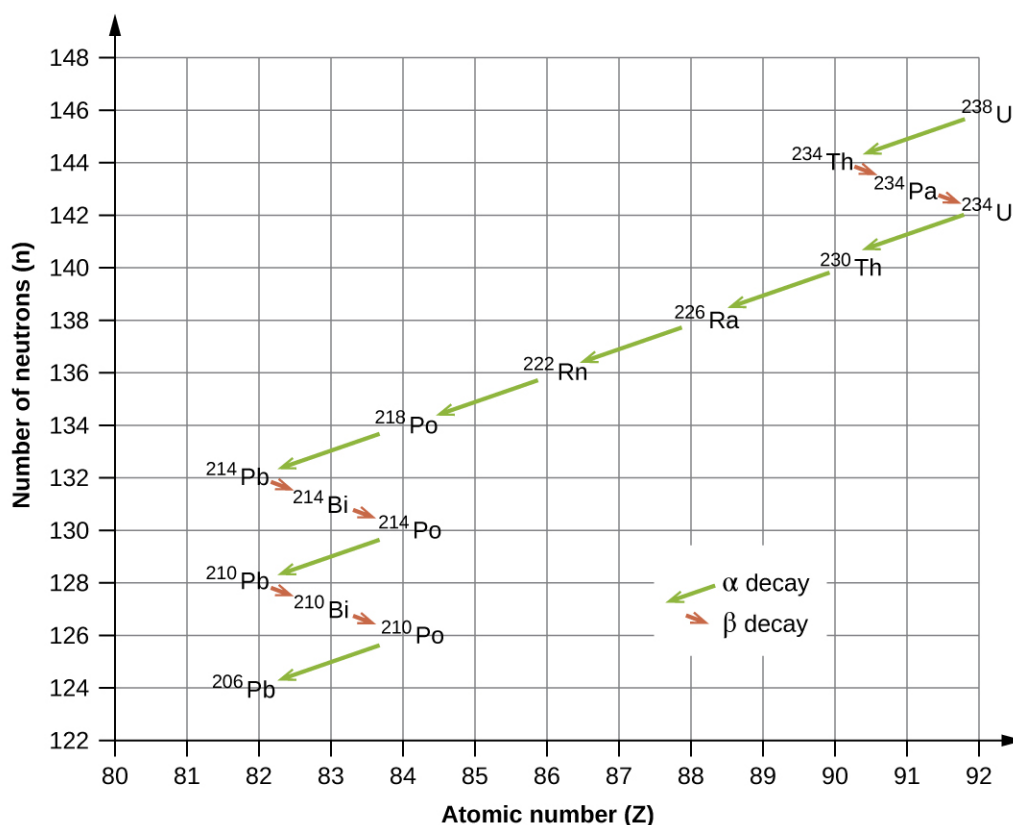


FIGURE 21.9 Uranium-238 undergoes a radioactive decay series consisting of 14 separate steps before producing stable lead-206. This series consists of eight α decays and six β decays.

Radioactive Half-Lives

Radioactive decay follows first-order kinetics. Since first-order reactions have already been covered in detail in the kinetics chapter, we will now apply those concepts to nuclear decay reactions. Each radioactive nuclide has a characteristic, constant **half-life** ($t_{1/2}$), the time required for half of the atoms in a sample to decay. An isotope's half-life allows us to determine how long a sample of a useful isotope will be available, and how long a sample of an undesirable or dangerous isotope must be stored before it decays to a low-enough radiation level that is no longer a problem.

For example, cobalt-60, an isotope that emits gamma rays used to treat cancer, has a half-life of 5.27 years ([Figure 21.10](#)). In a given cobalt-60 source, since half of the $^{60}_{27}\text{Co}$ nuclei decay every 5.27 years, both the amount of material and the intensity of the radiation emitted is cut in half every 5.27 years. (Note that for a given substance, the intensity of radiation that it produces is directly proportional to the rate of decay of the substance and the amount of the substance.) This is as expected for a process following first-order kinetics. Thus, a cobalt-60 source that is used for cancer treatment must be replaced regularly to continue to be effective.

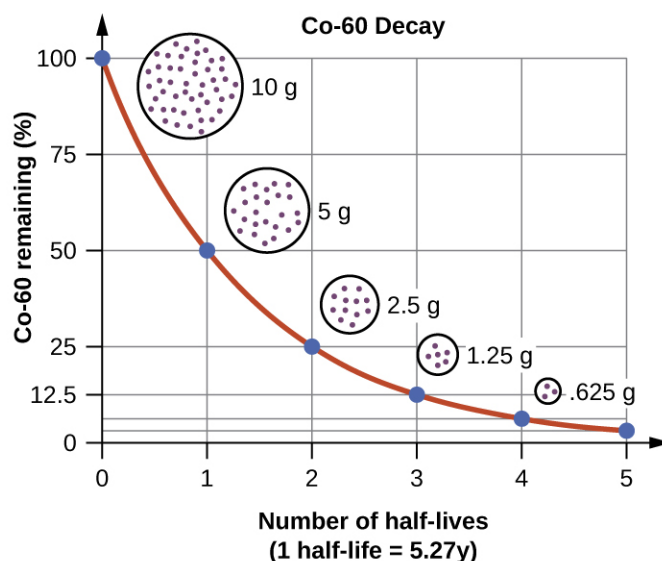


FIGURE 21.10 For cobalt-60, which has a half-life of 5.27 years, 50% remains after 5.27 years (one half-life), 25% remains after 10.54 years (two half-lives), 12.5% remains after 15.81 years (three half-lives), and so on.

Since nuclear decay follows first-order kinetics, we can adapt the mathematical relationships used for first-order chemical reactions. We generally substitute the number of nuclei, N , for the concentration. If the rate is stated in nuclear decays per second, we refer to it as the activity of the radioactive sample. The rate for radioactive decay is:

decay rate = λN with λ = the decay constant for the particular radioisotope

The decay constant, λ , which is the same as a rate constant discussed in the kinetics chapter. It is possible to express the decay constant in terms of the half-life, $t_{1/2}$:

$$\lambda = \frac{\ln 2}{t_{1/2}} = \frac{0.693}{t_{1/2}} \quad \text{or} \quad t_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

The first-order equations relating amount, N , and time are:

$$N_t = N_0 e^{-\lambda t} \quad \text{or} \quad t = -\frac{1}{\lambda} \ln \left(\frac{N_t}{N_0} \right)$$

where N_0 is the initial number of nuclei or moles of the isotope, and N_t is the number of nuclei/moles remaining at time t . [Example 21.5](#) applies these calculations to find the rates of radioactive decay for specific nuclides.



EXAMPLE 21.5

Rates of Radioactive Decay

$^{60}_{27}\text{Co}$ decays with a half-life of 5.27 years to produce $^{60}_{28}\text{Ni}$.

- What is the decay constant for the radioactive disintegration of cobalt-60?
- Calculate the fraction of a sample of the $^{60}_{27}\text{Co}$ isotope that will remain after 15 years.
- How long does it take for a sample of $^{60}_{27}\text{Co}$ to disintegrate to the extent that only 2.0% of the original amount remains?

Solution

(a) The value of the rate constant is given by:

$$\lambda = \frac{\ln 2}{t_{1/2}} = \frac{0.693}{5.27 \text{ y}} = 0.132 \text{ y}^{-1}$$

(b) The fraction of $^{60}_{27}\text{Co}$ that is left after time t is given by $\frac{N_t}{N_0}$. Rearranging the first-order relationship $N_t = N_0 e^{-\lambda t}$ to solve for this ratio yields:

$$\frac{N_t}{N_0} = e^{-\lambda t} = e^{-(0.132/\text{y})(15\text{y})} = 0.138$$

The fraction of $^{60}_{27}\text{Co}$ that will remain after 15.0 years is 0.138. Or put another way, 13.8% of the $^{60}_{27}\text{Co}$ originally present will remain after 15 years.

(c) 2.00% of the original amount of $^{60}_{27}\text{Co}$ is equal to $0.0200 \times N_0$. Substituting this into the equation for time for first-order kinetics, we have:

$$t = -\frac{1}{\lambda} \ln \left(\frac{N_t}{N_0} \right) = -\frac{1}{0.132 \text{ y}^{-1}} \ln \left(\frac{0.0200 \times N_0}{N_0} \right) = 29.6 \text{ y}$$

Check Your Learning

Radon-222, $^{222}_{86}\text{Rn}$, has a half-life of 3.823 days. How long will it take a sample of radon-222 with a mass of 0.750 g to decay into other elements, leaving only 0.100 g of radon-222?

Answer:

11.1 days

Because each nuclide has a specific number of nucleons, a particular balance of repulsion and attraction, and its own degree of stability, the half-lives of radioactive nuclides vary widely. For example: the half-life of $^{209}_{83}\text{Bi}$ is 1.9×10^{19} years; $^{239}_{94}\text{Ra}$ is 24,000 years; $^{222}_{86}\text{Rn}$ is 3.82 days; and element-111 (Rg for roentgenium) is 1.5×10^{-3} seconds. The half-lives of a number of radioactive isotopes important to medicine are shown in [Table 21.2](#), and others are listed in [Appendix M](#).

Half-lives of Radioactive Isotopes Important to Medicine

Type ¹	Decay Mode	Half-Life	Uses
F-18	β^+ decay	110. minutes	PET scans
Co-60	β decay, γ decay	5.27 years	cancer treatment
Tc-99m	γ decay	8.01 hours	scans of brain, lung, heart, bone
I-131	β decay	8.02 days	thyroid scans and treatment
Tl-201	electron capture	73 hours	heart and arteries scans; cardiac stress tests

TABLE 21.2

Radiometric Dating

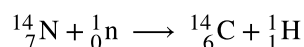
Several radioisotopes have half-lives and other properties that make them useful for purposes of “dating” the origin of objects such as archaeological artifacts, formerly living organisms, or geological formations. This process is **radiometric dating** and has been responsible for many breakthrough scientific discoveries about the geological history of the earth, the evolution of life, and the history of human civilization. We will explore some of the most common types of radioactive dating and how the particular isotopes work for each type.

1 The “m” in Tc-99m stands for “metastable,” indicating that this is an unstable, high-energy state of Tc-99. Metastable isotopes emit γ radiation to rid themselves of excess energy and become (more) stable.

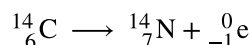
Radioactive Dating Using Carbon-14

The radioactivity of carbon-14 provides a method for dating objects that were a part of a living organism. This method of radiometric dating, which is also called **radiocarbon dating** or carbon-14 dating, is accurate for dating carbon-containing substances that are up to about 30,000 years old, and can provide reasonably accurate dates up to a maximum of about 50,000 years old.

Naturally occurring carbon consists of three isotopes: $^{12}_6\text{C}$, which constitutes about 99% of the carbon on earth; $^{13}_6\text{C}$, about 1% of the total; and trace amounts of $^{14}_6\text{C}$. Carbon-14 forms in the upper atmosphere by the reaction of nitrogen atoms with neutrons from cosmic rays in space:



All isotopes of carbon react with oxygen to produce CO_2 molecules. The ratio of $^{14}_6\text{CO}_2$ to $^{12}_6\text{CO}_2$ depends on the ratio of $^{14}_6\text{C}$ to $^{12}_6\text{C}$ in the atmosphere. The natural abundance of $^{14}_6\text{C}$ in the atmosphere is approximately 1 part per trillion; until recently, this has generally been constant over time, as seen in gas samples found trapped in ice. The incorporation of $^{14}_6\text{C}$ and $^{12}_6\text{C}$ into plants is a regular part of the photosynthesis process, which means that the $^{14}_6\text{C}$: $^{12}_6\text{C}$ ratio found in a living plant is the same as the $^{14}_6\text{C}$: $^{12}_6\text{C}$ ratio in the atmosphere. But when the plant dies, it no longer traps carbon through photosynthesis. Because $^{12}_6\text{C}$ is a stable isotope and does not undergo radioactive decay, its concentration in the plant does not change. However, carbon-14 decays by β emission with a half-life of 5730 years:



Thus, the $^{14}_6\text{C}$: $^{12}_6\text{C}$ ratio gradually decreases after the plant dies. The decrease in the ratio with time provides a measure of the time that has elapsed since the death of the plant (or other organism that ate the plant). [Figure 21.11](#) visually depicts this process.

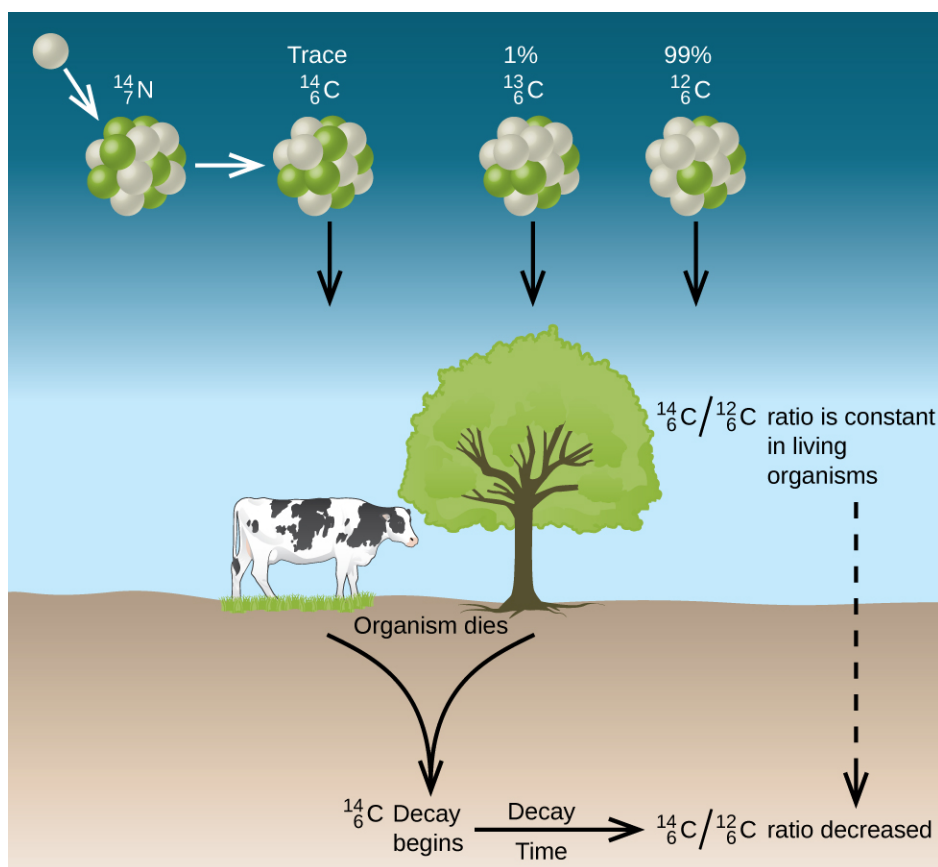


FIGURE 21.11 Along with stable carbon-12, radioactive carbon-14 is taken in by plants and animals, and remains at a constant level within them while they are alive. After death, the C-14 decays and the C-14:C-12 ratio in the remains decreases. Comparing this ratio to the

C-14:C-12 ratio in living organisms allows us to determine how long ago the organism lived (and died).

For example, with the half-life of $^{14}_6\text{C}$ being 5730 years, if the $^{14}_6\text{C}$: $^{12}_6\text{C}$ ratio in a wooden object found in an archaeological dig is half what it is in a living tree, this indicates that the wooden object is 5730 years old. Highly accurate determinations of $^{14}_6\text{C}$: $^{12}_6\text{C}$ ratios can be obtained from very small samples (as little as a milligram) by the use of a mass spectrometer.

LINK TO LEARNING

Visit this [website \(http://openstax.org/l/16phetradiom\)](http://openstax.org/l/16phetradiom) to perform simulations of radiometric dating.



EXAMPLE 21.6

Radiocarbon Dating

A tiny piece of paper (produced from formerly living plant matter) taken from the Dead Sea Scrolls has an activity of 10.8 disintegrations per minute per gram of carbon. If the initial C-14 activity was 13.6 disintegrations/min/g of C, estimate the age of the Dead Sea Scrolls.

Solution

The rate of decay (number of disintegrations/minute/gram of carbon) is proportional to the amount of radioactive C-14 left in the paper, so we can substitute the rates for the amounts, N , in the relationship:

$$t = -\frac{1}{\lambda} \ln \left(\frac{N_t}{N_0} \right) \rightarrow t = -\frac{1}{\lambda} \ln \left(\frac{\text{Rate}_t}{\text{Rate}_0} \right)$$

where the subscript 0 represents the time when the plants were cut to make the paper, and the subscript t represents the current time.

The decay constant can be determined from the half-life of C-14, 5730 years:

$$\lambda = \frac{\ln 2}{t_{1/2}} = \frac{0.693}{5730 \text{ y}} = 1.21 \times 10^{-4} \text{ y}^{-1}$$

Substituting and solving, we have:

$$t = -\frac{1}{\lambda} \ln \left(\frac{\text{Rate}_t}{\text{Rate}_0} \right) = -\frac{1}{1.21 \times 10^{-4} \text{ y}^{-1}} \ln \left(\frac{10.8 \text{ dis/min/g C}}{13.6 \text{ dis/min/g C}} \right) = 1910 \text{ y}$$

Therefore, the Dead Sea Scrolls are approximately 1900 years old (Figure 21.12).



FIGURE 21.12 Carbon-14 dating has shown that these pages from the Dead Sea Scrolls were written or copied on paper made from plants that died between 100 BC and AD 50.

Check Your Learning

More accurate dates of the reigns of ancient Egyptian pharaohs have been determined recently using plants that were preserved in their tombs. Samples of seeds and plant matter from King Tutankhamun's tomb have a C-14 decay rate of 9.07 disintegrations/min/g of C. How long ago did King Tut's reign come to an end?

Answer:

about 3350 years ago, or approximately 1340 BC

There have been some significant, well-documented changes to the $^{14}_6\text{C} : ^{12}_6\text{C}$ ratio. The accuracy of a straightforward application of this technique depends on the $^{14}_6\text{C} : ^{12}_6\text{C}$ ratio in a living plant being the same now as it was in an earlier era, but this is not always valid. Due to the increasing accumulation of CO_2 molecules (largely $^{12}_6\text{CO}_2$) in the atmosphere caused by combustion of fossil fuels (in which essentially all of the $^{14}_6\text{C}$ has decayed), the ratio of $^{14}_6\text{C} : ^{12}_6\text{C}$ in the atmosphere may be changing. This manmade increase in $^{12}_6\text{CO}_2$ in the atmosphere causes the $^{14}_6\text{C} : ^{12}_6\text{C}$ ratio to decrease, and this in turn affects the ratio in currently living organisms on the earth. Fortunately, however, we can use other data, such as tree dating via examination of annual growth rings, to calculate correction factors. With these correction factors, accurate dates can be determined. In general, radioactive dating only works for about 10 half-lives; therefore, the limit for carbon-14 dating is about 57,000 years.

Radioactive Dating Using Nuclides Other than Carbon-14

Radioactive dating can also use other radioactive nuclides with longer half-lives to date older events. For example, uranium-238 (which decays in a series of steps into lead-206) can be used for establishing the age of rocks (and the approximate age of the oldest rocks on earth). Since U-238 has a half-life of 4.5 billion years, it takes that amount of time for half of the original U-238 to decay into Pb-206. In a sample of rock that does not contain appreciable amounts of Pb-208, the most abundant isotope of lead, we can assume that lead was not present when the rock was formed. Therefore, by measuring and analyzing the ratio of U-238:Pb-206, we can determine the age of the rock. This assumes that all of the lead-206 present came from the decay of uranium-238. If there is additional lead-206 present, which is indicated by the presence of other lead isotopes in the sample, it is necessary to make an adjustment. Potassium-argon dating uses a similar method. K-40 decays by positron emission and electron capture to form Ar-40 with a half-life of 1.25 billion years. If a rock sample is crushed and the amount of Ar-40 gas that escapes is measured, determination of the Ar-40:K-40 ratio yields the age of the rock. Other methods, such as rubidium-strontium dating (Rb-87 decays into Sr-87 with a half-life of 48.8 billion years), operate on the same principle. To estimate the lower limit for the earth's age, scientists determine the age of various rocks and minerals, making the assumption that the earth is older than the oldest rocks and minerals in its crust. As of 2014, the oldest known rocks on earth are the Jack Hills zircons from Australia, found by uranium-lead dating to be almost 4.4 billion years old.



EXAMPLE 21.7

Radioactive Dating of Rocks

An igneous rock contains 9.58×10^{-5} g of U-238 and 2.51×10^{-5} g of Pb-206, and much, much smaller amounts of Pb-208. Determine the approximate time at which the rock formed.

Solution

The sample of rock contains very little Pb-208, the most common isotope of lead, so we can safely assume that all the Pb-206 in the rock was produced by the radioactive decay of U-238. When the rock formed, it contained all of the U-238 currently in it, plus some U-238 that has since undergone radioactive decay.

The amount of U-238 currently in the rock is:

$$9.58 \times 10^{-5} \text{ g U} \times \left(\frac{1 \text{ mol U}}{238 \text{ g U}} \right) = 4.03 \times 10^{-7} \text{ mol U}$$

Because when one mole of U-238 decays, it produces one mole of Pb-206, the amount of U-238 that has undergone radioactive decay since the rock was formed is:

$$2.51 \times 10^{-5} \text{ g Pb} \times \left(\frac{1 \text{ mol Pb}}{206 \text{ g Pb}} \right) \times \left(\frac{1 \text{ mol U}}{1 \text{ mol Pb}} \right) = 1.22 \times 10^{-7} \text{ mol U}$$

The total amount of U-238 originally present in the rock is therefore:

$$4.03 \times 10^{-7} \text{ mol} + 1.22 \times 10^{-7} \text{ mol} = 5.25 \times 10^{-7} \text{ mol U}$$

The amount of time that has passed since the formation of the rock is given by:

$$t = -\frac{1}{\lambda} \ln \left(\frac{N_t}{N_0} \right)$$

with N_0 representing the original amount of U-238 and N_t representing the present amount of U-238.

U-238 decays into Pb-206 with a half-life of 4.5×10^9 y, so the decay constant λ is:

$$\lambda = \frac{\ln 2}{t_{1/2}} = \frac{0.693}{4.5 \times 10^9 \text{ y}} = 1.54 \times 10^{-10} \text{ y}^{-1}$$

Substituting and solving, we have:

$$t = -\frac{1}{1.54 \times 10^{-10} \text{ y}^{-1}} \ln \left(\frac{4.03 \times 10^{-7} \text{ mol U}}{5.25 \times 10^{-7} \text{ mol U}} \right) = 1.7 \times 10^9 \text{ y}$$

Therefore, the rock is approximately 1.7 billion years old.

Check Your Learning

A sample of rock contains 6.14×10^{-4} g of Rb-87 and 3.51×10^{-5} g of Sr-87. Calculate the age of the rock. (The half-life of the β decay of Rb-87 is 4.7×10^{10} y.)

Answer:

$$3.7 \times 10^9 \text{ y}$$

21.4 Transmutation and Nuclear Energy

LEARNING OBJECTIVES

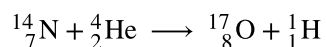
By the end of this section, you will be able to:

- Describe the synthesis of transuranium nuclides
- Explain nuclear fission and fusion processes
- Relate the concepts of critical mass and nuclear chain reactions
- Summarize basic requirements for nuclear fission and fusion reactors

After the discovery of radioactivity, the field of nuclear chemistry was created and developed rapidly during the early twentieth century. A slew of new discoveries in the 1930s and 1940s, along with World War II, combined to usher in the Nuclear Age in the mid-twentieth century. Scientists learned how to create new substances, and certain isotopes of certain elements were found to possess the capacity to produce unprecedented amounts of energy, with the potential to cause tremendous damage during war, as well as produce enormous amounts of power for society's needs during peace.

Synthesis of Nuclides

Nuclear transmutation is the conversion of one nuclide into another. It can occur by the radioactive decay of a nucleus, or the reaction of a nucleus with another particle. The first manmade nucleus was produced in Ernest Rutherford's laboratory in 1919 by a **transmutation** reaction, the bombardment of one type of nuclei with other nuclei or with neutrons. Rutherford bombarded nitrogen atoms with high-speed α particles from a natural radioactive isotope of radium and observed protons resulting from the reaction:



The ${}^{17}_8\text{O}$ and ${}^1_1\text{H}$ nuclei that are produced are stable, so no further (nuclear) changes occur.

To reach the kinetic energies necessary to produce transmutation reactions, devices called **particle accelerators** are used. These devices use magnetic and electric fields to increase the speeds of nuclear particles. In all accelerators, the particles move in a vacuum to avoid collisions with gas molecules. When neutrons are required for transmutation reactions, they are usually obtained from radioactive decay reactions or from various nuclear reactions occurring in nuclear reactors. The Chemistry in Everyday Life feature that follows discusses a famous particle accelerator that made worldwide news.

CHEMISTRY IN EVERYDAY LIFE

CERN Particle Accelerator

Located near Geneva, the CERN (“Conseil Européen pour la Recherche Nucléaire,” or European Council for Nuclear Research) Laboratory is the world’s premier center for the investigations of the fundamental particles that make up matter. It contains the 27-kilometer (17 mile) long, circular Large Hadron Collider (LHC), the largest particle accelerator in the world (Figure 21.13). In the LHC, particles are boosted to high energies and are then made to collide with each other or with stationary targets at nearly the speed of light. Superconducting electromagnets are used to produce a strong magnetic field that guides the particles around the ring. Specialized, purpose-built detectors observe and record the results of these collisions, which are then analyzed by CERN scientists using powerful computers.



FIGURE 21.13 A small section of the LHC is shown with workers traveling along it. (credit: Christophe Delaere)

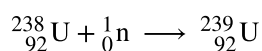
In 2012, CERN announced that experiments at the LHC showed the first observations of the Higgs boson, an elementary particle that helps explain the origin of mass in fundamental particles. This long-anticipated discovery made worldwide news and resulted in the awarding of the 2013 Nobel Prize in Physics to François Englert and Peter Higgs, who had predicted the existence of this particle almost 50 years previously.

LINK TO LEARNING

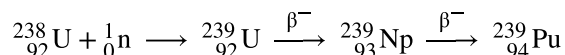
Famous physicist Brian Cox talks about his work on the Large Hadron Collider at CERN, providing an entertaining and engaging [tour \(http://openstax.org/l/16tedCERN\)](http://openstax.org/l/16tedCERN) of this massive project and the physics behind it.

View a short [video \(http://openstax.org/l/16CERNvideo\)](http://openstax.org/l/16CERNvideo) from CERN, describing the basics of how its particle accelerators work.

Prior to 1940, the heaviest-known element was uranium, whose atomic number is 92. Now, many artificial elements have been synthesized and isolated, including several on such a large scale that they have had a profound effect on society. One of these—element 93, neptunium (Np)—was first made in 1940 by McMillan and Abelson by bombarding uranium-238 with neutrons. The reaction creates unstable uranium-239, with a half-life of 23.5 minutes, which then decays into neptunium-239. Neptunium-239 is also radioactive, with a half-life of 2.36 days, and it decays into plutonium-239. The nuclear reactions are:



Plutonium is now mostly formed in nuclear reactors as a byproduct during the fission of U-235. Additional neutrons are released during this fission process (see the next section), some of which combine with U-238 nuclei to form uranium-239; this undergoes β decay to form neptunium-239, which in turn undergoes β decay to form plutonium-239 as illustrated in the preceding three equations. These processes are summarized in the equation:



Heavier isotopes of plutonium—Pu-240, Pu-241, and Pu-242—are also produced when lighter plutonium nuclei capture neutrons. Some of this highly radioactive plutonium is used to produce military weapons, and the rest presents a serious storage problem because they have half-lives from thousands to hundreds of thousands of years.

Although they have not been prepared in the same quantity as plutonium, many other synthetic nuclei have been produced. Nuclear medicine has developed from the ability to convert atoms of one type into other types of atoms. Radioactive isotopes of several dozen elements are currently used for medical applications. The radiation produced by their decay is used to image or treat various organs or portions of the body, among other uses.

The elements beyond element 92 (uranium) are called **transuranium elements**. As of this writing, 22 transuranium elements have been produced and officially recognized by IUPAC; several other elements have formation claims that are waiting for approval. Some of these elements are shown in [Table 21.3](#).

Preparation of Some of the Transuranium Elements

Name	Symbol	Atomic Number	Reaction
americium	Am	95	${}_{94}^{239}\text{Pu} + {}_0^1\text{n} \longrightarrow {}_{95}^{240}\text{Am} + {}_{-1}^0\text{e}$
curium	Cm	96	${}_{94}^{239}\text{Pu} + {}_2^4\text{He} \longrightarrow {}_{96}^{242}\text{Cm} + {}_0^1\text{n}$
californium	Cf	98	${}_{96}^{242}\text{Cm} + {}_2^4\text{He} \longrightarrow {}_{98}^{245}\text{Cf} + {}_0^1\text{n}$
einsteinium	Es	99	${}_{92}^{238}\text{U} + 15{}_0^1\text{n} \longrightarrow {}_{99}^{253}\text{Es} + 7{}_{-1}^0\text{e}$
mendelevium	Md	101	${}_{99}^{253}\text{Es} + {}_2^4\text{He} \longrightarrow {}_{101}^{256}\text{Md} + {}_0^1\text{n}$
nobelium	No	102	${}_{96}^{246}\text{Cm} + {}_6^{12}\text{C} \longrightarrow {}_{102}^{254}\text{No} + 4{}_0^1\text{n}$
rutherfordium	Rf	104	${}_{98}^{249}\text{Cf} + {}_6^{12}\text{C} \longrightarrow {}_{104}^{257}\text{Rf} + 4{}_0^1\text{n}$
seaborgium	Sg	106	${}_{82}^{206}\text{Pb} + {}_{24}^{54}\text{Cr} \longrightarrow {}_{106}^{257}\text{Sg} + 3{}_0^1\text{n}$ ${}_{98}^{249}\text{Cf} + {}_8^{18}\text{O} \longrightarrow {}_{106}^{263}\text{Sg} + 4{}_0^1\text{n}$
meitnerium	Mt	107	${}_{83}^{209}\text{Bi} + {}_{26}^{58}\text{Fe} \longrightarrow {}_{109}^{266}\text{Mt} + {}_0^1\text{n}$

TABLE 21.3

Nuclear Fission

Many heavier elements with smaller binding energies per nucleon can decompose into more stable elements that have intermediate mass numbers and larger binding energies per nucleon—that is, mass numbers and binding energies per nucleon that are closer to the “peak” of the binding energy graph near 56 (see [Figure 21.3](#)). Sometimes neutrons are also produced. This decomposition is called **fission**, the breaking of a large nucleus into smaller pieces. The breaking is rather random with the formation of a large number of different products. Fission usually does not occur naturally, but is induced by bombardment with neutrons. The first reported nuclear fission occurred in 1939 when three German scientists, Lise Meitner, Otto Hahn, and Fritz Strassman, bombarded uranium-235 atoms with slow-moving neutrons that split the U-238 nuclei into smaller fragments that consisted of several neutrons and elements near the middle of the periodic table. Since then, fission has been observed in many other isotopes, including most actinide isotopes that have an odd number of neutrons. A typical nuclear fission reaction is shown in [Figure 21.14](#).

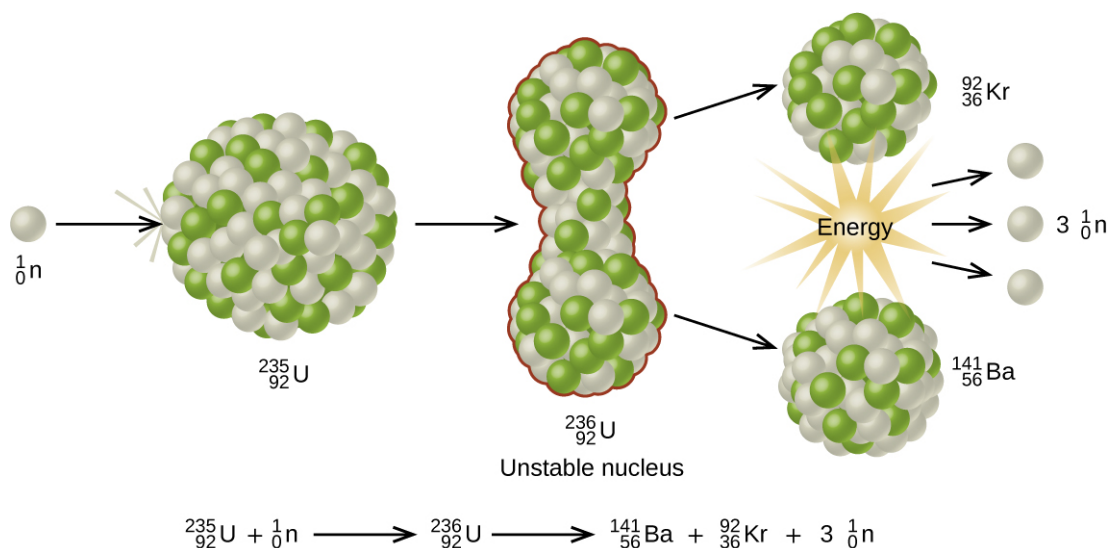
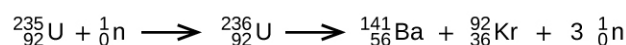
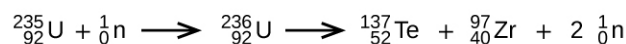
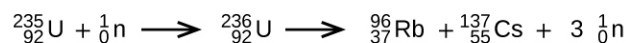
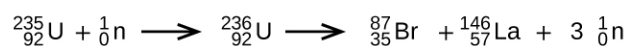
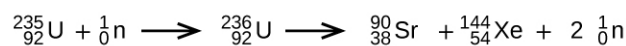
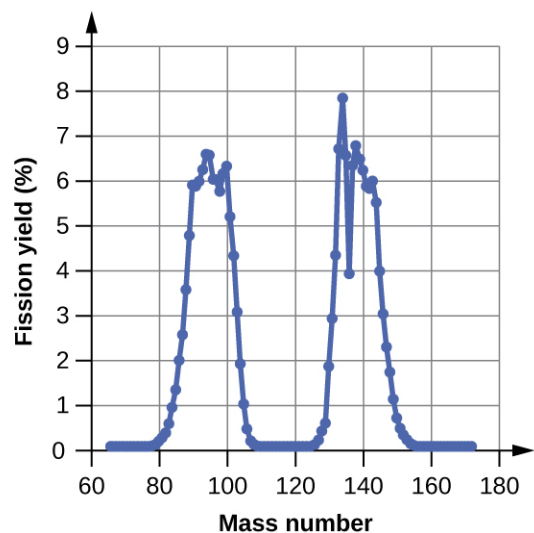


FIGURE 21.14 When a slow neutron hits a fissionable U-235 nucleus, it is absorbed and forms an unstable U-236 nucleus. The U-236 nucleus then rapidly breaks apart into two smaller nuclei (in this case, Ba-141 and Kr-92) along with several neutrons (usually two or three), and releases a very large amount of energy.

Among the products of Meitner, Hahn, and Strassman’s fission reaction were barium, krypton, lanthanum, and cerium, all of which have nuclei that are more stable than uranium-235. Since then, hundreds of different isotopes have been observed among the products of fissionable substances. A few of the many reactions that occur for U-235, and a graph showing the distribution of its fission products and their yields, are shown in [Figure 21.15](#). Similar fission reactions have been observed with other uranium isotopes, as well as with a variety of other isotopes such as those of plutonium.



(a)



(b)

FIGURE 21.15 (a) Nuclear fission of U-235 produces a range of fission products. (b) The larger fission products of U-235 are typically one isotope with a mass number around 85–105, and another isotope with a mass number that is about 50% larger, that is, about 130–150.

LINK TO LEARNING

View this [link \(http://openstax.org/l/16fission\)](http://openstax.org/l/16fission) to see a simulation of nuclear fission.

A tremendous amount of energy is produced by the fission of heavy elements. For instance, when one mole of U-235 undergoes fission, the products weigh about 0.2 grams less than the reactants; this “lost” mass is converted into a very large amount of energy, about 1.8×10^{10} kJ per mole of U-235. Nuclear fission reactions produce incredibly large amounts of energy compared to chemical reactions. The fission of 1 kilogram of uranium-235, for example, produces about 2.5 million times as much energy as is produced by burning 1 kilogram of coal.

As described earlier, when undergoing fission U-235 produces two “medium-sized” nuclei, and two or three neutrons. These neutrons may then cause the fission of other uranium-235 atoms, which in turn provide more neutrons that can cause fission of even more nuclei, and so on. If this occurs, we have a nuclear **chain reaction** (see [Figure 21.16](#)). On the other hand, if too many neutrons escape the bulk material without interacting with a nucleus, then no chain reaction will occur.

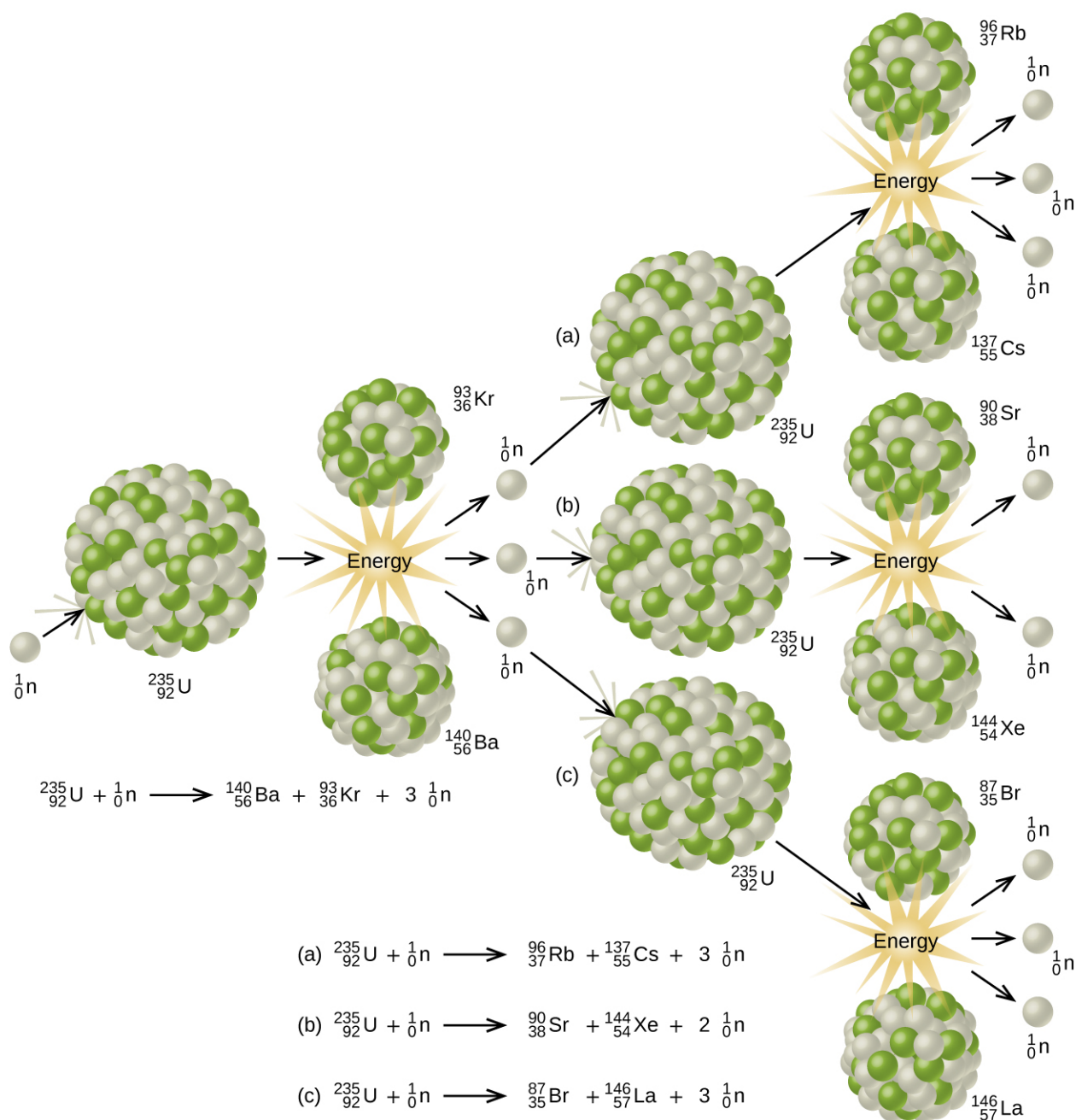


FIGURE 21.16 The fission of a large nucleus, such as U-235, produces two or three neutrons, each of which is capable of causing fission of another nucleus by the reactions shown. If this process continues, a nuclear chain reaction occurs.

Material that can sustain a nuclear fission chain reaction is said to be **fissile** or **fissionable**. (Technically, fissile material can undergo fission with neutrons of any energy, whereas fissionable material requires high-energy neutrons.) Nuclear fission becomes self-sustaining when the number of neutrons produced by fission equals or exceeds the number of neutrons absorbed by splitting nuclei plus the number that escape into the surroundings. The amount of a fissionable material that will support a self-sustaining chain reaction is a **critical mass**. An amount of fissionable material that cannot sustain a chain reaction is a **subcritical mass**. An amount of material in which there is an increasing rate of fission is known as a **supercritical mass**. The critical mass depends on the type of material: its purity, the temperature, the shape of the sample, and how the neutron reactions are controlled ([Figure 21.17](#)).

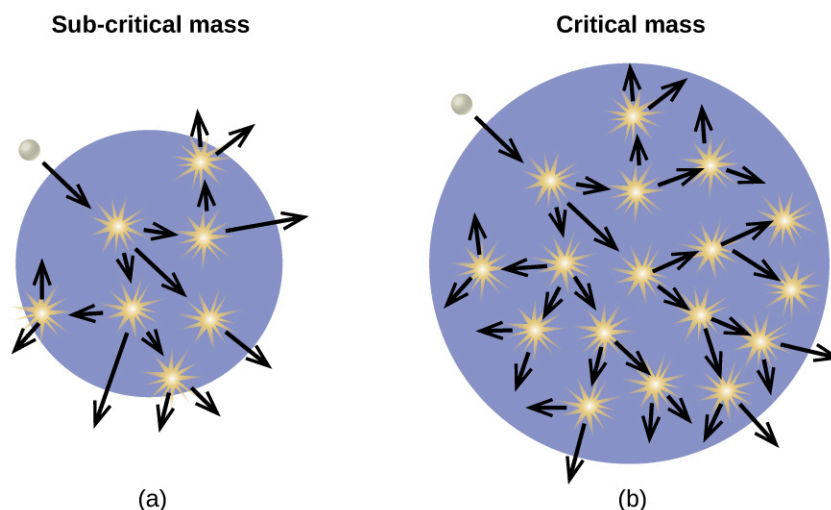
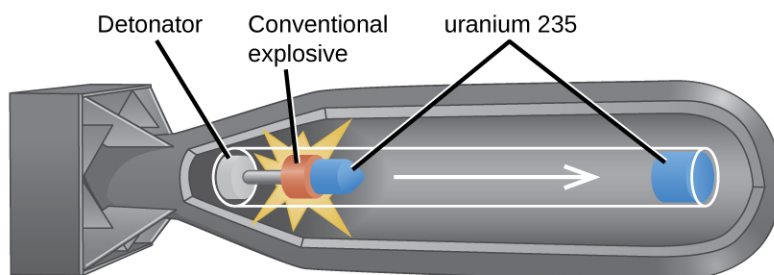
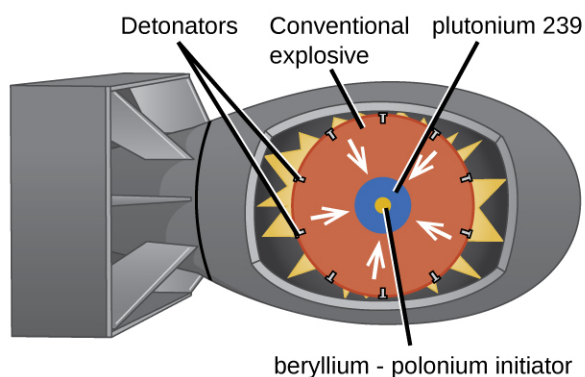
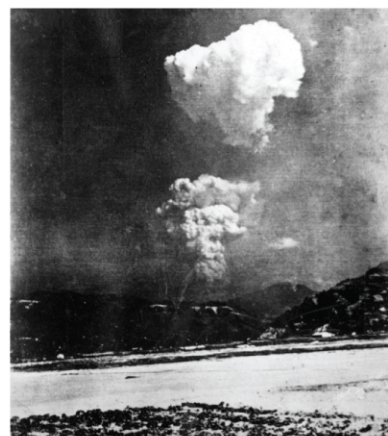


FIGURE 21.17 (a) In a subcritical mass, the fissile material is too small and allows too many neutrons to escape the material, so a chain reaction does not occur. (b) In a critical mass, a large enough number of neutrons in the fissile material induce fission to create a chain reaction.

An atomic bomb (Figure 21.18) contains several pounds of fissionable material, $^{235}_{92}\text{U}$ or $^{239}_{94}\text{Pu}$, a source of neutrons, and an explosive device for compressing it quickly into a small volume. When fissionable material is in small pieces, the proportion of neutrons that escape through the relatively large surface area is great, and a chain reaction does not take place. When the small pieces of fissionable material are brought together quickly to form a body with a mass larger than the critical mass, the relative number of escaping neutrons decreases, and a chain reaction and explosion result.



(a)



(b)



FIGURE 21.18 (a) The nuclear fission bomb that destroyed Hiroshima on August 6, 1945, consisted of two subcritical masses of U-235,

where conventional explosives were used to fire one of the subcritical masses into the other, creating the critical mass for the nuclear explosion. (b) The plutonium bomb that destroyed Nagasaki on August 9, 1945, consisted of a hollow sphere of plutonium that was rapidly compressed by conventional explosives. This led to a concentration of plutonium in the center that was greater than the critical mass necessary for the nuclear explosion.

Fission Reactors

Chain reactions of fissionable materials can be controlled and sustained without an explosion in a **nuclear reactor** (Figure 21.19). Any nuclear reactor that produces power via the fission of uranium or plutonium by bombardment with neutrons must have at least five components: nuclear fuel consisting of fissionable material, a nuclear moderator, reactor coolant, control rods, and a shield and containment system. We will discuss these components in greater detail later in the section. The reactor works by separating the fissionable nuclear material such that a critical mass cannot be formed, controlling both the flux and absorption of neutrons to allow shutting down the fission reactions. In a nuclear reactor used for the production of electricity, the energy released by fission reactions is trapped as thermal energy and used to boil water and produce steam. The steam is used to turn a turbine, which powers a generator for the production of electricity.

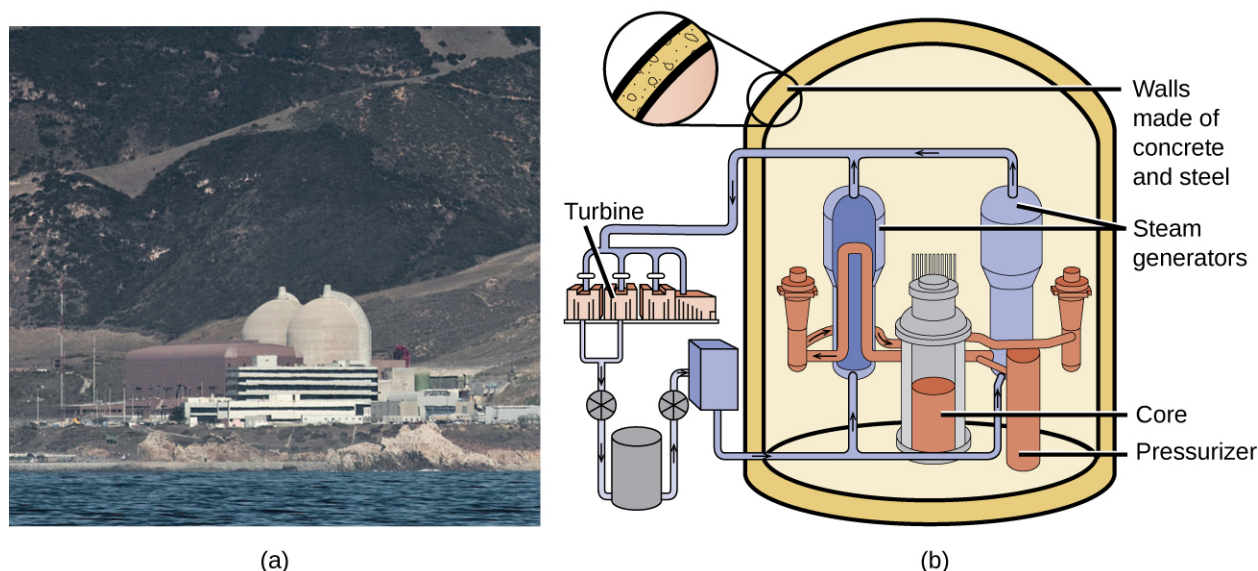


FIGURE 21.19 (a) The Diablo Canyon Nuclear Power Plant near San Luis Obispo is the only nuclear power plant currently in operation in California. The domes are the containment structures for the nuclear reactors, and the brown building houses the turbine where electricity is generated. Ocean water is used for cooling. (b) The Diablo Canyon uses a pressurized water reactor, one of a few different fission reactor designs in use around the world, to produce electricity. Energy from the nuclear fission reactions in the core heats water in a closed, pressurized system. Heat from this system produces steam that drives a turbine, which in turn produces electricity. (credit a: modification of work by "Mike" Michael L. Baird; credit b: modification of work by the Nuclear Regulatory Commission)

Nuclear Fuels

Nuclear fuel consists of a fissionable isotope, such as uranium-235, which must be present in sufficient quantity to provide a self-sustaining chain reaction. In the United States, uranium ores contain from 0.05–0.3% of the uranium oxide U_3O_8 ; the uranium in the ore is about 99.3% nonfissionable U-238 with only 0.7% fissionable U-235. Nuclear reactors require a fuel with a higher concentration of U-235 than is found in nature; it is normally enriched to have about 5% of uranium mass as U-235. At this concentration, it is not possible to achieve the supercritical mass necessary for a nuclear explosion. Uranium can be enriched by gaseous diffusion (the only method currently used in the US), using a gas centrifuge, or by laser separation.

In the gaseous diffusion enrichment plant where U-235 fuel is prepared, UF_6 (uranium hexafluoride) gas at low pressure moves through barriers that have holes just barely large enough for UF_6 to pass through. The slightly lighter $^{235}\text{UF}_6$ molecules diffuse through the barrier slightly faster than the heavier $^{238}\text{UF}_6$ molecules. This process is repeated through hundreds of barriers, gradually increasing the concentration of $^{235}\text{UF}_6$ to the level needed by the nuclear reactor. The basis for this process, Graham's law, is described in the chapter on gases. The enriched UF_6 gas is collected, cooled until it solidifies, and then taken to a fabrication facility where it is made into fuel assemblies. Each fuel assembly consists of fuel rods that contain many thimble-sized, ceramic-encased, enriched uranium (usually UO_2) fuel pellets. Modern nuclear reactors may contain as many as 10 million fuel pellets. The

amount of energy in each of these pellets is equal to that in almost a ton of coal or 150 gallons of oil.

Nuclear Moderators

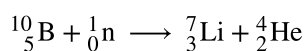
Neutrons produced by nuclear reactions move too fast to cause fission (refer back to [Figure 21.17](#)). They must first be slowed to be absorbed by the fuel and produce additional nuclear reactions. A **nuclear moderator** is a substance that slows the neutrons to a speed that is low enough to cause fission. Early reactors used high-purity graphite as a moderator. Modern reactors in the US exclusively use heavy water (${}^2_1\text{H}_2\text{O}$) or light water (ordinary H_2O), whereas some reactors in other countries use other materials, such as carbon dioxide, beryllium, or graphite.

Reactor Coolants

A nuclear **reactor coolant** is used to carry the heat produced by the fission reaction to an external boiler and turbine, where it is transformed into electricity. Two overlapping coolant loops are often used; this counteracts the transfer of radioactivity from the reactor to the primary coolant loop. All nuclear power plants in the US use water as a coolant. Other coolants include molten sodium, lead, a lead-bismuth mixture, or molten salts.

Control Rods

Nuclear reactors use **control rods** ([Figure 21.20](#)) to control the fission rate of the nuclear fuel by adjusting the number of slow neutrons present to keep the rate of the chain reaction at a safe level. Control rods are made of boron, cadmium, hafnium, or other elements that are able to absorb neutrons. Boron-10, for example, absorbs neutrons by a reaction that produces lithium-7 and alpha particles:



When control rod assemblies are inserted into the fuel element in the reactor core, they absorb a larger fraction of the slow neutrons, thereby slowing the rate of the fission reaction and decreasing the power produced. Conversely, if the control rods are removed, fewer neutrons are absorbed, and the fission rate and energy production increase. In an emergency, the chain reaction can be shut down by fully inserting all of the control rods into the nuclear core between the fuel rods.

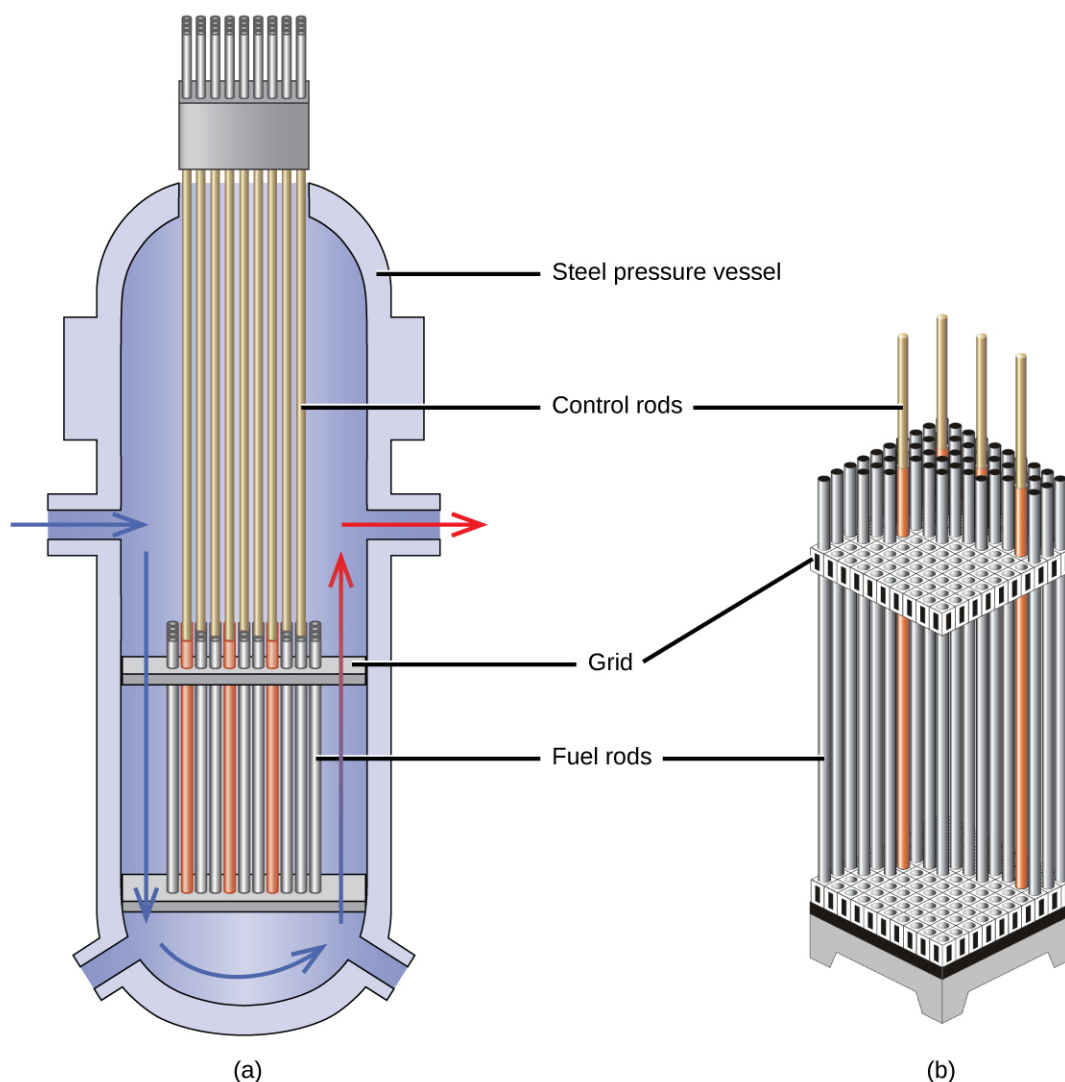


FIGURE 21.20 The nuclear reactor core shown in (a) contains the fuel and control rod assembly shown in (b). (credit: modification of work by E. Generalic, <http://glossary.periodni.com/glossary.php?en=control+rod>)

Shield and Containment System

During its operation, a nuclear reactor produces neutrons and other radiation. Even when shut down, the decay products are radioactive. In addition, an operating reactor is thermally very hot, and high pressures result from the circulation of water or another coolant through it. Thus, a reactor must withstand high temperatures and pressures, and must protect operating personnel from the radiation. Reactors are equipped with a **containment system** (or shield) that consists of three parts:

1. The reactor vessel, a steel shell that is 3–20-centimeters thick and, with the moderator, absorbs much of the radiation produced by the reactor
2. A main shield of 1–3 meters of high-density concrete
3. A personnel shield of lighter materials that protects operators from γ rays and X-rays

In addition, reactors are often covered with a steel or concrete dome that is designed to contain any radioactive materials might be released by a reactor accident.

LINK TO LEARNING

Click here to watch a [3-minute video](http://openstax.org/l/16nuclearactors) (<http://openstax.org/l/16nuclearactors>) from the Nuclear Energy Institute on

how nuclear reactors work.

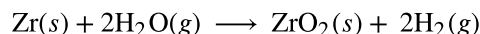
Nuclear power plants are designed in such a way that they cannot form a supercritical mass of fissionable material and therefore cannot create a nuclear explosion. But as history has shown, failures of systems and safeguards can cause catastrophic accidents, including chemical explosions and nuclear meltdowns (damage to the reactor core from overheating). The following Chemistry in Everyday Life feature explores three infamous meltdown incidents.

CHEMISTRY IN EVERYDAY LIFE

Nuclear Accidents

The importance of cooling and containment are amply illustrated by three major accidents that occurred with the nuclear reactors at nuclear power generating stations in the United States (Three Mile Island), the former Soviet Union (Chernobyl), and Japan (Fukushima).

In March 1979, the cooling system of the Unit 2 reactor at Three Mile Island Nuclear Generating Station in Pennsylvania failed, and the cooling water spilled from the reactor onto the floor of the containment building. After the pumps stopped, the reactors overheated due to the high radioactive decay heat produced in the first few days after the nuclear reactor shut down. The temperature of the core climbed to at least 2200 °C, and the upper portion of the core began to melt. In addition, the zirconium alloy cladding of the fuel rods began to react with steam and produced hydrogen:



The hydrogen accumulated in the confinement building, and it was feared that there was danger of an explosion of the mixture of hydrogen and air in the building. Consequently, hydrogen gas and radioactive gases (primarily krypton and xenon) were vented from the building. Within a week, cooling water circulation was restored and the core began to cool. The plant was closed for nearly 10 years during the cleanup process.

Although zero discharge of radioactive material is desirable, the discharge of radioactive krypton and xenon, such as occurred at the Three Mile Island plant, is among the most tolerable. These gases readily disperse in the atmosphere and thus do not produce highly radioactive areas. Moreover, they are noble gases and are not incorporated into plant and animal matter in the food chain. Effectively none of the heavy elements of the core of the reactor were released into the environment, and no cleanup of the area outside of the containment building was necessary (Figure 21.21).



(a)



(b)

FIGURE 21.21 (a) In this 2010 photo of Three Mile Island, the remaining structures from the damaged Unit 2 reactor are seen on the left, whereas the separate Unit 1 reactor, unaffected by the accident, continues generating power to this day (right). (b) President Jimmy Carter visited the Unit 2 control room a few days after the accident in 1979.

Another major nuclear accident involving a reactor occurred in April 1986, at the Chernobyl Nuclear Power Plant in Ukraine, which was still a part of the former Soviet Union. While operating at low power during an unauthorized experiment with some of its safety devices shut off, one of the reactors at the plant became unstable. Its chain reaction became uncontrollable and increased to a level far beyond what the reactor was

designed for. The steam pressure in the reactor rose to between 100 and 500 times the full power pressure and ruptured the reactor. Because the reactor was not enclosed in a containment building, a large amount of radioactive material spewed out, and additional fission products were released, as the graphite (carbon) moderator of the core ignited and burned. The fire was controlled, but over 200 plant workers and firefighters developed acute radiation sickness and at least 32 soon died from the effects of the radiation. It is predicted that about 4000 more deaths will occur among emergency workers and former Chernobyl residents from radiation-induced cancer and leukemia. The reactor has since been encapsulated in steel and concrete, a now-decaying structure known as the sarcophagus. Almost 30 years later, significant radiation problems still persist in the area, and Chernobyl largely remains a wasteland.

In 2011, the Fukushima Daiichi Nuclear Power Plant in Japan was badly damaged by a 9.0-magnitude earthquake and resulting tsunami. Three reactors up and running at the time were shut down automatically, and emergency generators came online to power electronics and coolant systems. However, the tsunami quickly flooded the emergency generators and cut power to the pumps that circulated coolant water through the reactors. High-temperature steam in the reactors reacted with zirconium alloy to produce hydrogen gas. The gas escaped into the containment building, and the mixture of hydrogen and air exploded. Radioactive material was released from the containment vessels as the result of deliberate venting to reduce the hydrogen pressure, deliberate discharge of coolant water into the sea, and accidental or uncontrolled events.

An evacuation zone around the damaged plant extended over 12.4 miles away, and an estimated 200,000 people were evacuated from the area. All 48 of Japan's nuclear power plants were subsequently shut down, remaining shuttered as of December 2014. Since the disaster, public opinion has shifted from largely favoring to largely opposing increasing the use of nuclear power plants, and a restart of Japan's atomic energy program is still stalled ([Figure 21.22](#)).



FIGURE 21.22 (a) After the accident, contaminated waste had to be removed, and (b) an evacuation zone was set up around the plant in areas that received heavy doses of radioactive fallout. (credit a: modification of work by “Live Action Hero”/Flickr)

The energy produced by a reactor fueled with enriched uranium results from the fission of uranium as well as from the fission of plutonium produced as the reactor operates. As discussed previously, the plutonium forms from the combination of neutrons and the uranium in the fuel. In any nuclear reactor, only about 0.1% of the mass of the fuel is converted into energy. The other 99.9% remains in the fuel rods as fission products and unused fuel. All of the fission products absorb neutrons, and after a period of several months to a few years, depending on the reactor, the fission products must be removed by changing the fuel rods. Otherwise, the concentration of these fission products would increase and absorb more neutrons until the reactor could no longer operate.

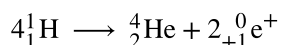
Spent fuel rods contain a variety of products, consisting of unstable nuclei ranging in atomic number from 25 to 60, some transuranium elements, including plutonium and americium, and unreacted uranium isotopes. The unstable nuclei and the transuranium isotopes give the spent fuel a dangerously high level of radioactivity. The long-lived isotopes require thousands of years to decay to a safe level. The ultimate fate of the nuclear reactor as a significant source of energy in the United States probably rests on whether or not a politically and scientifically satisfactory technique for processing and storing the components of spent fuel rods can be developed.

LINK TO LEARNING

Explore the information in this [link \(http://openstax.org/l/16wastemgmt\)](http://openstax.org/l/16wastemgmt) to learn about the approaches to nuclear waste management.

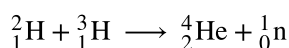
Nuclear Fusion and Fusion Reactors

The process of converting very light nuclei into heavier nuclei is also accompanied by the conversion of mass into large amounts of energy, a process called **fusion**. The principal source of energy in the sun is a net fusion reaction in which four hydrogen nuclei fuse and produce one helium nucleus and two positrons. This is a net reaction of a more complicated series of events:



A helium nucleus has a mass that is 0.7% less than that of four hydrogen nuclei; this lost mass is converted into energy during the fusion. This reaction produces about 3.6×10^{11} kJ of energy per mole of $\text{}^4_2\text{He}$ produced. This is somewhat larger than the energy produced by the nuclear fission of one mole of U-235 (1.8×10^{10} kJ), and over 3 million times larger than the energy produced by the (chemical) combustion of one mole of octane (5471 kJ).

It has been determined that the nuclei of the heavy isotopes of hydrogen, a deuteron, $\text{}^2_1\text{H}$ and a triton, $\text{}^3_1\text{H}$, undergo fusion at extremely high temperatures (thermonuclear fusion). They form a helium nucleus and a neutron:



This change proceeds with a mass loss of 0.0188 amu, corresponding to the release of 1.69×10^9 kilojoules per mole of $\text{}^4_2\text{He}$ formed. The very high temperature is necessary to give the nuclei enough kinetic energy to overcome the very strong repulsive forces resulting from the positive charges on their nuclei so they can collide.

Useful fusion reactions require very high temperatures for their initiation—about 15,000,000 K or more. At these temperatures, all molecules dissociate into atoms, and the atoms ionize, forming plasma. These conditions occur in an extremely large number of locations throughout the universe—stars are powered by fusion. Humans have already figured out how to create temperatures high enough to achieve fusion on a large scale in thermonuclear weapons. A thermonuclear weapon such as a hydrogen bomb contains a nuclear fission bomb that, when exploded, gives off enough energy to produce the extremely high temperatures necessary for fusion to occur.

Another much more beneficial way to create fusion reactions is in a **fusion reactor**, a nuclear reactor in which fusion reactions of light nuclei are controlled. Because no solid materials are stable at such high temperatures, mechanical devices cannot contain the plasma in which fusion reactions occur. Two techniques to contain plasma at the density and temperature necessary for a fusion reaction are currently the focus of intensive research efforts: containment by a magnetic field and by the use of focused laser beams ([Figure 21.23](#)). A number of large projects are working to attain one of the biggest goals in science: getting hydrogen fuel to ignite and produce more energy than the amount supplied to achieve the extremely high temperatures and pressures that are required for fusion. At the time of this writing, there are no self-sustaining fusion reactors operating in the world, although small-scale controlled fusion reactions have been run for very brief periods.

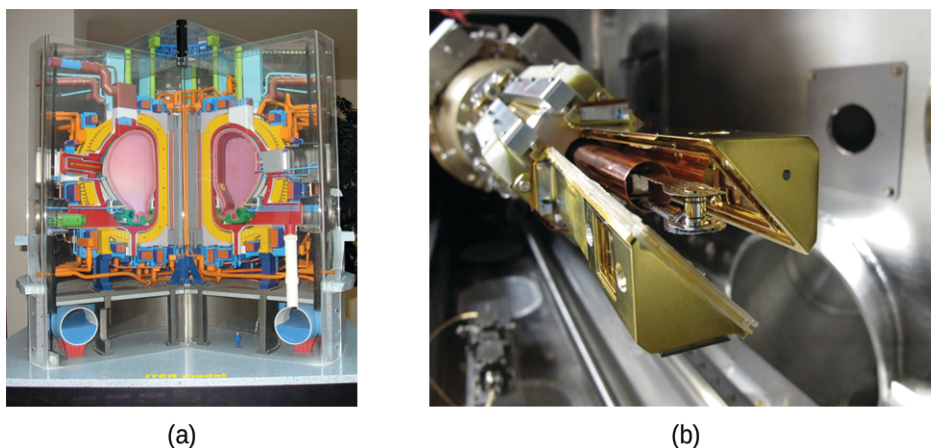


FIGURE 21.23 (a) This model is of the International Thermonuclear Experimental Reactor (ITER) reactor. Currently under construction in the south of France with an expected completion date of 2027, the ITER will be the world's largest experimental Tokamak nuclear fusion reactor with a goal of achieving large-scale sustained energy production. (b) In 2012, the National Ignition Facility at Lawrence Livermore National Laboratory briefly produced over 500,000,000,000 watts (500 terawatts, or 500 TW) of peak power and delivered 1,850,000 joules (1.85 MJ) of energy, the largest laser energy ever produced and 1000 times the power usage of the entire United States in any given moment. Although lasting only a few billionths of a second, the 192 lasers attained the conditions needed for nuclear fusion ignition. This image shows the target prior to the laser shot. (credit a: modification of work by Stephan Mosel)

21.5 Uses of Radioisotopes

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- List common applications of radioactive isotopes

Radioactive isotopes have the same chemical properties as stable isotopes of the same element, but they emit radiation, which can be detected. If we replace one (or more) atom(s) with radioisotope(s) in a compound, we can track them by monitoring their radioactive emissions. This type of compound is called a **radioactive tracer** (or **radioactive label**). Radioisotopes are used to follow the paths of biochemical reactions or to determine how a substance is distributed within an organism. Radioactive tracers are also used in many medical applications, including both diagnosis and treatment. They are used to measure engine wear, analyze the geological formation around oil wells, and much more.

Radioimmunoassays (RIA), for example, rely on radioisotopes to detect the presence and/or concentration of certain antigens. Developed by Rosalyn Sussman Yalow and Solomon Berson in the 1950s, the technique is known for extreme sensitivity, meaning that it can detect and measure very small quantities of a substance. Prior to its discovery, most similar detection relied on large enough quantities to produce visible outcomes. RIA revolutionized and expanded entire fields of study, most notably endocrinology, and is commonly used in narcotics detection, blood bank screening, early cancer screening, hormone measurement, and allergy diagnosis. Based on her significant contribution to medicine, Yalow received a Nobel Prize, making her the second woman to be awarded the prize for medicine.

Radioisotopes have revolutionized medical practice (see [Appendix M](#)), where they are used extensively. Over 10 million nuclear medicine procedures and more than 100 million nuclear medicine tests are performed annually in the United States. Four typical examples of radioactive tracers used in medicine are technetium-99 ($^{99}_{43}\text{Tc}$), thallium-201 ($^{201}_{81}\text{Tl}$), iodine-131 ($^{131}_{53}\text{I}$), and sodium-24 ($^{24}_{11}\text{Na}$). Damaged tissues in the heart, liver, and lungs absorb certain compounds of technetium-99 preferentially. After it is injected, the location of the technetium compound, and hence the damaged tissue, can be determined by detecting the γ rays emitted by the Tc-99 isotope. Thallium-201 ([Figure 21.24](#)) becomes concentrated in healthy heart tissue, so the two isotopes, Tc-99 and Tl-201, are used together to study heart tissue. Iodine-131 concentrates in the thyroid gland, the liver, and some parts of the brain. It can therefore be used to monitor goiter and treat thyroid conditions, such as Grave's disease, as well as liver and brain tumors. Salt solutions containing compounds of sodium-24 are injected into the bloodstream to help locate obstructions to the flow of blood.

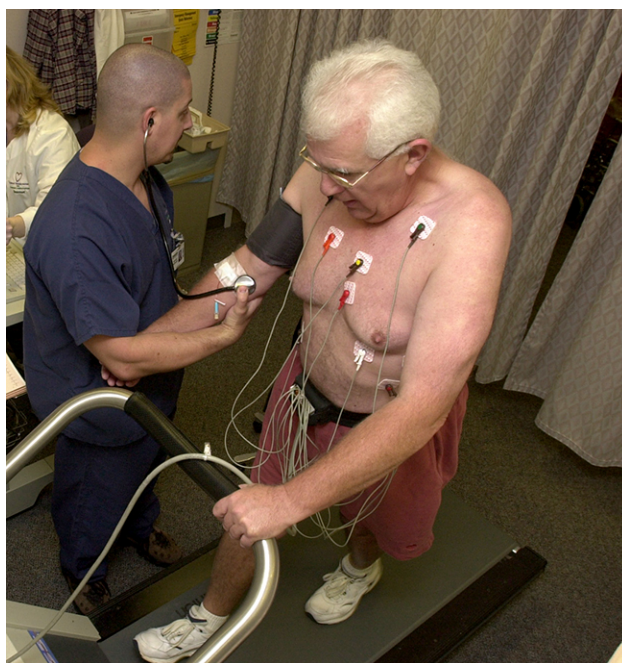


FIGURE 21.24 Administering thallium-201 to a patient and subsequently performing a stress test offer medical professionals an opportunity to visually analyze heart function and blood flow. (credit: modification of work by “BlueOctane”/Wikimedia Commons)

Radioisotopes used in medicine typically have short half-lives—for example, the ubiquitous Tc-99m has a half-life of 6.01 hours. This makes Tc-99m essentially impossible to store and prohibitively expensive to transport, so it is made on-site instead. Hospitals and other medical facilities use Mo-99 (which is primarily extracted from U-235 fission products) to generate Tc-99. Mo-99 undergoes β decay with a half-life of 66 hours, and the Tc-99 is then chemically extracted (Figure 21.25). The parent nuclide Mo-99 is part of a molybdate ion, MoO_4^{2-} ; when it decays, it forms the pertechnetate ion, TcO_4^- . These two water-soluble ions are separated by column chromatography, with the higher charge molybdate ion adsorbing onto the alumina in the column, and the lower charge pertechnetate ion passing through the column in the solution. A few micrograms of Mo-99 can produce enough Tc-99 to perform as many as 10,000 tests.

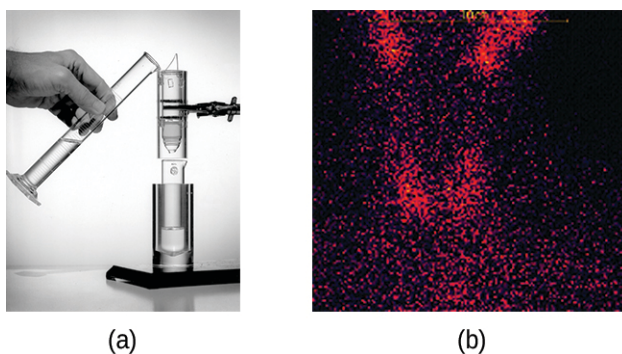


FIGURE 21.25 (a) The first Tc-99m generator (circa 1958) is used to separate Tc-99 from Mo-99. The MoO_4^{2-} is retained by the matrix in the column, whereas the TcO_4^- passes through and is collected. (b) Tc-99 was used in this scan of the neck of a patient with Grave's disease. The scan shows the location of high concentrations of Tc-99. (credit a: modification of work by the Department of Energy; credit b: modification of work by “MBq”/Wikimedia Commons)

Radioisotopes can also be used, typically in higher doses than as a tracer, as treatment. **Radiation therapy** is the use of high-energy radiation to damage the DNA of cancer cells, which kills them or keeps them from dividing (Figure 21.26). A cancer patient may receive **external beam radiation therapy** delivered by a machine outside the body, or **internal radiation therapy (brachytherapy)** from a radioactive substance that has been introduced into the body. Note that **chemotherapy** is similar to internal radiation therapy in that the cancer treatment is injected into the body, but differs in that chemotherapy uses chemical rather than radioactive substances to kill the cancer cells.

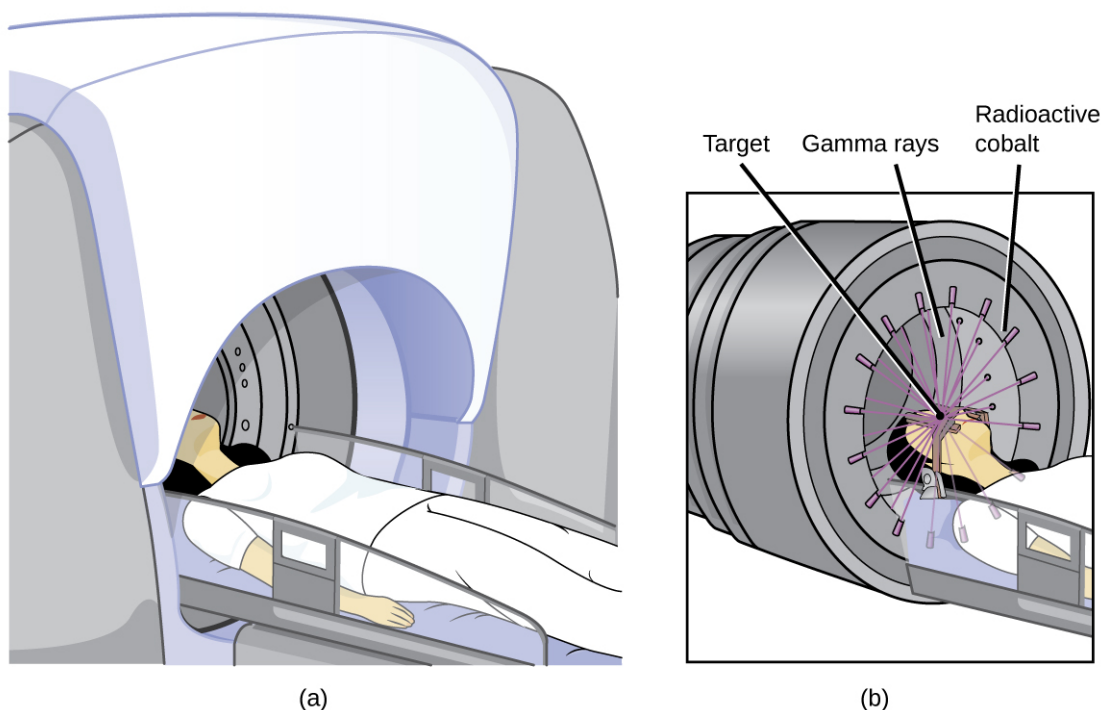
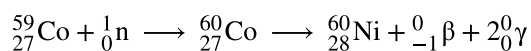


FIGURE 21.26 The cartoon in (a) shows a cobalt-60 machine used in the treatment of cancer. The diagram in (b) shows how the gantry of the Co-60 machine swings through an arc, focusing radiation on the targeted region (tumor) and minimizing the amount of radiation that passes through nearby regions.

Cobalt-60 is a synthetic radioisotope produced by the neutron activation of Co-59, which then undergoes β decay to form Ni-60, along with the emission of γ radiation. The overall process is:



The overall decay scheme for this is shown graphically in [Figure 21.27](#).

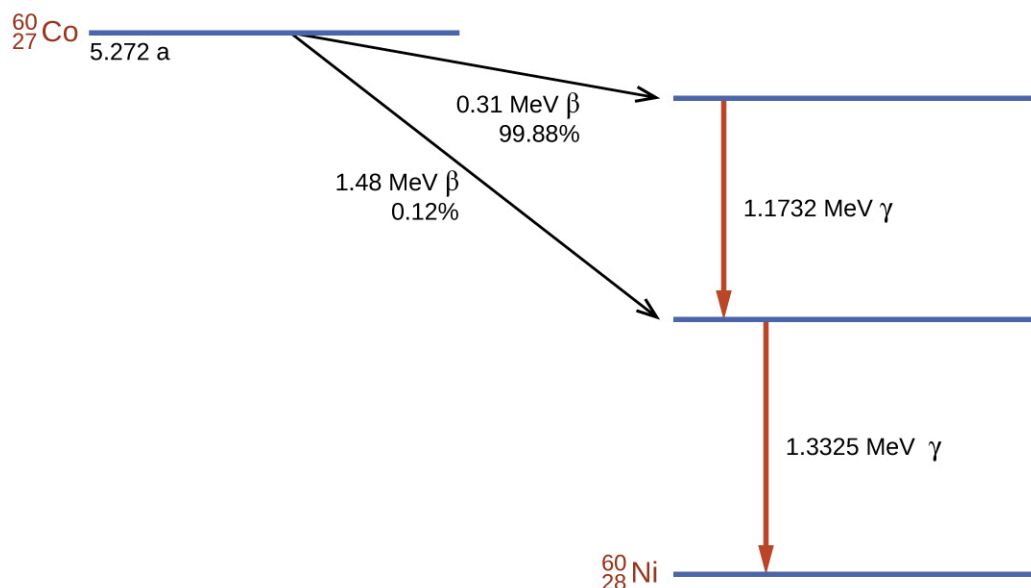
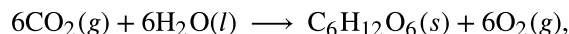


FIGURE 21.27 Co-60 undergoes a series of radioactive decays. The γ emissions are used for radiation therapy.

Radioisotopes are used in diverse ways to study the mechanisms of chemical reactions in plants and animals. These include labeling fertilizers in studies of nutrient uptake by plants and crop growth, investigations of digestive and milk-producing processes in cows, and studies on the growth and metabolism of animals and plants.

For example, the radioisotope C-14 was used to elucidate the details of how photosynthesis occurs. The overall

reaction is:



but the process is much more complex, proceeding through a series of steps in which various organic compounds are produced. In studies of the pathway of this reaction, plants were exposed to CO_2 containing a high concentration of $^{14}_6\text{C}$. At regular intervals, the plants were analyzed to determine which organic compounds contained carbon-14 and how much of each compound was present. From the time sequence in which the compounds appeared and the amount of each present at given time intervals, scientists learned more about the pathway of the reaction.

Commercial applications of radioactive materials are equally diverse ([Figure 21.28](#)). They include determining the thickness of films and thin metal sheets by exploiting the penetration power of various types of radiation. Flaws in metals used for structural purposes can be detected using high-energy gamma rays from cobalt-60 in a fashion similar to the way X-rays are used to examine the human body. In one form of pest control, flies are controlled by sterilizing male flies with γ radiation so that females breeding with them do not produce offspring. Many foods are preserved by radiation that kills microorganisms that cause the foods to spoil.



FIGURE 21.28 Common commercial uses of radiation include (a) X-ray examination of luggage at an airport and (b) preservation of food. (credit a: modification of work by the Department of the Navy; credit b: modification of work by the US Department of Agriculture)

Americium-241, an α emitter with a half-life of 458 years, is used in tiny amounts in ionization-type smoke detectors ([Figure 21.29](#)). The α emissions from Am-241 ionize the air between two electrode plates in the ionizing chamber. A battery supplies a potential that causes movement of the ions, thus creating a small electric current. When smoke enters the chamber, the movement of the ions is impeded, reducing the conductivity of the air. This causes a marked drop in the current, triggering an alarm.

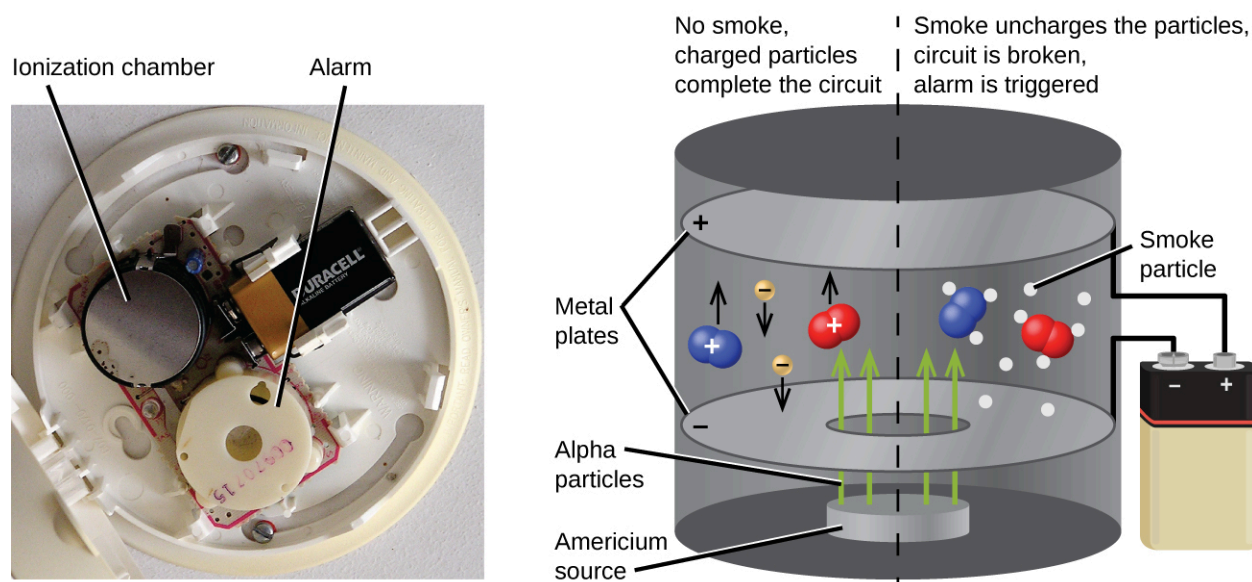


FIGURE 21.29 Inside a smoke detector, Am-241 emits α particles that ionize the air, creating a small electric current. During a fire, smoke particles impede the flow of ions, reducing the current and triggering an alarm. (credit a: modification of work by “Muffet”/Wikimedia Commons)

21.6 Biological Effects of Radiation

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the biological impact of ionizing radiation
- Define units for measuring radiation exposure
- Explain the operation of common tools for detecting radioactivity
- List common sources of radiation exposure in the US

The increased use of radioisotopes has led to increased concerns over the effects of these materials on biological systems (such as humans). All radioactive nuclides emit high-energy particles or electromagnetic waves. When this radiation encounters living cells, it can cause heating, break chemical bonds, or ionize molecules. The most serious biological damage results when these radioactive emissions fragment or ionize molecules. For example, alpha and beta particles emitted from nuclear decay reactions possess much higher energies than ordinary chemical bond energies. When these particles strike and penetrate matter, they produce ions and molecular fragments that are extremely reactive. The damage this does to biomolecules in living organisms can cause serious malfunctions in normal cell processes, taxing the organism's repair mechanisms and possibly causing illness or even death ([Figure 21.30](#)).

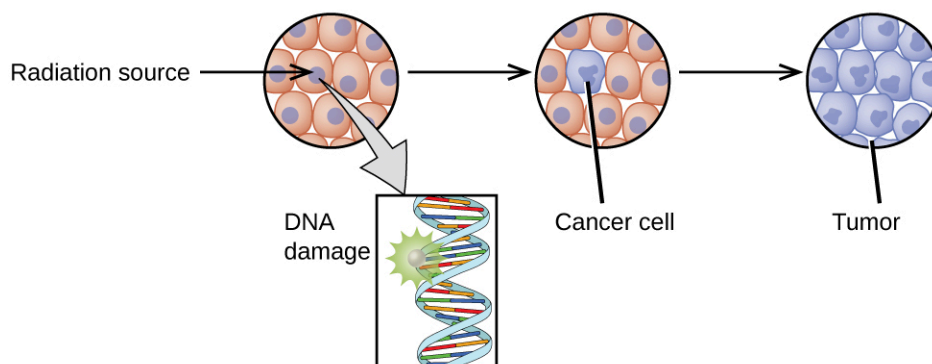


FIGURE 21.30 Radiation can harm biological systems by damaging the DNA of cells. If this damage is not properly repaired, the cells may divide in an uncontrolled manner and cause cancer.

Ionizing and Nonionizing Radiation

There is a large difference in the magnitude of the biological effects of **nonionizing radiation** (for example, light and microwaves) and **ionizing radiation**, emissions energetic enough to knock electrons out of molecules (for example, α and β particles, γ rays, X-rays, and high-energy ultraviolet radiation) (Figure 21.31).

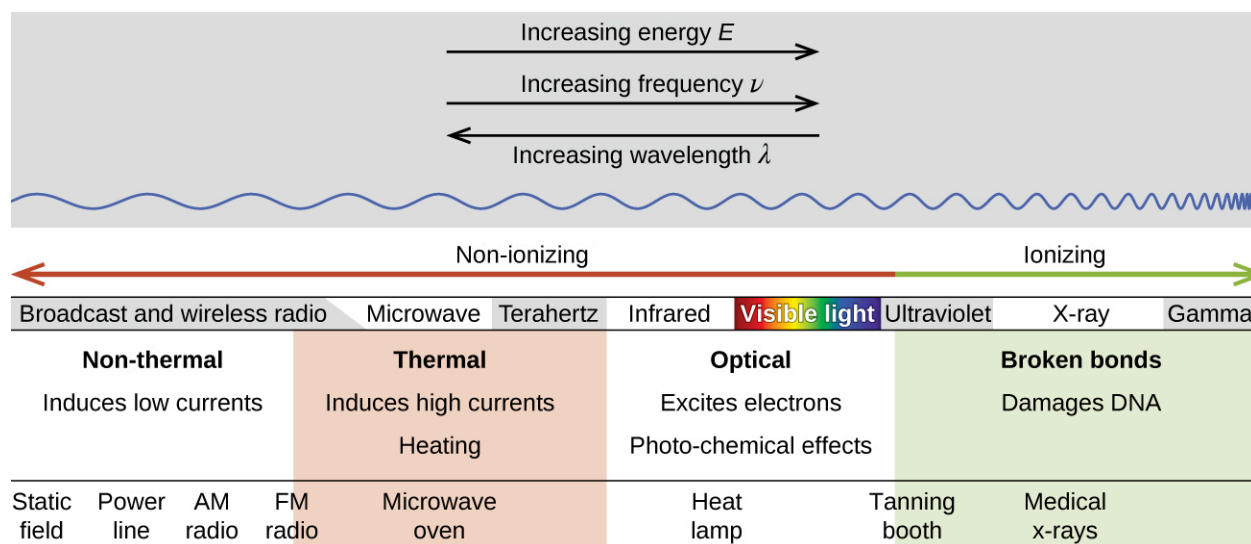
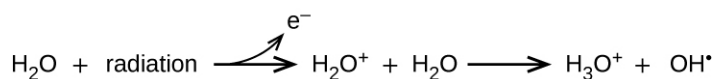


FIGURE 21.31 Lower frequency, lower-energy electromagnetic radiation is nonionizing, and higher frequency, higher-energy electromagnetic radiation is ionizing.

Energy absorbed from nonionizing radiation speeds up the movement of atoms and molecules, which is equivalent to heating the sample. Although biological systems are sensitive to heat (as we might know from touching a hot stove or spending a day at the beach in the sun), a large amount of nonionizing radiation is necessary before dangerous levels are reached. Ionizing radiation, however, may cause much more severe damage by breaking bonds or removing electrons in biological molecules, disrupting their structure and function. The damage can also be done indirectly, by first ionizing H_2O (the most abundant molecule in living organisms), which forms a H_2O^+ ion that reacts with water, forming a hydronium ion and a hydroxyl radical:



Because the hydroxyl radical has an unpaired electron, it is highly reactive. (This is true of any substance with unpaired electrons, known as a free radical.) This hydroxyl radical can react with all kinds of biological molecules (DNA, proteins, enzymes, and so on), causing damage to the molecules and disrupting physiological processes. Examples of direct and indirect damage are shown in Figure 21.32.

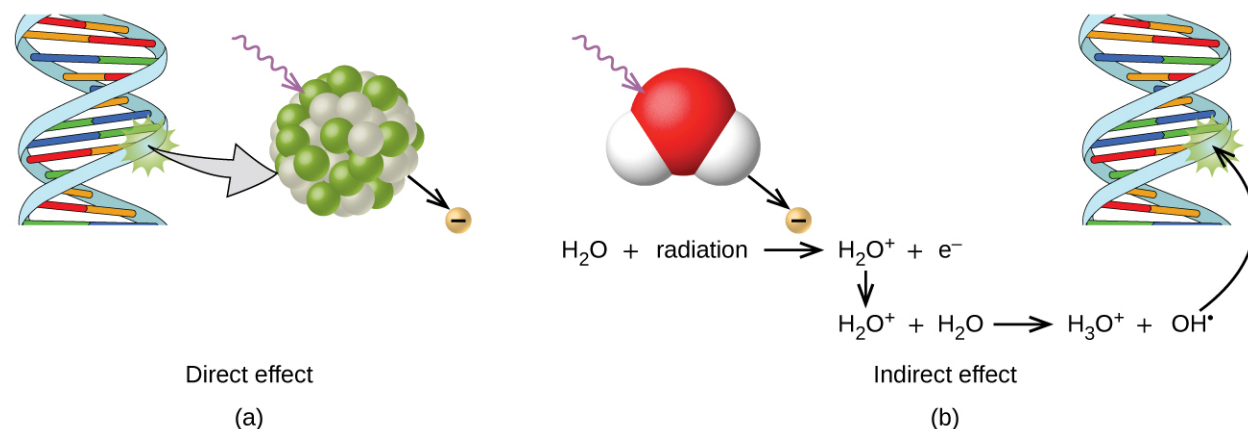


FIGURE 21.32 Ionizing radiation can (a) directly damage a biomolecule by ionizing it or breaking its bonds, or (b) create an H_2O^+ ion, which reacts with H_2O to form a hydroxyl radical, which in turn reacts with the biomolecule, causing damage indirectly.

Biological Effects of Exposure to Radiation

Radiation can harm either the whole body (somatic damage) or eggs and sperm (genetic damage). Its effects are more pronounced in cells that reproduce rapidly, such as the stomach lining, hair follicles, bone marrow, and embryos. This is why patients undergoing radiation therapy often feel nauseous or sick to their stomach, lose hair, have bone aches, and so on, and why particular care must be taken when undergoing radiation therapy during pregnancy.

Different types of radiation have differing abilities to pass through material ([Figure 21.33](#)). A very thin barrier, such as a sheet or two of paper, or the top layer of skin cells, usually stops alpha particles. Because of this, alpha particle sources are usually not dangerous if outside the body, but are quite hazardous if ingested or inhaled (see the Chemistry in Everyday Life feature on Radon Exposure). Beta particles will pass through a hand, or a thin layer of material like paper or wood, but are stopped by a thin layer of metal. Gamma radiation is very penetrating and can pass through a thick layer of most materials. Some high-energy gamma radiation is able to pass through a few feet of concrete. Certain dense, high atomic number elements (such as lead) can effectively attenuate gamma radiation with thinner material and are used for shielding. The ability of various kinds of emissions to cause ionization varies greatly, and some particles have almost no tendency to produce ionization. Alpha particles have about twice the ionizing power of fast-moving neutrons, about 10 times that of β particles, and about 20 times that of γ rays and X-rays.

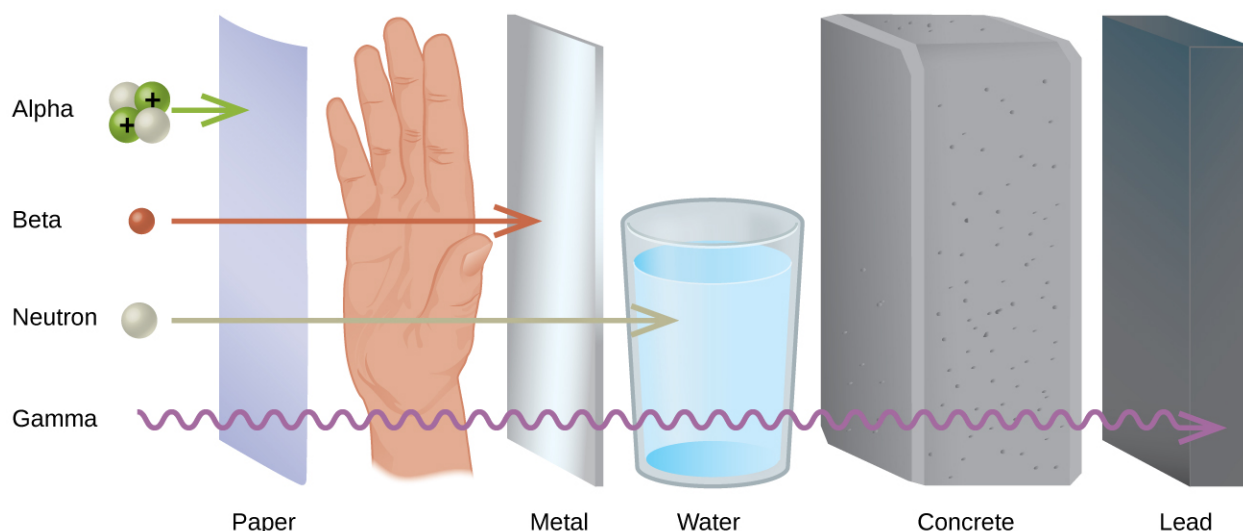


FIGURE 21.33 The ability of different types of radiation to pass through material is shown. From least to most penetrating, they are alpha < beta < neutron < gamma.

CHEMISTRY IN EVERYDAY LIFE

Radon Exposure

For many people, one of the largest sources of exposure to radiation is from radon gas (Rn-222). Radon-222 is an α emitter with a half-life of 3.82 days. It is one of the products of the radioactive decay series of U-238 ([Figure 21.9](#)), which is found in trace amounts in soil and rocks. The radon gas that is produced slowly escapes from the ground and gradually seeps into homes and other structures above. Since it is about eight times more dense than air, radon gas accumulates in basements and lower floors, and slowly diffuses throughout buildings ([Figure 21.34](#)).

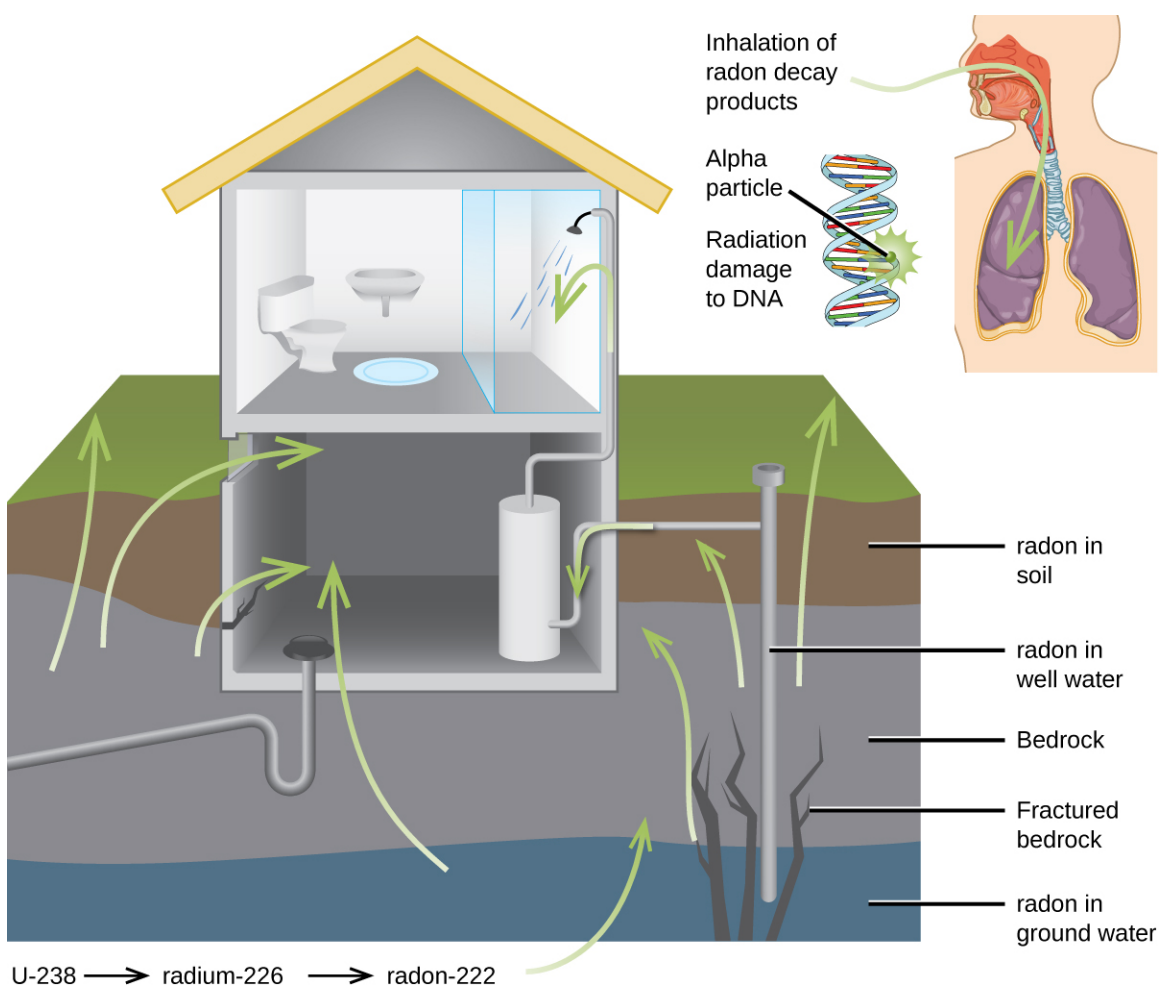


FIGURE 21.34 Radon-222 seeps into houses and other buildings from rocks that contain uranium-238, a radon emitter. The radon enters through cracks in concrete foundations and basement floors, stone or porous cinderblock foundations, and openings for water and gas pipes.

Radon is found in buildings across the country, with amounts depending on where you live. The average concentration of radon inside houses in the US (1.25 pCi/L) is about three times the levels found in outside air, and about one in six houses have radon levels high enough that remediation efforts to reduce the radon concentration are recommended. Exposure to radon increases one's risk of getting cancer (especially lung cancer), and high radon levels can be as bad for health as smoking a carton of cigarettes a day. Radon is the number one cause of lung cancer in nonsmokers and the second leading cause of lung cancer overall. Radon exposure is believed to cause over 20,000 deaths in the US per year.

Measuring Radiation Exposure

Several different devices are used to detect and measure radiation, including Geiger counters, scintillation counters (scintillators), and radiation dosimeters (Figure 21.35). Probably the best-known radiation instrument, the **Geiger counter** (also called the Geiger-Müller counter) detects and measures radiation. Radiation causes the ionization of the gas in a Geiger-Müller tube. The rate of ionization is proportional to the amount of radiation. A **scintillation counter** contains a scintillator—a material that emits light (luminesces) when excited by ionizing radiation—and a sensor that converts the light into an electric signal. **Radiation dosimeters** also measure ionizing radiation and are often used to determine personal radiation exposure. Commonly used types are electronic, film badge, thermoluminescent, and quartz fiber dosimeters.

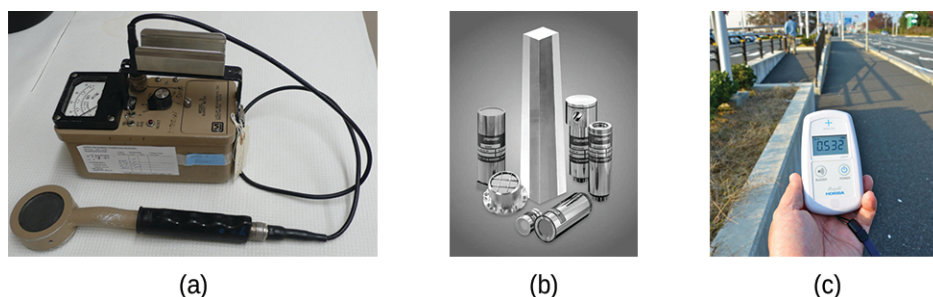
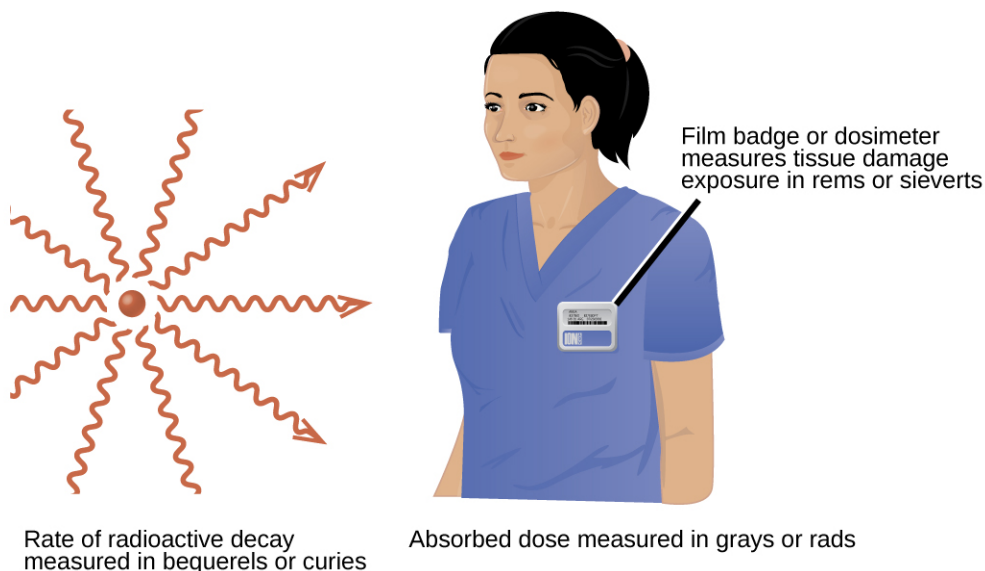


FIGURE 21.35 Devices such as (a) Geiger counters, (b) scintillators, and (c) dosimeters can be used to measure radiation. (credit c: modification of work by “osaMu”/Wikimedia commons)

A variety of units are used to measure various aspects of radiation ([Figure 21.36](#)). The SI unit for rate of radioactive decay is the **becquerel (Bq)**, with $1 \text{ Bq} = 1$ disintegration per second. The **curie (Ci)** and **millicurie (mCi)** are much larger units and are frequently used in medicine ($1 \text{ curie} = 1 \text{ Ci} = 3.7 \times 10^{10}$ disintegrations per second). The SI unit for measuring radiation dose is the **gray (Gy)**, with $1 \text{ Gy} = 1 \text{ J}$ of energy absorbed per kilogram of tissue. In medical applications, the **radiation absorbed dose (rad)** is more often used ($1 \text{ rad} = 0.01 \text{ Gy}$; 1 rad results in the absorption of 0.01 J/kg of tissue). The SI unit measuring tissue damage caused by radiation is the **sievert (Sv)**. This takes into account both the energy and the biological effects of the type of radiation involved in the radiation dose. The **roentgen equivalent for man (rem)** is the unit for radiation damage that is used most frequently in medicine ($100 \text{ rem} = 1 \text{ Sv}$). Note that the tissue damage units (rem or Sv) includes the energy of the radiation dose (rad or Gy) along with a biological factor referred to as the **RBE** (for **relative biological effectiveness**) that is an approximate measure of the relative damage done by the radiation. These are related by:

$$\text{number of rems} = \text{RBE} \times \text{number of rads}$$

with RBE approximately 10 for α radiation, 2(+) for protons and neutrons, and 1 for β and γ radiation.



Rate of radioactive decay
measured in becquerels or curies

Absorbed dose measured in grays or rads

FIGURE 21.36 Different units are used to measure the rate of emission from a radioactive source, the energy that is absorbed from the source, and the amount of damage the absorbed radiation does.

Units of Radiation Measurement

[Table 21.4](#) summarizes the units used for measuring radiation.

Units Used for Measuring Radiation

Measurement Purpose	Unit	Quantity Measured	Description
activity of source	becquerel (Bq)	radioactive decays or emissions	amount of sample that undergoes 1 decay/second
	curie (Ci)		amount of sample that undergoes 3.7×10^{10} decays/second
absorbed dose	gray (Gy)	energy absorbed per kg of tissue	1 Gy = 1 J/kg tissue
	radiation absorbed dose (rad)		1 rad = 0.01 J/kg tissue
biologically effective dose	sievert (Sv)	tissue damage	Sv = RBE $\times$ Gy
	roentgen equivalent for man (rem)		Rem = RBE $\times$ rad

TABLE 21.4

**EXAMPLE 21.8****Amount of Radiation**

Cobalt-60 ($t_{1/2} = 5.26$ y) is used in cancer therapy since the γ rays it emits can be focused in small areas where the cancer is located. A 5.00-g sample of Co-60 is available for cancer treatment.

(a) What is its activity in Bq?

(b) What is its activity in Ci?

Solution

The activity is given by:

$$\text{Activity} = \lambda N = \left(\frac{\ln 2}{t_{1/2}} \right) N = \left(\frac{\ln 2}{5.26 \text{ y}} \right) \times 5.00 \text{ g} = 0.659 \frac{\text{g}}{\text{y}} \text{ of Co-60 that decay}$$

And to convert this to decays per second:

$$0.659 \frac{\text{g}}{\text{y}} \times \frac{1 \text{ y}}{365 \text{ d}} \times \frac{1 \text{ d}}{24 \text{ h}} \times \frac{1 \text{ h}}{3600 \text{ s}} \times \frac{1 \text{ mol}}{59.9 \text{ g}} \times \frac{6.02 \times 10^{23} \text{ atoms}}{1 \text{ mol}} \times \frac{1 \text{ decay}}{1 \text{ atom}}$$

$$= 2.10 \times 10^{14} \frac{\text{decay}}{\text{s}}$$

(a) Since $1 \text{ Bq} = \frac{1 \text{ decay}}{\text{s}}$, the activity in Becquerel (Bq) is:

$$2.10 \times 10^{14} \frac{\text{decay}}{\text{s}} \times \left(\frac{1 \text{ Bq}}{1 \frac{\text{decay}}{\text{s}}} \right) = 2.10 \times 10^{14} \text{ Bq}$$

(b) Since $1 \text{ Ci} = \frac{3.7 \times 10^{10} \text{ decay}}{\text{s}}$, the activity in curie (Ci) is:

$$2.10 \times 10^{14} \frac{\text{decay}}{\text{s}} \times \left(\frac{1 \text{ Ci}}{3.7 \times 10^{11} \frac{\text{decay}}{\text{s}}} \right) = 5.7 \times 10^2 \text{ Ci}$$

Check Your Learning

Tritium is a radioactive isotope of hydrogen ($t_{1/2} = 12.32 \text{ y}$) that has several uses, including self-powered lighting, in which electrons emitted in tritium radioactive decay cause phosphorus to glow. Its nucleus contains one proton and two neutrons, and the atomic mass of tritium is 3.016 amu. What is the activity of a sample containing 1.00mg of tritium (a) in Bq and (b) in Ci?

Answer:

(a) $3.56 \times 10^{11} \text{ Bq}$; (b) 0.962 Ci

Effects of Long-term Radiation Exposure on the Human Body

The effects of radiation depend on the type, energy, and location of the radiation source, and the length of exposure. As shown in [Figure 21.37](#), the average person is exposed to background radiation, including cosmic rays from the sun and radon from uranium in the ground (see the Chemistry in Everyday Life feature on Radon Exposure); radiation from medical exposure, including CAT scans, radioisotope tests, X-rays, and so on; and small amounts of radiation from other human activities, such as airplane flights (which are bombarded by increased numbers of cosmic rays in the upper atmosphere), radioactivity from consumer products, and a variety of radionuclides that enter our bodies when we breathe (for example, carbon-14) or through the food chain (for example, potassium-40, strontium-90, and iodine-131).

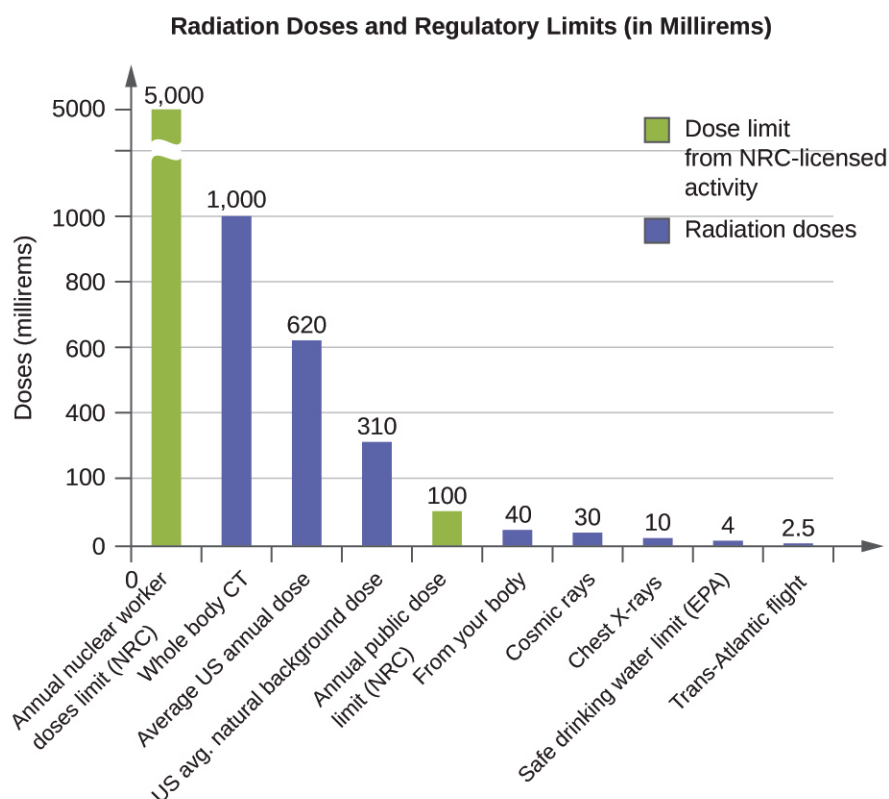


FIGURE 21.37 The total annual radiation exposure for a person in the US is about 620 mrem. The various sources and their relative amounts are shown in this bar graph. (source: U.S. Nuclear Regulatory Commission)

A short-term, sudden dose of a large amount of radiation can cause a wide range of health effects, from changes in blood chemistry to death. Short-term exposure to tens of rems of radiation will likely cause very noticeable symptoms or illness; a dose of about 500 rems is estimated to have a 50% probability of causing the death of the

victim within 30 days of exposure. Exposure to radioactive emissions has a cumulative effect on the body during a person's lifetime, which is another reason why it is important to avoid any unnecessary exposure to radiation. Health effects of short-term exposure to radiation are shown in [Table 21.5](#).

Health Effects of Radiation²

Exposure (rem)	Health Effect	Time to Onset (without treatment)
5–10	changes in blood chemistry	—
50	nausea	hours
55	fatigue	—
70	vomiting	—
75	hair loss	2–3 weeks
90	diarrhea	—
100	hemorrhage	—
400	possible death	within 2 months
1000	destruction of intestinal lining	—
	internal bleeding	—
	death	1–2 weeks
2000	damage to central nervous system	—
	loss of consciousness;	minutes
	death	hours to days

TABLE 21.5

It is impossible to avoid some exposure to ionizing radiation. We are constantly exposed to background radiation from a variety of natural sources, including cosmic radiation, rocks, medical procedures, consumer products, and even our own atoms. We can minimize our exposure by blocking or shielding the radiation, moving farther from the source, and limiting the time of exposure.

2 Source: US Environmental Protection Agency

Key Terms

alpha (α) decay loss of an alpha particle during radioactive decay

alpha particle (α or ${}^4_2\text{He}$ or ${}^4_2\alpha$) high-energy helium nucleus; a helium atom that has lost two electrons and contains two protons and two neutrons

antimatter particles with the same mass but opposite properties (such as charge) of ordinary particles

band of stability (also, belt of stability, zone of stability, or valley of stability) region of graph of number of protons versus number of neutrons containing stable (nonradioactive) nuclides

becquerel (Bq) SI unit for rate of radioactive decay; 1 Bq = 1 disintegration/s

beta (β) decay breakdown of a neutron into a proton, which remains in the nucleus, and an electron, which is emitted as a beta particle

beta particle (β or ${}^0_{-1}\text{e}$ or ${}^0_{-1}\beta$) high-energy electron

binding energy per nucleon total binding energy for the nucleus divided by the number of nucleons in the nucleus

chain reaction repeated fission caused when the neutrons released in fission bombard other atoms

chemotherapy similar to internal radiation therapy, but chemical rather than radioactive substances are introduced into the body to kill cancer cells

containment system (also, shield) a three-part structure of materials that protects the exterior of a nuclear fission reactor and operating personnel from the high temperatures, pressures, and radiation levels inside the reactor

control rod material inserted into the fuel assembly that absorbs neutrons and can be raised or lowered to adjust the rate of a fission reaction

critical mass amount of fissionable material that will support a self-sustaining (nuclear fission) chain reaction

curie (Ci) larger unit for rate of radioactive decay frequently used in medicine; 1 Ci = 3.7×10^{10} disintegrations/s

daughter nuclide nuclide produced by the radioactive decay of another nuclide; may be stable or may decay further

electron capture combination of a core electron with a proton to yield a neutron within the nucleus

electron volt (eV) measurement unit of nuclear binding energies, with 1 eV equaling the amount energy due to the moving an electron across an electric potential difference of 1 volt

external beam radiation therapy radiation delivered by a machine outside the body

fissile (or fissionable) when a material is capable of sustaining a nuclear fission reaction

fission splitting of a heavier nucleus into two or more lighter nuclei, usually accompanied by the conversion of mass into large amounts of energy

fusion combination of very light nuclei into heavier nuclei, accompanied by the conversion of mass into large amounts of energy

fusion reactor nuclear reactor in which fusion reactions of light nuclei are controlled

gamma (γ) emission decay of an excited-state nuclide accompanied by emission of a gamma ray

gamma ray (γ or ${}^0_0\gamma$) short wavelength, high-energy electromagnetic radiation that exhibits wave-particle duality

Geiger counter instrument that detects and measures radiation via the ionization produced in a Geiger-Müller tube

gray (Gy) SI unit for measuring radiation dose; 1 Gy = 1 J absorbed/kg tissue

half-life ($t_{1/2}$) time required for half of the atoms in a radioactive sample to decay

internal radiation therapy (also, brachytherapy) radiation from a radioactive substance introduced into the body to kill cancer cells

ionizing radiation radiation that can cause a molecule to lose an electron and form an ion

magic number nuclei with specific numbers of nucleons that are within the band of stability

mass defect difference between the mass of an atom and the summed mass of its constituent subatomic particles (or the mass “lost” when nucleons are brought together to form a nucleus)

mass-energy equivalence equation Albert Einstein’s relationship showing that mass and energy are equivalent

millicurie (mCi) larger unit for rate of radioactive decay frequently used in medicine; 1 Ci = 3.7×10^{10} disintegrations/s

nonionizing radiation radiation that speeds up the movement of atoms and molecules; it is equivalent to heating a sample, but is not energetic enough to cause the ionization of molecules

nuclear binding energy energy lost when an atom’s nucleons are bound together (or the energy needed to break a nucleus into its constituent protons and neutrons)

nuclear chemistry study of the structure of atomic nuclei and processes that change nuclear structure

nuclear fuel fissionable isotope present in sufficient quantities to provide a self-sustaining chain reaction

in a nuclear reactor

nuclear moderator substance that slows neutrons to a speed low enough to cause fission

nuclear reaction change to a nucleus resulting in changes in the atomic number, mass number, or energy state

nuclear reactor environment that produces energy via nuclear fission in which the chain reaction is controlled and sustained without explosion

nuclear transmutation conversion of one nuclide into another nuclide

nucleon collective term for protons and neutrons in a nucleus

nuclide nucleus of a particular isotope

parent nuclide unstable nuclide that changes spontaneously into another (daughter) nuclide

particle accelerator device that uses electric and magnetic fields to increase the kinetic energy of nuclei used in transmutation reactions

positron (${}^0_{+1}\beta$ or ${}^0_{+1}e$) antiparticle to the electron; it has identical properties to an electron, except for having the opposite (positive) charge

positron emission (also, β^+ decay) conversion of a proton into a neutron, which remains in the nucleus, and a positron, which is emitted

radiation absorbed dose (rad) SI unit for measuring radiation dose, frequently used in medical applications; $1 \text{ rad} = 0.01 \text{ Gy}$

radiation dosimeter device that measures ionizing radiation and is used to determine personal radiation exposure

radiation therapy use of high-energy radiation to damage the DNA of cancer cells, which kills them or keeps them from dividing

radioactive decay spontaneous decay of an unstable nuclide into another nuclide

radioactive decay series chains of successive disintegrations (radioactive decays) that ultimately lead to a stable end-product

radioactive tracer (also, radioactive label) radioisotope used to track or follow a substance by monitoring its radioactive emissions

radioactivity phenomenon exhibited by an unstable

nucleon that spontaneously undergoes change into a nucleon that is more stable; an unstable nucleon is said to be radioactive

radiocarbon dating highly accurate means of dating objects 30,000–50,000 years old that were derived from once-living matter; achieved by calculating the ratio of ${}^{14}_6\text{C}$: ${}^{12}_6\text{C}$ in the object vs. the ratio of ${}^{14}_6\text{C}$: ${}^{12}_6\text{C}$ in the present-day atmosphere

radioisotope isotope that is unstable and undergoes conversion into a different, more stable isotope

radiometric dating use of radioisotopes and their properties to date the formation of objects such as archeological artifacts, formerly living organisms, or geological formations

reactor coolant assembly used to carry the heat produced by fission in a reactor to an external boiler and turbine where it is transformed into electricity

relative biological effectiveness (RBE) measure of the relative damage done by radiation

roentgen equivalent man (rem) unit for radiation damage, frequently used in medicine; $100 \text{ rem} = 1 \text{ Sv}$

scintillation counter instrument that uses a scintillator—a material that emits light when excited by ionizing radiation—to detect and measure radiation

sievert (Sv) SI unit measuring tissue damage caused by radiation; takes into account energy and biological effects of radiation

strong nuclear force force of attraction between nucleons that holds a nucleus together

subcritical mass amount of fissionable material that cannot sustain a chain reaction; less than a critical mass

supercritical mass amount of material in which there is an increasing rate of fission

transmutation reaction bombardment of one type of nuclei with other nuclei or neutrons

transuranium element element with an atomic number greater than 92; these elements do not occur in nature

Key Equations

$$E = mc^2$$

$$\text{decay rate} = \lambda N$$

$$t_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

$$\text{rem} = \text{RBE} \times \text{rad}$$

$$\text{Sv} = \text{RBE} \times \text{Gy}$$

Summary

21.1 Nuclear Structure and Stability

An atomic nucleus consists of protons and neutrons, collectively called nucleons. Although protons repel each other, the nucleus is held tightly together by a short-range, but very strong, force called the strong nuclear force. A nucleus has less mass than the total mass of its constituent nucleons. This “missing” mass is the mass defect, which has been converted into the binding energy that holds the nucleus together according to Einstein’s mass-energy equivalence equation, $E = mc^2$. Of the many nuclides that exist, only a small number are stable. Nuclides with even numbers of protons or neutrons, or those with magic numbers of nucleons, are especially likely to be stable. These stable nuclides occupy a narrow band of stability on a graph of number of protons versus number of neutrons. The binding energy per nucleon is largest for the elements with mass numbers near 56; these are the most stable nuclei.

21.2 Nuclear Equations

Nuclei can undergo reactions that change their number of protons, number of neutrons, or energy state. Many different particles can be involved in nuclear reactions. The most common are protons, neutrons, positrons (which are positively charged electrons), alpha (α) particles (which are high-energy helium nuclei), beta (β) particles (which are high-energy electrons), and gamma (γ) rays (which compose high-energy electromagnetic radiation). As with chemical reactions, nuclear reactions are always balanced. When a nuclear reaction occurs, the total mass (number) and the total charge remain unchanged.

21.3 Radioactive Decay

Nuclei that have unstable n:p ratios undergo spontaneous radioactive decay. The most common types of radioactivity are α decay, β decay, γ emission, positron emission, and electron capture. Nuclear reactions also often involve γ rays, and some nuclei decay by electron capture. Each of these modes of decay leads to the formation of a new nucleus with a more stable n:p ratio. Some substances undergo radioactive decay series, proceeding through multiple decays before ending in a stable isotope. All nuclear decay processes follow first-order kinetics, and each radioisotope has its own characteristic half-life, the time that is required for half of its atoms to decay. Because of the large differences in stability among nuclides, there is a very wide range of half-lives of radioactive substances. Many of these substances have

found useful applications in medical diagnosis and treatment, determining the age of archaeological and geological objects, and more.

21.4 Transmutation and Nuclear Energy

It is possible to produce new atoms by bombarding other atoms with nuclei or high-speed particles. The products of these transmutation reactions can be stable or radioactive. A number of artificial elements, including technetium, astatine, and the transuranium elements, have been produced in this way.

Nuclear power as well as nuclear weapon detonations can be generated through fission (reactions in which a heavy nucleus is split into two or more lighter nuclei and several neutrons). Because the neutrons may induce additional fission reactions when they combine with other heavy nuclei, a chain reaction can result. Useful power is obtained if the fission process is carried out in a nuclear reactor. The conversion of light nuclei into heavier nuclei (fusion) also produces energy. At present, this energy has not been contained adequately and is too expensive to be feasible for commercial energy production.

21.5 Uses of Radioisotopes

Compounds known as radioactive tracers can be used to follow reactions, track the distribution of a substance, diagnose and treat medical conditions, and much more. Other radioactive substances are helpful for controlling pests, visualizing structures, providing fire warnings, and for many other applications. Hundreds of millions of nuclear medicine tests and procedures, using a wide variety of radioisotopes with relatively short half-lives, are performed every year in the US. Most of these radioisotopes have relatively short half-lives; some are short enough that the radioisotope must be made on-site at medical facilities. Radiation therapy uses high-energy radiation to kill cancer cells by damaging their DNA. The radiation used for this treatment may be delivered externally or internally.

21.6 Biological Effects of Radiation

We are constantly exposed to radiation from a variety of naturally occurring and human-produced sources. This radiation can affect living organisms. Ionizing radiation is the most harmful because it can ionize molecules or break chemical bonds, which damages the molecule and causes malfunctions in cell processes. It can also create reactive hydroxyl radicals that damage biological molecules and disrupt

physiological processes. Radiation can cause somatic or genetic damage, and is most harmful to rapidly reproducing cells. Types of radiation differ in their ability to penetrate material and damage tissue, with alpha particles the least penetrating but potentially most damaging and gamma rays the most penetrating.

Various devices, including Geiger counters, scintillators, and dosimeters, are used to detect and measure radiation, and monitor radiation exposure. We

use several units to measure radiation: becquerels or curies for rates of radioactive decay; gray or rads for energy absorbed; and rems or sieverts for biological effects of radiation. Exposure to radiation can cause a wide range of health effects, from minor to severe, and including death. We can minimize the effects of radiation by shielding with dense materials such as lead, moving away from the source, and limiting time of exposure.

Exercises

21.1 Nuclear Structure and Stability

- Write the following isotopes in hyphenated form (e.g., "carbon-14")
 - ${}^{24}_{11}\text{Na}$
 - ${}^{29}_{13}\text{Al}$
 - ${}^{73}_{36}\text{Kr}$
 - ${}^{194}_{77}\text{Ir}$
- Write the following isotopes in nuclide notation (e.g., " ${}^{14}_6\text{C}$ ")
 - oxygen-14
 - copper-70
 - tantalum-175
 - francium-217
- For the following isotopes that have missing information, fill in the missing information to complete the notation
 - ${}^{34}_{14}\text{X}$
 - ${}^{36}_{\text{X}}\text{P}$
 - ${}^{57}_{\text{X}}\text{Mn}$
 - ${}^{121}_{56}\text{X}$
- For each of the isotopes in [Exercise 21.1](#), determine the numbers of protons, neutrons, and electrons in a neutral atom of the isotope.
- Write the nuclide notation, including charge if applicable, for atoms with the following characteristics:
 - 25 protons, 20 neutrons, 24 electrons
 - 45 protons, 24 neutrons, 43 electrons
 - 53 protons, 89 neutrons, 54 electrons
 - 97 protons, 146 neutrons, 97 electrons
- Calculate the density of the ${}^{24}_{12}\text{Mg}$ nucleus in g/mL, assuming that it has the typical nuclear diameter of 1×10^{-13} cm and is spherical in shape.
- What are the two principal differences between nuclear reactions and ordinary chemical changes?
- The mass of the atom ${}^{23}_{11}\text{Na}$ is 22.9898 amu.
 - Calculate its binding energy per atom in millions of electron volts.
 - Calculate its binding energy per nucleon.
- Which of the following nuclei lie within the band of stability shown in [Figure 21.2](#)?
 - chlorine-37
 - calcium-40
 - ${}^{204}\text{Bi}$
 - ${}^{56}\text{Fe}$

- (e) ^{206}Pb
- (f) ^{211}Pb
- (g) ^{222}Rn
- (h) carbon-14

10. Which of the following nuclei lie within the band of stability shown in [Figure 21.2](#)?

- (a) argon-40
- (b) oxygen-16
- (c) ^{122}Ba
- (d) ^{58}Ni
- (e) ^{205}Tl
- (f) ^{210}Tl
- (g) ^{226}Ra
- (h) magnesium-24

21.2 Nuclear Equations

11. Write a brief description or definition of each of the following:

- (a) nucleon
- (b) α particle
- (c) β particle
- (d) positron
- (e) γ ray
- (f) nuclide
- (g) mass number
- (h) atomic number

12. Which of the various particles (α particles, β particles, and so on) that may be produced in a nuclear reaction are actually nuclei?

13. Complete each of the following equations by adding the missing species:

- (a) $^{27}_{13}\text{Al} + ^4_2\text{He} \longrightarrow ? + ^1_0\text{n}$
- (b) $^{239}_{94}\text{Pu} + ? \longrightarrow ^{242}_{96}\text{Cm} + ^1_0\text{n}$
- (c) $^{14}_7\text{N} + ^4_2\text{He} \longrightarrow ? + ^1_1\text{H}$
- (d) $^{235}_{92}\text{U} \longrightarrow ? + ^{135}_{55}\text{Cs} + 4^1_0\text{n}$

14. Complete each of the following equations:

- (a) $^7_3\text{Li} + ? \longrightarrow ^4_2\text{He}$
- (b) $^{14}_6\text{C} \longrightarrow ^{14}_7\text{N} + ?$
- (c) $^{27}_{13}\text{Al} + ^4_2\text{He} \longrightarrow ? + ^1_0\text{n}$
- (d) $^{250}_{96}\text{Cm} \longrightarrow ? + ^{98}_{38}\text{Sr} + 4^1_0\text{n}$

15. Write a balanced equation for each of the following nuclear reactions:

- (a) the production of ^{17}O from ^{14}N by α particle bombardment
- (b) the production of ^{14}C from ^{14}N by neutron bombardment
- (c) the production of ^{233}Th from ^{232}Th by neutron bombardment
- (d) the production of ^{239}U from ^{238}U by ^2_1H bombardment

16. Technetium-99 is prepared from ^{98}Mo . Molybdenum-98 combines with a neutron to give molybdenum-99, an unstable isotope that emits a β particle to yield an excited form of technetium-99, represented as $^{99}\text{Tc}^*$. This excited nucleus relaxes to the ground state, represented as ^{99}Tc , by emitting a γ ray. The ground state of ^{99}Tc then emits a β particle. Write the equations for each of these nuclear reactions.

17. The mass of the atom $^{19}_9\text{F}$ is 18.99840 amu.

- (a) Calculate its binding energy per atom in millions of electron volts.
- (b) Calculate its binding energy per nucleon.

18. For the reaction $^{14}_6\text{C} \longrightarrow ^{14}_7\text{N} + ?$, if 100.0 g of carbon reacts, what volume of nitrogen gas (N_2) is produced at 273K and 1 atm?

21.3 Radioactive Decay

19. What are the types of radiation emitted by the nuclei of radioactive elements?
20. What changes occur to the atomic number and mass of a nucleus during each of the following decay scenarios?
- an α particle is emitted
 - a β particle is emitted
 - γ radiation is emitted
 - a positron is emitted
 - an electron is captured
21. What is the change in the nucleus that results from the following decay scenarios?
- emission of a β particle
 - emission of a β^+ particle
 - capture of an electron
22. Many nuclides with atomic numbers greater than 83 decay by processes such as electron emission. Explain the observation that the emissions from these unstable nuclides also normally include α particles.
23. Why is electron capture accompanied by the emission of an X-ray?
24. Explain, in terms of [Figure 21.2](#), how unstable heavy nuclides (atomic number > 83) may decompose to form nuclides of greater stability (a) if they are below the band of stability and (b) if they are above the band of stability.
25. Which of the following nuclei is most likely to decay by positron emission? Explain your choice.
- chromium-53
 - manganese-51
 - iron-59
26. The following nuclei do not lie in the band of stability. How would they be expected to decay? Explain your answer.
- $^{34}_{15}\text{P}$
 - $^{239}_{92}\text{U}$
 - $^{38}_{20}\text{Ca}$
 - ^3_1H
 - $^{245}_{94}\text{Pu}$
27. The following nuclei do not lie in the band of stability. How would they be expected to decay?
- $^{28}_{15}\text{P}$
 - $^{235}_{92}\text{U}$
 - $^{37}_{20}\text{Ca}$
 - ^9_3Li
 - $^{245}_{96}\text{Cm}$
28. Predict by what mode(s) of spontaneous radioactive decay each of the following unstable isotopes might proceed:
- ^6_2He
 - $^{60}_{30}\text{Zn}$
 - $^{235}_{91}\text{Pa}$
 - $^{241}_{94}\text{Np}$
 - ^{18}F

- (f) ^{129}Ba
 (g) ^{237}Pu

29. Write a nuclear reaction for each step in the formation of $^{218}_{84}\text{Po}$ from $^{238}_{92}\text{U}$, which proceeds by a series of decay reactions involving the step-wise emission of α , β , β , α , α , α , α particles, in that order.
30. Write a nuclear reaction for each step in the formation of $^{208}_{82}\text{Pb}$ from $^{228}_{90}\text{Th}$, which proceeds by a series of decay reactions involving the step-wise emission of α , α , α , α , β , β , α particles, in that order.
31. Define the term half-life and illustrate it with an example.
32. A 1.00×10^{-6} -g sample of nobelium, $^{254}_{102}\text{No}$, has a half-life of 55 seconds after it is formed. What is the percentage of $^{254}_{102}\text{No}$ remaining at the following times?
 - (a) 5.0 min after it forms
 - (b) 1.0 h after it forms
33. ^{239}Pu is a nuclear waste byproduct with a half-life of 24,000 y. What fraction of the ^{239}Pu present today will be present in 1000 y?
34. The isotope ^{208}Tl undergoes β decay with a half-life of 3.1 min.
 - (a) What isotope is produced by the decay?
 - (b) How long will it take for 99.0% of a sample of pure ^{208}Tl to decay?
 - (c) What percentage of a sample of pure ^{208}Tl remains un-decayed after 1.0 h?
35. If 1.000 g of $^{226}_{88}\text{Ra}$ produces 0.0001 mL of the gas $^{222}_{86}\text{Rn}$ at STP (standard temperature and pressure) in 24 h, what is the half-life of ^{226}Ra in years?
36. The isotope $^{90}_{38}\text{Sr}$ is one of the extremely hazardous species in the residues from nuclear power generation. The strontium in a 0.500-g sample diminishes to 0.393 g in 10.0 y. Calculate the half-life.
37. Technetium-99 is often used for assessing heart, liver, and lung damage because certain technetium compounds are absorbed by damaged tissues. It has a half-life of 6.0 h. Calculate the rate constant for the decay of $^{99}_{43}\text{Tc}$.
38. What is the age of mummified primate skin that contains 8.25% of the original quantity of ^{14}C ?
39. A sample of rock was found to contain 8.23 mg of rubidium-87 and 0.47 mg of strontium-87.
 - (a) Calculate the age of the rock if the half-life of the decay of rubidium by β emission is 4.7×10^{10} y.
 - (b) If some $^{87}_{38}\text{Sr}$ was initially present in the rock, would the rock be younger, older, or the same age as the age calculated in (a)? Explain your answer.
40. A laboratory investigation shows that a sample of uranium ore contains 5.37 mg of $^{238}_{92}\text{U}$ and 2.52 mg of $^{206}_{82}\text{Pb}$. Calculate the age of the ore. The half-life of $^{238}_{92}\text{U}$ is 4.5×10^9 yr.
41. Plutonium was detected in trace amounts in natural uranium deposits by Glenn Seaborg and his associates in 1941. They proposed that the source of this ^{239}Pu was the capture of neutrons by ^{238}U nuclei. Why is this plutonium not likely to have been trapped at the time the solar system formed 4.7×10^9 years ago?
42. A ^7_4Be atom (mass = 7.0169 amu) decays into a ^7_3Li atom (mass = 7.0160 amu) by electron capture. How much energy (in millions of electron volts, MeV) is produced by this reaction?
43. A ^8_5B atom (mass = 8.0246 amu) decays into a ^8_4Be atom (mass = 8.0053 amu) by loss of a β^+ particle (mass = 0.00055 amu) or by electron capture. How much energy (in millions of electron volts) is produced by this reaction?
44. Isotopes such as ^{26}Al (half-life: 7.2×10^5 years) are believed to have been present in our solar system as it formed, but have since decayed and are now called extinct nuclides.
 - (a) ^{26}Al decays by β^+ emission or electron capture. Write the equations for these two nuclear transformations.

- (b) The earth was formed about 4.7×10^9 (4.7 billion) years ago. How old was the earth when 99.999999% of the ^{26}Al originally present had decayed?
45. Write a balanced equation for each of the following nuclear reactions:
- bismuth-212 decays into polonium-212
 - beryllium-8 and a positron are produced by the decay of an unstable nucleus
 - neptunium-239 forms from the reaction of uranium-238 with a neutron and then spontaneously converts into plutonium-239
 - strontium-90 decays into yttrium-90
46. Write a balanced equation for each of the following nuclear reactions:
- mercury-180 decays into platinum-176
 - zirconium-90 and an electron are produced by the decay of an unstable nucleus
 - thorium-232 decays and produces an alpha particle and a radium-228 nucleus, which decays into actinium-228 by beta decay
 - neon-19 decays into fluorine-19

21.4 Transmutation and Nuclear Energy

47. Write the balanced nuclear equation for the production of the following transuranium elements:
- berkelium-244, made by the reaction of Am-241 and He-4
 - fermium-254, made by the reaction of Pu-239 with a large number of neutrons
 - lawrencium-257, made by the reaction of Cf-250 and B-11
 - dubnium-260, made by the reaction of Cf-249 and N-15
48. How does nuclear fission differ from nuclear fusion? Why are both of these processes exothermic?
49. Both fusion and fission are nuclear reactions. Why is a very high temperature required for fusion, but not for fission?
50. Cite the conditions necessary for a nuclear chain reaction to take place. Explain how it can be controlled to produce energy, but not produce an explosion.
51. Describe the components of a nuclear reactor.
52. In usual practice, both a moderator and control rods are necessary to operate a nuclear chain reaction safely for the purpose of energy production. Cite the function of each and explain why both are necessary.
53. Describe how the potential energy of uranium is converted into electrical energy in a nuclear power plant.
54. The mass of a hydrogen atom (^1_1H) is 1.007825 amu; that of a tritium atom (^3_1H) is 3.01605 amu; and that of an α particle is 4.00150 amu. How much energy in kilojoules per mole of ^4_2He produced is released by the following fusion reaction: $^1_1\text{H} + ^3_1\text{H} \longrightarrow ^4_2\text{He}$.

21.5 Uses of Radioisotopes

55. How can a radioactive nuclide be used to show that the equilibrium:
- $$\text{AgCl}(s) \rightleftharpoons \text{Ag}^+(aq) + \text{Cl}^-(aq)$$
- is a dynamic equilibrium?
56. Technetium-99m has a half-life of 6.01 hours. If a patient injected with technetium-99m is safe to leave the hospital once 75% of the dose has decayed, when is the patient allowed to leave?
57. Iodine that enters the body is stored in the thyroid gland from which it is released to control growth and metabolism. The thyroid can be imaged if iodine-131 is injected into the body. In larger doses, I-131 is also used as a means of treating cancer of the thyroid. I-131 has a half-life of 8.70 days and decays by β^- emission.
- Write an equation for the decay.
 - How long will it take for 95.0% of a dose of I-131 to decay?

21.6 Biological Effects of Radiation

- 58.** If a hospital were storing radioisotopes, what is the minimum containment needed to protect against:
- (a) cobalt-60 (a strong γ emitter used for irradiation)
 - (b) molybdenum-99 (a beta emitter used to produce technetium-99 for imaging)
- 59.** Based on what is known about Radon-222's primary decay method, why is inhalation so dangerous?
- 60.** Given specimens uranium-232 ($t_{1/2} = 68.9$ y) and uranium-233 ($t_{1/2} = 159,200$ y) of equal mass, which one would have greater activity and why?
- 61.** A scientist is studying a 2.234 g sample of thorium-229 ($t_{1/2} = 7340$ y) in a laboratory.
- (a) What is its activity in Bq?
 - (b) What is its activity in Ci?
- 62.** Given specimens neon-24 ($t_{1/2} = 3.38$ min) and bismuth-211 ($t_{1/2} = 2.14$ min) of equal mass, which one would have greater activity and why?

APPENDIX A

The Periodic Table

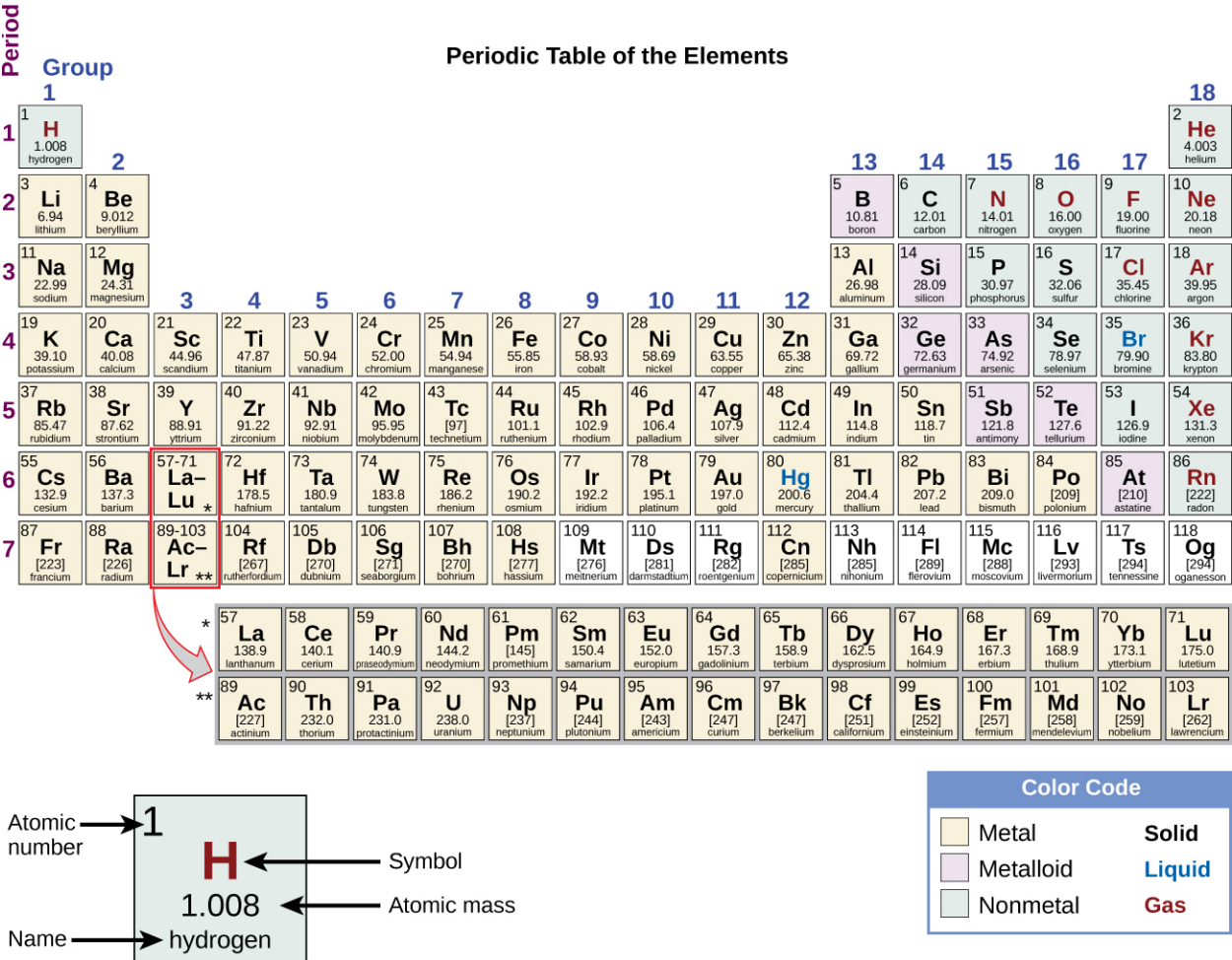


FIGURE A1

APPENDIX B

Essential Mathematics

Exponential Arithmetic

Exponential notation is used to express very large and very small numbers as a product of two numbers. The first number of the product, the *digit term*, is usually a number not less than 1 and not equal to or greater than 10. The second number of the product, the *exponential term*, is written as 10 with an exponent. Some examples of exponential notation are:

$$\begin{aligned} 1000 &= 1 \times 10^3 \\ 100 &= 1 \times 10^2 \\ 10 &= 1 \times 10^1 \\ 1 &= 1 \times 10^0 \\ 0.1 &= 1 \times 10^{-1} \\ 0.001 &= 1 \times 10^{-3} \\ 2386 &= 2.386 \times 1000 = 2.386 \times 10^3 \\ 0.123 &= 1.23 \times 0.1 = 1.23 \times 10^{-1} \end{aligned}$$

The power (exponent) of 10 is equal to the number of places the decimal is shifted to give the digit number. The exponential method is particularly useful notation for very large and very small numbers. For example, $1,230,000,000 = 1.23 \times 10^9$, and $0.00000000036 = 3.6 \times 10^{-10}$.

Addition of Exponentials

Convert all numbers to the same power of 10, add the digit terms of the numbers, and if appropriate, convert the digit term back to a number between 1 and 10 by adjusting the exponential term.



EXAMPLE B1

Adding Exponentials

Add 5.00×10^{-5} and 3.00×10^{-3} .

Solution

$$\begin{aligned} 3.00 \times 10^{-3} &= 300 \times 10^{-5} \\ (5.00 \times 10^{-5}) + (300 \times 10^{-5}) &= 305 \times 10^{-5} = 3.05 \times 10^{-3} \end{aligned}$$

Subtraction of Exponentials

Convert all numbers to the same power of 10, take the difference of the digit terms, and if appropriate, convert the digit term back to a number between 1 and 10 by adjusting the exponential term.



EXAMPLE B2

Subtracting Exponentials

Subtract 4.0×10^{-7} from 5.0×10^{-6} .

Solution

$$4.0 \times 10^{-7} = 0.40 \times 10^{-6}$$

$$(5.0 \times 10^{-6}) - (0.40 \times 10^{-6}) = 4.6 \times 10^{-6}$$

Multiplication of Exponentials

Multiply the digit terms in the usual way and add the exponents of the exponential terms.

**EXAMPLE B3****Multiplying Exponentials**

Multiply 4.2×10^{-8} by 2.0×10^3 .

Solution

$$(4.2 \times 10^{-8}) \times (2.0 \times 10^3) = (4.2 \times 2.0) \times 10^{(-8)+(3)} = 8.4 \times 10^{-5}$$

Division of Exponentials

Divide the digit term of the numerator by the digit term of the denominator and subtract the exponents of the exponential terms.

**EXAMPLE B4****Dividing Exponentials**

Divide 3.6×10^{-5} by 6.0×10^{-4} .

Solution

$$\frac{3.6 \times 10^{-5}}{6.0 \times 10^{-4}} = \left(\frac{3.6}{6.0} \right) \times 10^{(-5)-(-4)} = 0.60 \times 10^{-1} = 6.0 \times 10^{-2}$$

Squaring of Exponentials

Square the digit term in the usual way and multiply the exponent of the exponential term by 2.

**EXAMPLE B5****Squaring Exponentials**

Square the number 4.0×10^{-6} .

Solution

$$(4.0 \times 10^{-6})^2 = 4 \times 4 \times 10^{2 \times (-6)} = 16 \times 10^{-12} = 1.6 \times 10^{-11}$$

Cubing of Exponentials

Cube the digit term in the usual way and multiply the exponent of the exponential term by 3.

**EXAMPLE B6**

Cubing Exponentials

Cube the number 2×10^4 .

Solution

$$(2 \times 10^4)^3 = 2 \times 2 \times 2 \times 10^{3 \times 4} = 8 \times 10^{12}$$

Taking Square Roots of Exponentials

If necessary, decrease or increase the exponential term so that the power of 10 is evenly divisible by 2. Extract the square root of the digit term and divide the exponential term by 2.

**EXAMPLE B7****Finding the Square Root of Exponentials**

Find the square root of 1.6×10^{-7} .

Solution

$$\begin{aligned} 1.6 \times 10^{-7} &= 16 \times 10^{-8} \\ \sqrt{16 \times 10^{-8}} &= \sqrt{16} \times \sqrt{10^{-8}} = \sqrt{16} \times 10^{-\frac{8}{2}} = 4.0 \times 10^{-4} \end{aligned}$$

Significant Figures

A beekeeper reports that he has 525,341 bees. The last three figures of the number are obviously inaccurate, for during the time the keeper was counting the bees, some of them died and others hatched; this makes it quite difficult to determine the exact number of bees. It would have been more reasonable if the beekeeper had reported the number 525,000. In other words, the last three figures are not significant, except to set the position of the decimal point. Their exact values have no useful meaning in this situation. When reporting quantities, use only as many significant figures as the accuracy of the measurement warrants.

The importance of significant figures lies in their application to fundamental computation. In addition and subtraction, the sum or difference should contain as many digits to the right of the decimal as that in the least certain of the numbers used in the computation (indicated by underscoring in the following example).

**EXAMPLE B8****Addition and Subtraction with Significant Figures**

Add 4.383 g and 0.0023 g.

Solution

$$\begin{array}{r} 4.\underline{383} \text{ g} \\ 0.\underline{0023} \text{ g} \\ \hline 4.\underline{385} \text{ g} \end{array}$$

In multiplication and division, the product or quotient should contain no more digits than that in the factor containing the least number of significant figures.

**EXAMPLE B9**

Multiplication and Division with Significant Figures

Multiply 0.6238 by 6.6.

Solution

$$0.6238 \times 6.6 = 4.1$$

When rounding numbers, increase the retained digit by 1 if it is followed by a number larger than 5 (“round up”). Do not change the retained digit if the digits that follow are less than 5 (“round down”). If the retained digit is followed by 5, round up if the retained digit is odd, or round down if it is even (after rounding, the retained digit will thus always be even).

The Use of Logarithms and Exponential Numbers

The common logarithm of a number (log) is the power to which 10 must be raised to equal that number. For example, the common logarithm of 100 is 2, because 10 must be raised to the second power to equal 100. Additional examples follow.

Logarithms and Exponential Numbers

Number	Number Expressed Exponentially	Common Logarithm
1000	10^3	3
10	10^1	1
1	10^0	0
0.1	10^{-1}	-1
0.001	10^{-3}	-3

TABLE B1

What is the common logarithm of 60? Because 60 lies between 10 and 100, which have logarithms of 1 and 2, respectively, the logarithm of 60 is 1.7782; that is,

$$60 = 10^{1.7782}$$

The common logarithm of a number less than 1 has a negative value. The logarithm of 0.03918 is -1.4069, or

$$0.03918 = 10^{-1.4069} = \frac{1}{10^{1.4069}}$$

To obtain the common logarithm of a number, use the *log* button on your calculator. To calculate a number from its logarithm, take the inverse log of the logarithm, or calculate 10^x (where x is the logarithm of the number).

The natural logarithm of a number (ln) is the power to which e must be raised to equal the number; e is the constant 2.7182818. For example, the natural logarithm of 10 is 2.303; that is,

$$10 = e^{2.303} = 2.7182818^{2.303}$$

To obtain the natural logarithm of a number, use the *ln* button on your calculator. To calculate a number from its natural logarithm, enter the natural logarithm and take the inverse ln of the natural logarithm, or calculate e^x (where x is the natural logarithm of the number).

Logarithms are exponents; thus, operations involving logarithms follow the same rules as operations involving exponents.

1. The logarithm of a product of two numbers is the sum of the logarithms of the two numbers.

$$\log xy = \log x + \log y, \text{ and } \ln xy = \ln x + \ln y$$

2. The logarithm of the number resulting from the division of two numbers is the difference between the logarithms of the two numbers.

$$\log \frac{x}{y} = \log x - \log y, \text{ and } \ln \frac{x}{y} = \ln x - \ln y$$

3. The logarithm of a number raised to an exponent is the product of the exponent and the logarithm of the number.

$$\log x^n = n \log x \text{ and } \ln x^n = n \ln x$$

The Solution of Quadratic Equations

Mathematical functions of this form are known as second-order polynomials or, more commonly, quadratic functions.

$$ax^2 + bx + c = 0$$

The solution or roots for any quadratic equation can be calculated using the following formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



EXAMPLE B10

Solving Quadratic Equations

Solve the quadratic equation $3x^2 + 13x - 10 = 0$.

Solution

Substituting the values $a = 3$, $b = 13$, $c = -10$ in the formula, we obtain

$$\begin{aligned} x &= \frac{-13 \pm \sqrt{(13)^2 - 4 \times 3 \times (-10)}}{2 \times 3} \\ x &= \frac{-13 \pm \sqrt{169 + 120}}{6} = \frac{-13 \pm \sqrt{289}}{6} = \frac{-13 \pm 17}{6} \end{aligned}$$

The two roots are therefore

$$x = \frac{-13 + 17}{6} = \frac{2}{3} \text{ and } x = \frac{-13 - 17}{6} = -5$$

Quadratic equations constructed on physical data always have real roots, and of these real roots, often only those having positive values are of any significance.

Two-Dimensional (x-y) Graphing

The relationship between any two properties of a system can be represented graphically by a two-dimensional data plot. Such a graph has two axes: a horizontal one corresponding to the independent variable, or the variable whose value is being controlled (x), and a vertical axis corresponding to the dependent variable, or the variable whose value is being observed or measured (y).

When the value of y is changing as a function of x (that is, different values of x correspond to different values of y), a graph of this change can be plotted or sketched. The graph can be produced by using specific values for (x, y) data pairs.

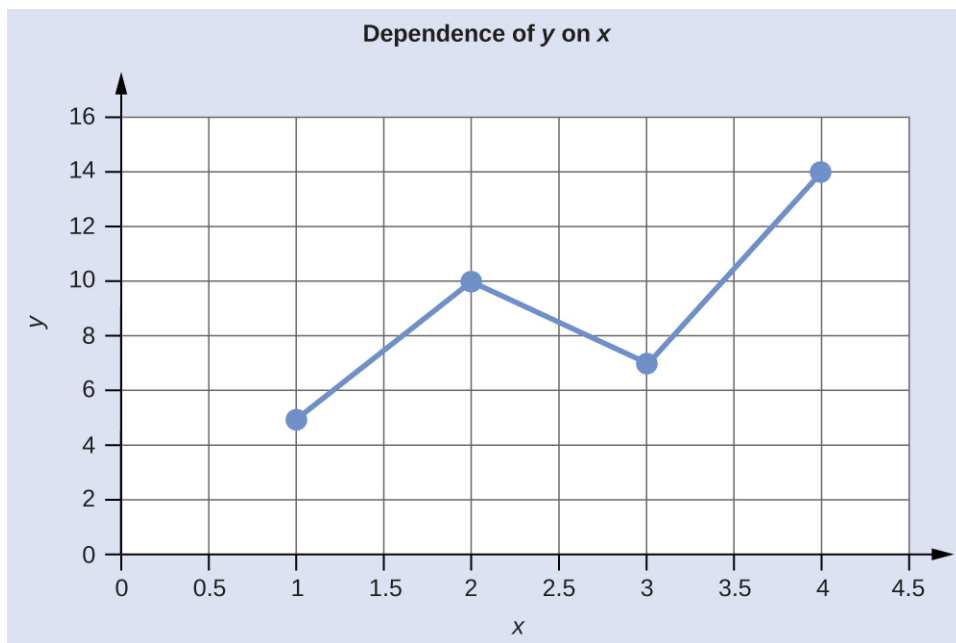


EXAMPLE B11

Graphing the Dependence of y on x

x	y
1	5
2	10
3	7
4	14

This table contains the following points: $(1,5)$, $(2,10)$, $(3,7)$, and $(4,14)$. Each of these points can be plotted on a graph and connected to produce a graphical representation of the dependence of y on x .



If the function that describes the dependence of y on x is known, it may be used to compute x,y data pairs that may subsequently be plotted.



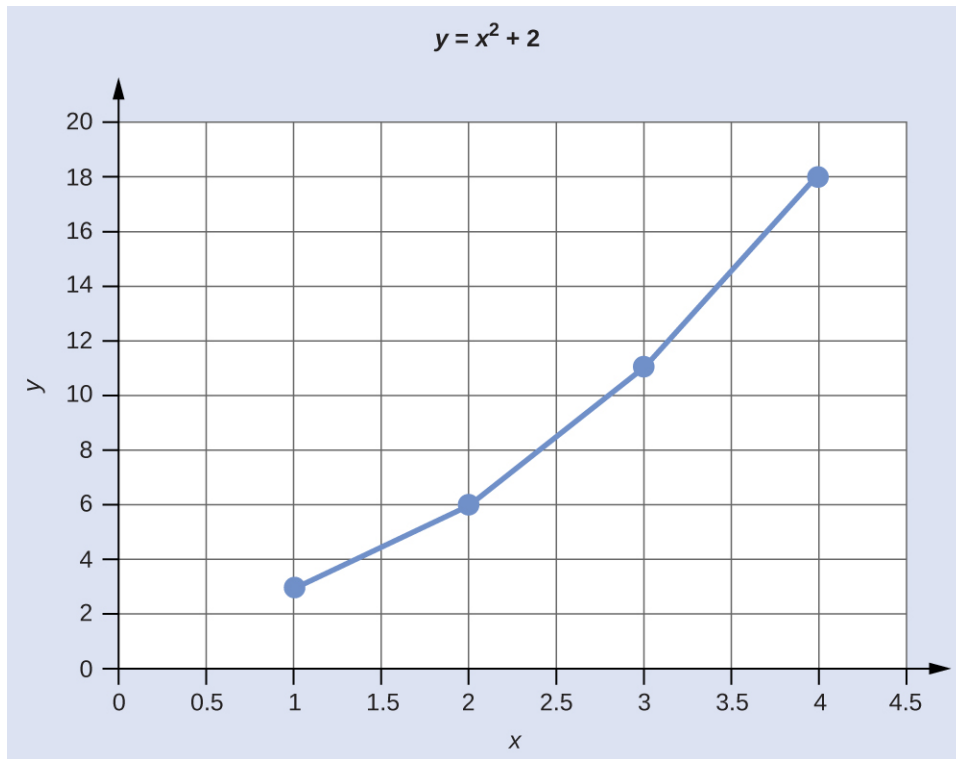
EXAMPLE B12

Plotting Data Pairs

If we know that $y = x^2 + 2$, we can produce a table of a few (x,y) values and then plot the line based on the data shown here.

x	$y = x^2 + 2$
1	3
2	6

x	$y = x^2 + 2$
3	11
4	18



APPENDIX C

Units and Conversion Factors

Units of Length

meter (m)	= 39.37 inches (in.) = 1.094 yards (yd)
centimeter (cm)	= 0.01 m (exact, definition)
millimeter (mm)	= 0.001 m (exact, definition)
kilometer (km)	= 1000 m (exact, definition)
angstrom (Å)	= 10^{-8} cm (exact, definition) = 10^{-10} m (exact, definition)
yard (yd)	= 0.9144 m
inch (in.)	= 2.54 cm (exact, definition)
mile (US)	= 1.60934 km

TABLE C1

Units of Volume

liter (L)	= 0.001 m ³ (exact, definition) = 1000 cm ³ (exact, definition) = 1.057 (US) quarts
milliliter (mL)	= 0.001 L (exact, definition) = 1 cm ³ (exact, definition)
microliter (μL)	= 10^{-6} L (exact, definition) = 10^{-3} cm ³ (exact, definition)
liquid quart (US)	= 32 (US) liquid ounces (exact, definition) = 0.25 (US) gallon (exact, definition) = 0.9463 L
dry quart	= 1.1012 L
cubic foot (US)	= 28.316 L

TABLE C2

Units of Mass

gram (g)	= 0.001 kg (exact, definition)
milligram (mg)	= 0.001 g (exact, definition)
kilogram (kg)	= 1000 g (exact, definition) = 2.205 lb ¹
ton (metric)	= 1000 kg (exact, definition) = 2204.62 lb
ounce (oz)	= 28.35 g
pound (lb)	= 0.4535924 kg
ton (short)	= 2000 lb (exact, definition) = 907.185 kg
ton (long)	= 2240 lb (exact, definition) = 1.016 metric ton

TABLE C3

Units of Energy

4.184 joule (J)	= 1 thermochemical calorie (cal)
1 thermochemical calorie (cal)	= 4.184×10^7 erg
erg	= 10^{-7} J (exact, definition)
electron-volt (eV)	= 1.60218×10^{-19} J = 23.061 kcal mol ⁻¹
liter•atmosphere	= 24.217 cal = 101.325 J (exact, definition)
nutritional calorie (Cal)	= 1000 cal (exact, definition) = 4184 J
British thermal unit (BTU)	= 1054.804 J ²

TABLE C4

Units of Pressure

torr	= 1 mm Hg (exact, definition)
pascal (Pa)	= N m ⁻² (exact, definition) = kg m ⁻¹ s ⁻² (exact, definition)

TABLE C5

1 Strictly speaking, the ounce and pound are units of weight, W (a force equal to the product of mass and gravitational acceleration, $W = mg$). The conversion relations in this table are commonly used to equate masses and weight assuming a nominal value for g at the surface of the earth.

2 BTU is the amount of energy needed to heat one pound of water by one degree Fahrenheit. Therefore, the exact relationship of BTU to joules and other energy units depends on the temperature at which BTU is measured. 59 °F (15 °C) is the most widely used reference temperature for BTU definition in the United States. At this temperature, the conversion factor is the one provided in this table.

atmosphere (atm)	= 760 mm Hg (exact, definition) = 760 torr (exact, definition) = 101,325 N m ⁻² (exact, definition) = 101,325 Pa (exact, definition)
bar	= 10 ⁵ Pa (exact, definition) = 10 ⁵ kg m ⁻¹ s ⁻² (exact, definition)

TABLE C5

APPENDIX D

Fundamental Physical Constants

Fundamental Physical Constants

Name and Symbol	Value
atomic mass unit (amu)	$1.6605402 \times 10^{-27} \text{ kg}$
Avogadro's number	$6.02214076 \times 10^{23} \text{ mol}^{-1}$
Boltzmann's constant (k)	$1.380649 \times 10^{-23} \text{ J K}^{-1}$
charge-to-mass ratio for electron (e/m_e)	$1.75881962 \times 10^{11} \text{ C kg}^{-1}$
fundamental unit of charge (e)	$1.602176634 \times 10^{-19} \text{ C}$
electron rest mass (m_e)	$9.1093897 \times 10^{-31} \text{ kg}$
Faraday's constant (F)	$9.6485309 \times 10^4 \text{ C mol}^{-1}$
gas constant (R)	$8.205784 \times 10^{-2} \text{ L atm mol}^{-1} \text{ K}^{-1} = 8.314510 \text{ J mol}^{-1} \text{ K}^{-1}$
molar volume of an ideal gas, 1 atm, 0 °C	$22.41409 \text{ L mol}^{-1}$
molar volume of an ideal gas, 1 bar, 0 °C	$22.71108 \text{ L mol}^{-1}$
neutron rest mass (m_n)	$1.6749274 \times 10^{-27} \text{ kg}$
Planck's constant (h)	$6.62607015 \times 10^{-34} \text{ J s}$
proton rest mass (m_p)	$1.6726231 \times 10^{-27} \text{ kg}$
Rydberg constant (R)	$1.0973731534 \times 10^7 \text{ m}^{-1} = 2.1798736 \times 10^{-18} \text{ J}$
speed of light (in vacuum) (c)	$2.99792458 \times 10^8 \text{ m s}^{-1}$

TABLE D1

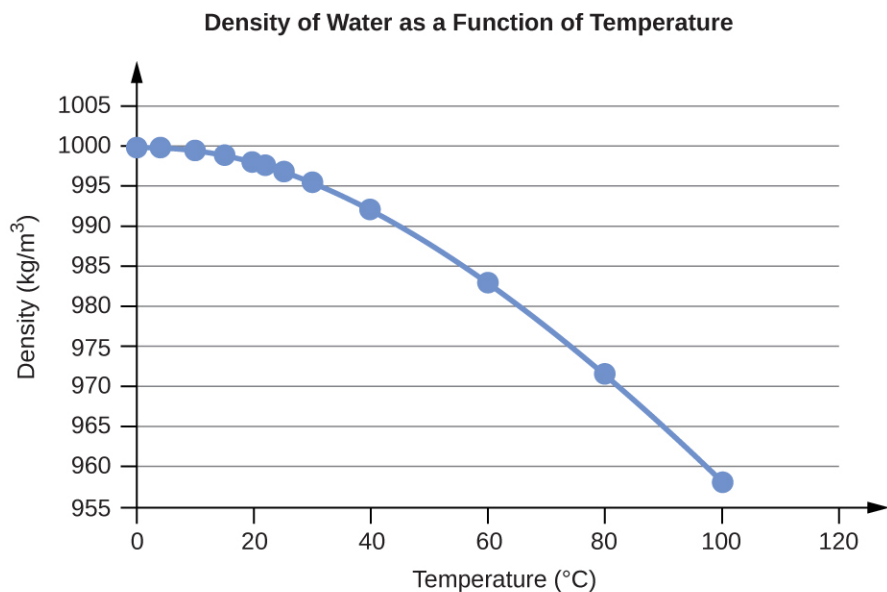
APPENDIX E

Water Properties

Water Density (g/mL) at Different Temperatures (°C)

Temperature	Density (g/mL)
0	0.9998395
4	0.9999720 (density maximum)
10	0.9997026
15	0.9991026
20	0.9982071
22	0.9977735
25	0.9970479
30	0.9956502
40	0.9922
60	0.9832
80	0.9718
100	0.9584

TABLE E1



Water Vapor Pressure at Different Temperatures (°C)

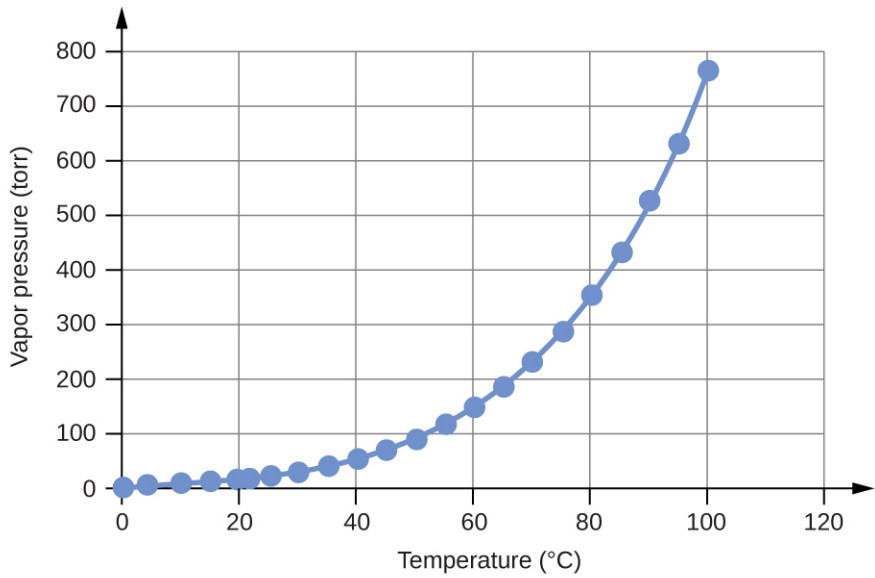
Temperature	Vapor Pressure (torr)	Vapor Pressure (Pa)
0	4.6	613.2812
4	6.1	813.2642
10	9.2	1226.562
15	12.8	1706.522
20	17.5	2333.135
22	19.8	2639.776
25	23.8	3173.064
30	31.8	4239.64
35	42.2	5626.188
40	55.3	7372.707
45	71.9	9585.852
50	92.5	12332.29
55	118.0	15732
60	149.4	19918.31

TABLE E2

Temperature	Vapor Pressure (torr)	Vapor Pressure (Pa)
65	187.5	24997.88
70	233.7	31157.35
75	289.1	38543.39
80	355.1	47342.64
85	433.6	57808.42
90	525.8	70100.71
95	633.9	84512.82
100	760.0	101324.7

TABLE E2

Vapor Pressure as a Function of Temperature



Water K_w and pK_w at Different Temperatures (°C)

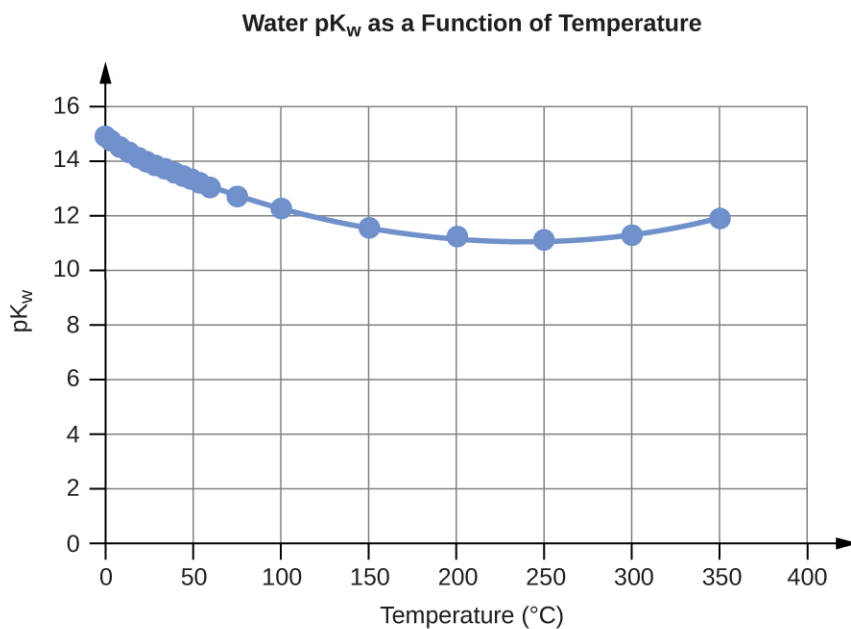
Temperature	$K_w \cdot 10^{-14}$	pK_w ¹
0	0.112	14.95
5	0.182	14.74
10	0.288	14.54

TABLE E3

1 $pK_w = -\log_{10}(K_w)$

Temperature	$K_w \cdot 10^{-14}$	pK_w
15	0.465	14.33
20	0.671	14.17
25	0.991	14.00
30	1.432	13.84
35	2.042	13.69
40	2.851	13.55
45	3.917	13.41
50	5.297	13.28
55	7.080	13.15
60	9.311	13.03
75	19.95	12.70
100	56.23	12.25

TABLE E3



Specific Heat Capacity for Water

$$C^{\circ}(\text{H}_2\text{O}(l)) = 4.184 \text{ J}\cdot\text{g}^{-1}\cdot^{\circ}\text{C}^{-1}$$

$$C^{\circ}(\text{H}_2\text{O}(s)) = 2.093 \text{ J}\cdot\text{g}^{-1}\cdot^{\circ}\text{C}^{-1}$$

$$C^{\circ}(\text{H}_2\text{O}(g)) = 1.864 \text{ J}\cdot\text{g}^{-1}\cdot^{\circ}\text{C}^{-1}$$

TABLE E4

Standard Water Melting and Boiling Temperatures and Enthalpies of the Transitions

	Temperature (K)	ΔH (kJ/mol)
melting	273.15	6.088
boiling	373.15	40.656 (44.016 at 298 K)

TABLE E5

Water Cryoscopic (Freezing Point Depression) and Ebullioscopic (Boiling Point Elevation) Constants

$$K_f = 1.86^{\circ}\text{C}\cdot\text{kg}\cdot\text{mol}^{-1} \text{ (cryoscopic constant)}$$

$$K_b = 0.51^{\circ}\text{C}\cdot\text{kg}\cdot\text{mol}^{-1} \text{ (ebullioscopic constant)}$$

TABLE E6

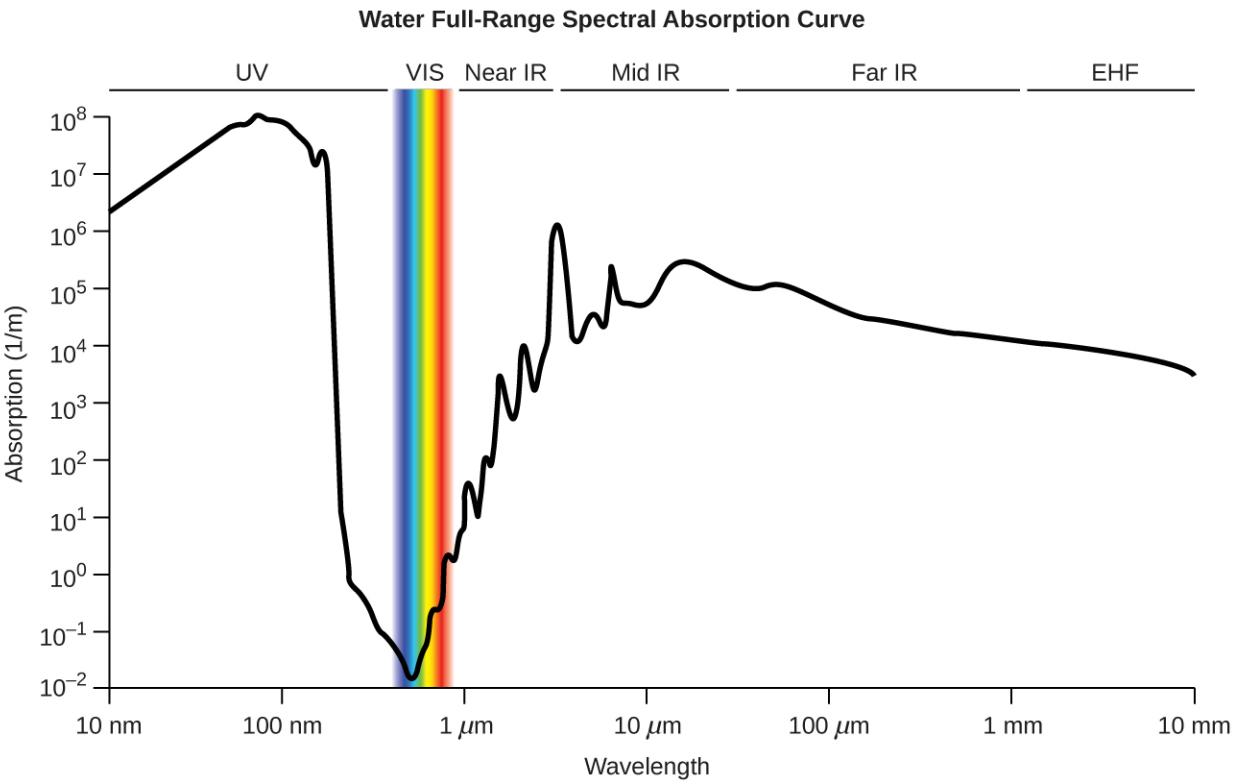


FIGURE E1 The plot shows the extent of light absorption versus wavelength for water. Absorption is reported in reciprocal meters and corresponds to the inverse of the distance light may travel through water before its intensity is diminished by $1/e$ (~37%).

APPENDIX F

Composition of Commercial Acids and Bases

Composition of Commercial Acids and Bases

Acid or Base ¹	Density (g/mL) ²	Percentage by Mass	Molarity
acetic acid, glacial	1.05	99.5%	17.4
aqueous ammonia ³	0.90	28%	14.8
hydrochloric acid	1.18	36%	11.6
nitric acid	1.42	71%	16.0
perchloric acid	1.67	70%	11.65
phosphoric acid	1.70	85%	14.7
sodium hydroxide	1.53	50%	19.1
sulfuric acid	1.84	96%	18.0

TABLE F1

1 Acids and bases are commercially available as aqueous solutions. This table lists properties (densities and concentrations) of common acid and base solutions. Nominal values are provided in cases where the manufacturer cites a range of concentrations and densities.

2 This column contains specific gravity data. In the case of this table, specific gravity is the ratio of density of a substance to the density of pure water at the same conditions. Specific gravity is often cited on commercial labels.

3 This solution is sometimes called “ammonium hydroxide,” although this term is not chemically accurate.

APPENDIX G

Standard Thermodynamic Properties for Selected Substances

Standard Thermodynamic Properties for Selected Substances

Substance	ΔH_f° (kJ mol ⁻¹)	ΔG_f° (kJ mol ⁻¹)	S° (J K ⁻¹ mol ⁻¹)
aluminum			
Al(s)	0	0	28.3
Al(g)	324.4	285.7	164.54
Al ³⁺ (aq)	-531	-485	-321.7
Al ₂ O ₃ (s)	-1676	-1582	50.92
AlF ₃ (s)	-1510.4	-1425	66.5
AlCl ₃ (s)	-704.2	-628.8	110.67
AlCl ₃ ·6H ₂ O(s)	-2691.57	-2269.40	376.56
Al ₂ S ₃ (s)	-724.0	-492.4	116.9
Al ₂ (SO ₄) ₃ (s)	-3445.06	-3506.61	239.32
antimony			
Sb(s)	0	0	45.69
Sb(g)	262.34	222.17	180.16
Sb ₄ O ₆ (s)	-1440.55	-1268.17	220.92
SbCl ₃ (g)	-313.8	-301.2	337.80
SbCl ₅ (g)	-394.34	-334.29	401.94
Sb ₂ S ₃ (s)	-174.89	-173.64	182.00
SbCl ₃ (s)	-382.17	-323.72	184.10
SbOCl(s)	-374.0	—	—
arsenic			

TABLE G1

Substance	ΔH_f° (kJ mol ⁻¹)	ΔG_f° (kJ mol ⁻¹)	S° (J K ⁻¹ mol ⁻¹)
As(s)	0	0	35.1
As(g)	302.5	261.0	174.21
As ₄ (g)	143.9	92.4	314
As ₄ O ₆ (s)	-1313.94	-1152.52	214.22
As ₂ O ₅ (s)	-924.87	-782.41	105.44
AsCl ₃ (g)	-261.50	-248.95	327.06
As ₂ S ₃ (s)	-169.03	-168.62	163.59
AsH ₃ (g)	66.44	68.93	222.78
H ₃ AsO ₄ (s)	-906.3	—	—
barium			
Ba(s)	0	0	62.5
Ba(g)	180	146	170.24
Ba ²⁺ (aq)	-537.6	-560.8	9.6
BaO(s)	-548.0	-520.3	72.1
BaCl ₂ (s)	-855.0	-806.7	123.7
BaSO ₄ (s)	-1473.2	-1362.3	132.2
beryllium			
Be(s)	0	0	9.50
Be(g)	324.3	286.6	136.27
BeO(s)	-609.4	-580.1	13.8
bismuth			
Bi(s)	0	0	56.74
Bi(g)	207.1	168.2	187.00
Bi ₂ O ₃ (s)	-573.88	-493.7	151.5
BiCl ₃ (s)	-379.07	-315.06	176.98

TABLE G1

Substance	ΔH_f° (kJ mol ⁻¹)	ΔG_f° (kJ mol ⁻¹)	S° (J K ⁻¹ mol ⁻¹)
Bi ₂ S ₃ (s)	-143.1	-140.6	200.4
boron			
B(s)	0	0	5.86
B(g)	565.0	521.0	153.4
B ₂ O ₃ (s)	-1273.5	-1194.3	53.97
B ₂ H ₆ (g)	36.4	87.6	232.1
H ₃ BO ₃ (s)	-1094.33	-968.92	88.83
BF ₃ (g)	-1136.0	-1119.4	254.4
BCl ₃ (g)	-403.8	-388.7	290.1
B ₃ N ₃ H ₆ (l)	-540.99	-392.79	199.58
HBO ₂ (s)	-794.25	-723.41	37.66
bromine			
Br ₂ (l)	0	0	152.23
Br ₂ (g)	30.91	3.142	245.5
Br(g)	111.88	82.429	175.0
Br ⁻ (aq)	-120.9	-102.82	80.71
BrF ₃ (g)	-255.60	-229.45	292.42
HBr(g)	-36.3	-53.43	198.7
cadmium			
Cd(s)	0	0	51.76
Cd(g)	112.01	77.41	167.75
Cd ²⁺ (aq)	-75.90	-77.61	-73.2
CdO(s)	-258.2	-228.4	54.8
CdCl ₂ (s)	-391.5	-343.9	115.3
CdSO ₄ (s)	-933.3	-822.7	123.0

TABLE G1

Substance	ΔH_f° (kJ mol ⁻¹)	ΔG_f° (kJ mol ⁻¹)	S° (J K ⁻¹ mol ⁻¹)
CdS(s)	-161.9	-156.5	64.9
calcium			
Ca(s)	0	0	41.6
Ca(g)	178.2	144.3	154.88
Ca ²⁺ (aq)	-542.96	-553.04	-55.2
CaO(s)	-634.9	-603.3	38.1
Ca(OH) ₂ (s)	-985.2	-897.5	83.4
CaSO ₄ (s)	-1434.5	-1322.0	106.5
CaSO ₄ ·2H ₂ O(s)	-2022.63	-1797.45	194.14
CaCO ₃ (s) (calcite)	-1220.0	-1081.4	110.0
CaSO ₃ ·H ₂ O(s)	-1752.68	-1555.19	184.10
carbon			
C(s) (graphite)	0	0	5.740
C(s) (diamond)	1.89	2.90	2.38
C(g)	716.681	671.2	158.1
CO(g)	-110.52	-137.15	197.7
CO ₂ (g)	-393.51	-394.36	213.8
CO ₃ ²⁻ (aq)	-677.1	-527.8	-56.9
CH ₄ (g)	-74.6	-50.5	186.3
CH ₃ OH(l)	-239.2	-166.6	126.8
CH ₃ OH(g)	-201.0	-162.3	239.9
CCl ₄ (l)	-128.2	-62.5	214.4
CCl ₄ (g)	-95.7	-58.2	309.7
CHCl ₃ (l)	-134.1	-73.7	201.7
CHCl ₃ (g)	-103.14	-70.34	295.71

TABLE G1

Substance	ΔH_f° (kJ mol ⁻¹)	ΔG_f° (kJ mol ⁻¹)	S° (J K ⁻¹ mol ⁻¹)
CS ₂ (l)	89.70	65.27	151.34
CS ₂ (g)	116.9	66.8	238.0
C ₂ H ₂ (g)	227.4	209.2	200.9
C ₂ H ₄ (g)	52.4	68.4	219.3
C ₂ H ₆ (g)	-84.0	-32.0	229.2
CH ₃ CO ₂ H(l)	-484.3	-389.9	159.8
CH ₃ CO ₂ H(g)	-434.84	-376.69	282.50
C ₂ H ₅ OH(l)	-277.6	-174.8	160.7
C ₂ H ₅ OH(g)	-234.8	-167.9	281.6
HCO ₃ ⁻ (aq)	-691.11	-587.06	95
C ₃ H ₈ (g)	-103.8	-23.4	270.3
C ₆ H ₆ (g)	82.927	129.66	269.2
C ₆ H ₆ (l)	49.1	124.50	173.4
CH ₂ Cl ₂ (l)	-124.2	-63.2	177.8
CH ₂ Cl ₂ (g)	-95.4	-65.90	270.2
CH ₃ Cl(g)	-81.9	-60.2	234.6
C ₂ H ₅ Cl(l)	-136.52	-59.31	190.79
C ₂ H ₅ Cl(g)	-112.17	-60.39	276.00
C ₂ N ₂ (g)	308.98	297.36	241.90
HCN(l)	108.9	125.0	112.8
HCN(g)	135.5	124.7	201.8
cesium			
Cs ⁺ (aq)	-248	-282.0	133
chlorine			
Cl ₂ (g)	0	0	223.1

TABLE G1

Substance	ΔH_f° (kJ mol ⁻¹)	ΔG_f° (kJ mol ⁻¹)	S° (J K ⁻¹ mol ⁻¹)
Cl(g)	121.3	105.70	165.2
Cl ⁻ (aq)	-167.2	-131.2	56.5
ClF(g)	-54.48	-55.94	217.78
ClF ₃ (g)	-158.99	-118.83	281.50
Cl ₂ O(g)	80.3	97.9	266.2
Cl ₂ O ₇ (l)	238.1	—	—
Cl ₂ O ₇ (g)	272.0	—	—
HCl(g)	-92.307	-95.299	186.9
HClO ₄ (l)	-40.58	—	—
chromium			
Cr(s)	0	0	23.77
Cr(g)	396.6	351.8	174.50
CrO ₄ ²⁻ (aq)	-881.2	-727.8	50.21
Cr ₂ O ₇ ²⁻ (aq)	-1490.3	-1301.1	261.9
Cr ₂ O ₃ (s)	-1139.7	-1058.1	81.2
CrO ₃ (s)	-589.5	—	—
(NH ₄) ₂ Cr ₂ O ₇ (s)	-1806.7	—	—
cobalt			
Co(s)	0	0	30.0
Co ²⁺ (aq)	-67.4	-51.5	-155
Co ³⁺ (aq)	92	134	-305.0
CoO(s)	-237.9	-214.2	52.97
Co ₃ O ₄ (s)	-910.02	-794.98	114.22
Co(NO ₃) ₂ (s)	-420.5	—	—
copper			

TABLE G1

Substance	ΔH_f° (kJ mol ⁻¹)	ΔG_f° (kJ mol ⁻¹)	S° (J K ⁻¹ mol ⁻¹)
Cu(s)	0	0	33.15
Cu(g)	338.32	298.58	166.38
Cu ⁺ (aq)	51.9	50.2	-26
Cu ²⁺ (aq)	64.77	65.49	-99.6
CuO(s)	-157.3	-129.7	42.63
Cu ₂ O(s)	-168.6	-146.0	93.14
CuS(s)	-53.1	-53.6	66.5
Cu ₂ S(s)	-79.5	-86.2	120.9
CuSO ₄ (s)	-771.36	-662.2	109.2
Cu(NO ₃) ₂ (s)	-302.9	—	—
fluorine			
F ₂ (g)	0	0	202.8
F(g)	79.4	62.3	158.8
F ⁻ (aq)	-332.6	-278.8	-13.8
F ₂ O(g)	24.7	41.9	247.43
HF(g)	-273.3	-275.4	173.8
hydrogen			
H ₂ (g)	0	0	130.7
H(g)	217.97	203.26	114.7
H ⁺ (aq)	0	0	0
OH ⁻ (aq)	-230.0	-157.2	-10.75
H ₃ O ⁺ (aq)	-285.8		69.91
H ₂ O(l)	-285.83	-237.1	70.0
H ₂ O(g)	-241.82	-228.59	188.8
H ₂ O ₂ (l)	-187.78	-120.35	109.6

TABLE G1

Substance	ΔH_f° (kJ mol ⁻¹)	ΔG_f° (kJ mol ⁻¹)	S° (J K ⁻¹ mol ⁻¹)
H ₂ O ₂ (g)	-136.3	-105.6	232.7
HF(g)	-273.3	-275.4	173.8
HCl(g)	-92.307	-95.299	186.9
HBr(g)	-36.3	-53.43	198.7
HI(g)	26.48	1.70	206.59
H ₂ S(g)	-20.6	-33.4	205.8
H ₂ Se(g)	29.7	15.9	219.0
HNO ₃	-206.64	—	—
iodine			
I ₂ (s)	0	0	116.14
I ₂ (g)	62.438	19.3	260.7
I(g)	106.84	70.2	180.8
I ⁻ (aq)	-55.19	-51.57	11.13
IF(g)	95.65	-118.49	236.06
ICl(g)	17.78	-5.44	247.44
IBr(g)	40.84	3.72	258.66
IF ₇ (g)	-943.91	-818.39	346.44
HI(g)	26.48	1.70	206.59
iron			
Fe(s)	0	0	27.3
Fe(g)	416.3	370.7	180.5
Fe ²⁺ (aq)	-89.1	-78.90	-137.7
Fe ³⁺ (aq)	-48.5	-4.7	-315.9
Fe ₂ O ₃ (s)	-824.2	-742.2	87.40
Fe ₃ O ₄ (s)	-1118.4	-1015.4	146.4

TABLE G1

Substance	ΔH_f° (kJ mol ⁻¹)	ΔG_f° (kJ mol ⁻¹)	S° (J K ⁻¹ mol ⁻¹)
Fe(CO) ₅ (l)	-774.04	-705.42	338.07
Fe(CO) ₅ (g)	-733.87	-697.26	445.18
FeCl ₂ (s)	-341.79	-302.30	117.95
FeCl ₃ (s)	-399.49	-334.00	142.3
FeO(s)	-272.0	-255.2	60.75
Fe(OH) ₂ (s)	-569.0	-486.5	88.
Fe(OH) ₃ (s)	-823.0	-696.5	106.7
FeS(s)	-100.0	-100.4	60.29
Fe ₃ C(s)	25.10	20.08	104.60
lead			
Pb(s)	0	0	64.81
Pb(g)	195.2	162.	175.4
Pb ²⁺ (aq)	-1.7	-24.43	10.5
PbO(s) (yellow)	-217.32	-187.89	68.70
PbO(s) (red)	-218.99	-188.93	66.5
Pb(OH) ₂ (s)	-515.9	—	—
PbS(s)	-100.4	-98.7	91.2
Pb(NO ₃) ₂ (s)	-451.9	—	—
PbO ₂ (s)	-277.4	-217.3	68.6
PbCl ₂ (s)	-359.4	-314.1	136.0
lithium			
Li(s)	0	0	29.1
Li(g)	159.3	126.6	138.8
Li ⁺ (aq)	-278.5	-293.3	13.4
LiH(s)	-90.5	-68.3	20.0

TABLE G1

Substance	ΔH_f° (kJ mol ⁻¹)	ΔG_f° (kJ mol ⁻¹)	S° (J K ⁻¹ mol ⁻¹)
Li(OH)(s)	-487.5	-441.5	42.8
LiF(s)	-616.0	-587.5	35.7
Li ₂ CO ₃ (s)	-1216.04	-1132.19	90.17
magnesium			
Mg ²⁺ (aq)	-466.9	-454.8	-138.1
manganese			
Mn(s)	0	0	32.0
Mn(g)	280.7	238.5	173.7
Mn ²⁺ (aq)	-220.8	-228.1	-73.6
MnO(s)	-385.2	-362.9	59.71
MnO ₂ (s)	-520.03	-465.1	53.05
Mn ₂ O ₃ (s)	-958.97	-881.15	110.46
Mn ₃ O ₄ (s)	-1378.83	-1283.23	155.64
MnO ₄ ⁻ (aq)	-541.4	-447.2	191.2
MnO ₄ ²⁻ (aq)	-653.0	-500.7	59
mercury			
Hg(l)	0	0	75.9
Hg(g)	61.4	31.8	175.0
Hg ²⁺ (aq)		164.8	
Hg ²⁺ (aq)	172.4	153.9	84.5
HgO(s) (red)	-90.83	-58.5	70.29
HgO(s) (yellow)	-90.46	-58.43	71.13
HgCl ₂ (s)	-224.3	-178.6	146.0
Hg ₂ Cl ₂ (s)	-265.4	-210.7	191.6
HgS(s) (red)	-58.16	-50.6	82.4

TABLE G1

Substance	ΔH_f° (kJ mol ⁻¹)	ΔG_f° (kJ mol ⁻¹)	S° (J K ⁻¹ mol ⁻¹)
HgS(s) (black)	-53.56	-47.70	88.28
HgSO ₄ (s)	-707.51	-594.13	0.00
nickel			
Ni ²⁺ (aq)	-64.0	-46.4	-159
nitrogen			
N ₂ (g)	0	0	191.6
N(g)	472.704	455.5	153.3
NO(g)	90.25	87.6	210.8
NO ₂ (g)	33.2	51.30	240.1
N ₂ O(g)	81.6	103.7	220.0
N ₂ O ₃ (g)	83.72	139.41	312.17
NO ₃ ⁻ (aq)	-205.0	-108.7	146.4
N ₂ O ₄ (g)	11.1	99.8	304.4
N ₂ O ₅ (g)	11.3	115.1	355.7
NH ₃ (g)	-45.9	-16.5	192.8
NH ₄ ⁺ (aq)	-132.5	-79.31	113.4
N ₂ H ₄ (l)	50.63	149.43	121.21
N ₂ H ₄ (g)	95.4	159.4	238.5
NH ₄ NO ₃ (s)	-365.56	-183.87	151.08
NH ₄ Cl(s)	-314.43	-202.87	94.6
NH ₄ Br(s)	-270.8	-175.2	113.0
NH ₄ I(s)	-201.4	-112.5	117.0
NH ₄ NO ₂ (s)	-256.5	—	—
HNO ₃ (l)	-174.1	-80.7	155.6
HNO ₃ (g)	-133.9	-73.5	266.9

TABLE G1

Substance	ΔH_f° (kJ mol ⁻¹)	ΔG_f° (kJ mol ⁻¹)	S° (J K ⁻¹ mol ⁻¹)
HNO ₃ (aq)	-207.4	-110.5	146
oxygen			
O ₂ (g)	0	0	205.2
O(g)	249.17	231.7	161.1
O ₃ (g)	142.7	163.2	238.9
phosphorus			
P ₄ (s)	0	0	164.4
P ₄ (g)	58.91	24.4	280.0
P(g)	314.64	278.25	163.19
PH ₃ (g)	5.4	13.5	210.2
PCl ₃ (g)	-287.0	-267.8	311.78
PCl ₅ (g)	-374.9	-305.0	364.4
P ₄ O ₆ (s)	-1640.1	—	—
P ₄ O ₁₀ (s)	-2984.0	-2697.0	228.86
PO ₄ ³⁻ (aq)	-1277	-1019	-222
HPO ₃ (s)	-948.5	—	—
HPO ₄ ²⁻ (aq)	-1292.1	-1089.3	-33
H ₂ PO ₄ ²⁻ (aq)	-1296.3	-1130.4	90.4
H ₃ PO ₂ (s)	-604.6	—	—
H ₃ PO ₃ (s)	-964.4	—	—
H ₃ PO ₄ (s)	-1279.0	-1119.1	110.50
H ₃ PO ₄ (l)	-1266.9	-1124.3	110.5
H ₄ P ₂ O ₇ (s)	-2241.0	—	—
POCl ₃ (l)	-597.1	-520.8	222.5
POCl ₃ (g)	-558.5	-512.9	325.5

TABLE G1

Substance	ΔH_f° (kJ mol ⁻¹)	ΔG_f° (kJ mol ⁻¹)	S° (J K ⁻¹ mol ⁻¹)
potassium			
K(s)	0	0	64.7
K(g)	89.0	60.5	160.3
K ⁺ (aq)	-252.4	-283.3	102.5
KF(s)	-576.27	-537.75	66.57
KCl(s)	-436.5	-408.5	82.6
rubidium			
Rb ⁺ (aq)	-246	-282.2	124
silicon			
Si(s)	0	0	18.8
Si(g)	450.0	405.5	168.0
SiO ₂ (s)	-910.7	-856.3	41.5
SiH ₄ (g)	34.3	56.9	204.6
H ₂ SiO ₃ (s)	-1188.67	-1092.44	133.89
H ₄ SiO ₄ (s)	-1481.14	-1333.02	192.46
SiF ₄ (g)	-1615.0	-1572.8	282.8
SiCl ₄ (l)	-687.0	-619.8	239.7
SiCl ₄ (g)	-662.75	-622.58	330.62
SiC(s, <i>beta cubic</i>)	-73.22	-70.71	16.61
SiC(s, <i>alpha hexagonal</i>)	-71.55	-69.04	16.48
silver			
Ag(s)	0	0	42.55
Ag(g)	284.9	246.0	172.89
Ag ⁺ (aq)	105.6	77.11	72.68
Ag ₂ O(s)	-31.05	-11.20	121.3

TABLE G1

Substance	ΔH_f° (kJ mol ⁻¹)	ΔG_f° (kJ mol ⁻¹)	S° (J K ⁻¹ mol ⁻¹)
AgCl(s)	-127.0	-109.8	96.3
Ag ₂ S(s)	-32.6	-40.7	144.0
sodium			
Na(s)	0	0	51.3
Na(g)	107.5	77.0	153.7
Na ⁺ (aq)	-240.1	-261.9	59
Na ₂ O(s)	-414.2	-375.5	75.1
NaCl(s)	-411.2	-384.1	72.1
strontium			
Sr ²⁺ (aq)	-545.8	-557.3	-32.6
sulfur			
S ₈ (s) (rhombic)	0	0	256.8
S(g)	278.81	238.25	167.82
S ²⁻ (aq)	41.8	83.7	22
SO ₂ (g)	-296.83	-300.1	248.2
SO ₃ (g)	-395.72	-371.06	256.76
SO ₄ ²⁻ (aq)	-909.3	-744.5	20.1
S ₂ O ₃ ²⁻ (aq)	-648.5	-522.5	67
H ₂ S(g)	-20.6	-33.4	205.8
HS ⁻ (aq)	-17.7	12.6	61.1
H ₂ SO ₄ (l)	-813.989	-690.00	156.90
HSO ₄ ⁻ (aq)	-885.75	-752.87	126.9
H ₂ S ₂ O ₇ (s)	-1273.6	—	—
SF ₄ (g)	-728.43	-684.84	291.12
SF ₆ (g)	-1220.5	-1116.5	291.5

TABLE G1

Substance	ΔH_f° (kJ mol ⁻¹)	ΔG_f° (kJ mol ⁻¹)	S° (J K ⁻¹ mol ⁻¹)
SCl ₂ (l)	-50	—	—
SCl ₂ (g)	-19.7	—	—
S ₂ Cl ₂ (l)	-59.4	—	—
S ₂ Cl ₂ (g)	-19.50	-29.25	319.45
SOCl ₂ (g)	-212.55	-198.32	309.66
SOCl ₂ (l)	-245.6	—	—
SO ₂ Cl ₂ (l)	-394.1	—	—
SO ₂ Cl ₂ (g)	-354.80	-310.45	311.83
tin			
Sn(s)	0	0	51.2
Sn(g)	301.2	266.2	168.5
SnO(s)	-285.8	-256.9	56.5
SnO ₂ (s)	-577.6	-515.8	49.0
SnCl ₄ (l)	-511.3	-440.1	258.6
SnCl ₄ (g)	-471.5	-432.2	365.8
titanium			
Ti(s)	0	0	30.7
Ti(g)	473.0	428.4	180.3
TiO ₂ (s)	-944.0	-888.8	50.6
TiCl ₄ (l)	-804.2	-737.2	252.4
TiCl ₄ (g)	-763.2	-726.3	353.2
tungsten			
W(s)	0	0	32.6
W(g)	849.4	807.1	174.0
WO ₃ (s)	-842.9	-764.0	75.9

TABLE G1

Substance	ΔH_f° (kJ mol ⁻¹)	ΔG_f° (kJ mol ⁻¹)	S° (J K ⁻¹ mol ⁻¹)
zinc			
Zn(s)	0	0	41.6
Zn(g)	130.73	95.14	160.98
Zn ²⁺ (aq)	-153.9	-147.1	-112.1
ZnO(s)	-350.5	-320.5	43.7
ZnCl ₂ (s)	-415.1	-369.43	111.5
ZnS(s)	-206.0	-201.3	57.7
ZnSO ₄ (s)	-982.8	-871.5	110.5
ZnCO ₃ (s)	-812.78	-731.57	82.42
complexes			
[Co(NH ₃) ₄ (NO ₂) ₂]NO ₃ , <i>cis</i>	-898.7	—	—
[Co(NH ₃) ₄ (NO ₂) ₂]NO ₃ , <i>trans</i>	-896.2	—	—
NH ₄ [Co(NH ₃) ₂ (NO ₂) ₄]	-837.6	—	—
[Co(NH ₃) ₆][Co(NH ₃) ₂ (NO ₂) ₄] ₃	-2733.0	—	—
[Co(NH ₃) ₄ Cl ₂]Cl, <i>cis</i>	-874.9	—	—
[Co(NH ₃) ₄ Cl ₂]Cl, <i>trans</i>	-877.4	—	—
[Co(en) ₂ (NO ₂) ₂]NO ₃ , <i>cis</i>	-689.5	—	—
[Co(en) ₂ Cl ₂]Cl, <i>cis</i>	-681.2	—	—
[Co(en) ₂ Cl ₂]Cl, <i>trans</i>	-677.4	—	—
[Co(en) ₃](ClO ₄) ₃	-762.7	—	—
[Co(en) ₃]Br ₂	-595.8	—	—
[Co(en) ₃]I ₂	-475.3	—	—
[Co(en) ₃]I ₃	-519.2	—	—
[Co(NH ₃) ₆](ClO ₄) ₃	-1034.7	-221.1	615
[Co(NH ₃) ₅ NO ₂](NO ₃) ₂	-1088.7	-412.9	331

TABLE G1

Substance	ΔH_f° (kJ mol ⁻¹)	ΔG_f° (kJ mol ⁻¹)	S° (J K ⁻¹ mol ⁻¹)
[Co(NH ₃) ₆](NO ₃) ₃	-1282.0	-524.5	448
[Co(NH ₃) ₅ Cl]Cl ₂	-1017.1	-582.5	366.1
[Pt(NH ₃) ₄]Cl ₂	-725.5	—	—
[Ni(NH ₃) ₆]Cl ₂	-994.1	—	—
[Ni(NH ₃) ₆]Br ₂	-923.8	—	—
[Ni(NH ₃) ₆]I ₂	-808.3	—	—

TABLE G1

APPENDIX H

Ionization Constants of Weak Acids

Ionization Constants of Weak Acids

Acid	Formula	K_a at 25 °C	Lewis Structure
acetic	$\text{CH}_3\text{CO}_2\text{H}$	1.8×10^{-5}	
arsenic	H_3AsO_4	5.5×10^{-3}	
	H_2AsO_4^-	1.7×10^{-7}	
	HAsO_4^{2-}	3.0×10^{-12}	
arsenous	H_3AsO_3	5.1×10^{-10}	
boric	H_3BO_3	5.4×10^{-10}	
carbonic	H_2CO_3	4.3×10^{-7}	
	HCO_3^-	4.7×10^{-11}	
formic	HCO_2H	1.8×10^{-4}	
hydrazoic	HN_3	2.5×10^{-5}	
hydrocyanic	HCN	4.9×10^{-10}	

TABLE H1

Acid	Formula	K_a at 25 °C	Lewis Structure
hydrofluoric	HF	6.4×10^{-4}	
hydrogen peroxide	H ₂ O ₂	2.4×10^{-12}	
hydrogen selenide	H ₂ Se	1.29×10^{-4}	
	HSe ⁻	1×10^{-12}	
hydrogen sulfate ion	HSO ₄ ⁻	1.2×10^{-2}	
hydrogen sulfide	H ₂ S	8.9×10^{-8}	
	HS ⁻	1.0×10^{-19}	
hydrogen telluride	H ₂ Te	2.3×10^{-3}	
	HTe ⁻	1.6×10^{-11}	
hypobromous	HBrO	2.8×10^{-9}	
hypochlorous	HClO	2.9×10^{-8}	
isocyanic	HCNO	2×10^{-4}	
nitrous	HNO ₂	4.6×10^{-4}	
oxalic	H ₂ C ₂ O ₄	6.0×10^{-2}	
	HC ₂ O ₄ ⁻	6.1×10^{-5}	
phosphoric	H ₃ PO ₄	7.5×10^{-3}	
	H ₂ PO ₄ ⁻	6.2×10^{-8}	
	HPO ₄ ²⁻	4.2×10^{-13}	

TABLE H1

Acid	Formula	K_a at 25 °C	Lewis Structure
phosphorous	H_3PO_3	5×10^{-2}	
	H_2PO_3^-	2.0×10^{-7}	
sulfurous	H_2SO_3	1.6×10^{-2}	
	HSO_3^-	6.4×10^{-8}	

TABLE H1

APPENDIX I

Ionization Constants of Weak Bases

Ionization Constants of Weak Bases

Base	Lewis Structure	K_b at 25 °C
ammonia	$\begin{array}{c} \text{H} - \ddot{\text{N}} - \text{H} \\ \\ \text{H} \end{array}$	1.8×10^{-5}
dimethylamine	$\begin{array}{ccccc} & \text{H} & & \text{H} & & \text{H} \\ & & & & & \\ \text{H} - & \text{C} & - & \text{N} & - & \text{C} & - \text{H} \\ & & & \ddot{\text{N}} & & \\ & \text{H} & & & & \text{H} \end{array}$	5.9×10^{-4}
methylamine	$\begin{array}{ccc} & \text{H} & \\ & & \\ \text{H} - & \text{C} & - \ddot{\text{N}} - \text{H} \\ & & \\ & \text{H} & \text{H} \end{array}$	4.4×10^{-4}
phenylamine (aniline)	$\begin{array}{c} \text{H} \\ \\ \text{H} - \text{C} - \text{C} = \text{C} - \text{H} \\ \quad \\ \text{H} - \text{C} \quad \text{C} = \text{C} - \text{H} \\ \\ \text{H} - \text{N} - \text{H} \\ \\ \text{H} \end{array}$	4.3×10^{-10}
trimethylamine	$\begin{array}{ccccc} & \text{H} & & & \\ & & & & \\ \text{H} - & \text{C} & - & \text{H} & \\ & & & & \\ \text{H} - & \text{C} & - & \text{N} & - \text{C} - \text{H} \\ & \ddot{\text{N}} & & & \\ \text{H} & & & \text{H} & \end{array}$	6.3×10^{-5}

TABLE I1

APPENDIX J

Solubility Products

Solubility Products

Substance	K_{sp} at 25 °C
aluminum	
$\text{Al}(\text{OH})_3$	2×10^{-32}
barium	
BaCO_3	1.6×10^{-9}
$\text{BaC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$	1.1×10^{-7}
BaSO_4	2.3×10^{-8}
BaCrO_4	8.5×10^{-11}
BaF_2	2.4×10^{-5}
$\text{Ba}(\text{OH})_2 \cdot 8\text{H}_2\text{O}$	5.0×10^{-3}
$\text{Ba}_3(\text{PO}_4)_2$	6×10^{-39}
$\text{Ba}_3(\text{AsO}_4)_2$	1.1×10^{-13}
bismuth	
$\text{BiO}(\text{OH})$	4×10^{-10}
BiOCl	1.8×10^{-31}
Bi_2S_3	1×10^{-97}
cadmium	
$\text{Cd}(\text{OH})_2$	5.9×10^{-15}
CdS	1.0×10^{-28}
CdCO_3	5.2×10^{-12}
calcium	

TABLE J1

Substance	K_{sp} at 25 °C
Ca(OH)_2	1.3×10^{-6}
CaCO_3	8.7×10^{-9}
$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$	6.1×10^{-5}
$\text{CaC}_2\text{O}_4 \cdot \text{H}_2\text{O}$	1.96×10^{-9}
$\text{Ca}_3(\text{PO}_4)_2$	1.3×10^{-32}
CaHPO_4	7×10^{-7}
CaF_2	4.0×10^{-11}
chromium	
Cr(OH)_3	6.7×10^{-31}
cobalt	
Co(OH)_2	2.5×10^{-16}
$\text{CoS}(\alpha)$	5×10^{-22}
$\text{CoS}(\beta)$	3×10^{-26}
CoCO_3	1.4×10^{-13}
Co(OH)_3	2.5×10^{-43}
copper	
CuCl	1.2×10^{-6}
CuBr	6.27×10^{-9}
CuI	1.27×10^{-12}
CuSCN	1.6×10^{-11}
Cu_2S	2.5×10^{-48}
Cu(OH)_2	2.2×10^{-20}
CuS	8.5×10^{-45}
CuCO_3	2.5×10^{-10}
iron	

TABLE J1

Substance	K_{sp} at 25 °C
Fe(OH)_2	1.8×10^{-15}
FeCO_3	2.1×10^{-11}
FeS	3.7×10^{-19}
Fe(OH)_3	4×10^{-38}
lead	
Pb(OH)_2	1.2×10^{-15}
PbF_2	4×10^{-8}
PbCl_2	1.6×10^{-5}
PbBr_2	4.6×10^{-6}
PbI_2	1.4×10^{-8}
PbCO_3	1.5×10^{-15}
PbS	7×10^{-29}
PbCrO_4	2×10^{-16}
PbSO_4	1.3×10^{-8}
$\text{Pb}_3(\text{PO}_4)_2$	1×10^{-54}
magnesium	
Mg(OH)_2	8.9×10^{-12}
$\text{MgCO}_3 \cdot 3\text{H}_2\text{O}$	$ca 1 \times 10^{-5}$
MgNH_4PO_4	3×10^{-13}
MgF_2	6.4×10^{-9}
MgC_2O_4	7×10^{-7}
manganese	
Mn(OH)_2	2×10^{-13}
MnCO_3	8.8×10^{-11}
MnS	2.3×10^{-13}

TABLE J1

Substance	K_{sp} at 25 °C
mercury	
$\text{Hg}_2\text{O} \cdot \text{H}_2\text{O}$	3.6×10^{-26}
Hg_2Cl_2	1.1×10^{-18}
Hg_2Br_2	1.3×10^{-22}
Hg_2I_2	4.5×10^{-29}
Hg_2CO_3	9×10^{-15}
Hg_2SO_4	7.4×10^{-7}
Hg_2S	1.0×10^{-47}
Hg_2CrO_4	2×10^{-9}
HgS	1.6×10^{-54}
nickel	
$\text{Ni}(\text{OH})_2$	1.6×10^{-16}
NiCO_3	1.4×10^{-7}
$\text{NiS}(\alpha)$	4×10^{-20}
$\text{NiS}(\beta)$	1.3×10^{-25}
potassium	
KClO_4	1.05×10^{-2}
K_2PtCl_6	7.48×10^{-6}
$\text{KHC}_4\text{H}_4\text{O}_6$	3×10^{-4}
silver	
$\frac{1}{2}\text{Ag}_2\text{O}(\text{Ag}^+ + \text{OH}^-)$	2×10^{-8}
AgCl	1.6×10^{-10}
AgBr	5.0×10^{-13}
AgI	1.5×10^{-16}
AgCN	1.2×10^{-16}

TABLE J1

Substance	K_{sp} at 25 °C
AgSCN	1.0×10^{-12}
Ag ₂ S	1.6×10^{-49}
Ag ₂ CO ₃	8.1×10^{-12}
Ag ₂ CrO ₄	9.0×10^{-12}
Ag ₄ Fe(CN) ₆	1.55×10^{-41}
Ag ₂ SO ₄	1.2×10^{-5}
Ag ₃ PO ₄	1.8×10^{-18}
strontium	
Sr(OH) ₂ ·8H ₂ O	3.2×10^{-4}
SrCO ₃	7×10^{-10}
SrCrO ₄	3.6×10^{-5}
SrSO ₄	3.2×10^{-7}
SrC ₂ O ₄ ·H ₂ O	4×10^{-7}
thallium	
TlCl	1.7×10^{-4}
TlSCN	1.6×10^{-4}
Tl ₂ S	6×10^{-22}
Tl(OH) ₃	6.3×10^{-46}
tin	
Sn(OH) ₂	3×10^{-27}
SnS	1×10^{-26}
Sn(OH) ₄	1.0×10^{-57}
zinc	
ZnCO ₃	2×10^{-10}

TABLE J1

APPENDIX K

Formation Constants for Complex Ions

Formation Constants for Complex Ions

Equilibrium	K_f
$\text{Al}^{3+} + 6\text{F}^- \rightleftharpoons [\text{AlF}_6]^{3-}$	7×10^{19}
$\text{Cd}^{2+} + 4\text{NH}_3 \rightleftharpoons [\text{Cd}(\text{NH}_3)_4]^{2+}$	1.3×10^7
$\text{Cd}^{2+} + 4\text{CN}^- \rightleftharpoons [\text{Cd}(\text{CN})_4]^{2-}$	3×10^{18}
$\text{Co}^{2+} + 6\text{NH}_3 \rightleftharpoons [\text{Co}(\text{NH}_3)_6]^{2+}$	1.3×10^5
$\text{Co}^{3+} + 6\text{NH}_3 \rightleftharpoons [\text{Co}(\text{NH}_3)_6]^{3+}$	2.3×10^{33}
$\text{Cu}^+ + 2\text{CN}^- \rightleftharpoons [\text{Cu}(\text{CN})_2]^-$	1.0×10^{16}
$\text{Cu}^{2+} + 4\text{NH}_3 \rightleftharpoons [\text{Cu}(\text{NH}_3)_4]^{2+}$	1.7×10^{13}
$\text{Fe}^{2+} + 6\text{CN}^- \rightleftharpoons [\text{Fe}(\text{CN})_6]^{4-}$	1.5×10^{35}
$\text{Fe}^{3+} + 6\text{CN}^- \rightleftharpoons [\text{Fe}(\text{CN})_6]^{3-}$	2×10^{43}
$\text{Fe}^{3+} + 6\text{SCN}^- \rightleftharpoons [\text{Fe}(\text{SCN})_6]^{3-}$	3.2×10^3
$\text{Hg}^{2+} + 4\text{Cl}^- \rightleftharpoons [\text{HgCl}_4]^{2-}$	1.1×10^{16}
$\text{Ni}^{2+} + 6\text{NH}_3 \rightleftharpoons [\text{Ni}(\text{NH}_3)_6]^{2+}$	2.0×10^8
$\text{Ag}^+ + 2\text{Cl}^- \rightleftharpoons [\text{AgCl}_2]^-$	1.8×10^5
$\text{Ag}^+ + 2\text{CN}^- \rightleftharpoons [\text{Ag}(\text{CN})_2]^-$	1×10^{21}
$\text{Ag}^+ + 2\text{NH}_3 \rightleftharpoons [\text{Ag}(\text{NH}_3)_2]^+$	1.7×10^7
$\text{Zn}^{2+} + 4\text{CN}^- \rightleftharpoons [\text{Zn}(\text{CN})_4]^{2-}$	2.1×10^{19}
$\text{Zn}^{2+} + 4\text{OH}^- \rightleftharpoons [\text{Zn}(\text{OH})_4]^{2-}$	2×10^{15}
$\text{Fe}^{3+} + \text{SCN}^- \rightleftharpoons [\text{Fe}(\text{SCN})]^{2+}$	8.9×10^2

TABLE K1

Equilibrium	K_f
$\text{Ag}^+ + 4\text{SCN}^- \rightleftharpoons [\text{Ag}(\text{SCN})_4]^{3-}$	1.2×10^{10}
$\text{Pb}^{2+} + 4\text{I}^- \rightleftharpoons [\text{PbI}_4]^{2-}$	3.0×10^4
$\text{Pt}^{2+} + 4\text{Cl}^- \rightleftharpoons [\text{PtCl}_4]^{2-}$	1×10^{16}
$\text{Cu}^{2+} + 4\text{CN}^- \rightleftharpoons [\text{Cu}(\text{CN})_4]^{2-}$	1.0×10^{25}
$\text{Co}^{2+} + 4\text{SCN}^- \rightleftharpoons [\text{Co}(\text{SCN})_4]^{2-}$	1×10^3

TABLE K1

APPENDIX L

Standard Electrode (Half-Cell) Potentials

Standard Electrode (Half-Cell) Potentials

Half-Reaction	E° (V)
$\text{Ag}^+ + \text{e}^- \longrightarrow \text{Ag}$	+0.7996
$\text{AgCl} + \text{e}^- \longrightarrow \text{Ag} + \text{Cl}^-$	+0.22233
$[\text{Ag}(\text{CN})_2]^- + \text{e}^- \longrightarrow \text{Ag} + 2\text{CN}^-$	-0.31
$\text{Ag}_2\text{CrO}_4 + 2\text{e}^- \longrightarrow 2\text{Ag} + \text{CrO}_4^{2-}$	+0.45
$[\text{Ag}(\text{NH}_3)_2]^+ + \text{e}^- \longrightarrow \text{Ag} + 2\text{NH}_3$	+0.373
$[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-} + \text{e}^- \longrightarrow \text{Ag} + 2\text{S}_2\text{O}_3^{2-}$	+0.017
$[\text{AlF}_6]^{3-} + 3\text{e}^- \longrightarrow \text{Al} + 6\text{F}^-$	-2.07
$\text{Al}^{3+} + 3\text{e}^- \longrightarrow \text{Al}$	-1.662
$\text{Am}^{3+} + 3\text{e}^- \longrightarrow \text{Am}$	-2.048
$\text{Au}^{3+} + 3\text{e}^- \longrightarrow \text{Au}$	+1.498
$\text{Au}^+ + \text{e}^- \longrightarrow \text{Au}$	+1.692
$\text{Ba}^{2+} + 2\text{e}^- \longrightarrow \text{Ba}$	-2.912
$\text{Be}^{2+} + 2\text{e}^- \longrightarrow \text{Be}$	-1.847
$\text{Br}_2(\text{aq}) + 2\text{e}^- \longrightarrow 2\text{Br}^-$	+1.0873
$\text{Ca}^{2+} + 2\text{e}^- \longrightarrow \text{Ca}$	-2.868
$\text{Ce}^3 + 3\text{e}^- \longrightarrow \text{Ce}$	-2.483
$\text{Ce}^{4+} + \text{e}^- \longrightarrow \text{Ce}^{3+}$	+1.61
$\text{Cd}^{2+} + 2\text{e}^- \longrightarrow \text{Cd}$	-0.4030
$[\text{Cd}(\text{CN})_4]^{2-} + 2\text{e}^- \longrightarrow \text{Cd} + 4\text{CN}^-$	-1.09

TABLE L1

Half-Reaction	E° (V)
$[\text{Cd}(\text{NH}_3)_4]^{2+} + 2e^- \rightarrow \text{Cd} + 4\text{NH}_3$	-0.61
$\text{CdS} + 2e^- \rightarrow \text{Cd} + \text{S}^{2-}$	-1.17
$\text{Cl}_2 + 2e^- \rightarrow 2\text{Cl}^-$	+1.35827
$\text{ClO}_4^- + \text{H}_2\text{O} + 2e^- \rightarrow \text{ClO}_3^- + 2\text{OH}^-$	+0.36
$\text{ClO}_3^- + \text{H}_2\text{O} + 2e^- \rightarrow \text{ClO}_2^- + 2\text{OH}^-$	+0.33
$\text{ClO}_2^- + \text{H}_2\text{O} + 2e^- \rightarrow \text{ClO}^- + 2\text{OH}^-$	+0.66
$\text{ClO}^- + \text{H}_2\text{O} + 2e^- \rightarrow \text{Cl}^- + 2\text{OH}^-$	+0.89
$\text{ClO}_4^- + 2\text{H}_3\text{O}^+ + 2e^- \rightarrow \text{ClO}_3^- + 3\text{H}_2\text{O}$	+1.189
$\text{ClO}_3^- + 3\text{H}_3\text{O}^+ + 2e^- \rightarrow \text{HClO}_2 + 4\text{H}_2\text{O}$	+1.21
$\text{HClO} + \text{H}_3\text{O}^+ + 2e^- \rightarrow \text{Cl}^- + 2\text{H}_2\text{O}$	+1.482
$\text{HClO} + \text{H}_3\text{O}^+ + e^- \rightarrow \frac{1}{2}\text{Cl}_2 + 2\text{H}_2\text{O}$	+1.611
$\text{HClO}_2 + 2\text{H}_3\text{O}^+ + 2e^- \rightarrow \text{HClO} + 3\text{H}_2\text{O}$	+1.628
$\text{Co}^{3+} + e^- \rightarrow \text{Co}^{2+}$ (2 mol // H_2SO_4)	+1.83
$\text{Co}^{2+} + 2e^- \rightarrow \text{Co}$	-0.28
$[\text{Co}(\text{NH}_3)_6]^{3+} + e^- \rightarrow [\text{Co}(\text{NH}_3)_6]^{2+}$	+0.1
$\text{Co}(\text{OH})_3 + e^- \rightarrow \text{Co}(\text{OH})_2 + \text{OH}^-$	+0.17
$\text{Cr}^3 + 3e^- \rightarrow \text{Cr}$	-0.744
$\text{Cr}^{3+} + e^- \rightarrow \text{Cr}^{2+}$	-0.407
$\text{Cr}^{2+} + 2e^- \rightarrow \text{Cr}$	-0.913
$[\text{Cu}(\text{CN})_2]^- + e^- \rightarrow \text{Cu} + 2\text{CN}^-$	-0.43
$\text{CrO}_4^{2-} + 4\text{H}_2\text{O} + 3e^- \rightarrow \text{Cr}(\text{OH})_3 + 5\text{OH}^-$	-0.13
$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}_3\text{O}^+ + 6e^- \rightarrow 2\text{Cr}^{3+} + 21\text{H}_2\text{O}$	+1.232
$[\text{Cr}(\text{OH})_4]^- + 3e^- \rightarrow \text{Cr} + 4\text{OH}^-$	-1.2
$\text{Cr}(\text{OH})_3 + 3e^- \rightarrow \text{Cr} + 3\text{OH}^-$	-1.48

TABLE L1

Half-Reaction	E° (V)
$\text{Cu}^{2+} + \text{e}^- \longrightarrow \text{Cu}^+$	+0.153
$\text{Cu}^{2+} + 2\text{e}^- \longrightarrow \text{Cu}$	+0.337
$\text{Cu}^+ + \text{e}^- \longrightarrow \text{Cu}$	+0.521
$\text{F}_2 + 2\text{e}^- \longrightarrow 2\text{F}^-$	+2.866
$\text{Fe}^{2+} + 2\text{e}^- \longrightarrow \text{Fe}$	-0.447
$\text{Fe}^{3+} + \text{e}^- \longrightarrow \text{Fe}^{2+}$	+0.771
$[\text{Fe}(\text{CN})_6]^{3-} + \text{e}^- \longrightarrow [\text{Fe}(\text{CN})_6]^{4-}$	+0.36
$\text{Fe}(\text{OH})_2 + 2\text{e}^- \longrightarrow \text{Fe} + 2\text{OH}^-$	-0.88
$\text{FeS} + 2\text{e}^- \longrightarrow \text{Fe} + \text{S}^{2-}$	-1.01
$\text{Ga}^{3+} + 3\text{e}^- \longrightarrow \text{Ga}$	-0.549
$\text{Gd}^{3+} + 3\text{e}^- \longrightarrow \text{Gd}$	-2.279
$\frac{1}{2}\text{H}_2 + \text{e}^- \longrightarrow \text{H}^-$	-2.23
$2\text{H}_2\text{O} + 2\text{e}^- \longrightarrow \text{H}_2 + 2\text{OH}^-$	-0.8277
$\text{H}_2\text{O}_2 + 2\text{H}_3\text{O}^+ + 2\text{e}^- \longrightarrow 4\text{H}_2\text{O}$	+1.776
$2\text{H}_3\text{O}^+ + 2\text{e}^- \longrightarrow \text{H}_2 + 2\text{H}_2\text{O}$	0.00
$\text{HO}_2^- + \text{H}_2\text{O} + 2\text{e}^- \longrightarrow 3\text{OH}^-$	+0.878
$\text{Hf}^{4+} + 4\text{e}^- \longrightarrow \text{Hf}$	-1.55
$\text{Hg}^{2+} + 2\text{e}^- \longrightarrow \text{Hg}$	+0.851
$2\text{Hg}^{2+} + 2\text{e}^- \longrightarrow \text{Hg}_2^{2+}$	+0.92
$\text{Hg}_2^{2+} + 2\text{e}^- \longrightarrow 2\text{Hg}$	+0.7973
$[\text{HgBr}_4]^{2-} + 2\text{e}^- \longrightarrow \text{Hg} + 4\text{Br}^-$	+0.21
$\text{Hg}_2\text{Cl}_2 + 2\text{e}^- \longrightarrow 2\text{Hg} + 2\text{Cl}^-$	+0.26808
$[\text{Hg}(\text{CN})_4]^{2-} + 2\text{e}^- \longrightarrow \text{Hg} + 4\text{CN}^-$	-0.37
$[\text{HgI}_4]^{2-} + 2\text{e}^- \longrightarrow \text{Hg} + 4\text{I}^-$	-0.04

TABLE L1

Half-Reaction	E° (V)
$\text{HgS} + 2\text{e}^- \longrightarrow \text{Hg} + \text{S}^{2-}$	-0.70
$\text{I}_2 + 2\text{e}^- \longrightarrow 2\text{I}^-$	+0.5355
$\text{In}^{3+} + 3\text{e}^- \longrightarrow \text{In}$	-0.3382
$\text{K}^+ + \text{e}^- \longrightarrow \text{K}$	-2.931
$\text{La}^{3+} + 3\text{e}^- \longrightarrow \text{La}$	-2.52
$\text{Li}^+ + \text{e}^- \longrightarrow \text{Li}$	-3.04
$\text{Lu}^{3+} + 3\text{e}^- \longrightarrow \text{Lu}$	-2.28
$\text{Mg}^{2+} + 2\text{e}^- \longrightarrow \text{Mg}$	-2.372
$\text{Mn}^{2+} + 2\text{e}^- \longrightarrow \text{Mn}$	-1.185
$\text{MnO}_2 + 2\text{H}_2\text{O} + 2\text{e}^- \longrightarrow \text{Mn(OH)}_2 + 2\text{OH}^-$	-0.05
$\text{MnO}_4^- + 2\text{H}_2\text{O} + 3\text{e}^- \longrightarrow \text{MnO}_2 + 4\text{OH}^-$	+0.558
$\text{MnO}_2 + 4\text{H}^+ + 2\text{e}^- \longrightarrow \text{Mn}^{2+} + 2\text{H}_2\text{O}$	+1.23
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \longrightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	+1.507
$\text{Na}^+ + \text{e}^- \longrightarrow \text{Na}$	-2.71
$\text{Nd}^{3+} + 3\text{e}^- \longrightarrow \text{Nd}$	-2.323
$\text{Ni}^{2+} + 2\text{e}^- \longrightarrow \text{Ni}$	-0.257
$[\text{Ni}(\text{NH}_3)_6]^{2+} + 2\text{e}^- \longrightarrow \text{Ni} + 6\text{NH}_3$	-0.49
$\text{NiO}_2 + 4\text{H}^+ + 2\text{e}^- \longrightarrow \text{Ni}^{2+} + 2\text{H}_2\text{O}$	+1.593
$\text{NiO}_2 + 2\text{H}_2\text{O} + 2\text{e}^- \longrightarrow \text{Ni(OH)}_2 + 2\text{OH}^-$	+0.49
$\text{NiS} + 2\text{e}^- \longrightarrow \text{Ni} + \text{S}^{2-}$	+0.76
$\text{NO}_3^- + 4\text{H}^+ + 3\text{e}^- \longrightarrow \text{NO} + 2\text{H}_2\text{O}$	+0.957
$\text{NO}_3^- + 3\text{H}^+ + 2\text{e}^- \longrightarrow \text{HNO}_2 + \text{H}_2\text{O}$	+0.92
$\text{NO}_3^- + \text{H}_2\text{O} + 2\text{e}^- \longrightarrow \text{NO}_2^- + 2\text{OH}^-$	+0.10
$\text{Np}^{3+} + 3\text{e}^- \longrightarrow \text{Np}$	-1.856

TABLE L1

Half-Reaction	E° (V)
$\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \longrightarrow 4\text{OH}^-$	+0.401
$\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{H}_2\text{O}_2$	+0.695
$\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \longrightarrow 2\text{H}_2\text{O}$	+1.229
$\text{Pb}^{2+} + 2\text{e}^- \longrightarrow \text{Pb}$	-0.1262
$\text{PbO}_2 + \text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^- \longrightarrow \text{PbSO}_4 + 2\text{H}_2\text{O}$	+1.69
$\text{PbS} + 2\text{e}^- \longrightarrow \text{Pb} + \text{S}^{2-}$	-0.95
$\text{PbSO}_4 + 2\text{e}^- \longrightarrow \text{Pb} + \text{SO}_4^{2-}$	-0.3505
$\text{Pd}^{2+} + 2\text{e}^- \longrightarrow \text{Pd}$	+0.987
$[\text{PdCl}_4]^{2-} + 2\text{e}^- \longrightarrow \text{Pd} + 4\text{Cl}^-$	+0.591
$\text{Pt}^{2+} + 2\text{e}^- \longrightarrow \text{Pt}$	+1.20
$[\text{PtBr}_4]^{2-} + 2\text{e}^- \longrightarrow \text{Pt} + 4\text{Br}^-$	+0.58
$[\text{PtCl}_4]^{2-} + 2\text{e}^- \longrightarrow \text{Pt} + 4\text{Cl}^-$	+0.755
$[\text{PtCl}_6]^{2-} + 2\text{e}^- \longrightarrow [\text{PtCl}_4]^{2-} + 2\text{Cl}^-$	+0.68
$\text{Pu}^3 + 3\text{e}^- \longrightarrow \text{Pu}$	-2.03
$\text{Ra}^{2+} + 2\text{e}^- \longrightarrow \text{Ra}$	-2.92
$\text{Rb}^+ + \text{e}^- \longrightarrow \text{Rb}$	-2.98
$[\text{RhCl}_6]^{3-} + 3\text{e}^- \longrightarrow \text{Rh} + 6\text{Cl}^-$	+0.44
$\text{S} + 2\text{e}^- \longrightarrow \text{S}^{2-}$	-0.47627
$\text{S} + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{H}_2\text{S}$	+0.142
$\text{Sc}^{3+} + 3\text{e}^- \longrightarrow \text{Sc}$	-2.09
$\text{Se} + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{H}_2\text{Se}$	-0.399
$[\text{SiF}_6]^{2-} + 4\text{e}^- \longrightarrow \text{Si} + 6\text{F}^-$	-1.2
$\text{SiO}_3^{2-} + 3\text{H}_2\text{O} + 4\text{e}^- \longrightarrow \text{Si} + 6\text{OH}^-$	-1.697
$\text{SiO}_2 + 4\text{H}^+ + 4\text{e}^- \longrightarrow \text{Si} + 2\text{H}_2\text{O}$	-0.86

TABLE L1

Half-Reaction	E° (V)
$\text{Sm}^{3+} + 3\text{e}^- \longrightarrow \text{Sm}$	-2.304
$\text{Sn}^{4+} + 2\text{e}^- \longrightarrow \text{Sn}^{2+}$	+0.151
$\text{Sn}^{2+} + 2\text{e}^- \longrightarrow \text{Sn}$	-0.1375
$[\text{SnF}_6]^{2-} + 4\text{e}^- \longrightarrow \text{Sn} + 6\text{F}^-$	-0.25
$\text{SnS} + 2\text{e}^- \longrightarrow \text{Sn} + \text{S}^{2-}$	-0.94
$\text{Sr}^{2+} + 2\text{e}^- \longrightarrow \text{Sr}$	-2.89
$\text{TeO}_2 + 4\text{H}^+ + 4\text{e}^- \longrightarrow \text{Te} + 2\text{H}_2\text{O}$	+0.593
$\text{Th}^{4+} + 4\text{e}^- \longrightarrow \text{Th}$	-1.90
$\text{Ti}^{2+} + 2\text{e}^- \longrightarrow \text{Ti}$	-1.630
$\text{U}^{3+} + 3\text{e}^- \longrightarrow \text{U}$	-1.79
$\text{V}^{2+} + 2\text{e}^- \longrightarrow \text{V}$	-1.19
$\text{Y}^{3+} + 3\text{e}^- \longrightarrow \text{Y}$	-2.37
$\text{Zn}^{2+} + 2\text{e}^- \longrightarrow \text{Zn}$	-0.7618
$[\text{Zn}(\text{CN})_4]^{2-} + 2\text{e}^- \longrightarrow \text{Zn} + 4\text{CN}^-$	-1.26
$[\text{Zn}(\text{NH}_3)_4]^{2+} + 2\text{e}^- \longrightarrow \text{Zn} + 4\text{NH}_3$	-1.04
$\text{Zn}(\text{OH})_2 + 2\text{e}^- \longrightarrow \text{Zn} + 2\text{OH}^-$	-1.245
$[\text{Zn}(\text{OH})_4]^{2-} + 2\text{e}^- \longrightarrow \text{Zn} + 4\text{OH}^-$	-1.199
$\text{ZnS} + 2\text{e}^- \longrightarrow \text{Zn} + \text{S}^{2-}$	-1.40
$\text{Zr}^{4+} + 4\text{e}^- \longrightarrow \text{Zr}$	-1.539

TABLE L1

APPENDIX M

Half-Lives for Several Radioactive Isotopes

Half-Lives for Several Radioactive Isotopes

Isotope	Half-Life ¹	Type of Emission ²	Isotope	Half-Life ³	Type of Emission ⁴
$^{14}_6\text{C}$	5730 y	(β^-)	$^{210}_{83}\text{Bi}$	5.01 d	(β^-)
$^{13}_7\text{N}$	9.97 m	(β^+)	$^{212}_{83}\text{Bi}$	60.55 m	(α or β^-)
$^{15}_9\text{F}$	4.1×10^{-22} s	(p)	$^{210}_{84}\text{Po}$	138.4 d	(α)
$^{24}_{11}\text{Na}$	15.00 h	(β^-)	$^{212}_{84}\text{Po}$	3×10^{-7} s	(α)
$^{32}_{15}\text{P}$	14.29 d	(β^-)	$^{216}_{84}\text{Po}$	0.15 s	(α)
$^{40}_{19}\text{K}$	1.27×10^9 y	(β or $E.C.$)	$^{218}_{84}\text{Po}$	3.05 m	(α)
$^{49}_{26}\text{Fe}$	0.08 s	(β^+)	$^{215}_{85}\text{At}$	1.0×10^{-4} s	(α)
$^{60}_{26}\text{Fe}$	2.6×10^6 y	(β^-)	$^{218}_{85}\text{At}$	1.6 s	(α)
$^{60}_{27}\text{Co}$	5.27 y	(β^-)	$^{220}_{86}\text{Rn}$	55.6 s	(α)
$^{87}_{37}\text{Rb}$	4.7×10^{10} y	(β^-)	$^{222}_{86}\text{Rn}$	3.82 d	(α)
$^{90}_{38}\text{Sr}$	29 y	(β^-)	$^{224}_{88}\text{Ra}$	3.66 d	(α)
$^{115}_{49}\text{In}$	5.1×10^{15} y	(β^-)	$^{226}_{88}\text{Ra}$	1600 y	(α)
$^{131}_{53}\text{I}$	8.040 d	(β^-)	$^{228}_{88}\text{Ra}$	5.75 y	(β^-)
$^{142}_{58}\text{Ce}$	5×10^{15} y	(α)	$^{228}_{89}\text{Ac}$	6.13 h	(β^-)
$^{208}_{81}\text{Tl}$	3.07 m	(β^-)	$^{228}_{90}\text{Th}$	1.913 y	(α)
$^{210}_{82}\text{Pb}$	22.3 y	(β^-)	$^{232}_{90}\text{Th}$	1.4×10^{10} y	(α)
$^{212}_{82}\text{Pb}$	10.6 h	(β^-)	$^{233}_{90}\text{Th}$	22 m	(β^-)
$^{214}_{82}\text{Pb}$	26.8 m	(β^-)	$^{234}_{90}\text{Th}$	24.10 d	(β^-)

TABLE M1

1 y = years, d = days, h = hours, m = minutes, s = seconds

2 E.C. = electron capture, S.F. = Spontaneous fission

Isotope	Half-Life ³	Type of Emission ⁴	Isotope	Half-Life ³	Type of Emission ⁴
$^{206}_{83}\text{Bi}$	6.243 d	(<i>E. C.</i>)	$^{233}_{91}\text{Pa}$	27 d	(β^-)
$^{233}_{92}\text{U}$	1.59×10^5 y	(α)	$^{242}_{96}\text{Cm}$	162.8 d	(α)
$^{234}_{92}\text{U}$	2.45×10^5 y	(α)	$^{243}_{97}\text{Bk}$	4.5 h	(α or <i>E. C.</i>)
$^{235}_{92}\text{U}$	7.03×10^8 y	(α)	$^{253}_{99}\text{Es}$	20.47 d	(α)
$^{238}_{92}\text{U}$	4.47×10^9 y	(α)	$^{254}_{100}\text{Fm}$	3.24 h	(α or <i>S. F.</i>)
$^{239}_{92}\text{U}$	23.54 m	(β^-)	$^{255}_{100}\text{Fm}$	20.1 h	(α)
$^{239}_{93}\text{Np}$	2.3 d	(β^-)	$^{256}_{101}\text{Md}$	76 m	(α or <i>E. C.</i>)
$^{239}_{94}\text{Pu}$	2.407×10^4 y	(α)	$^{254}_{102}\text{No}$	55 s	(α)
$^{239}_{94}\text{Pu}$	6.54×10^3 y	(α)	$^{257}_{103}\text{Lr}$	0.65 s	(α)
$^{241}_{94}\text{Pu}$	14.4 y	(α or β^-)	$^{260}_{105}\text{Ha}$	1.5 s	(α or <i>S. F.</i>)
$^{241}_{95}\text{Am}$	432.2 y	(α)	$^{263}_{106}\text{Sg}$	0.8 s	(α or <i>S. F.</i>)

TABLE M1

³ y = years, d = days, h = hours, m = minutes, s = seconds

⁴ *E.C.* = electron capture, *S.F.* = Spontaneous fission

ANSWER KEY

Chapter 1

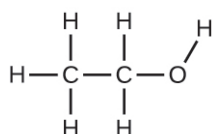
1. Place a glass of water outside. It will freeze if the temperature is below 0 °C.
3. (a) law (states a consistently observed phenomenon, can be used for prediction); (b) theory (a widely accepted explanation of the behavior of matter); (c) hypothesis (a tentative explanation, can be investigated by experimentation)
5. (a) symbolic, microscopic; (b) macroscopic; (c) symbolic, macroscopic; (d) microscopic
7. Macroscopic. The heat required is determined from macroscopic properties.
9. Liquids can change their shape (flow); solids can't. Gases can undergo large volume changes as pressure changes; liquids do not. Gases flow and change volume; solids do not.
11. The mixture can have a variety of compositions; a pure substance has a definite composition. Both have the same composition from point to point.
13. Molecules of elements contain only one type of atom; molecules of compounds contain two or more types of atoms. They are similar in that both are comprised of two or more atoms chemically bonded together.
15. Answers will vary. Sample answer: Gatorade contains water, sugar, dextrose, citric acid, salt, sodium chloride, monopotassium phosphate, and sucrose acetate isobutyrate.
17. (a) element; (b) element; (c) compound; (d) mixture; (e) compound; (f) compound; (g) compound; (h) mixture
19. In each case, a molecule consists of two or more combined atoms. They differ in that the types of atoms change from one substance to the next.
21. Gasoline (a mixture of compounds), oxygen, and to a lesser extent, nitrogen are consumed. Carbon dioxide and water are the principal products. Carbon monoxide and nitrogen oxides are produced in lesser amounts.
23. (a) Increased as it would have combined with oxygen in the air thus increasing the amount of matter and therefore the mass. (b) 0.9 g
25. (a) 200.0 g; (b) The mass of the container and contents would decrease as carbon dioxide is a gaseous product and would leave the container. (c) 102.3 g
27. (a) physical; (b) chemical; (c) chemical; (d) physical; (e) physical
29. physical
31. The value of an extensive property depends upon the amount of matter being considered, whereas the value of an intensive property is the same regardless of the amount of matter being considered.
33. Being extensive properties, both mass and volume are directly proportional to the amount of substance under study. Dividing one extensive property by another will in effect "cancel" this dependence on amount, yielding a ratio that is independent of amount (an intensive property).
35. about a yard
37. (a) kilograms; (b) meters; (c) meters/second; (d) kilograms/cubic meter; (e) kelvin; (f) square meters; (g) cubic meters
39. (a) centi-, $\times 10^{-2}$; (b) deci-, $\times 10^{-1}$; (c) Giga-, $\times 10^9$; (d) kilo-, $\times 10^3$; (e) milli-, $\times 10^{-3}$; (f) nano-, $\times 10^{-9}$; (g) pico-, $\times 10^{-12}$; (h) tera-, $\times 10^{12}$
41. (a) $m = 18.58 \text{ g}$, $V = 5.7 \text{ mL}$; (b) $d = 3.3 \text{ g/mL}$; (c) malachite (basic copper carbonate, $d = 3.25\text{--}4.10 \text{ g/mL}$).
43. (a) displaced water volume = 2.8 mL; (b) displaced water mass = 2.8 g; (c) The block mass is 2.76 g, essentially equal to the mass of displaced water (2.8 g) and consistent with Archimedes' principle of buoyancy.
45. (a) 7.04×10^2 ; (b) 3.344×10^{-2} ; (c) 5.479×10^2 ; (d) 2.2086×10^4 ; (e) 1.00000×10^3 ; (f) 6.51×10^{-8} ; (g) 7.157×10^{-3}
47. (a) exact; (b) exact; (c) uncertain; (d) exact; (e) uncertain; (f) uncertain
49. (a) two; (b) three; (c) five; (d) four; (e) six; (f) two; (g) five
51. (a) 0.44; (b) 9.0; (c) 27; (d) 140; (e) 1.5×10^{-3} ; (f) 0.44
53. (a) 2.15×10^5 ; (b) 4.2×10^6 ; (c) 2.08; (d) 0.19; (e) 27,440; (f) 43.0
55. (a) Archer X; (b) Archer W; (c) Archer Y

57. (a) $\frac{1.0936 \text{ yd}}{1 \text{ m}}$; (b) $\frac{0.94635 \text{ L}}{1 \text{ qt}}$; (c) $\frac{2.2046 \text{ lb}}{1 \text{ kg}}$
59. $\frac{2.0 \text{ L}}{67.6 \text{ fl oz}} = \frac{0.030 \text{ L}}{1 \text{ fl oz}}$
Only two significant figures are justified.
61. 68–71 cm; 400–450 g
63. 355 mL
65. $8 \times 10^{-4} \text{ cm}$
67. yes; weight = 89.4 kg
69. $5.0 \times 10^{-3} \text{ mL}$
71. (a) $1.3 \times 10^{-4} \text{ kg}$; (b) $2.32 \times 10^8 \text{ kg}$; (c) $5.23 \times 10^{-12} \text{ m}$; (d) $8.63 \times 10^{-5} \text{ kg}$; (e) $3.76 \times 10^{-1} \text{ m}$; (f) $5.4 \times 10^{-5} \text{ m}$; (g) $1 \times 10^{12} \text{ s}$; (h) $2.7 \times 10^{-11} \text{ s}$; (i) $1.5 \times 10^{-4} \text{ K}$
73. 45.4 L
75. $1.0160 \times 10^3 \text{ kg}$
77. (a) 394 ft; (b) 5.9634 km; (c) 6.0×10^2 ; (d) 2.64 L; (e) $5.1 \times 10^{18} \text{ kg}$; (f) 14.5 kg; (g) 325 mg
79. 0.46 m; 1.5 ft/cubit
81. Yes, the acid's volume is 123 mL.
83. 62.6 in (about 5 ft 3 in.) and 101 lb
85. (a) $3.81 \text{ cm} \times 8.89 \text{ cm} \times 2.44 \text{ m}$; (b) 40.6 cm
87. 2.70 g/cm^3
89. (a) 81.6 g; (b) 17.6 g
91. (a) 5.1 mL; (b) 37 L
93. 5371 °F, 3239 K
95. -23 °C, 250 K
97. -33.4 °C, 239.8 K
99. 113 °F

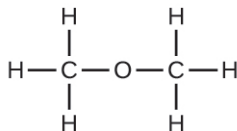
Chapter 2

- The starting materials consist of one green sphere and two blue spheres. The products consist of two green spheres and two blue spheres. This violates Dalton's postulate that atoms are not created during a chemical change, but are merely redistributed.
- This statement violates Dalton's fourth postulate: In a given compound, the numbers of atoms of each type (and thus also the percentage) always have the same ratio.
- Dalton originally thought that all atoms of a particular element had identical properties, including mass. Thus, the concept of isotopes, in which an element has different masses, was a violation of the original idea. To account for the existence of isotopes, the second postulate of his atomic theory was modified to state that atoms of the same element must have identical chemical properties.
- Both are subatomic particles that reside in an atom's nucleus. Both have approximately the same mass. Protons are positively charged, whereas neutrons are uncharged.
- (a) The Rutherford atom has a small, positively charged nucleus, so most α particles will pass through empty space far from the nucleus and be undeflected. Those α particles that pass near the nucleus will be deflected from their paths due to positive-positive repulsion. The more directly toward the nucleus the α particles are headed, the larger the deflection angle will be. (b) Higher-energy α particles that pass near the nucleus will still undergo deflection, but the faster they travel, the less the expected angle of deflection. (c) If the nucleus is smaller, the positive charge is smaller and the expected deflections are smaller—both in terms of how closely the α particles pass by the nucleus undeflected and the angle of deflection. If the nucleus is larger, the positive charge is larger and the expected deflections are larger—more α particles will be deflected, and the deflection angles will be larger. (d) The paths followed by the α particles match the predictions from (a), (b), and (c).
- (a) $^{133}\text{Cs}^+$; (b) $^{127}\text{I}^-$; (c) $^{31}\text{P}^{3-}$; (d) $^{57}\text{Co}^{3+}$
- (a) Carbon-12, ^{12}C ; (b) This atom contains six protons and six neutrons. There are six electrons in a neutral ^{12}C atom. The net charge of such a neutral atom is zero, and the mass number is 12. (c) The preceding answers are correct. (d) The atom will be stable since C-12 is a stable isotope of carbon. (e) The preceding answer is correct. Other answers for this exercise are possible if a different element or isotope is chosen.

15. (a) Lithium-6 contains three protons, three neutrons, and three electrons. The isotope symbol is ${}^6\text{Li}$ or ${}^6_3\text{Li}$. (b) ${}^6\text{Li}^+$ or ${}^6_3\text{Li}^+$
17. (a) Iron, 26 protons, 24 electrons, and 32 neutrons; (b) iodine, 53 protons, 54 electrons, and 74 neutrons
19. (a) 3 protons, 3 electrons, 4 neutrons; (b) 52 protons, 52 electrons, 73 neutrons; (c) 47 protons, 47 electrons, 62 neutrons; (d) 7 protons, 7 electrons, 8 neutrons; (e) 15 protons, 15 electrons, 16 neutrons
21. Let us use neon as an example. Since there are three isotopes, there is no way to be sure to accurately predict the abundances to make the total of 20.18 amu average atomic mass. Let us guess that the abundances are 9% Ne-22, 91% Ne-20, and only a trace of Ne-21. The average mass would be 20.18 amu. Checking the nature's mix of isotopes shows that the abundances are 90.48% Ne-20, 9.25% Ne-22, and 0.27% Ne-21, so our guessed amounts have to be slightly adjusted.
23. 79.90 amu
25. Turkey source: 20.3% (of 10.0129 amu isotope); US source: 19.1% (of 10.0129 amu isotope)
27. The symbol for the element oxygen, O, represents both the element and one atom of oxygen. A molecule of oxygen, O_2 , contains two oxygen atoms; the subscript 2 in the formula must be used to distinguish the diatomic molecule from two single oxygen atoms.
29. (a) molecular CO_2 , empirical CO_2 ; (b) molecular C_2H_2 , empirical CH; (c) molecular C_2H_4 , empirical CH_2 ; (d) molecular H_2SO_4 , empirical H_2SO_4
31. (a) $\text{C}_4\text{H}_5\text{N}_2\text{O}$; (b) $\text{C}_{12}\text{H}_{22}\text{O}_{11}$; (c) HO; (d) CH_2O ; (e) $\text{C}_3\text{H}_4\text{O}_3$
33. (a) CH_2O ; (b) $\text{C}_2\text{H}_4\text{O}$
35. (a) ethanol



(b) methoxymethane, more commonly known as dimethyl ether



(c) These molecules have the same chemical composition (types and number of atoms) but different chemical structures. They are structural isomers.

37. (a) metal, inner transition metal; (b) nonmetal, representative element; (c) metal, representative element; (d) nonmetal, representative element; (e) metal, transition metal; (f) metal, inner transition metal; (g) metal, transition metal; (h) nonmetal, representative element; (i) nonmetal, representative element; (j) metal, representative element
39. (a) He; (b) Be; (c) Li; (d) O
41. (a) krypton, Kr; (b) calcium, Ca; (c) fluorine, F; (d) tellurium, Te
43. (a) ${}^{23}_{11}\text{Na}$; (b) ${}^{129}_{54}\text{Xe}$; (c) ${}^{73}_{33}\text{As}$; (d) ${}^{226}_{88}\text{Ra}$
45. Ionic: KCl, MgCl_2 ; Covalent: NCl_3 , ICl, PCl_5 , CCl_4
47. (a) covalent; (b) ionic, Ba^{2+} , O^{2-} ; (c) ionic, NH_4^+ , CO_3^{2-} ; (d) ionic, Sr^{2+} , H_2PO_4^- ; (e) covalent; (f) ionic, Na^+ , O^{2-}
49. (a) CaS; (b) $(\text{NH}_4)_2\text{SO}_4$; (c) AlBr_3 ; (d) Na_2HPO_4 ; (e) $\text{Mg}_3(\text{PO}_4)_2$
51. (a) cesium chloride; (b) barium oxide; (c) potassium sulfide; (d) beryllium chloride; (e) hydrogen bromide; (f) aluminum fluoride
53. (a) RbBr; (b) MgSe; (c) Na_2O ; (d) CaCl_2 ; (e) HF; (f) GaP; (g) AlBr_3 ; (h) $(\text{NH}_4)_2\text{SO}_4$
55. (a) ClO_2 ; (b) N_2O_4 ; (c) K_3P ; (d) Ag_2S ; (e) $\text{AlF}_3 \cdot 3\text{H}_2\text{O}$; (f) SiO_2
57. (a) chromium(III) oxide; (b) iron(II) chloride; (c) chromium(VI) oxide; (d) titanium(IV) chloride; (e) cobalt(II) chloride hexahydrate; (f) molybdenum(IV) sulfide
59. (a) K_3PO_4 ; (b) CuSO_4 ; (c) CaCl_2 ; (d) TiO_2 ; (e) NH_4NO_3 ; (f) NaHSO_4
61. (a) manganese(IV) oxide; (b) mercury(I) chloride; (c) iron(III) nitrate; (d) titanium(IV) chloride; (e) copper(II) bromide

Chapter 3

1. (a) 12.01 amu; (b) 12.01 amu; (c) 144.12 amu; (d) 60.05 amu
3. (a) 123.896 amu; (b) 18.015 amu; (c) 164.086 amu; (d) 60.052 amu; (e) 342.297 amu
5. (a) 56.107 amu; (b) 54.091 amu; (c) 199.9976 amu; (d) 97.9950 amu
7. Use the molecular formula to find the molar mass; to obtain the number of moles, divide the mass of compound by the molar mass of the compound expressed in grams.
9. Formic acid. Its formula has twice as many oxygen atoms as the other two compounds (one each). Therefore, 0.60 mol of formic acid would be equivalent to 1.20 mol of a compound containing a single oxygen atom.
11. The two masses have the same numerical value, but the units are different: The molecular mass is the mass of 1 molecule while the molar mass is the mass of 6.022×10^{23} molecules.
13. (a) 256.48 g/mol; (b) 72.150 g mol⁻¹; (c) 378.103 g mol⁻¹; (d) 58.080 g mol⁻¹; (e) 180.158 g mol⁻¹
15. (a) 197.382 g mol⁻¹; (b) 257.163 g mol⁻¹; (c) 194.193 g mol⁻¹; (d) 60.056 g mol⁻¹; (e) 306.464 g mol⁻¹
17. (a) 0.819 g; (b) 307 g; (c) 0.23 g; (d) 1.235×10^6 g (1235 kg); (e) 765 g
19. (a) 99.41 g; (b) 2.27 g; (c) 3.5 g; (d) 222 kg; (e) 160.1 g
21. (a) 9.60 g; (b) 19.2 g; (c) 28.8 g
23. zirconium: 2.038×10^{23} atoms; 30.87 g; silicon: 2.038×10^{23} atoms; 9.504 g; oxygen: 8.151×10^{23} atoms; 21.66 g
25. AlPO₄: 1.000 mol, or 26.98 g Al; Al₂Cl₆: 1.994 mol, or 53.74 g Al; Al₂S₃: 3.00 mol, or 80.94 g Al; The Al₂S₃ sample thus contains the greatest mass of Al.
27. 3.113×10^{25} C atoms
29. 0.865 servings, or about 1 serving.
31. 20.0 g H₂O represents the least number of molecules since it has the least number of moles.
33. (a) % N = 82.24%, % H = 17.76%; (b) % Na = 29.08%, % S = 40.56%, % O = 30.36%; (c) % Ca²⁺ = 38.76%
35. % NH₃ = 38.2%
37. (a) CS₂; (b) CH₂O
39. C₆H₆
41. Mg₃Si₂H₃O₈ (empirical formula), Mg₆Si₄H₆O₁₆ (molecular formula)
43. C₁₅H₁₅N₃
45. We need to know the number of moles of sulfuric acid dissolved in the solution and the volume of the solution.
47. (a) 0.679 M; (b) 1.00 M; (c) 0.06998 M; (d) 1.75 M; (e) 0.070 M; (f) 6.6 M
49. (a) determine the number of moles of glucose in 0.500 L of solution; determine the molar mass of glucose; determine the mass of glucose from the number of moles and its molar mass; (b) 27 g
51. (a) 37.0 mol H₂SO₄, 3.63×10^3 g H₂SO₄; (b) 3.8×10^{-7} mol NaCN, 1.9×10^{-5} g NaCN; (c) 73.2 mol H₂CO, 2.20 kg H₂CO; (d) 5.9×10^{-7} mol FeSO₄, 8.9×10^{-5} g FeSO₄
53. (a) Determine the molar mass of KMnO₄; determine the number of moles of KMnO₄ in the solution; from the number of moles and the volume of solution, determine the molarity; (b) 1.15×10^{-3} M
55. (a) 5.04×10^{-3} M; (b) 0.499 M; (c) 9.92 M; (d) 1.1×10^{-3} M
57. 0.025 M
59. 0.5000 L
61. 1.9 mL
63. (a) 0.125 M; (b) 0.04888 M; (c) 0.206 M; (d) 0.0056 M
65. 11.9 M
67. 1.6 L
69. (a) The dilution equation can be used, appropriately modified to accommodate mass-based concentration units: %mass₁ × mass₁ = %mass₂ × mass₂. This equation can be rearranged to isolate mass₁ and the given quantities substituted into this equation. (b) 58.8 g
71. 114 g
73. 1.75×10^{-3} M
75. 95 mg/dL
77. 2.38×10^{-4} mol
79. 0.29 M

Chapter 4

- An equation is balanced when the same number of each element is represented on the reactant and product sides. Equations must be balanced to accurately reflect the law of conservation of matter.
- (a) $\text{PCl}_5(s) + \text{H}_2\text{O}(l) \longrightarrow \text{POCl}_3(l) + 2\text{HCl}(aq)$; (b) $3\text{Cu}(s) + 8\text{HNO}_3(aq) \longrightarrow 3\text{Cu}(\text{NO}_3)_2(aq) + 4\text{H}_2\text{O}(l) + 2\text{NO}(g)$; (c) $\text{H}_2(g) + \text{I}_2(s) \longrightarrow 2\text{HI}(s)$; (d) $4\text{Fe}(s) + 3\text{O}_2(g) \longrightarrow 2\text{Fe}_2\text{O}_3(s)$; (e) $2\text{Na}(s) + 2\text{H}_2\text{O}(l) \longrightarrow 2\text{NaOH}(aq) + \text{H}_2(g)$; (f) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7(s) \longrightarrow \text{Cr}_2\text{O}_3(s) + \text{N}_2(g) + 4\text{H}_2\text{O}(g)$; (g) $\text{P}_4(s) + 6\text{Cl}_2(g) \longrightarrow 4\text{PCl}_3(l)$; (h) $\text{PtCl}_4(s) \longrightarrow \text{Pt}(s) + 2\text{Cl}_2(g)$
- (a) $\text{CaCO}_3(s) \longrightarrow \text{CaO}(s) + \text{CO}_2(g)$; (b) $2\text{C}_4\text{H}_{10}(g) + 13\text{O}_2(g) \longrightarrow 8\text{CO}_2(g) + 10\text{H}_2\text{O}(g)$; (c) $\text{MgCl}_2(aq) + 2\text{NaOH}(aq) \longrightarrow \text{Mg}(\text{OH})_2(s) + 2\text{NaCl}(aq)$; (d) $2\text{H}_2\text{O}(g) + 2\text{Na}(s) \longrightarrow 2\text{NaOH}(s) + \text{H}_2(g)$
- (a) $\text{Ba}(\text{NO}_3)_2, \text{KClO}_3$; (b) $2\text{KClO}_3(s) \longrightarrow 2\text{KCl}(s) + 3\text{O}_2(g)$; (c) $2\text{Ba}(\text{NO}_3)_2(s) \longrightarrow 2\text{BaO}(s) + 2\text{N}_2(g) + 5\text{O}_2(g)$; (d) $2\text{Mg}(s) + \text{O}_2(g) \longrightarrow 2\text{MgO}(s)$; $4\text{Al}(s) + 3\text{O}_2(g) \longrightarrow 2\text{Al}_2\text{O}_3(s)$; $4\text{Fe}(s) + 3\text{O}_2(g) \longrightarrow 2\text{Fe}_2\text{O}_3(s)$
- (a) $4\text{HF}(aq) + \text{SiO}_2(s) \longrightarrow \text{SiF}_4(g) + 2\text{H}_2\text{O}(l)$; (b) complete ionic equation: $2\text{Na}^+(aq) + 2\text{F}^-(aq) + \text{Ca}^{2+}(aq) + 2\text{Cl}^-(aq) \longrightarrow \text{CaF}_2(s) + 2\text{Na}^+(aq) + 2\text{Cl}^-(aq)$, net ionic equation: $2\text{F}^-(aq) + \text{Ca}^{2+}(aq) \longrightarrow \text{CaF}_2(s)$
- (a)

$$2\text{K}^+(aq) + \text{C}_2\text{O}_4^{2-}(aq) + \text{Ba}^{2+}(aq) + 2\text{OH}^-(aq) \longrightarrow 2\text{K}^+(aq) + 2\text{OH}^-(aq) + \text{BaC}_2\text{O}_4(s) \quad (\text{complete})$$

$$\text{Ba}^{2+}(aq) + \text{C}_2\text{O}_4^{2-}(aq) \longrightarrow \text{BaC}_2\text{O}_4(s) \quad (\text{net})$$
 (b)

$$\text{Pb}^{2+}(aq) + 2\text{NO}_3^-(aq) + 2\text{H}^+(aq) + \text{SO}_4^{2-}(aq) \longrightarrow \text{PbSO}_4(s) + 2\text{H}^+(aq) + 2\text{NO}_3^-(aq) \quad (\text{complete})$$

$$\text{Pb}^{2+}(aq) + \text{SO}_4^{2-}(aq) \longrightarrow \text{PbSO}_4(s) \quad (\text{net})$$
 (c) $\text{CaCO}_3(s) + 2\text{H}^+(aq) + \text{SO}_4^{2-}(aq) \longrightarrow \text{CaSO}_4(s) + \text{CO}_2(g) + \text{H}_2\text{O}(l) \quad (\text{complete})$

$$\text{CaCO}_3(s) + 2\text{H}^+(aq) + \text{SO}_4^{2-}(aq) \longrightarrow \text{CaSO}_4(s) + \text{CO}_2(g) + \text{H}_2\text{O}(l) \quad (\text{net})$$
- (a) oxidation-reduction (addition); (b) acid-base (neutralization); (c) oxidation-reduction (combustion)
- It is an oxidation-reduction reaction because the oxidation state of the silver changes during the reaction.
- (a) H +1, P +5, O -2; (b) Al +3, H +1, O -2; (c) Se +4, O -2; (d) K +1, N +3, O -2; (e) In +3, S -2; (f) P +3, O -2
- (a) acid-base; (b) oxidation-reduction: Na is oxidized, H^+ is reduced; (c) oxidation-reduction: Mg is oxidized, Cl_2 is reduced; (d) acid-base; (e) oxidation-reduction: P^{3-} is oxidized, O_2 is reduced; (f) acid-base
- (a) $2\text{HCl}(g) + \text{Ca}(\text{OH})_2(s) \longrightarrow \text{CaCl}_2(s) + 2\text{H}_2\text{O}(l)$; (b) $\text{Sr}(\text{OH})_2(aq) + 2\text{HNO}_3(aq) \longrightarrow \text{Sr}(\text{NO}_3)_2(aq) + 2\text{H}_2\text{O}(l)$
- (a) $2\text{Al}(s) + 3\text{F}_2(g) \longrightarrow 2\text{AlF}_3(s)$; (b) $2\text{Al}(s) + 3\text{CuBr}_2(aq) \longrightarrow 3\text{Cu}(s) + 2\text{AlBr}_3(aq)$; (c) $\text{P}_4(s) + 5\text{O}_2(g) \longrightarrow \text{P}_4\text{O}_{10}(s)$; (d) $\text{Ca}(s) + 2\text{H}_2\text{O}(l) \longrightarrow \text{Ca}(\text{OH})_2(aq) + \text{H}_2(g)$
- (a) $\text{Mg}(\text{OH})_2(s) + 2\text{HClO}_4(aq) \longrightarrow \text{Mg}^{2+}(aq) + 2\text{ClO}_4^-(aq) + 2\text{H}_2\text{O}(l)$; (b) $\text{SO}_3(g) + 2\text{H}_2\text{O}(l) \longrightarrow \text{H}_3\text{O}^+(aq) + \text{HSO}_4^-(aq)$, (a solution of H_2SO_4); (c) $\text{SrO}(s) + \text{H}_2\text{SO}_4(l) \longrightarrow \text{SrSO}_4(s) + \text{H}_2\text{O}$
- $\text{H}_2(g) + \text{F}_2(g) \longrightarrow 2\text{HF}(g)$
- $2\text{NaBr}(aq) + \text{Cl}_2(g) \longrightarrow 2\text{NaCl}(aq) + \text{Br}_2(l)$
- $2\text{LiOH}(aq) + \text{CO}_2(g) \longrightarrow \text{Li}_2\text{CO}_3(aq) + \text{H}_2\text{O}(l)$
- (a) $\text{Ca}(\text{OH})_2(s) + \text{H}_2\text{S}(g) \longrightarrow \text{CaS}(s) + 2\text{H}_2\text{O}(l)$; (b) $\text{Na}_2\text{CO}_3(aq) + \text{H}_2\text{S}(g) \longrightarrow \text{Na}_2\text{S}(aq) + \text{CO}_2(g) + \text{H}_2\text{O}(l)$
- (a) step 1: $\text{N}_2(g) + 3\text{H}_2(g) \longrightarrow 2\text{NH}_3(g)$, step 2: $\text{NH}_3(g) + \text{HNO}_3(aq) \longrightarrow \text{NH}_4\text{NO}_3(aq) \longrightarrow \text{NH}_4\text{NO}_3(s)$ (after drying); (b) $\text{H}_2(g) + \text{Br}_2(l) \longrightarrow 2\text{HBr}(g)$; (c) $\text{Zn}(s) + \text{S}(s) \longrightarrow \text{ZnS}(s)$ and $\text{ZnS}(s) + 2\text{HCl}(aq) \longrightarrow \text{ZnCl}_2(aq) + \text{H}_2\text{S}(g)$
- (a) $\text{Sn}^{4+}(aq) + 2\text{e}^- \longrightarrow \text{Sn}^{2+}(aq)$, (b) $[\text{Ag}(\text{NH}_3)_2]^+(aq) + \text{e}^- \longrightarrow \text{Ag}(s) + 2\text{NH}_3(aq)$; (c) $\text{Hg}_2\text{Cl}_2(s) + 2\text{e}^- \longrightarrow 2\text{Hg}(l) + 2\text{Cl}^-(aq)$; (d) $2\text{H}_2\text{O}(l) \longrightarrow \text{O}_2(g) + 4\text{H}^+(aq) + 4\text{e}^-$; (e) $6\text{H}_2\text{O}(l) + 2\text{IO}_3^-(aq) + 10\text{e}^- \longrightarrow \text{I}_2(s) + 12\text{OH}^-(aq)$; (f)

- $\text{H}_2\text{O}(l) + \text{SO}_3^{2-}(aq) \longrightarrow \text{SO}_4^{2-}(aq) + 2\text{H}^+(aq) + 2\text{e}^-$; (g)
 $8\text{H}^+(aq) + \text{MnO}_4^-(aq) + 5\text{e}^- \longrightarrow \text{Mn}^{2+}(aq) + 4\text{H}_2\text{O}(l)$; (h)
 $\text{Cl}^-(aq) + 6\text{OH}^-(aq) \longrightarrow \text{ClO}_3^-(aq) + 3\text{H}_2\text{O}(l) + 6\text{e}^-$
39. (a) $\text{Sn}^{2+}(aq) + 2\text{Cu}^{2+}(aq) \longrightarrow \text{Sn}^{4+}(aq) + 2\text{Cu}^+(aq)$; (b)
 $\text{H}_2\text{S}(g) + \text{Hg}_2^{2+}(aq) + 2\text{H}_2\text{O}(l) \longrightarrow 2\text{Hg}(l) + \text{S}(s) + 2\text{H}_3\text{O}^+(aq)$; (c)
 $5\text{CN}^-(aq) + 2\text{ClO}_2(aq) + 3\text{H}_2\text{O}(l) \longrightarrow 5\text{CNO}^-(aq) + 2\text{Cl}^-(aq) + 2\text{H}_3\text{O}^+(aq)$; (d)
 $\text{Fe}^{2+}(aq) + \text{Ce}^{4+}(aq) \longrightarrow \text{Fe}^{3+}(aq) + \text{Ce}^{3+}(aq)$; (e)
 $2\text{HBrO}(aq) + 2\text{H}_2\text{O}(l) \longrightarrow 2\text{H}_3\text{O}^+(aq) + 2\text{Br}^-(aq) + \text{O}_2(g)$
41. (a) $2\text{MnO}_4^-(aq) + 3\text{NO}_2^-(aq) + \text{H}_2\text{O}(l) \longrightarrow 2\text{MnO}_2(s) + 3\text{NO}_3^-(aq) + 2\text{OH}^-(aq)$; (b)
 $3\text{MnO}_4^{2-}(aq) + 2\text{H}_2\text{O}(l) \longrightarrow 2\text{MnO}_4^-(aq) + 4\text{OH}^-(aq) + \text{MnO}_2(s)$ (in base); (c)
 $\text{Br}_2(l) + \text{SO}_2(g) + 2\text{H}_2\text{O}(l) \longrightarrow 4\text{H}^+(aq) + 2\text{Br}^-(aq) + \text{SO}_4^{2-}(aq)$
43. (a) 0.435 mol Na, 0.217 mol Cl_2 , 15.4 g Cl_2 ; (b) 0.005780 mol HgO, 2.890×10^{-3} mol O_2 , 9.248×10^{-2} g O_2 ;
(c) 8.00 mol NaNO_3 , 6.8×10^2 g NaNO_3 ; (d) 1665 mol CO_2 , 73.3 kg CO_2 ; (e) 18.86 mol CuO, 2.330 kg CuCO_3 ;
(f) 0.4580 mol $\text{C}_2\text{H}_4\text{Br}_2$, 86.05 g $\text{C}_2\text{H}_4\text{Br}_2$
45. (a) 0.0686 mol Mg, 1.67 g Mg; (b) 2.701×10^{-3} mol O_2 , 0.08644 g O_2 ; (c) 6.43 mol MgCO_3 , 542 g MgCO_3 (d)
768 mol H_2O , 13.8 kg H_2O ; (e) 16.31 mol BaO_2 , 2762 g BaO_2 ; (f) 0.207 mol C_2H_4 , 5.81 g C_2H_4
47. (a) volume HCl solution $\longrightarrow$ mol HCl $\longrightarrow$ mol GaCl_3 ; (b) 1.25 mol GaCl_3 , 2.2×10^2 g GaCl_3
49. (a) 5.337×10^{22} molecules; (b) 10.41 g $\text{Zn}(\text{CN})_2$
51. $\text{SiO}_2 + 3\text{C} \longrightarrow \text{SiC} + 2\text{CO}$, 4.50 kg SiO_2
53. 5.00×10^3 kg
55. 1.28×10^5 g CO_2
57. 161.4 mL KI solution
59. 176 g TiO_2
61. The limiting reactant is Cl_2 .
63. Percent yield = 31%
65. g $\text{CCl}_4 \longrightarrow$ mol $\text{CCl}_4 \longrightarrow$ mol $\text{CCl}_2\text{F}_2 \longrightarrow$ g CCl_2F_2 , percent yield = 48.3%
67. percent yield = 91.3%
69. Convert mass of ethanol to moles of ethanol; relate the moles of ethanol to the moles of ether produced using the stoichiometry of the balanced equation. Convert moles of ether to grams; divide the actual grams of ether (determined through the density) by the theoretical mass to determine the percent yield; 87.6%
71. The conversion needed is mol Cr $\longrightarrow$ mol H_3PO_4 . Then compare the amount of Cr to the amount of acid present. Cr is the limiting reactant.
73. $\text{Na}_2\text{C}_2\text{O}_4$ is the limiting reactant. percent yield = 86.56%
75. Only four molecules can be made.
77. This amount cannot be weighted by ordinary balances and is worthless.
79. 3.4×10^{-3} M H_2SO_4
81. 9.6×10^{-3} M Cl^-
83. 22.4%
85. The empirical formula is BH_3 . The molecular formula is B_2H_6 .
87. 49.6 mL
89. 13.64 mL
91. 0.0122 M
93. 34.99 mL KOH
95. The empirical formula is WCl_4 .

Chapter 5

- The temperature of 1 gram of burning wood is approximately the same for both a match and a bonfire. This is an intensive property and depends on the material (wood). However, the overall amount of produced heat depends on the amount of material; this is an extensive property. The amount of wood in a bonfire is much greater than that in a match; the total amount of produced heat is also much greater, which is why we can sit around a bonfire to stay warm, but a match would not provide enough heat to keep us from getting cold.
- Heat capacity refers to the heat required to raise the temperature of the mass of the substance 1 degree;

specific heat refers to the heat required to raise the temperature of 1 gram of the substance 1 degree. Thus, heat capacity is an extensive property, and specific heat is an intensive one.

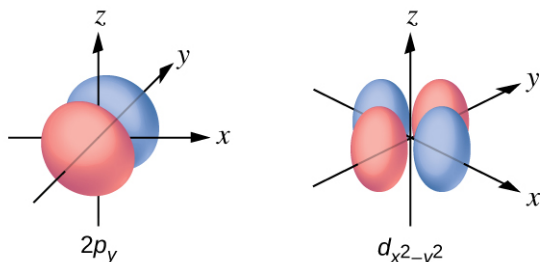
5. (a) 47.6 J/°C; 11.38 cal °C⁻¹; (b) 407 J/°C; 97.3 cal °C⁻¹
7. 1310 J; 313 cal
9. 7.15 °C
11. (a) 0.390 J/g °C; (b) Copper is a likely candidate.
13. We assume that the density of water is 1.0 g/cm³ (1 g/mL) and that it takes as much energy to keep the water at 85 °F as to heat it from 72 °F to 85 °F. We also assume that only the water is going to be heated. Energy required = 7.47 kWh
15. lesser; more heat would be lost to the coffee cup and the environment and so ΔT for the water would be lesser and the calculated q would be lesser
17. greater, since taking the calorimeter's heat capacity into account will compensate for the thermal energy transferred to the solution from the calorimeter; this approach includes the calorimeter itself, along with the solution, as "surroundings": $q_{\text{rxn}} = -(q_{\text{solution}} + q_{\text{calorimeter}})$; since both q_{solution} and $q_{\text{calorimeter}}$ are negative, including the latter term (q_{rxn}) will yield a greater value for the heat of the dissolution
19. The temperature of the coffee will drop 1 degree.
21. 5.7×10^2 kJ
23. 38.5 °C
25. -2.2 kJ; The heat produced shows that the reaction is exothermic.
27. 1.4 kJ
29. 22.6. Since the mass and the heat capacity of the solution is approximately equal to that of the water, the two-fold increase in the amount of water leads to a two-fold decrease of the temperature change.
31. 11.7 kJ
33. 30%
35. 0.24 g
37. 1.4×10^2 Calories
39. The enthalpy change of the indicated reaction is for exactly 1 mol HCl and 1 mol NaOH; the heat in the example is produced by 0.0500 mol HCl and 0.0500 mol NaOH.
41. 25 kJ mol⁻¹
43. 81 kJ mol⁻¹
45. 5204.4 kJ
47. 1.83×10^{-2} mol
49. -802 kJ mol⁻¹
51. 15.5 kJ/°C
53. 7.43 g
55. Yes.
57. 459.6 kJ
59. -494 kJ/mol
61. 44.01 kJ/mol
63. -394 kJ
65. 265 kJ
67. 90.3 kJ/mol
69. (a) -1615.0 kJ mol⁻¹; (b) -484.3 kJ mol⁻¹; (c) 164.2 kJ; (d) -232.1 kJ
71. -54.04 kJ mol⁻¹
73. -2660 kJ mol⁻¹
75. -66.4 kJ
77. -122.8 kJ
79. 3.7 kg
81. On the assumption that the best rocket fuel is the one that gives off the most heat, B₂H₆ is the prime candidate.
83. -88.2 kJ
85. (a) $\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \longrightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{l})$; (b) 330 L air; (c) -104.5 kJ mol⁻¹; (d) 75.4 °C

Chapter 6

1. The spectrum consists of colored lines, at least one of which (probably the brightest) is red.
3. 3.15 m
5. 3.233×10^{-19} J; 2.018 eV
7. $\nu = 4.568 \times 10^{14} \text{ s}^{-1}$; $\lambda = 656.3 \text{ nm}$; Energy $\text{mol}^{-1} = 1.823 \times 10^5 \text{ J mol}^{-1}$; red
9. (a) $\lambda = 8.69 \times 10^{-7} \text{ m}$; $E = 2.29 \times 10^{-19} \text{ J}$; (b) $\lambda = 4.59 \times 10^{-7} \text{ m}$; $E = 4.33 \times 10^{-19} \text{ J}$; The color of (a) is red; (b) is blue.
11. $E = 9.502 \times 10^{-15} \text{ J}$; $\nu = 1.434 \times 10^{19} \text{ s}^{-1}$
13. Red: 660 nm; $4.54 \times 10^{14} \text{ Hz}$; $3.01 \times 10^{-19} \text{ J}$. Green: 520 nm; $5.77 \times 10^{14} \text{ Hz}$; $3.82 \times 10^{-19} \text{ J}$. Blue: 440 nm; $6.81 \times 10^{14} \text{ Hz}$; $4.51 \times 10^{-19} \text{ J}$. Somewhat different numbers are also possible.
15. $5.49 \times 10^{14} \text{ s}^{-1}$; no
17. Quantized energy means that the electrons can possess only certain discrete energy values; values between those quantized values are not permitted.
19. 2.856 eV
21. $-8.716 \times 10^{-18} \text{ J}$
23. $-3.405 \times 10^{-20} \text{ J}$
25. 33.9 Å
27. $1.471 \times 10^{-17} \text{ J}$
29. Both involve a relatively heavy nucleus with electrons moving around it, although strictly speaking, the Bohr model works only for one-electron atoms or ions. According to classical mechanics, the Rutherford model predicts a miniature “solar system” with electrons moving about the nucleus in circular or elliptical orbits that are confined to planes. If the requirements of classical electromagnetic theory that electrons in such orbits would emit electromagnetic radiation are ignored, such atoms would be stable, having constant energy and angular momentum, but would not emit any visible light (contrary to observation). If classical electromagnetic theory is applied, then the Rutherford atom would emit electromagnetic radiation of continually increasing frequency (contrary to the observed discrete spectra), thereby losing energy until the atom collapsed in an absurdly short time (contrary to the observed long-term stability of atoms). The Bohr model retains the classical mechanics view of circular orbits confined to planes having constant energy and angular momentum, but restricts these to quantized values dependent on a single quantum number, n . The orbiting electron in Bohr’s model is assumed not to emit any electromagnetic radiation while moving about the nucleus in its stationary orbits, but the atom can emit or absorb electromagnetic radiation when the electron changes from one orbit to another. Because of the quantized orbits, such “quantum jumps” will produce discrete spectra, in agreement with observations.
31. Both models have a central positively charged nucleus with electrons moving about the nucleus in accordance with the Coulomb electrostatic potential. The Bohr model *assumes* that the electrons move in circular orbits that have quantized energies, angular momentum, and radii that are specified by a single quantum number, $n = 1, 2, 3, \dots$, but this quantization is an ad hoc assumption made by Bohr to incorporate quantization into an essentially classical mechanics description of the atom. Bohr also assumed that electrons orbiting the nucleus normally do not emit or absorb electromagnetic radiation, but do so when the electron switches to a different orbit. In the quantum mechanical model, the electrons do not move in precise orbits (such orbits violate the Heisenberg uncertainty principle) and, instead, a probabilistic interpretation of the electron’s position at any given instant is used, with a mathematical function ψ called a wavefunction that can be used to determine the electron’s spatial probability distribution. These wavefunctions, or orbitals, are three-dimensional stationary waves that can be specified by three quantum numbers that arise naturally from their underlying mathematics (no ad hoc assumptions required): the principal quantum number, n (the same one used by Bohr), which specifies shells such that orbitals having the same n all have the same energy and approximately the same spatial extent; the angular momentum quantum number l , which is a measure of the orbital’s angular momentum and corresponds to the orbitals’ general shapes, as well as specifying subshells such that orbitals having the same l (and n) all have the same energy; and the orientation quantum number m , which is a measure of the z component of the angular momentum and corresponds to the orientations of the orbitals. The Bohr model gives the same expression for the energy as the quantum mechanical expression and, hence, both properly account for hydrogen’s discrete spectrum (an example of getting the right answers for the wrong reasons, something that many chemistry students can sympathize with), but gives the wrong

expression for the angular momentum (Bohr orbits necessarily all have non-zero angular momentum, but some quantum orbitals [*s* orbitals] can have zero angular momentum).

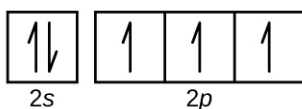
33. *n* determines the general range for the value of energy and the probable distances that the electron can be from the nucleus. *l* determines the shape of the orbital. *m_l* determines the orientation of the orbitals of the same *l* value with respect to one another. *m_s* determines the spin of an electron.
35. (a) 2*p*; (b) 4*d*; (c) 6*s*
37. (a) 3*d*; (b) 1*s*; (c) 4*f*
- 39.



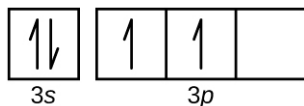
41. (a) *x*. 2, *y*. 2, *z*. 2; (b) *x*. 1, *y*. 3, *z*. 0; (c) *x*. 4 0 0 $\frac{1}{2}$, *y*. 2 1 0 $\frac{1}{2}$, *z*. 3 2 0 $\frac{1}{2}$; (d) *x*. 1, *y*. 2, *z*. 3; (e) *x*. *l* = 0, *m_l* = 0, *y*. *l* = 1, *m_l* = -1, 0, or +1, *z*. *l* = 2, *m_l* = -2, -1, 0, +1, +2
43. 12
- 45.

<i>n</i>	<i>l</i>	<i>m_l</i>	<i>s</i>
4	0	0	$+\frac{1}{2}$
4	0	0	$-\frac{1}{2}$
4	1	-1	$+\frac{1}{2}$
4	1	0	$+\frac{1}{2}$
4	1	+1	$+\frac{1}{2}$
4	1	-1	$-\frac{1}{2}$

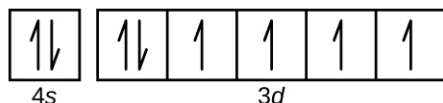
47. For example, Na⁺: 1*s*²2*s*²2*p*⁶; Ca²⁺: 1*s*²2*s*²2*p*⁶3*s*²3*p*⁶; Sn²⁺: 1*s*²2*s*²2*p*⁶3*s*²3*p*⁶3*d*¹⁰4*s*²4*p*⁶4*d*¹⁰5*s*²; F⁻: 1*s*²2*s*²2*p*⁶; O²⁻: 1*s*²2*s*²2*p*⁶; Cl⁻: 1*s*²2*s*²2*p*⁶3*s*²3*p*⁶.
49. (a) 1*s*²2*s*²2*p*³; (b) 1*s*²2*s*²2*p*⁶3*s*²3*p*²; (c) 1*s*²2*s*²2*p*⁶3*s*²3*p*⁶4*s*²3*d*⁶; (d) 1*s*²2*s*²2*p*⁶3*s*²3*p*⁶4*s*²3*d*¹⁰4*p*⁶5*s*²4*d*¹⁰5*p*⁴; (e) 1*s*²2*s*²2*p*⁶3*s*²3*p*⁶4*s*²3*d*¹⁰4*p*⁶5*s*²4*d*¹⁰5*p*⁶6*s*²4*f*⁹
51. The charge on the ion.
53. (a)



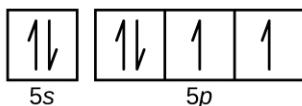
(b)



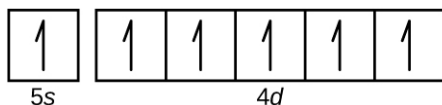
(c)



(d)



(e)



55. Zr

57. Rb^+ , Se^{2-}

59. Although both (b) and (c) are correct, (e) encompasses both and is the best answer.

61. K

63. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^6 6s^2 4f^{14} 5d^{10}$ 65. Co has 27 protons, 27 electrons, and 33 neutrons: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$. I has 53 protons, 53 electrons, and 78 neutrons: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^2 5p^5$.

67. Cl

69. O

71. $\text{Rb} < \text{Li} < \text{N} < \text{F}$

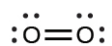
73. 15 (5A)

75. $\text{Mg} < \text{Ca} < \text{Rb} < \text{Cs}$ 77. $\text{Si}^{4+} < \text{Al}^{3+} < \text{Ca}^{2+} < \text{K}^+$ 79. Se, As^- 81. $\text{Mg}^{2+} < \text{K}^+ < \text{Br}^- < \text{As}^{3-}$ 83. O, IE_1

85. Ra

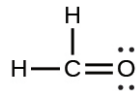
Chapter 7

- The protons in the nucleus do not change during normal chemical reactions. Only the outer electrons move. Positive charges form when electrons are lost.
- P, I, Cl, and O would form anions because they are nonmetals. Mg, In, Cs, Pb, and Co would form cations because they are metals.
- (a) P^{3-} ; (b) Mg^{2+} ; (c) Al^{3+} ; (d) O^{2-} ; (e) Cl^- ; (f) Cs^+
- (a) $[\text{Ar}]4s^2 3d^{10} 4p^6$; (b) $[\text{Kr}]4d^{10} 5s^2 5p^6$ (c) $1s^2$ (d) $[\text{Kr}]4d^{10}$; (e) $[\text{He}]2s^2 2p^6$; (f) $[\text{Ar}]3d^{10}$; (g) $1s^2$ (h) $[\text{He}]2s^2 2p^6$ (i) $[\text{Kr}]4d^{10} 5s^2$ (j) $[\text{Ar}]3d^7$ (k) $[\text{Ar}]3d^6$, (l) $[\text{Ar}]3d^{10} 4s^2$
- (a) $1s^2 2s^2 2p^6 3s^2 3p^1$; Al^{3+} : $1s^2 2s^2 2p^6$; (b) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^5$; $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$; (c) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 5s^2$; Sr^{2+} : $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$; (d) $1s^2 2s^1$; Li^+ : $1s^2$; (e) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^3$; $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$; (f) $1s^2 2s^2 2p^6 3s^2 3p^4$; $1s^2 2s^2 2p^6 3s^2 3p^6$
- NaCl consists of discrete ions arranged in a crystal lattice, not covalently bonded molecules.
- ionic: (b), (d), (e), (g), and (i); covalent: (a), (c), (f), (h), (j), and (k)
- (a) Cl; (b) O; (c) O; (d) S; (e) N; (f) P; (g) N
- (a) H, C, N, O, F; (b) H, I, Br, Cl, F; (c) H, P, S, O, F; (d) Na, Al, H, P, O; (e) Ba, H, As, N, O
- N, O, F, and Cl
- (a) HF; (b) CO; (c) OH; (d) PCl; (e) NH; (f) PO; (g) CN

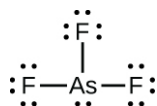


In this case, the Lewis structure is inadequate to depict the fact that experimental studies have shown two unpaired electrons in each oxygen molecule.

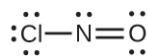
(b)



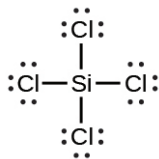
(c)



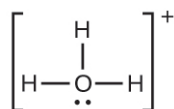
(d)



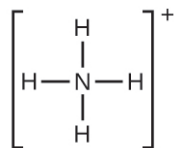
(e)



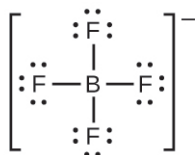
(f)



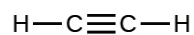
(g)



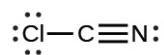
(h)



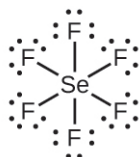
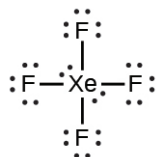
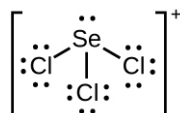
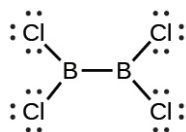
(i)



(j)

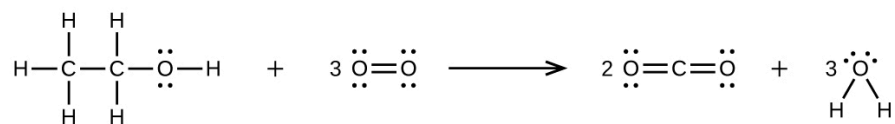
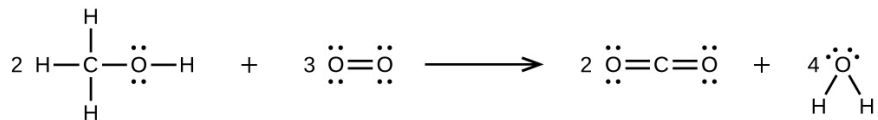


(k)

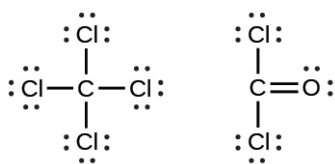
31. (a) SeF_6 :(b) XeF_4 :(c) SeCl_3^+ :(d) Cl_2BBCl_2 :

33. Two valence electrons per Pb atom are transferred to Cl atoms; the resulting Pb^{2+} ion has a $6s^2$ valence shell configuration. Two of the valence electrons in the HCl molecule are shared, and the other six are located on the Cl atom as lone pairs of electrons.

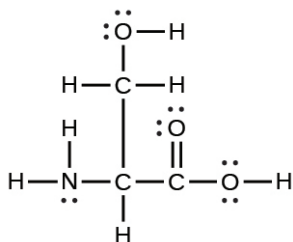
35.



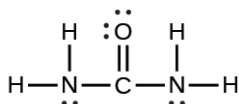
37.



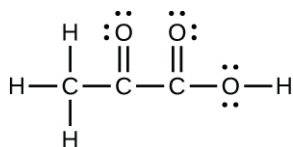
39. (a)



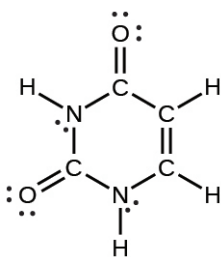
(b)



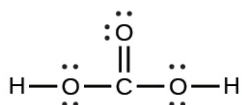
(c)



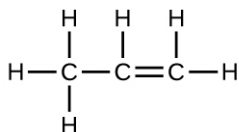
(d)



(e)



41.

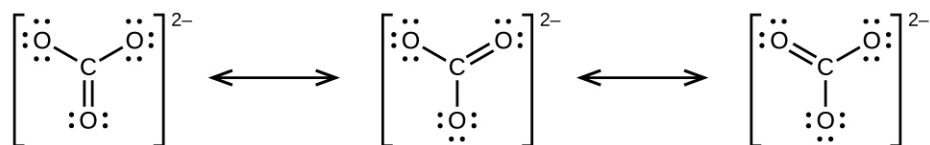


43. Each bond includes a sharing of electrons between atoms. Two electrons are shared in a single bond; four electrons are shared in a double bond; and six electrons are shared in a triple bond.

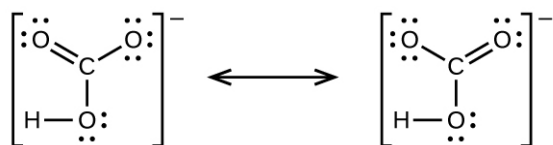
45. (a)



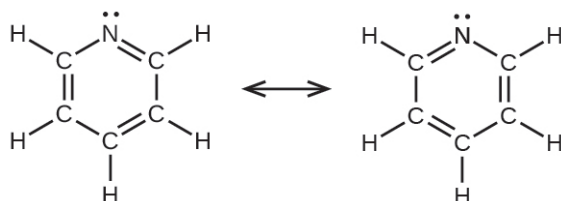
(b)



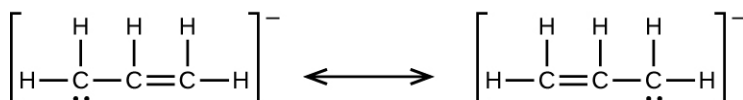
(c)



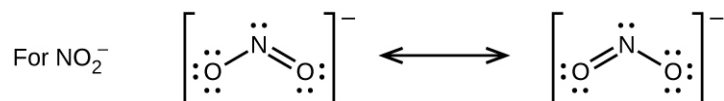
(d)



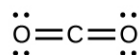
(e)



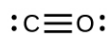
47.



49. (a)



(b)

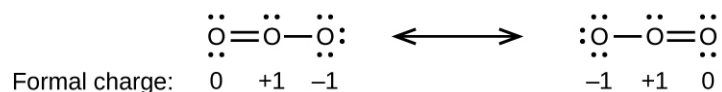


CO has the strongest carbon-oxygen bond because there is a triple bond joining C and O. CO_2 has double bonds.

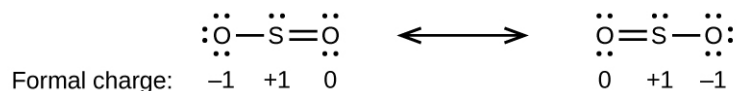
51. (a) H: 0, Cl: 0; (b) C: 0, F: 0; (c) P: 0, Cl 0; (d) P: 0, F: 0

53. Cl in Cl_2 : 0; Cl in BeCl_2 : 0; Cl in ClF_5 : 0

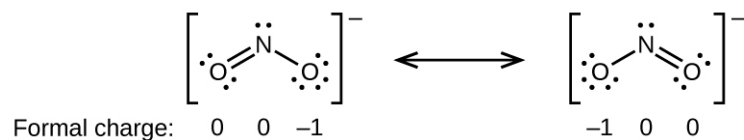
55. (a)



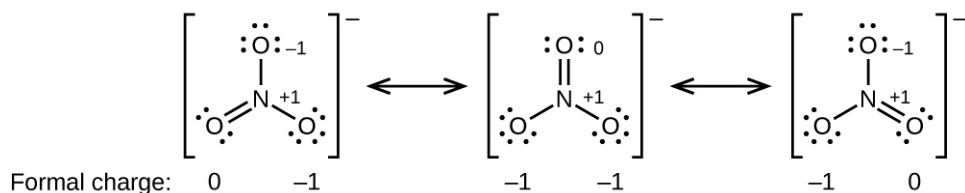
(b)



(c)

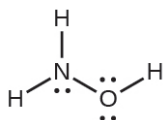
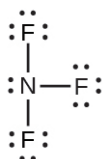


(d)

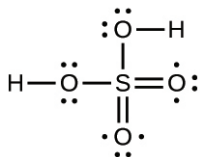


57. HOCl

59. The structure that gives zero formal charges is consistent with the actual structure:

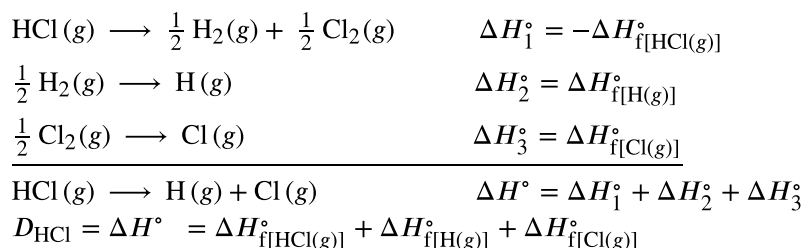
61. NF_3 ;

63.

65. (a) -114 kJ ; (b) 30 kJ ; (c) -1055 kJ

67. The greater bond energy is in the figure on the left. It is the more stable form.

69.

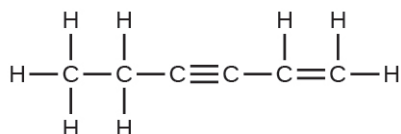


$$= -(-92.307 \text{ kJ}) + 217.97 \text{ kJ} + 121.3 \text{ kJ}$$

$$= 431.6 \text{ kJ}$$

71. The S–F bond in SF₄ is stronger.

73.



The C–C single bonds are longest.

75. (a) When two electrons are removed from the valence shell, the Ca radius loses the outermost energy level and reverts to the lower $n = 3$ level, which is much smaller in radius. (b) The +2 charge on calcium pulls the oxygen much closer compared with K, thereby increasing the lattice energy relative to a less charged ion. (c) Removal of the 4s electron in Ca requires more energy than removal of the 4s electron in K because of the stronger attraction of the nucleus and the extra energy required to break the pairing of the electrons. The second ionization energy for K requires that an electron be removed from a lower energy level, where the attraction is much stronger from the nucleus for the electron. In addition, energy is required to unpair two electrons in a full orbital. For Ca, the second ionization potential requires removing only a lone electron in the exposed outer energy level. (d) In Al, the removed electron is relatively unprotected and unpaired in a p orbital. The higher energy for Mg mainly reflects the unpairing of the 2s electron.
77. (d)
79. 4008 kJ/mol; both ions in MgO have twice the charge of the ions in LiF; the bond length is very similar and both have the same structure; a quadrupling of the energy is expected based on the equation for lattice energy
81. (a) Na₂O; Na⁺ has a smaller radius than K⁺; (b) BaS; Ba has a larger charge than K; (c) BaS; Ba and S have larger charges; (d) BaS; S has a larger charge
83. (e)
85. The placement of the two sets of unpaired electrons in water forces the bonds to assume a tetrahedral arrangement, and the resulting HOH molecule is bent. The HBeH molecule (in which Be has only two electrons to bond with the two electrons from the hydrogens) must have the electron pairs as far from one another as possible and is therefore linear.
87. Space must be provided for each pair of electrons whether they are in a bond or are present as lone pairs. Electron-pair geometry considers the placement of all electrons. Molecular structure considers only the bonding-pair geometry.
89. As long as the polar bonds are compensated (for example, two identical atoms are found directly across the central atom from one another), the molecule can be nonpolar.
91. (a) Both the electron geometry and the molecular structure are octahedral. (b) Both the electron geometry and the molecular structure are trigonal bipyramid. (c) Both the electron geometry and the molecular structure are linear. (d) Both the electron geometry and the molecular structure are trigonal planar.
93. (a) electron-pair geometry: octahedral, molecular structure: square pyramidal; (b) electron-pair geometry: tetrahedral, molecular structure: bent; (c) electron-pair geometry: octahedral, molecular structure: square planar; (d) electron-pair geometry: tetrahedral, molecular structure: trigonal pyramidal; (e) electron-pair

geometry: trigonal bipyramidal, molecular structure: seesaw; (f) electron-pair geometry: tetrahedral, molecular structure: bent (109°)

95. (a) electron-pair geometry: trigonal planar, molecular structure: bent (120°); (b) electron-pair geometry: linear, molecular structure: linear; (c) electron-pair geometry: trigonal planar, molecular structure: trigonal planar; (d) electron-pair geometry: tetrahedral, molecular structure: trigonal pyramidal; (e) electron-pair geometry: tetrahedral, molecular structure: tetrahedral; (f) electron-pair geometry: trigonal bipyramidal, molecular structure: seesaw; (g) electron-pair geometry: tetrahedral, molecular structure: trigonal pyramidal

97. All of these molecules and ions contain polar bonds. Only ClF_5 , ClO_2^- , PCl_3 , SeF_4 , and PH_2^- have dipole moments.

99. SeS_2 , CCl_2F_2 , PCl_3 , and ClNO all have dipole moments.

101. P

103. nonpolar

105. (a) tetrahedral; (b) trigonal pyramidal; (c) bent (109°); (d) trigonal planar; (e) bent ($<120^\circ$); (f) bent (109°); (g) CH_3CCH tetrahedral, CH_3CCH linear; (h) tetrahedral; (i) H_2CCCH_2 linear; H_2CCCH_2 trigonal planar

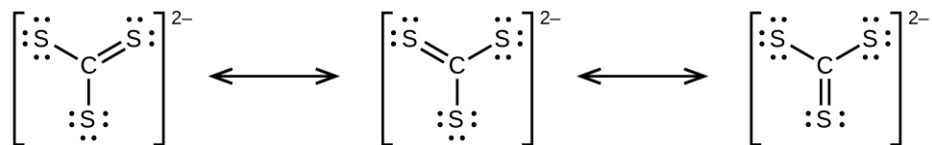
107.

$\text{B} \text{---} \text{A} \text{---} \text{B}$ CO_2 , linear

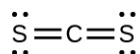
$\text{B} \text{---} \overset{\cdot\cdot}{\underset{\cdot\cdot}{\text{A}}} \text{---} \text{B}$ H_2O , bent with an approximately 109° angle

$\text{B} \text{---} \overset{\cdot\cdot}{\underset{\cdot\cdot}{\text{A}}} \text{---} \text{B}$ SO_2 , bent with an approximately 120° angle

109. (a)



(b)

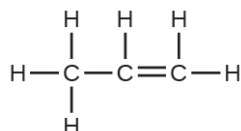


(c)



(d) CS_3^{2-} includes three regions of electron density (all are bonds with no lone pairs); the shape is trigonal planar; CS_2 has only two regions of electron density (all bonds with no lone pairs); the shape is linear

111. The Lewis structure is made from three units, but the atoms must be rearranged:

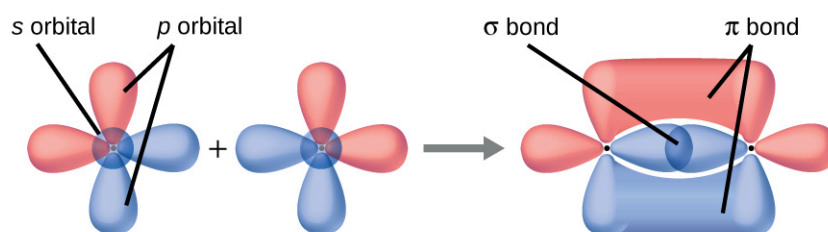


113. The molecular dipole points away from the hydrogen atoms.

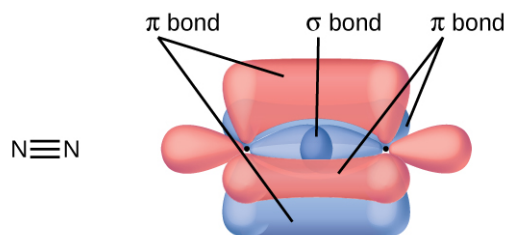
115. The structures are very similar. In the model mode, each electron group occupies the same amount of space, so the bond angle is shown as 109.5° . In the "real" mode, the lone pairs are larger, causing the hydrogens to be compressed. This leads to the smaller angle of 104.5° .

Chapter 8

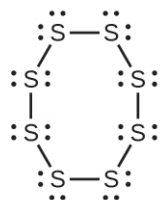
1. Similarities: Both types of bonds result from overlap of atomic orbitals on adjacent atoms and contain a maximum of two electrons. Differences: σ bonds are stronger and result from end-to-end overlap and all single bonds are σ bonds; π bonds between the same two atoms are weaker because they result from side-by-side overlap, and multiple bonds contain one or more π bonds (in addition to a σ bond).
3. The specific average bond distance is the distance with the lowest energy. At distances less than the bond distance, the positive charges on the two nuclei repel each other, and the overall energy increases.
5. Bonding: One σ bond and one π bond. The s orbitals are filled and do not overlap. The p orbitals overlap along the axis to form a σ bond and side-by-side to form the π bond.



7. No, two of the p orbitals (one on each N) will be oriented end-to-end and will form a σ bond.



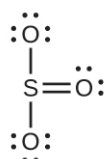
9. Hybridization is introduced to explain the geometry of bonding orbitals in valence bond theory.
11. There are no d orbitals in the valence shell of carbon.
13. trigonal planar, sp^2 ; trigonal pyramidal (one lone pair on A) sp^3 ; T-shaped (two lone pairs on A sp^3d , or (three lone pairs on A) sp^3d^2
15. (a) Each S has a bent (109°) geometry, sp^3

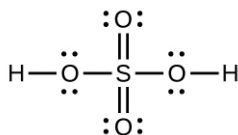


- (b) Bent (120°), sp^2

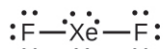


- (c) Trigonal planar, sp^2

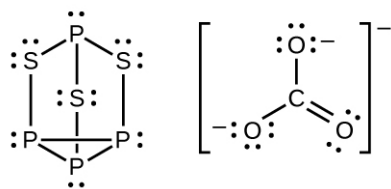


(d) Tetrahedral, sp^3 17. (a) XeF_2

(b)

(c) linear (d) sp^3d

19. (a)



(b) P atoms, trigonal pyramidal; S atoms, bent, with two lone pairs; Cl atoms, trigonal pyramidal; (c) Hybridization about P, S, and Cl is, in all cases, sp^3 ; (d) Oxidation states P +1, S $-1\frac{1}{3}$, Cl +5, O -2. Formal charges: P 0; S 0; Cl +2; O -1

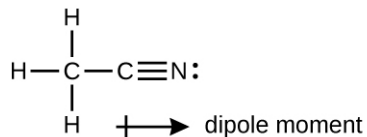
21.



Phosphorus and nitrogen can form sp^3 hybrids to form three bonds and hold one lone pair in PF_3 and NF_3 , respectively. However, nitrogen has no valence d orbitals, so it cannot form a set of sp^3d hybrid orbitals to bind five fluorine atoms in NF_5 . Phosphorus has d orbitals and can bind five fluorine atoms with sp^3d hybrid orbitals in PF_5 .

23. A triple bond consists of one σ bond and two π bonds. A σ bond is stronger than a π bond due to greater overlap.

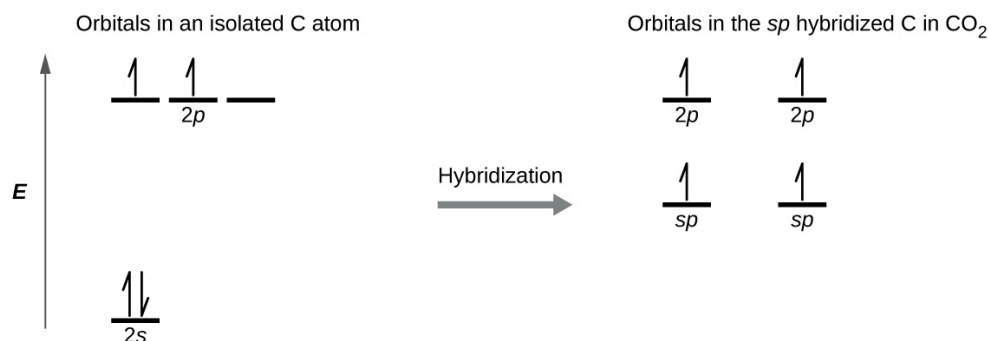
25. (a)



(b) The terminal carbon atom uses sp^3 hybrid orbitals, while the central carbon atom is sp hybridized. (c) Each of the two π bonds is formed by overlap of a $2p$ orbital on carbon and a nitrogen $2p$ orbital.

27. (a) sp^2 ; (b) sp ; (c) sp^2 ; (d) sp^3 ; (e) sp^3 ; (f) sp^3d ; (g) sp^3 29. (a) sp^2 , delocalized; (b) sp , localized; (c) sp^2 , delocalized; (d) sp^3 , delocalized

31.



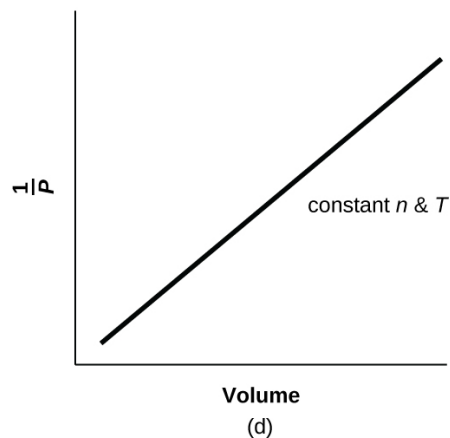
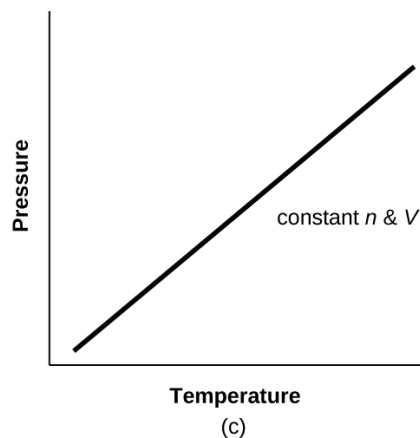
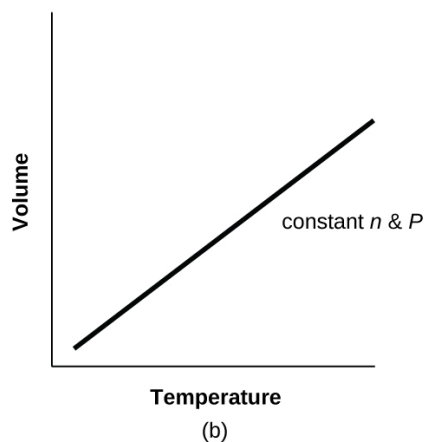
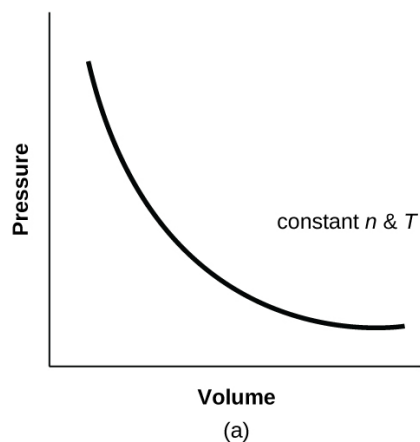
Each of the four electrons is in a separate orbital and overlaps with an electron on an oxygen atom.

33. (a) Similarities: Both are bonding orbitals that can contain a maximum of two electrons. Differences: σ orbitals are end-to-end combinations of atomic orbitals, whereas π orbitals are formed by side-by-side overlap of orbitals. (b) Similarities: Both are quantum-mechanical constructs that represent the probability of finding the electron about the atom or the molecule. Differences: ψ for an atomic orbital describes the behavior of only one electron at a time based on the atom. For a molecule, ψ represents a mathematical combination of atomic orbitals. (c) Similarities: Both are orbitals that can contain two electrons. Differences: Bonding orbitals result in holding two or more atoms together. Antibonding orbitals have the effect of destabilizing any bonding that has occurred.
35. An odd number of electrons can never be paired, regardless of the arrangement of the molecular orbitals. It will always be paramagnetic.
37. Bonding orbitals have electron density in close proximity to more than one nucleus. The interaction between the bonding positively charged nuclei and negatively charged electrons stabilizes the system.
39. The pairing of the two bonding electrons lowers the energy of the system relative to the energy of the nonbonded electrons.
41. (a) H_2 bond order = 1, H_2^+ bond order = 0.5, H_2^- bond order = 0.5, strongest bond is H_2 ; (b) O_2 bond order = 2, O_2^{2+} bond order = 3; O_2^{2-} bond order = 1, strongest bond is O_2^{2+} ; (c) Li_2 bond order = 1, Be_2^+ bond order = 0.5, Be_2 bond order = 0, strongest bond is Li_2 ; (d) F_2 bond order = 1, F_2^+ bond order = 1.5, F_2^- bond order = 0.5, strongest bond is F_2^+ ; (e) N_2 bond order = 3, N_2^+ bond order = 2.5, N_2^- bond order = 2.5, strongest bond is N_2
43. (a) H_2 ; (b) N_2 ; (c) O; (d) C_2 ; (e) B_2
45. Yes, fluorine is a smaller atom than Li, so atoms in the 2s orbital are closer to the nucleus and more stable.
47. 2+
49. N_2 has s-p mixing, so the π orbitals are the last filled in N_2^{2+} . O_2 does not have s-p mixing, so the σ_p orbital fills before the π orbitals.

Chapter 9

- The cutting edge of a knife that has been sharpened has a smaller surface area than a dull knife. Since pressure is force per unit area, a sharp knife will exert a higher pressure with the same amount of force and cut through material more effectively.
- Lying down distributes your weight over a larger surface area, exerting less pressure on the ice compared to standing up. If you exert less pressure, you are less likely to break through thin ice.
- 0.809 atm; 82.0 kPa
- 2.2×10^2 kPa
- Earth: 14.7 lb in^{-2} ; Venus: $1.30 \times 10^3 \text{ lb in}^{-2}$
- (a) 101.5 kPa; (b) 51 torr drop
- (a) 264 torr; (b) 35,200 Pa; (c) 0.352 bar
- (a) 623 mm Hg; (b) 0.820 atm; (c) 83.1 kPa
- With a closed-end manometer, no change would be observed, since the vaporized liquid would contribute equal, opposing pressures in both arms of the manometer tube. However, with an open-ended manometer, a higher pressure reading of the gas would be obtained than expected, since $P_{\text{gas}} = P_{\text{atm}} + P_{\text{vol liquid}}$.

19. As the bubbles rise, the pressure decreases, so their volume increases as suggested by Boyle's law.
 21. (a) The number of particles in the gas increases as the volume increases. (b) temperature, pressure
 23. The curve would be farther to the right and higher up, but the same basic shape.
 25. About 12.5 L
 27. 3.40×10^3 torr
 29. 12.1 L
 31. 217 L
 33. 8.190×10^{-2} mol; 5.553 g
 35. (a) 7.24×10^{-2} g; (b) 23.1 g; (c) 1.5×10^{-4} g
 37. 5561 L
 39. 46.4 g
 41. For a gas exhibiting ideal behavior:



43. (a) 1.85 L CCl_2F_2 ; (b) 4.66 L $\text{CH}_3\text{CH}_2\text{F}$
 45. 0.644 atm
 47. The pressure decreases by a factor of 3.
 49. 4.64 g L^{-1}
 51. 38.8 g
 53. 72.0 g mol^{-1}
 55. 88.1 g mol^{-1} ; PF_3
 57. 141 atm, 107,000 torr, 14,300 kPa
 59. CH_4 : 276 kPa; C_2H_6 : 27 kPa; C_3H_8 : 3.4 kPa
 61. Yes
 63. 740 torr
 65. (a) Determine the moles of HgO that decompose; using the chemical equation, determine the moles of O_2

produced by decomposition of this amount of HgO ; and determine the volume of O_2 from the moles of O_2 , temperature, and pressure. (b) 0.308 L

67. (a) Determine the molar mass of CCl_2F_2 . From the balanced equation, calculate the moles of H_2 needed for the complete reaction. From the ideal gas law, convert moles of H_2 into volume. (b) 3.72×10^3 L
69. (a) Balance the equation. Determine the grams of CO_2 produced and the number of moles. From the ideal gas law, determine the volume of gas. (b) 7.43×10^5 L

71. 42.00 L

73. (a) 18.0 L; (b) 0.533 atm

75. 10.57 L O_2

77. 5.40×10^5 L

79. XeF_4

81. 4.2 hours

83. Effusion can be defined as the process by which a gas escapes through a pinhole into a vacuum. Graham's law states that with a mixture of two gases A and B: $\left(\frac{\text{rate A}}{\text{rate B}}\right) = \left(\frac{\text{molar mass of B}}{\text{molar mass of A}}\right)^{1/2}$. Both A and B are in the same container at the same temperature, and therefore will have the same kinetic energy:

$$\text{KE}_A = \text{KE}_B \quad \text{KE} = \frac{1}{2} m v^2$$

$$\text{Therefore, } \frac{1}{2} m_A v_A^2 = \frac{1}{2} m_B v_B^2$$

$$\frac{v_A^2}{v_B^2} = \frac{m_B}{m_A}$$

$$\left(\frac{v_A^2}{v_B^2}\right)^{1/2} = \left(\frac{m_B}{m_A}\right)^{1/2}$$

$$\frac{v_A}{v_B} = \left(\frac{m_B}{m_A}\right)^{1/2}$$

85. F_2 , N_2O , Cl_2 , H_2S

87. 1.4; 1.2

89. 51.7 cm

91. Yes. At any given instant, there are a range of values of molecular speeds in a sample of gas. Any single molecule can speed up or slow down as it collides with other molecules. The average speed of all the molecules is constant at constant temperature.

93. H_2O . Cooling slows the speeds of the He atoms, causing them to behave as though they were heavier.

95. (a) The pressure of the gas remains constant. (b) The average kinetic energy doubles. (c) The root mean square speed increases to $\sqrt{2}$ times its initial value; u_{rms} is proportional to $\sqrt{\text{KE}_{\text{avg}}}$.

97. (a) equal; (b) less than; (c) 29.48 g mol^{-1} ; (d) 1.0966 g L^{-1} ; (e) 0.129 g/L; (f) 4.01×10^5 g; net lifting capacity = 384 lb; (g) 270 L; (h) 39.1 kJ min^{-1}

99. Gases C, E, and F

101. The gas behavior most like an ideal gas will occur under the conditions that minimize the chances of significant interactions between the gaseous atoms/molecules, namely, low pressures (fewer atoms/molecules per unit volume) and high temperatures (greater kinetic energies of atoms/molecules make them less susceptible to attractive forces). The conditions described in (b), high temperature and low pressure, are therefore most likely to yield ideal gas behavior.

103. SF_6

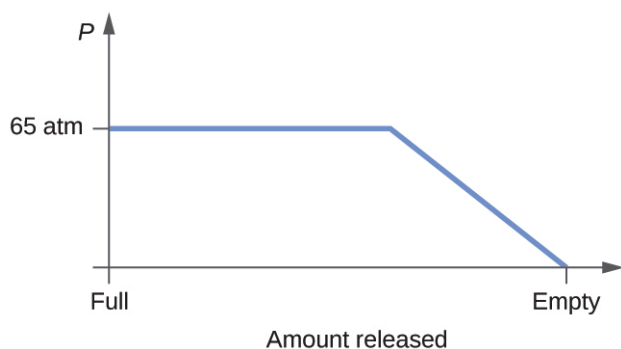
105. (a) A straight horizontal line at 1.0; (b) When real gases are at low pressures and high temperatures, they behave close enough to ideal gases that they are approximated as such; however, in some cases, we see that at a high pressure and temperature, the ideal gas approximation breaks down and is significantly different from the pressure calculated by the ideal gas equation. (c) The greater the compressibility, the more the volume matters. At low pressures, the correction factor for intermolecular attractions is more significant, and the effect of the volume of the gas molecules on Z would be a small lowering compressibility. At higher pressures, the effect of the volume of the gas molecules themselves on Z would increase compressibility (see [Figure 9.35](#)). (d) Once again, at low pressures, the effect of intermolecular attractions on Z would be more important than the correction factor for the volume of the gas molecules themselves, though perhaps still small. At higher pressures and low temperatures, the effect of intermolecular attractions would be larger. See

Figure 9.35. (e) Low temperatures

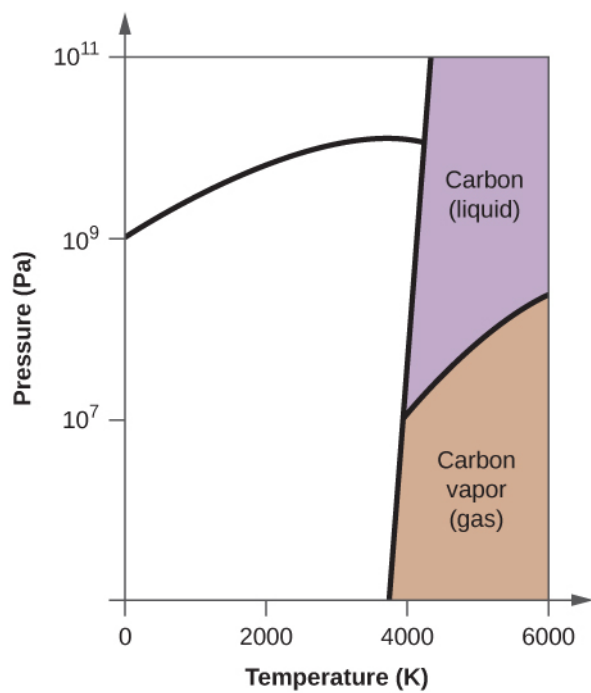
Chapter 10

1. Liquids and solids are similar in that they are matter composed of atoms, ions, or molecules. They are incompressible and have similar densities that are both much larger than those of gases. They are different in that liquids have no fixed shape, and solids are rigid.
3. They are similar in that the atoms or molecules are free to move from one position to another. They differ in that the particles of a liquid are confined to the shape of the vessel in which they are placed. In contrast, a gas will expand without limit to fill the space into which it is placed.
5. All atoms and molecules will condense into a liquid or solid in which the attractive forces exceed the kinetic energy of the molecules, at sufficiently low temperature.
7. (a) Dispersion forces occur as an atom develops a temporary dipole moment when its electrons are distributed asymmetrically about the nucleus. This structure is more prevalent in large atoms such as argon or radon. A second atom can then be distorted by the appearance of the dipole in the first atom. The electrons of the second atom are attracted toward the positive end of the first atom, which sets up a dipole in the second atom. The net result is rapidly fluctuating, temporary dipoles that attract one another (e.g., Ar). (b) A dipole-dipole attraction is a force that results from an electrostatic attraction of the positive end of one polar molecule for the negative end of another polar molecule (e.g., ICI molecules attract one another by dipole-dipole interaction). (c) Hydrogen bonds form whenever a hydrogen atom is bonded to one of the more electronegative atoms, such as a fluorine, oxygen, or nitrogen atom. The electrostatic attraction between the partially positive hydrogen atom in one molecule and the partially negative atom in another molecule gives rise to a strong dipole-dipole interaction called a hydrogen bond (e.g., $\text{HF} \cdots \text{HF}$).
9. The London forces typically increase as the number of electrons increase.
11. (a) $\text{SiH}_4 < \text{HCl} < \text{H}_2\text{O}$; (b) $\text{F}_2 < \text{Cl}_2 < \text{Br}_2$; (c) $\text{CH}_4 < \text{C}_2\text{H}_6 < \text{C}_3\text{H}_8$; (d) $\text{N}_2 < \text{O}_2 < \text{NO}$
13. Only rather small dipole-dipole interactions from C-H bonds are available to hold *n*-butane in the liquid state. Chloroethane, however, has rather large dipole interactions because of the Cl-C bond; the interaction, therefore, is stronger, leading to a higher boiling point.
15. -85°C . Water has stronger hydrogen bonds, so it melts at a higher temperature.
17. The hydrogen bond between two hydrogen fluoride molecules is stronger than that between two water molecules because the electronegativity of F is greater than that of O. Consequently, the partial negative charge on F is greater than that on O. The hydrogen bond between the partially positive H and the larger partially negative F will be stronger than that formed between H and O.
19. H-bonding is the principle IMF holding the protein strands together. The H-bonding is between the $\text{N}-\text{H}$ and $\text{C}=\text{O}$.
21. (a) hydrogen bonding, dipole-dipole attraction, and dispersion forces; (b) dispersion forces; (c) dipole-dipole attraction and dispersion forces
23. The water molecules have strong intermolecular forces of hydrogen bonding. The water molecules are thus attracted strongly to one another and exhibit a relatively large surface tension, forming a type of "skin" at its surface. This skin can support a bug or paper clip if gently placed on the water.
25. Temperature has an effect on intermolecular forces: The higher the temperature, the greater the kinetic energies of the molecules and the greater the extent to which their intermolecular forces are overcome, and so the more fluid (less viscous) the liquid. The lower the temperature, the less the intermolecular forces are overcome, and so the less viscous the liquid.
27. (a) As the water reaches higher temperatures, the increased kinetic energies of its molecules are more effective in overcoming hydrogen bonding, and so its surface tension decreases. Surface tension and intermolecular forces are directly related. (b) The same trend in viscosity is seen as in surface tension, and for the same reason.
29. $1.7 \times 10^{-4} \text{ m}$
31. The heat is absorbed by the ice, providing the energy required to partially overcome intermolecular attractive forces in the solid and causing a phase transition to liquid water. The solution remains at 0°C until all the ice is melted. Only the amount of water existing as ice changes until the ice disappears. Then the temperature of the water can rise.
33. We can see the amount of liquid in an open container decrease and we can smell the vapor of some liquids.

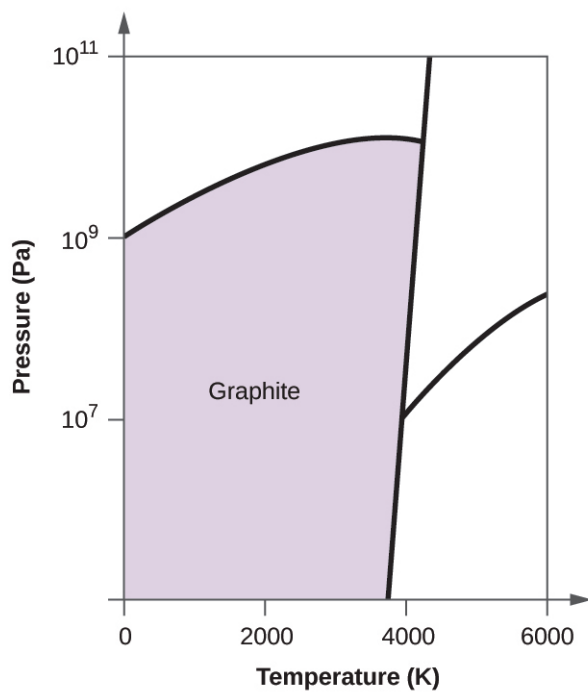
35. The vapor pressure of a liquid decreases as the strength of its intermolecular forces increases.
37. As the temperature increases, the average kinetic energy of the molecules of gasoline increases and so a greater fraction of molecules have sufficient energy to escape from the liquid than at lower temperatures.
39. They are equal when the pressure of gas above the liquid is exactly 1 atm.
41. approximately 95 °C
43. (a) At 5000 feet, the atmospheric pressure is lower than at sea level, and water will therefore boil at a lower temperature. This lower temperature will cause the physical and chemical changes involved in cooking the egg to proceed more slowly, and a longer time is required to fully cook the egg. (b) As long as the air surrounding the body contains less water vapor than the maximum that air can hold at that temperature, perspiration will evaporate, thereby cooling the body by removing the heat of vaporization required to vaporize the water.
45. Dispersion forces increase with molecular mass or size. As the number of atoms composing the molecules in this homologous series increases, so does the extent of intermolecular attraction via dispersion forces and, consequently, the energy required to overcome these forces and vaporize the liquids.
47. The boiling point of CS_2 is higher than that of CO_2 partially because of the higher molecular weight of CS_2 ; consequently, the attractive forces are stronger in CS_2 . It would be expected, therefore, that the heat of vaporization would be greater than that of 9.8 kJ/mol for CO_2 . A value of 28 kJ/mol would seem reasonable. A value of -8.4 kJ/mol would indicate a release of energy upon vaporization, which is clearly implausible.
49. The thermal energy (heat) needed to evaporate the liquid is removed from the skin.
51. 1125 kJ
53. (a) 13.0 kJ; (b) It is likely that the heat of vaporization will have a larger magnitude since in the case of vaporization the intermolecular interactions have to be completely overcome, while melting weakens or destroys only some of them.
55. At low pressures and 0.005 °C, the water is a gas. As the pressure increases to 4.6 torr, the water becomes a solid; as the pressure increases still more, it becomes a liquid. At 40 °C, water at low pressure is a vapor; at pressures higher than about 75 torr, it converts into a liquid. At -40 °C, water goes from a gas to a solid as the pressure increases above very low values.
57. (a) gas; (b) gas; (c) gas; (d) gas; (e) solid; (f) gas
- 59.



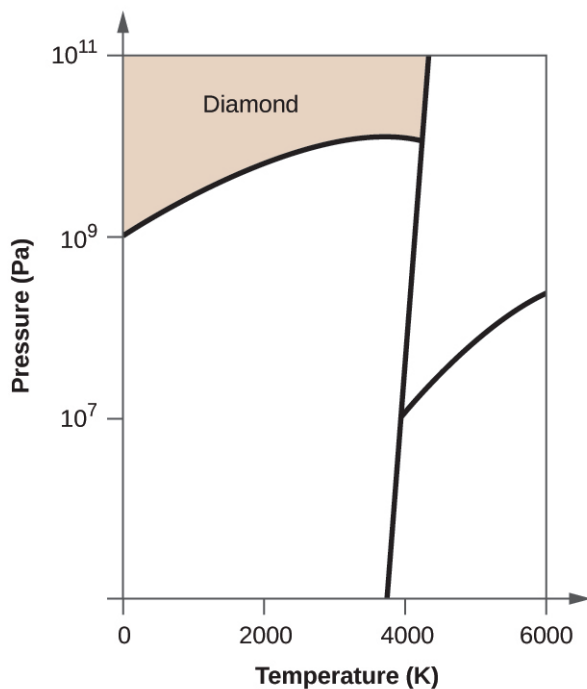
61. Yes, ice will sublime, although it may take it several days. Ice has a small vapor pressure, and some ice molecules form gas and escape from the ice crystals. As time passes, more and more solid converts to gas until eventually the clothes are dry.
63. (a)



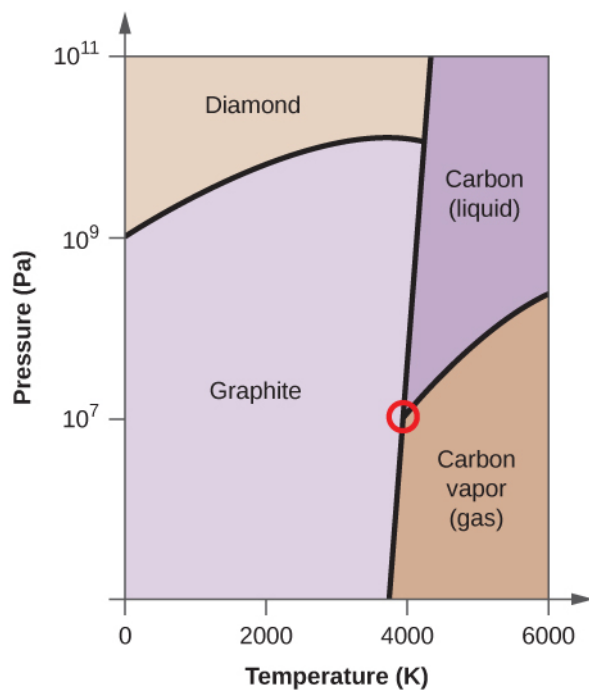
(b)



(c)



(d)



(e) liquid phase (f) sublimation

65. (e) molecular crystals

67. Ice has a crystalline structure stabilized by hydrogen bonding. These intermolecular forces are of comparable strength and thus require the same amount of energy to overcome. As a result, ice melts at a single temperature and not over a range of temperatures. The various, very large molecules that compose butter experience varied van der Waals attractions of various strengths that are overcome at various temperatures,

and so the melting process occurs over a wide temperature range.

69. (a) ionic; (b) covalent network; (c) molecular; (d) metallic; (e) covalent network; (f) molecular; (g) molecular; (h) ionic; (i) ionic
71. X = ionic; Y = metallic; Z = covalent network
73. (b) metallic solid
75. The structure of this low-temperature form of iron (below 910 °C) is body-centered cubic. There is one-eighth atom at each of the eight corners of the cube and one atom in the center of the cube.
77. eight
79. 12
81. (a) 1.370 Å; (b) 19.26 g/cm
83. (a) 2.176 Å; (b) 3.595 g/cm³
85. The crystal structure of Si shows that it is less tightly packed (coordination number 4) in the solid than Al (coordination number 12).
87. In a closest-packed array, two tetrahedral holes exist for each anion. If only half the tetrahedral holes are occupied, the numbers of anions and cations are equal. The formula for cadmium sulfide is CdS.
89. Co₃O₄
91. In a simple cubic array, only one cubic hole can be occupied by a cation for each anion in the array. The ratio of thallium to iodide must be 1:1; therefore, the formula for thallium is TlI.
93. 59.95%; The oxidation number of titanium is +4.
95. Both ions are close in size: Mg, 0.65; Li, 0.60. This similarity allows the two to interchange rather easily. The difference in charge is generally compensated by the switch of Si⁴⁺ for Al³⁺.
97. Mn₂O₃
99. 1.48 Å
101. 2.874 Å
103. 20.2°
105. 1.74×10^4 eV

Chapter 11

- A solution can vary in composition, while a compound cannot vary in composition. Solutions are homogeneous at the molecular level, while other mixtures are heterogeneous.
- (a) The process is endothermic as the solution is consuming heat. (b) Attraction between the K⁺ and NO₃[−] ions is stronger than between the ions and water molecules (the ion-ion interactions have a lower, more negative energy). Therefore, the dissolution process increases the energy of the molecular interactions, and it consumes the thermal energy of the solution to make up for the difference. (c) No, an ideal solution is formed with no appreciable heat release or consumption.
- (a) ion-dipole forces; (b) dipole-dipole forces; (c) dispersion forces; (d) dispersion forces; (e) hydrogen bonding
- Heat is released when the total intermolecular forces (IMFs) between the solute and solvent molecules are stronger than the total IMFs in the pure solute and in the pure solvent: Breaking weaker IMFs and forming stronger IMFs releases heat. Heat is absorbed when the total IMFs in the solution are weaker than the total of those in the pure solute and in the pure solvent: Breaking stronger IMFs and forming weaker IMFs absorbs heat.
- Crystals of NaCl dissolve in water, a polar liquid with a very large dipole moment, and the individual ions become strongly solvated. Hexane is a nonpolar liquid with a dipole moment of zero and, therefore, does not significantly interact with the ions of the NaCl crystals.
- (a) Fe(NO₃)₃ is a strong electrolyte, thus it should completely dissociate into Fe³⁺ and NO₃[−] ions. Therefore, (z) best represents the solution. (b) $\text{Fe}(\text{NO}_3)_3(s) \rightarrow \text{Fe}^{3+}(aq) + 3\text{NO}_3^{-}(aq)$
- (a) high conductivity (solute is an ionic compound that will dissociate when dissolved); (b) high conductivity (solute is a strong acid and will ionize completely when dissolved); (c) nonconductive (solute is a covalent compound, neither acid nor base, unreactive towards water); (d) low conductivity (solute is a weak base and will partially ionize when dissolved)
- (a) ion-dipole; (b) hydrogen bonds; (c) dispersion forces; (d) dipole-dipole attractions; (e) dispersion forces
- The solubility of solids usually decreases upon cooling a solution, while the solubility of gases usually

decreases upon heating.

19. 40%
21. 2.8 g
23. 2.9 atm
25. 102 L HCl
27. The strength of the bonds between like molecules is stronger than the strength between unlike molecules. Therefore, some regions will exist in which the water molecules will exclude oil molecules and other regions will exist in which oil molecules will exclude water molecules, forming a heterogeneous region.
29. Both form homogeneous solutions; their boiling point elevations are the same, as are their lowering of vapor pressures. Osmotic pressure and the lowering of the freezing point are also the same for both solutions.
31. (a) Find number of moles of HNO_3 and H_2O in 100 g of the solution. Find the mole fractions for the components. (b) The mole fraction of HNO_3 is 0.378. The mole fraction of H_2O is 0.622.
33. (a) $X_{\text{Na}_2\text{CO}_3} = 0.0119$; $X_{\text{H}_2\text{O}} = 0.988$; (b) $X_{\text{NH}_4\text{NO}_3} = 0.0928$; $X_{\text{H}_2\text{O}} = 0.907$; (c) $X_{\text{Cl}_2} = 0.192$; $X_{\text{CH}_2\text{Cl}_2} = 0.808$; (d) $X_{\text{C}_5\text{H}_9\text{N}} = 0.00426$; $X_{\text{CHCl}_3} = 0.997$
35. In a 1 *M* solution, the mole is contained in exactly 1 L of solution. In a 1 *m* solution, the mole is contained in exactly 1 kg of solvent.
37. (a) Determine the molar mass of HNO_3 . Determine the number of moles of acid in the solution. From the number of moles and the mass of solvent, determine the molality. (b) 33.7 *m*
39. (a) 6.70×10^{-1} *m*; (b) 5.67 *m*; (c) 2.8 *m*; (d) 0.0358 *m*
41. 1.08 *m*
43. (a) Determine the molar mass of sucrose; determine the number of moles of sucrose in the solution; convert the mass of solvent to units of kilograms; from the number of moles and the mass of solvent, determine the molality; determine the difference between the boiling point of water and the boiling point of the solution; determine the new boiling point. (b) 100.5 °C
45. (a) Determine the molar mass of sucrose; determine the number of moles of sucrose in the solution; convert the mass of solvent to units of kilograms; from the number of moles and the mass of solvent, determine the molality; determine the difference between the freezing temperature of water and the freezing temperature of the solution; determine the new freezing temperature. (b) -1.8 °C
47. (a) Determine the molar mass of $\text{Ca}(\text{NO}_3)_2$; determine the number of moles of $\text{Ca}(\text{NO}_3)_2$ in the solution; determine the number of moles of ions in the solution; determine the molarity of ions, then the osmotic pressure. (b) 2.67 atm
49. (a) Determine the molal concentration from the change in boiling point and K_b ; determine the moles of solute in the solution from the molal concentration and mass of solvent; determine the molar mass from the number of moles and the mass of solute. (b) 2.1×10^2 g mol⁻¹
51. No. Pure benzene freezes at 5.5 °C, and so the observed freezing point of this solution is depressed by $\Delta T_f = 5.5 - 0.4 = 5.1$ °C. The value computed, assuming no ionization of HCl, is $\Delta T_f = (1.0 \text{ m})(5.14 \text{ °C}/m) = 5.1$ °C. Agreement of these values supports the assumption that HCl is not ionized.
53. 144 g mol⁻¹
55. 0.870 °C
57. S_8
59. 1.39×10^4 g mol⁻¹
61. 54 g
63. 100.26 °C
65. (a) $X_{\text{CH}_3\text{OH}} = 0.590$; $X_{\text{C}_2\text{H}_5\text{OH}} = 0.410$; (b) Vapor pressures are: CH_3OH : 55 torr; $\text{C}_2\text{H}_5\text{OH}$: 18 torr; (c) CH_3OH : 0.75; $\text{C}_2\text{H}_5\text{OH}$: 0.25
67. The ions and compounds present in the water in the beef lower the freezing point of the beef below -1 °C.
69. $\Delta bp = K_b m = (1.20 \text{ °C}/m) \left(\frac{9.41 \text{ g} \times \frac{1 \text{ mol Hg Cl}_2}{271.496 \text{ g}}}{0.03275 \text{ kg}} \right) = 1.27 \text{ °C}$

The observed change equals the theoretical change; therefore, no dissociation occurs.

71.

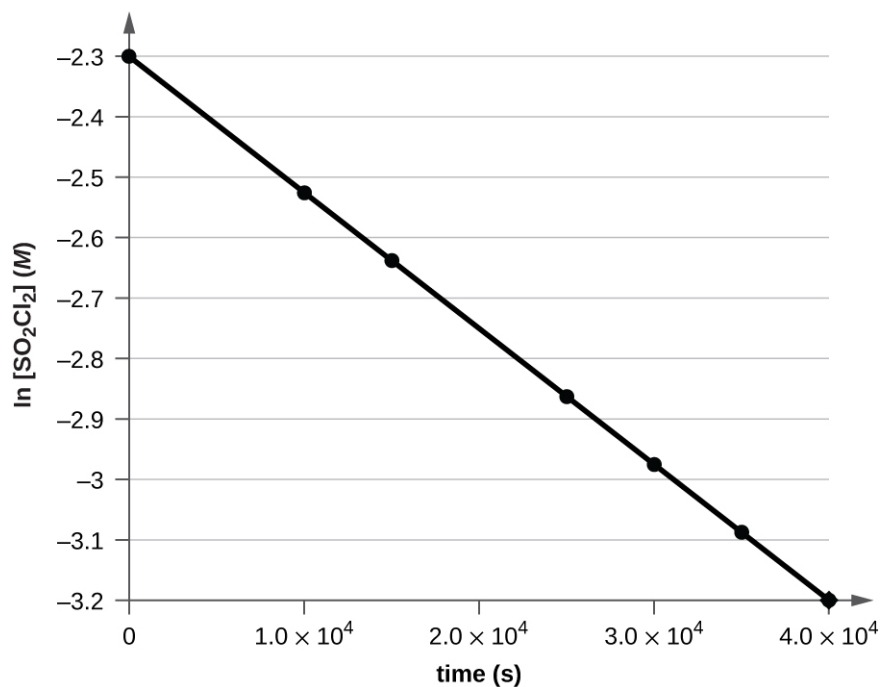
Colloidal System	Dispersed Phase	Dispersion Medium
starch dispersion	starch	water
smoke	solid particles	air
fog	water	air
pearl	water	calcium carbonate (CaCO_3)
whipped cream	air	cream
floating soap	air	soap
jelly	fruit juice	pectin gel
milk	butterfat	water
ruby	chromium(III) oxide (Cr_2O_3)	aluminum oxide (Al_2O_3)

73. Colloidal dispersions consist of particles that are much bigger than the solutes of typical solutions. Colloidal particles are either very large molecules or aggregates of smaller species that usually are big enough to scatter light. Colloids are homogeneous on a macroscopic (visual) scale, while solutions are homogeneous on a microscopic (molecular) scale.
75. If they are placed in an electrolytic cell, dispersed particles will move toward the electrode that carries a charge opposite to their own charge. At this electrode, the charged particles will be neutralized and will coagulate as a precipitate.

Chapter 12

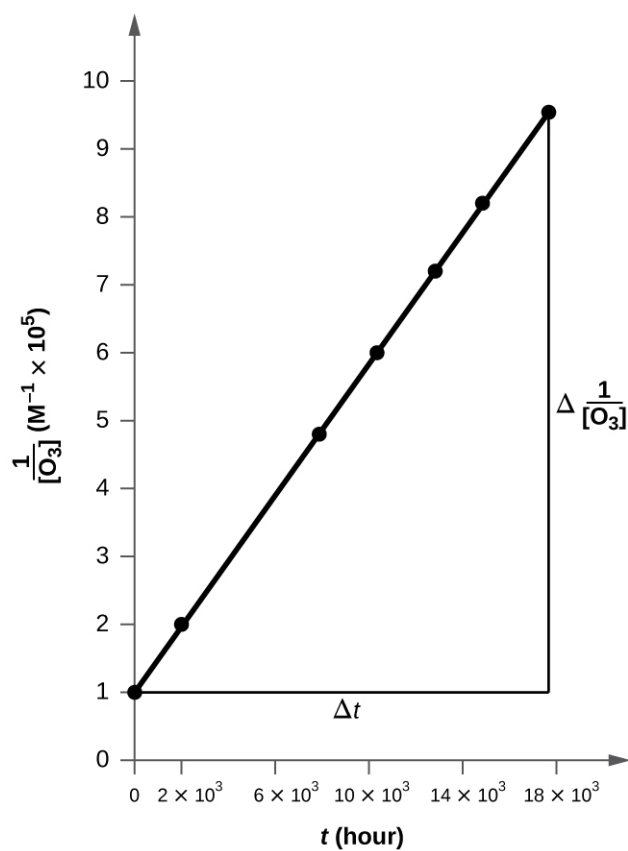
- The instantaneous rate is the rate of a reaction at any particular point in time, a period of time that is so short that the concentrations of reactants and products change by a negligible amount. The initial rate is the instantaneous rate of reaction as it starts (as product just begins to form). Average rate is the average of the instantaneous rates over a time period.
- rate = $+\frac{1}{2} \frac{\Delta[\text{ClF}_3]}{\Delta t} = -\frac{\Delta[\text{Cl}_2]}{\Delta t} = -\frac{1}{3} \frac{\Delta[\text{F}_2]}{\Delta t}$
- (a) average rate, 0 – 10 s = $0.0375 \text{ mol L}^{-1} \text{ s}^{-1}$; average rate, 10 – 20 s = $0.0265 \text{ mol L}^{-1} \text{ s}^{-1}$; (b) instantaneous rate, 15 s = $0.023 \text{ mol L}^{-1} \text{ s}^{-1}$; (c) average rate for B formation = $0.0188 \text{ mol L}^{-1} \text{ s}^{-1}$; instantaneous rate for B formation = $0.012 \text{ mol L}^{-1} \text{ s}^{-1}$
- Higher molarity increases the rate of the reaction. Higher temperature increases the rate of the reaction. Smaller pieces of magnesium metal will react more rapidly than larger pieces because more reactive surface exists.
- (a) Depending on the angle selected, the atom may take a long time to collide with the molecule and, when a collision does occur, it may not result in the breaking of the bond and the forming of the other. (b) Particles of reactant must come into contact with each other before they can react.
- (a) very slow; (b) As the temperature is increased, the reaction proceeds at a faster rate. The amount of reactants decreases, and the amount of products increases. After a while, there is a roughly equal amount of BC, AB, and C in the mixture and a slight excess of A.
- (a) 2; (b) 1
- (a) The process reduces the rate by a factor of 4. (b) Since CO does not appear in the rate law, the rate is not affected.
- $4.3 \times 10^{-5} \text{ mol/L/s}$
- $7.9 \times 10^{-13} \text{ mol/L/year}$

21. rate = k ; $k = 2.0 \times 10^{-2} \text{ mol L}^{-1} \text{ h}^{-1}$ (about $0.9 \text{ g L}^{-1} \text{ h}^{-1}$ for the average male); The reaction is zero order.
23. rate = $k[\text{NOCl}]^2$; $k = 8.0 \times 10^{-8} \text{ L/mol/h}$; second order
25. rate = $k[\text{NO}]^2[\text{Cl}_2]$; $k = 9.1 \text{ L}^2 \text{ mol}^{-2} \text{ h}^{-1}$; second order in NO; first order in Cl_2
27. (a) The rate law is second order in A and is written as rate = $k[A]^2$. (b) $k = 7.88 \times 10^{-3} \text{ L mol}^{-1} \text{ s}^{-1}$
29. (a) $2.5 \times 10^{-4} \text{ mol/L/min}$
31. rate = $k[\text{I}^-][\text{OCl}^-]$; $k = 6.1 \times 10^{-2} \text{ L mol}^{-1} \text{ s}^{-1}$
33. Plotting a graph of $\ln[\text{SO}_2\text{Cl}_2]$ versus t reveals a linear trend; therefore we know this is a first-order reaction:



$$k = 2.20 \times 10^{-5} \text{ s}^{-1}$$

34.



The plot is nicely linear, so the reaction is second order. $k = 50.1 \text{ L mol}^{-1} \text{ h}^{-1}$

36. 14.3 d

38. $8.3 \times 10^7 \text{ s}$

40. 0.826 s

42. The reaction is first order. $k = 1.0 \times 10^7 \text{ L mol}^{-1} \text{ min}^{-1}$ 44. $1.16 \times 10^3 \text{ s}$; 20% remains

46. 252 days

48.

$[A]_0 \text{ (M)}$	$k \times 10^3 \text{ (s}^{-1}\text{)}$
4.88	2.45
3.52	2.51
2.29	2.53
1.81	2.58
5.33	2.36
4.05	2.47

$[A]_0$ (M)	$k \times 10^3$ (s ⁻¹)
2.95	2.48
1.72	2.43

50. The reactants either may be moving too slowly to have enough kinetic energy to exceed the activation energy for the reaction, or the orientation of the molecules when they collide may prevent the reaction from occurring.
52. The activation energy is the minimum amount of energy necessary to form the activated complex in a reaction. It is usually expressed as the energy necessary to form one mole of activated complex.
54. After finding k at several different temperatures, a plot of $\ln k$ versus $\frac{1}{T}$, gives a straight line with the slope $\frac{-E_a}{R}$ from which E_a may be determined.
56. (a) 4-times faster (b) 128-times faster
58. $3.9 \times 10^{15} \text{ s}^{-1}$
60. 43.0 kJ/mol
62. 177 kJ/mol
64. $E_a = 108 \text{ kJ}$; $A = 2.0 \times 10^8 \text{ s}^{-1}$; $k = 3.2 \times 10^{-10} \text{ s}^{-1}$; (b) $1.81 \times 10^8 \text{ h}$ or $7.6 \times 10^6 \text{ day}$; (c) Assuming that the reaction is irreversible simplifies the calculation because we do not have to account for any reactant that, having been converted to product, returns to the original state.
66. The A atom has enough energy to react with BC; however, the different angles at which it bounces off of BC without reacting indicate that the orientation of the molecule is an important part of the reaction kinetics and determines whether a reaction will occur.
68. No. In general, for the overall reaction, we cannot predict the effect of changing the concentration without knowing the rate law. Yes. If the reaction is an elementary reaction, then doubling the concentration of A doubles the rate.
70. $\text{Rate} = k[A][B]^2$; $\text{Rate} = k[A]^3$
72. (a) $\text{Rate}_1 = k[\text{O}_3]$; (b) $\text{Rate}_2 = k[\text{O}_3][\text{Cl}];$ (c) $\text{Rate}_3 = k[\text{ClO}][\text{O}];$ (d) $\text{Rate}_2 = k[\text{O}_3][\text{NO}];$ (e) $\text{Rate}_3 = k[\text{NO}_2][\text{O}]$
74. (a) Doubling $[\text{H}_2]$ doubles the rate. $[\text{H}_2]$ must enter the rate law to the first power. Doubling $[\text{NO}]$ increases the rate by a factor of 4. $[\text{NO}]$ must enter the rate law to the second power. (b) $\text{Rate} = k[\text{NO}]^2[\text{H}_2];$ (c) $k = 5.0 \times 10^3 \text{ mol}^2 \text{ L}^{-2} \text{ min}^{-1};$ (d) 0.0050 mol/L; (e) Step II is the rate-determining step. If step I gives N_2O_2 in adequate amount, steps 1 and 2 combine to give $2\text{NO} + \text{H}_2 \longrightarrow \text{H}_2\text{O} + \text{N}_2\text{O}$. This reaction corresponds to the observed rate law. Combine steps 1 and 2 with step 3, which occurs by supposition in a rapid fashion, to give the appropriate stoichiometry.
76. The general mode of action for a catalyst is to provide a mechanism by which the reactants can unite more readily by taking a path with a lower reaction energy. The rates of both the forward and the reverse reactions are increased, leading to a faster achievement of equilibrium.
78. (a) Chlorine atoms are a catalyst because they react in the second step but are regenerated in the third step. Thus, they are not used up, which is a characteristic of catalysts. (b) NO is a catalyst for the same reason as in part (a).
80. The lowering of the transition state energy indicates the effect of a catalyst. (a) B; (b) B
82. The energy needed to go from the initial state to the transition state is (a) 10 kJ; (b) 10 kJ.
84. Both diagrams describe two-step, exothermic reactions, but with different changes in enthalpy, suggesting the diagrams depict two different overall reactions.

Chapter 13

- The reaction can proceed in both the forward and reverse directions.
- When a system has reached equilibrium, no further changes in the reactant and product concentrations occur; the forward and reverse reactions continue to proceed, but at equal rates.
- Not necessarily. A system at equilibrium is characterized by *constant* reactant and product concentrations, but the values of the reactant and product concentrations themselves need not be equal.

7. Equilibrium cannot be established between the liquid and the gas phase if the top is removed from the bottle because the system is not closed; one of the components of the equilibrium, the Br_2 vapor, would escape from the bottle until all liquid disappeared. Thus, more liquid would evaporate than can condense back from the gas phase to the liquid phase.
9. (a) $K_c = [\text{Ag}^+][\text{Cl}^-] < 1$. AgCl is insoluble; thus, the concentrations of ions are much less than 1 M; (b) $K_c = \frac{1}{[\text{Pb}^{2+}][\text{Cl}^-]^2} > 1$ because PbCl_2 is insoluble and formation of the solid will reduce the concentration of ions to a low level (< 1 M).
11. Since $K_c = \frac{[\text{C}_6\text{H}_6]}{[\text{C}_2\text{H}_2]^3}$, a value of $K_c \approx 10$ means that C_6H_6 predominates over C_2H_2 . In such a case, the reaction would be commercially feasible if the rate to equilibrium is suitable.
13. $K_c > 1$
15. (a) $Q_c = \frac{[\text{CH}_3\text{Cl}][\text{HCl}]}{[\text{CH}_4][\text{Cl}_2]}$; (b) $Q_c = \frac{[\text{NO}]^2}{[\text{N}_2][\text{O}_2]}$; (c) $Q_c = \frac{[\text{SO}_3]^2}{[\text{SO}_2]^2[\text{O}_2]}$; (d) $Q_c = [\text{SO}_2]$; (e) $Q_c = \frac{1}{[\text{P}_4][\text{O}_2]^5}$; (f) $Q_c = \frac{[\text{Br}]^2}{[\text{Br}_2]}$; (g) $Q_c = \frac{[\text{CO}_2]}{[\text{CH}_4][\text{O}_2]^2}$; (h) $Q_c = [\text{H}_2\text{O}]^5$
17. (a) Q_c 25 proceeds left; (b) Q_P 0.22 proceeds right; (c) Q_c undefined proceeds left; (d) Q_P 1.00 proceeds right; (e) Q_P 0 proceeds right; (f) Q_c 4 proceeds left
19. The system will shift toward the reactants to reach equilibrium.
21. (a) homogenous; (b) homogenous; (c) homogenous; (d) heterogeneous; (e) heterogeneous; (f) homogenous; (g) heterogeneous; (h) heterogeneous
23. This situation occurs in (a) and (b).
25. (a) $K_P = 1.6 \times 10^{-4}$; (b) $K_P = 50.2$; (c) $K_c = 5.31 \times 10^{-39}$; (d) $K_c = 4.60 \times 10^{-3}$
27. $K_P = P_{\text{H}_2\text{O}} = 0.042$.
29. $Q_c = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]}$
31. The amount of CaCO_3 must be so small that P_{CO_2} is less than K_P when the CaCO_3 has completely decomposed. In other words, the starting amount of CaCO_3 cannot completely generate the full P_{CO_2} required for equilibrium.
33. The change in enthalpy may be used. If the reaction is exothermic, the heat produced can be thought of as a product. If the reaction is endothermic the heat added can be thought of as a reactant. Additional heat would shift an exothermic reaction back to the reactants but would shift an endothermic reaction to the products. Cooling an exothermic reaction causes the reaction to shift toward the product side; cooling an endothermic reaction would cause it to shift to the reactants' side.
34. No, it is not at equilibrium. Because the system is not confined, products continuously escape from the region of the flame; reactants are also added continuously from the burner and surrounding atmosphere.
36. Add N_2 ; add H_2 ; decrease the container volume; heat the mixture.
38. (a) T increase = shift right, V decrease = shift left; (b) T increase = shift right, V = no effect; (c) T increase = shift left, V decrease = shift left; (d) T increase = shift left, V decrease = shift right.
40. (a) $K_c = \frac{[\text{CH}_3\text{OH}]}{[\text{H}_2]^2[\text{CO}]}$; (b) $[\text{H}_2]$ increases, $[\text{CO}]$ decreases, $[\text{CH}_3\text{OH}]$ increases; (c), $[\text{H}_2]$ increases, $[\text{CO}]$ decreases, $[\text{CH}_3\text{OH}]$ decreases; (d), $[\text{H}_2]$ increases, $[\text{CO}]$ increases, $[\text{CH}_3\text{OH}]$ increases; (e), $[\text{H}_2]$ increases, $[\text{CO}]$ increases, $[\text{CH}_3\text{OH}]$ decreases; (f), no changes.
42. (a) $K_c = \frac{[\text{CO}][\text{H}_2]}{[\text{H}_2\text{O}]}$; (b) $[\text{H}_2\text{O}]$ no change, $[\text{CO}]$ no change, $[\text{H}_2]$ no change; (c) $[\text{H}_2\text{O}]$ decreases, $[\text{CO}]$ decreases, $[\text{H}_2]$ decreases; (d) $[\text{H}_2\text{O}]$ increases, $[\text{CO}]$ increases, $[\text{H}_2]$ decreases; (e) $[\text{H}_2\text{O}]$ decreases, $[\text{CO}]$ increases, $[\text{H}_2]$ increases. In (b), (c), (d), and (e), the mass of carbon will change, but its concentration (activity) will not change.
44. Only (b)
46. Add NaCl or some other salt that produces Cl^- to the solution. Cooling the solution forces the equilibrium to the right, precipitating more $\text{AgCl}(s)$.
48. Though the solution is saturated, the dynamic nature of the solubility equilibrium means the opposing processes of solid dissolution and precipitation continue to occur (just at equal rates, meaning the dissolved

ion concentrations and the amount of undissolved solid remain constant). The radioactive Ag^+ ions detected in the solution phase come from dissolution of the added solid, and their presence is countered by precipitation of nonradioactive Ag^+ .

50. $K_c = \frac{[\text{C}]^2}{[\text{A}][\text{B}]^2}$. $[\text{A}] = 0.1 \text{ M}$, $[\text{B}] = 0.1 \text{ M}$, $[\text{C}] = 1 \text{ M}$; and $[\text{A}] = 0.01$, $[\text{B}] = 0.250$, $[\text{C}] = 0.791$.
52. $K_c = 6.00 \times 10^{-2}$
54. $K_c = 0.50$
56. $K_P = 1.9 \times 10^3$
58. $K_P = 3.06$
60. (a) $-2x$, $+2x$; (b) $\frac{4}{3}x$, $-\frac{2}{3}x$, $-2x$; (c) $-2x$, $3x$; (d) x , $-x$, $-3x$; (e) $+x$; (f) $-\frac{1}{4}x$
62. Activities of pure crystalline solids equal 1 and are constant; however, the mass of Ni does change.
64. $[\text{NH}_3] = 9.1 \times 10^{-2} \text{ M}$
66. $P_{\text{BrCl}} = 4.9 \times 10^{-2} \text{ atm}$
68. $[\text{CO}] = 2.04 \times 10^{-4} \text{ M}$
70. $P_{\text{H}_2\text{O}} = 3.64 \times 10^{-3} \text{ atm}$
72. Calculate Q based on the calculated concentrations and see if it is equal to K_c . Because Q does equal 4.32, the system must be at equilibrium.
74. (a) $[\text{NO}_2] = 1.17 \times 10^{-3} \text{ M}$; $[\text{N}_2\text{O}_4] = 0.128 \text{ M}$; (b) The assumption that x is negligibly small compared to 0.129 is confirmed by comparing the initial concentration of the N_2O_4 to its concentration at equilibrium (they differ by just 1 in the least significant digit's place).
76. (a) $[\text{H}_2\text{S}] = 0.810 \text{ atm}$, $[\text{H}_2] = 0.014 \text{ atm}$, $[\text{S}_2] = 0.0072 \text{ atm}$; (b) The assumption that $2x$ is negligibly small compared to 0.824 is confirmed by comparing the initial concentration of the H_2S to its concentration at equilibrium (0.824 atm versus 0.810 atm, a difference of less than 2%).
78. $[\text{PCl}_5] = 1.80 \text{ M}$; $[\text{Cl}_2] = 0.195 \text{ M}$; $[\text{PCl}_3] = 0.195 \text{ M}$.
79. 507 g
81. 330 g
84. (a) 0.33 mol. (b) $[\text{CO}_2] = 0.50 \text{ M}$. Added H_2 forms some water as a result of a shift to the left after H_2 is added.
86. (a) $K_c = \frac{[\text{NH}_3]^4[\text{O}_2]^7}{[\text{NO}_2]^4[\text{H}_2\text{O}]^6}$. (b) $[\text{NH}_3]$ must increase for Q_c to reach K_c . (c) The increase in system volume would lower the partial pressures of all reactants (including NO_2). (d) $P_{\text{O}_2} = 49 \text{ torr}$
88. $P_{\text{N}_2\text{O}_3} = 1.90 \text{ atm}$ and $P_{\text{NO}} = P_{\text{NO}_2} = 1.90 \text{ atm}$

Chapter 14

- One example for NH_3 as a conjugate acid: $\text{NH}_2^- + \text{H}^+ \rightarrow \text{NH}_3$; as a conjugate base: $\text{NH}_4^+ (\text{aq}) + \text{OH}^- (\text{aq}) \rightarrow \text{NH}_3 (\text{aq}) + \text{H}_2\text{O} (\text{l})$
- (a) $\text{H}_3\text{O}^+ (\text{aq}) \rightarrow \text{H}^+ (\text{aq}) + \text{H}_2\text{O} (\text{l})$; (b) $\text{HCl} (\text{aq}) \rightarrow \text{H}^+ (\text{aq}) + \text{Cl}^- (\text{aq})$; (c) $\text{NH}_3 (\text{aq}) \rightarrow \text{H}^+ (\text{aq}) + \text{NH}_2^- (\text{aq})$; (d) $\text{CH}_3\text{CO}_2\text{H} (\text{aq}) \rightarrow \text{H}^+ (\text{aq}) + \text{CH}_3\text{CO}_2^- (\text{aq})$; (e) $\text{NH}_4^+ (\text{aq}) \rightarrow \text{H}^+ (\text{aq}) + \text{NH}_3 (\text{aq})$; (f) $\text{HSO}_4^- (\text{aq}) \rightarrow \text{H}^+ (\text{aq}) + \text{SO}_4^{2-} (\text{aq})$
- (a) $\text{H}_2\text{O} (\text{l}) + \text{H}^+ (\text{aq}) \rightarrow \text{H}_3\text{O}^+ (\text{aq})$; (b) $\text{OH}^- (\text{aq}) + \text{H}^+ (\text{aq}) \rightarrow \text{H}_2\text{O} (\text{l})$; (c) $\text{NH}_3 (\text{aq}) + \text{H}^+ (\text{aq}) \rightarrow \text{NH}_4^+ (\text{aq})$; (d) $\text{CN}^- (\text{aq}) + \text{H}^+ (\text{aq}) \rightarrow \text{HCN} (\text{aq})$; (e) $\text{S}^{2-} (\text{aq}) + \text{H}^+ (\text{aq}) \rightarrow \text{HS}^- (\text{aq})$; (f) $\text{H}_2\text{PO}_4^- (\text{aq}) + \text{H}^+ (\text{aq}) \rightarrow \text{H}_3\text{PO}_4 (\text{aq})$
- (a) H_2O , O^{2-} ; (b) H_3O^+ , OH^- ; (c) H_2CO_3 , CO_3^{2-} ; (d) NH_4^+ , NH_2^- ; (e) H_2SO_4 , SO_4^{2-} ; (f) H_3O_2^+ , HO_2^- ; (g) H_2S ; S^{2-} ; (h) $\text{H}_6\text{N}_2^{2+}$, H_4N_2
- The labels are Brønsted-Lowry acid = BA; its conjugate base = CB; Brønsted-Lowry base = BB; its conjugate acid = CA. (a) HNO_3 (BA), H_2O (BB), H_3O^+ (CA), NO_3^- (CB); (b) CN^- (BB), H_2O (BA), HCN (CA), OH^- (CB); (c) H_2SO_4 (BA), Cl^- (BB), HCl (CA), HSO_4^- (CB); (d) HSO_4^- (BA), OH^- (BB), SO_4^{2-} (CB), H_2O (CA); (e) O^{2-} (BB), H_2O (BA), OH^- (CB and CA); (f) $[\text{Cu}(\text{H}_2\text{O})_3(\text{OH})]^+$ (BB), $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ (BA), $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ (CA), $[\text{Al}(\text{H}_2\text{O})_5(\text{OH})]^{2+}$ (CB); (g) H_2S (BA), NH_2^- (BB), HS^- (CB), NH_3 (CA)
- Amphiprotic species may either gain or lose a proton in a chemical reaction, thus acting as a base or an acid. An example is H_2O . As an acid: $\text{H}_2\text{O} (\text{aq}) + \text{NH}_3 (\text{aq}) \rightleftharpoons \text{NH}_4^+ (\text{aq}) + \text{OH}^- (\text{aq})$. As a base: $\text{H}_2\text{O} (\text{aq}) + \text{HCl} (\text{aq}) \rightleftharpoons \text{H}_3\text{O}^+ (\text{aq}) + \text{Cl}^- (\text{aq})$
- amphiprotic: (a) $\text{NH}_3 + \text{H}_3\text{O}^+ \rightarrow \text{NH}_4\text{OH} + \text{H}_2\text{O}$, $\text{NH}_3 + \text{OCH}_3^- \rightarrow \text{NH}_2^- + \text{CH}_3\text{OH}$; (b)

- $\text{HPO}_4^{2-} + \text{OH}^- \longrightarrow \text{PO}_4^{3-} + \text{H}_2\text{O}$, $\text{HPO}_4^{2-} + \text{HClO}_4 \longrightarrow \text{H}_2\text{PO}_4^- + \text{ClO}_4^-$; not amphiprotic: (c) Br^- ; (d) NH_4^+ ; (e) AsO_4^{3-}
15. In a neutral solution $[\text{H}_3\text{O}^+] = [\text{OH}^-]$. At 40 °C, $[\text{H}_3\text{O}^+] = [\text{OH}^-] = (2.910 \times 10^{-14})^{1/2} = 1.7 \times 10^{-7}$.
17. $x = 3.051 \times 10^{-7} \text{ M} = [\text{H}_3\text{O}^+] = [\text{OH}^-]$; $\text{pH} = -\log 3.051 \times 10^{-7} = -(-6.5156) = 6.5156$; $\text{pOH} = \text{pH} = 6.5156$
19. (a) $\text{pH} = 3.587$; $\text{pOH} = 10.413$; (b) $\text{pOH} = 0.68$; $\text{pH} = 13.32$; (c) $\text{pOH} = 3.85$; $\text{pH} = 10.15$; (d) $\text{pOH} = -0.40$; $\text{pH} = 14.4$
21. $[\text{H}_3\text{O}^+] = 3.0 \times 10^{-7} \text{ M}$; $[\text{OH}^-] = 3.3 \times 10^{-8} \text{ M}$
23. $[\text{H}_3\text{O}^+] = 1 \times 10^{-2} \text{ M}$; $[\text{OH}^-] = 1 \times 10^{-12} \text{ M}$
25. $[\text{OH}^-] = 3.1 \times 10^{-12} \text{ M}$
27. The salt ionizes in solution, but the anion slightly reacts with water to form the weak acid. This reaction also forms OH^- , which causes the solution to be basic.
29. $[\text{H}_2\text{O}] > [\text{CH}_3\text{CO}_2\text{H}] > [\text{H}_3\text{O}^+] \approx [\text{CH}_3\text{CO}_2^-] > [\text{OH}^-]$
31. The oxidation state of the sulfur in H_2SO_4 is greater than the oxidation state of the sulfur in H_2SO_3 .
33. $\text{Mg}(\text{OH})_2(\text{s}) + 2\text{HCl}(\text{aq}) \longrightarrow \text{Mg}^{2+}(\text{aq}) + 2\text{Cl}^-(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$
 BB BA CB CA
35. $K_a = 2.3 \times 10^{-11}$
37. The stronger base or stronger acid is the one with the larger K_b or K_a , respectively. In these two examples, they are $(\text{CH}_3)_2\text{NH}$ and CH_3NH_3^+ .
39. triethylamine
41. (a) HSO_4^- ; higher electronegativity of the central ion. (b) H_2O ; NH_3 is a base and water is neutral, or decide on the basis of K_a values. (c) HI ; PH_3 is weaker than HCl ; HCl is weaker than HI . Thus, PH_3 is weaker than HI . (d) PH_3 ; in binary compounds of hydrogen with nonmetals, the acidity increases for the element lower in a group. (e) HBr ; in a period, the acidity increases from left to right; in a group, it increases from top to bottom. Br is to the left and below S , so HBr is the stronger acid.
43. (a) $\text{NaHSeO}_3 < \text{NaHSO}_3 < \text{NaHSO}_4$; in polyoxy acids, the more electronegative central element— S , in this case—forms the stronger acid. The larger number of oxygen atoms on the central atom (giving it a higher oxidation state) also creates a greater release of hydrogen atoms, resulting in a stronger acid. As a salt, the acidity increases in the same manner. (b) $\text{ClO}_2^- < \text{BrO}_2^- < \text{IO}_2^-$; the basicity of the anions in a series of acids will be the opposite of the acidity in their oxyacids. The acidity increases as the electronegativity of the central atom increases. Cl is more electronegative than Br , and I is the least electronegative of the three. (c) $\text{HOI} < \text{HOBr} < \text{HOCl}$; in a series of the same form of oxyacids, the acidity increases as the electronegativity of the central atom increases. Cl is more electronegative than Br , and I is the least electronegative of the three. (d) $\text{HOCl} < \text{HOClO} < \text{HOClO}_2 < \text{HOClO}_3$; in a series of oxyacids of the same central element, the acidity increases as the number of oxygen atoms increases (or as the oxidation state of the central atom increases). (e) $\text{HTe}^- < \text{HS}^- \ll \text{PH}_2^- < \text{NH}_2^-$; PH_2^- and NH_2^- are anions of weak bases, so they act as strong bases toward H^+ . HTe^- and HS^- are anions of weak acids, so they have less basic character. In a periodic group, the more electronegative element has the more basic anion. (f) $\text{BrO}_4^- < \text{BrO}_3^- < \text{BrO}_2^- < \text{BrO}^-$; with a larger number of oxygen atoms (that is, as the oxidation state of the central ion increases), the corresponding acid becomes more acidic and the anion consequently less basic.
45. $[\text{H}_2\text{O}] > [\text{C}_6\text{H}_4\text{OH}(\text{CO}_2\text{H})] > [\text{H}^+] > [\text{C}_6\text{H}_4\text{OH}(\text{CO}_2^-)] \gg [\text{C}_6\text{H}_4\text{O}(\text{CO}_2\text{H})^-] > [\text{OH}^-]$
47. 1. Assume that the change in initial concentration of the acid as the equilibrium is established can be neglected, so this concentration can be assumed constant and equal to the initial value of the total acid concentration. 2. Assume we can neglect the contribution of water to the equilibrium concentration of H_3O^+ .
48. (b) The addition of HCl
50. (a) Adding HCl will add H_3O^+ ions, which will then react with the OH^- ions, lowering their concentration. The equilibrium will shift to the right, increasing the concentration of HNO_2 , and decreasing the concentration of NO_2^- ions. (b) Adding HNO_2 increases the concentration of HNO_2 and shifts the equilibrium to the left, increasing the concentration of NO_2^- ions and decreasing the concentration of OH^- ions. (c) Adding NaOH adds OH^- ions, which shifts the equilibrium to the left, increasing the concentration of NO_2^- ions and decreasing the concentrations of HNO_2 . (d) Adding NaCl has no effect on the concentrations of the ions. (e) Adding KNO_2 adds NO_2^- ions and shifts the equilibrium to the right, increasing the HNO_2 and OH^- ion concentrations.
52. This is a case in which the solution contains a mixture of acids of different ionization strengths. In solution,

the HCO_2H exists primarily as HCO_2H molecules because the ionization of the weak acid is suppressed by the strong acid. Therefore, the HCO_2H contributes a negligible amount of hydronium ions to the solution. The stronger acid, HCl , is the dominant producer of hydronium ions because it is completely ionized. In such a solution, the stronger acid determines the concentration of hydronium ions, and the ionization of the weaker acid is fixed by the $[\text{H}_3\text{O}^+]$ produced by the stronger acid.

54. (a) $K_b = 1.8 \times 10^{-5}$; (b) $K_a = 4.5 \times 10^{-4}$; (c) $K_b = 6.4 \times 10^{-5}$; (d) $K_a = 5.6 \times 10^{-10}$

56. $K_a = 1.2 \times 10^{-2}$

58. (a) $K_b = 4.3 \times 10^{-12}$ (b) $K_a = 1.6 \times 10^{-10}$ (c) $K_b = 5.9 \times 10^{-8}$ (d) $K_b = 4.2 \times 10^{-3}$ (e) $K_b = 2.3 \times 10^{-5}$ (f) $K_b = 6.3 \times 10^{-13}$

60. (a) $\frac{[\text{H}_3\text{O}^+][\text{ClO}^-]}{[\text{HClO}]} = \frac{(x)(x)}{(0.0092-x)} \approx \frac{(x)(x)}{0.0092} = 2.9 \times 10^{-8}$

Solving for x gives $1.63 \times 10^{-5} M$. This value is less than 5% of 0.0092, so the assumption that it can be neglected is valid. Thus, the concentrations of solute species at equilibrium are:

$$[\text{H}_3\text{O}^+] = [\text{ClO}^-] = 1.6 \times 10^{-5} M$$

$$[\text{HClO}] = 0.0092 M$$

$$[\text{OH}^-] = 6.1 \times 10^{-10} M;$$

(b) $\frac{[\text{C}_6\text{H}_5\text{NH}_3^+][\text{OH}^-]}{[\text{C}_6\text{H}_5\text{NH}_2]} = \frac{(x)(x)}{(0.0784-x)} \approx \frac{(x)(x)}{0.0784} = 4.3 \times 10^{-10}$

Solving for x gives $5.81 \times 10^{-6} M$. This value is less than 5% of 0.0784, so the assumption that it can be neglected is valid. Thus, the concentrations of solute species at equilibrium are:

$$[\text{C}_6\text{H}_5\text{NH}_3^+] = [\text{OH}^-] = 5.8 \times 10^{-6} M$$

$$[\text{C}_6\text{H}_5\text{NH}_2] = 0.0784 M$$

$$[\text{H}_3\text{O}^+] = 1.7 \times 10^{-9} M;$$

(c) $\frac{[\text{H}_3\text{O}^+][\text{CN}^-]}{[\text{HCN}]} = \frac{(x)(x)}{(0.0810-x)} \approx \frac{(x)(x)}{0.0810} = 4.9 \times 10^{-10}$

Solving for x gives $6.30 \times 10^{-6} M$. This value is less than 5% of 0.0810, so the assumption that it can be neglected is valid. Thus, the concentrations of solute species at equilibrium are:

$$[\text{H}_3\text{O}^+] = [\text{CN}^-] = 6.3 \times 10^{-6} M$$

$$[\text{HCN}] = 0.0810 M$$

$$[\text{OH}^-] = 1.6 \times 10^{-9} M;$$

(d) $\frac{[(\text{CH}_3)_3\text{NH}^+][\text{OH}^-]}{[(\text{CH}_3)_3\text{N}]} = \frac{(x)(x)}{(0.11-x)} \approx \frac{(x)(x)}{0.11} = 6.3 \times 10^{-5}$

Solving for x gives $2.63 \times 10^{-3} M$. This value is less than 5% of 0.11, so the assumption that it can be neglected is valid. Thus, the concentrations of solute species at equilibrium are:

$$[(\text{CH}_3)_3\text{NH}^+] = [\text{OH}^-] = 2.6 \times 10^{-3} M$$

$$[(\text{CH}_3)_3\text{N}] = 0.11 M$$

$$[\text{H}_3\text{O}^+] = 3.8 \times 10^{-12} M;$$

(e) $\frac{[\text{Fe}(\text{H}_2\text{O})_5(\text{OH})^+][\text{H}_3\text{O}^+]}{[\text{Fe}(\text{H}_2\text{O})_6^{2+}]} = \frac{(x)(x)}{(0.120-x)} \approx \frac{(x)(x)}{0.120} = 1.6 \times 10^{-7}$

Solving for x gives $1.39 \times 10^{-4} M$. This value is less than 5% of 0.120, so the assumption that it can be neglected is valid. Thus, the concentrations of solute species at equilibrium are:

$$[\text{Fe}(\text{H}_2\text{O})_5(\text{OH})^+] = [\text{H}_3\text{O}^+] = 1.4 \times 10^{-4} M$$

$$[\text{Fe}(\text{H}_2\text{O})_6^{2+}] = 0.120 M$$

$$[\text{OH}^-] = 7.2 \times 10^{-11} M$$

62. $\text{pH} = 2.41$

64. $[\text{C}_{10}\text{H}_{14}\text{N}_2] = 0.049 M$; $[\text{C}_{10}\text{H}_{14}\text{N}_2\text{H}^+] = 1.9 \times 10^{-4} M$; $[\text{C}_{10}\text{H}_{14}\text{N}_2\text{H}_2^{2+}] = 1.4 \times 10^{-11} M$; $[\text{OH}^-] = 1.9 \times 10^{-4} M$; $[\text{H}_3\text{O}^+] = 5.3 \times 10^{-11} M$

66. $K_a = 1.2 \times 10^{-2}$

68. $K_b = 1.77 \times 10^{-5}$

70. (a) acidic; (b) basic; (c) acidic; (d) neutral

72. $[\text{H}_3\text{O}^+]$ and $[\text{HCO}_3^-]$ are practically equal

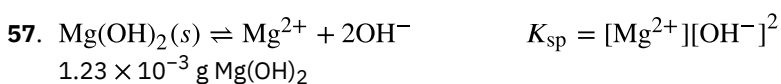
74. $[\text{C}_6\text{H}_4(\text{CO}_2\text{H})_2] = 7.2 \times 10^{-3} M$, $[\text{C}_6\text{H}_4(\text{CO}_2\text{H})(\text{CO}_2^-)] = [\text{H}_3\text{O}^+] = 2.8 \times 10^{-3} M$, $[\text{C}_6\text{H}_4(\text{CO}_2)_2^{2-}] = 3.9 \times 10^{-6} M$, $[\text{OH}^-] = 3.6 \times 10^{-12} M$

76. (a) $K_{a2} = 1.5 \times 10^{-11}$;
 (b) $K_b = 4.3 \times 10^{-12}$;
 (c) $\frac{[\text{Te}^{2-}][\text{H}_3\text{O}^+]}{[\text{HTe}^-]} = \frac{(x)(0.0141+x)}{(0.0141-x)} \approx \frac{(x)(0.0141)}{0.0141} = 1.5 \times 10^{-11}$
 Solving for x gives $1.5 \times 10^{-11} \text{ M}$. Therefore, compared with 0.014 M , this value is negligible ($1.1 \times 10^{-7}\%$).
78. Excess H_3O^+ is removed primarily by the reaction: $\text{H}_3\text{O}^+(aq) + \text{H}_2\text{PO}_4^-(aq) \longrightarrow \text{H}_3\text{PO}_4(aq) + \text{H}_2\text{O}(l)$
 Excess base is removed by the reaction: $\text{OH}^-(aq) + \text{H}_3\text{PO}_4(aq) \longrightarrow \text{H}_2\text{PO}_4^-(aq) + \text{H}_2\text{O}(l)$
80. $[\text{H}_3\text{O}^+] = 1.5 \times 10^{-4} \text{ M}$
82. $[\text{OH}^-] = 4.2 \times 10^{-4} \text{ M}$
84. (a) The added HCl will increase the concentration of H_3O^+ slightly, which will react with CH_3CO_2^- and produce $\text{CH}_3\text{CO}_2\text{H}$ in the process. Thus, $[\text{CH}_3\text{CO}_2^-]$ decreases and $[\text{CH}_3\text{CO}_2\text{H}]$ increases. (b) The added KCH_3CO_2 will increase the concentration of $[\text{CH}_3\text{CO}_2^-]$ which will react with H_3O^+ and produce $\text{CH}_3\text{CO}_2\text{H}$ in the process. Thus, $[\text{H}_3\text{O}^+]$ decreases slightly and $[\text{CH}_3\text{CO}_2\text{H}]$ increases. (c) The added NaCl will have no effect on the concentration of the ions. (d) The added KOH will produce OH^- ions, which will react with the H_3O^+ , thus reducing $[\text{H}_3\text{O}^+]$. Some additional $\text{CH}_3\text{CO}_2\text{H}$ will dissociate, producing $[\text{CH}_3\text{CO}_2^-]$ ions in the process. Thus, $[\text{CH}_3\text{CO}_2\text{H}]$ decreases slightly and $[\text{CH}_3\text{CO}_2^-]$ increases. (e) The added $\text{CH}_3\text{CO}_2\text{H}$ will increase its concentration, causing more of it to dissociate and producing more $[\text{CH}_3\text{CO}_2^-]$ and H_3O^+ in the process. Thus, $[\text{H}_3\text{O}^+]$ increases slightly and $[\text{CH}_3\text{CO}_2^-]$ increases.
86. pH = 8.95
88. 37 g (0.27 mol)
90. (a) pH = 5.222; (b) The solution is acidic. (c) pH = 5.220
92. At the equivalence point in the titration of a weak base with a strong acid, the resulting solution is slightly acidic due to the presence of the conjugate acid. Thus, pick an indicator that changes color in the acidic range and brackets the pH at the equivalence point. Methyl orange is a good example.
94. (a) pH = 2.50; (b) pH = 4.01; (c) pH = 5.60; (d) pH = 8.35; (e) pH = 11.08

Chapter 15

1. (a)
 $\text{AgI}(s) \rightleftharpoons \text{Ag}^+(aq) + \text{I}^-(aq)$
- (b)
 $\text{CaCO}_3(s) \rightleftharpoons \text{Ca}^{2+}(aq) + \text{CO}_3^{2-}(aq)$
- (c)
 $\text{Mg}(\text{OH})_2(s) \rightleftharpoons \text{Mg}^{2+}(aq) + 2\text{OH}^-(aq)$
- (d)
 $\text{Mg}_3(\text{PO}_4)_2(s) \rightleftharpoons 3\text{Mg}^{2+}(aq) + 2\text{PO}_4^{3-}(aq)$
- (e)
 $\text{Ca}_5(\text{PO}_4)_3\text{OH}(s) \rightleftharpoons 5\text{Ca}^{2+}(aq) + 3\text{PO}_4^{3-}(aq) + \text{OH}^-(aq)$
3. There is no change. A solid has an activity of 1 whether there is a little or a lot.
5. The solubility of silver bromide at the new temperature must be known. Normally the solubility increases and some of the solid silver bromide will dissolve.
7. CaF_2 , MnCO_3 , and ZnS

9. (a) $\text{LaF}_3(s) \rightleftharpoons \text{La}^{3+}(aq) + 3\text{F}^{-}(aq)$ $K_{\text{sp}} = [\text{La}^{3+}][\text{F}^{-}]^3$;
 (b) $\text{CaCO}_3(s) \rightleftharpoons \text{Ca}^{2+}(aq) + \text{CO}_3^{2-}(aq)$ $K_{\text{sp}} = [\text{Ca}^{2+}][\text{CO}_3^{2-}]$;
 (c) $\text{Ag}_2\text{SO}_4(s) \rightleftharpoons 2\text{Ag}^{+}(aq) + \text{SO}_4^{2-}(aq)$ $K_{\text{sp}} = [\text{Ag}^{+}]^2[\text{SO}_4^{2-}]$;
 (d) $\text{Pb}(\text{OH})_2(s) \rightleftharpoons \text{Pb}^{2+}(aq) + 2\text{OH}^{-}(aq)$ $K_{\text{sp}} = [\text{Pb}^{2+}][\text{OH}^{-}]^2$
11. (a) 1.77×10^{-7} ; (b) 1.6×10^{-6} ; (c) 2.2×10^{-9} ; (d) 7.91×10^{-22}
13. (a) $2 \times 10^{-2} \text{ M}$; (b) $1.5 \times 10^{-3} \text{ M}$; (c) $2.27 \times 10^{-9} \text{ M}$; (d) $2.2 \times 10^{-10} \text{ M}$
15. (a) $6.4 \times 10^{-9} \text{ M} = [\text{Ag}^{+}]$, $[\text{Cl}^{-}] = 0.025 \text{ M}$. Check: $\frac{6.4 \times 10^{-9} \text{ M}}{0.025 \text{ M}} \times 100\% = 2.6 \times 10^{-5}\%$, an insignificant change;
 (b) $2.2 \times 10^{-5} \text{ M} = [\text{Ca}^{2+}]$, $[\text{F}^{-}] = 0.0013 \text{ M}$. Check: $\frac{2.26 \times 10^{-5} \text{ M}}{0.00133 \text{ M}} \times 100\% = 1.70\%$. This value is less than 5% and can be ignored.
 (c) $0.2238 \text{ M} = [\text{SO}_4^{2-}]$; $[\text{Ag}^{+}] = 7.4 \times 10^{-3} \text{ M}$. Check: $\frac{3.7 \times 10^{-3}}{0.2238} \times 100\% = 1.64 \times 10^{-2}\%$; the condition is satisfied.
 (d) $[\text{OH}^{-}] = 2.8 \times 10^{-3} \text{ M}$; $5.7 \times 10^{-12} \text{ M} = [\text{Zn}^{2+}]$. Check: $\frac{5.7 \times 10^{-12}}{2.8 \times 10^{-3}} \times 100\% = 2.0 \times 10^{-7}\%$; x is less than 5% of $[\text{OH}^{-}]$ and is, therefore, negligible.
17. (a) The change in initial concentration of common ion may not be neglected (must solve quadratic equation)
 $[\text{Ti}^{+}] = 0.031 \text{ M}$
 $[\text{Cl}^{-}] = 0.0056 \text{ M}$ (b) The change in initial concentration of common ion may be neglected.
 $[\text{Ba}^{2+}] = 2.4 \times 10^{-4} \text{ M}$
 $[\text{F}^{-}] = 0.313 \text{ M}$;
 (c) The change in initial concentration of common ion may be neglected.
 $[\text{Mg}^{2+}] = 0.0244 \text{ M}$
 $[\text{C}_2\text{O}_4^{2-}] = 2.9 \times 10^{-5} \text{ M}$
 (d) The change in initial concentration of common ion may be neglected.
 $[\text{Ca}^{2+}] = 5.18 \times 10^{-4} \text{ M}$
 $[\text{C}_2\text{O}_4^{2-}] = 0.051 \text{ M}$
19. The changes in concentration are greater than 5% and thus exceed the maximum value for disregarding the change.
21. $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ is the most soluble Ca salt in mol/L, and it is also the most soluble Ca salt in g/L.
23. $7.8 \times 10^{-3} \text{ M} = [\text{SO}_4^{2-}] = [\text{Ca}^{2+}]$; Since this concentration is higher than $2.60 \times 10^{-3} \text{ M}$, "gyp" water does not meet the standards.
25. Mass $(\text{CaSO}_4 \cdot 2\text{H}_2\text{O}) = 0.72 \text{ g}$
27. (a) $[\text{Ag}^{+}] = [\text{I}^{-}] = 1.3 \times 10^{-5} \text{ M}$; (b) $[\text{Ag}^{+}] = 2.88 \times 10^{-2} \text{ M}$, $[\text{SO}_4^{2-}] = 1.44 \times 10^{-2} \text{ M}$; (c) $[\text{Mn}^{2+}] = 3.7 \times 10^{-5} \text{ M}$, $[\text{OH}^{-}] = 7.4 \times 10^{-5} \text{ M}$; (d) $[\text{Sr}^{2+}] = 4.3 \times 10^{-2} \text{ M}$, $[\text{OH}^{-}] = 8.6 \times 10^{-2} \text{ M}$; (e) $[\text{Mg}^{2+}] = 1.3 \times 10^{-4} \text{ M}$, $[\text{OH}^{-}] = 2.6 \times 10^{-4} \text{ M}$.
29. (a) 1.45×10^{-4} ; (b) 8.2×10^{-55} ; (c) 1.35×10^{-4} ; (d) 1.18×10^{-5} ; (e) 1.08×10^{-10}
31. (a) CaCO_3 does precipitate. (b) The compound does not precipitate. (c) The compound does not precipitate. (d) The compound precipitates.
33. $1.42 \times 10^{-9} \text{ M}$
35. $9.2 \times 10^{-13} \text{ M}$
37. $[\text{Ag}^{+}] = 1.8 \times 10^{-3} \text{ M}$
39. 6.3×10^{-4}
41. (a) 2.25 L; (b) $7.2 \times 10^{-7} \text{ g}$
43. 100% of it is dissolved
45. (a) Hg_2^{2+} and Cu^{2+} : Add SO_4^{2-} . (b) SO_4^{2-} and Cl^{-} : Add Ba^{2+} . (c) Hg^{2+} and Co^{2+} : Add S^{2-} . (d) Zn^{2+} and Sr^{2+} : Add OH^{-} until $[\text{OH}^{-}] = 0.050 \text{ M}$. (e) Ba^{2+} and Mg^{2+} : Add SO_4^{2-} . (f) CO_3^{2-} and OH^{-} : Add Ba^{2+} .
47. AgI will precipitate first.
49. $1.5 \times 10^{-12} \text{ M}$
51. 3.99 kg
53. (a) 3.1×10^{-11} ; (b) $[\text{Cu}^{2+}] = 2.6 \times 10^{-3}$; $[\text{IO}_3^{-}] = 5.3 \times 10^{-3}$
55. $9.3 \times 10^{-6} \text{ g Pb}(\text{OH})_2$



59. MnCO_3 will form first since it has the smallest K_{sp} value among these homologous compounds and is therefore the least soluble. $\text{MgCO}_3 \cdot 3\text{H}_2\text{O}$ will be the last to precipitate since it has the largest K_{sp} value and is the most soluble. K_{sp} value.

62. when the amount of solid is so small that a saturated solution is not produced

64. $8.2 \times 10^{-5} M$

66. 5×10^{23}

68.

	$[\text{Cd}(\text{CN})_4^{2-}]$	$[\text{CN}^-]$	$[\text{Cd}^{2+}]$
Initial concentration (M)	0.250	0	0
Equilibrium (M)	$0.250 - x$	$4x$	x

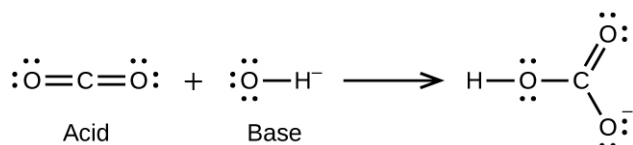
$[\text{Cd}^{2+}] = 5.0 \times 10^{-5} M$; $[\text{CN}^-] = 2.0 \times 10^{-4} M$

70. $[\text{Co}^{3+}] = 3.0 \times 10^{-6} M$; $[\text{NH}_3] = 1.8 \times 10^{-5} M$

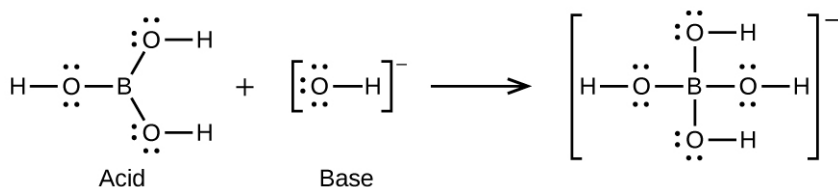
72. 1.3 g

74. 0.79 g

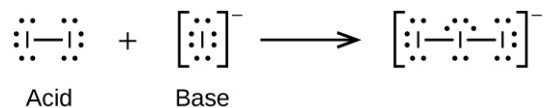
76. (a)



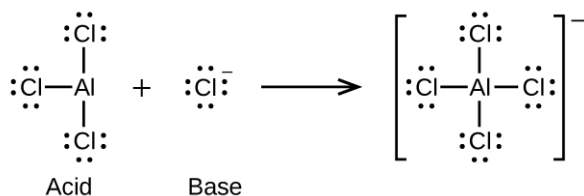
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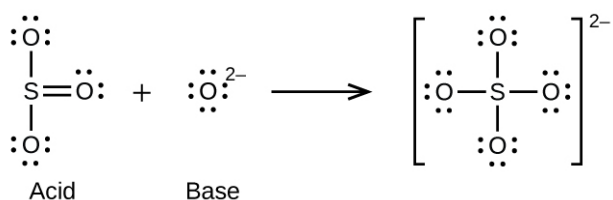
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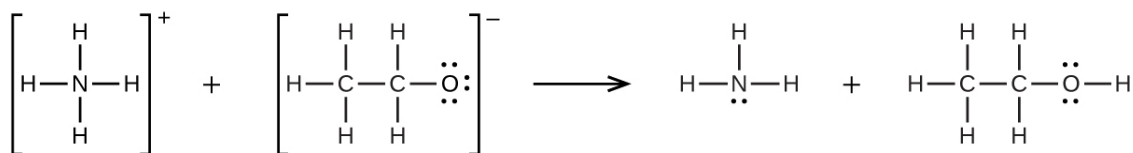
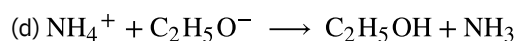
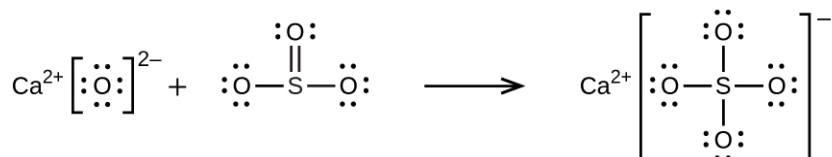
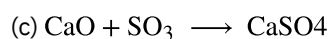
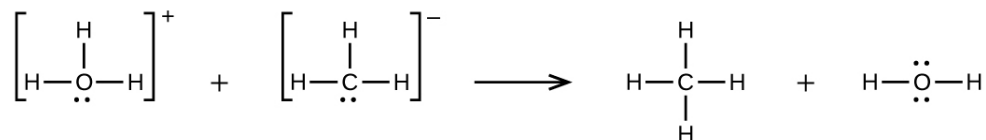
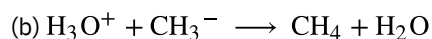
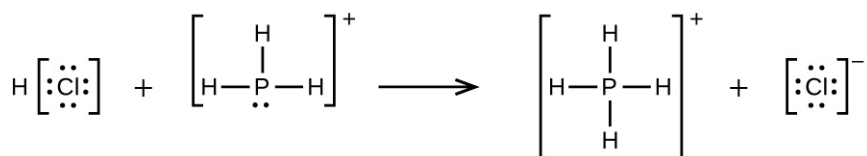
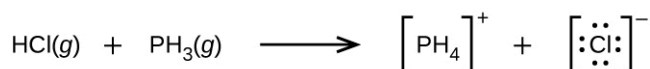
(d)



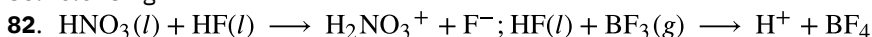
(e)



78. (a)



80. 0.0281 g



84. (a) $\text{H}_3\text{BO}_3 + \text{H}_2\text{O} \longrightarrow \text{H}_4\text{BO}_4^- + \text{H}^+$; (b) The electronic and molecular shapes are the same—both tetrahedral. (c) The tetrahedral structure is consistent with sp^3 hybridization.

86. 0.014 M

88. $7.2 \times 10^{-15} M$ 90. $4.4 \times 10^{-22} M$ 93. $[\text{OH}^-] = 4.5 \times 10^{-6}$; $[\text{Al}^{3+}] = 2 \times 10^{-16}$ (molar solubility)95. $K_{sp} = [\text{Ba}^{2+}][\text{SO}_4^{2-}] = [\text{Ba}^{2+}](0.049) = 1.08 \times 10^{-10}$; $[\text{Ba}^{2+}] = 2.2 \times 10^{-9}$; (molar solubility)97. $[\text{OH}^-] = 7.6 \times 10^{-3} M$; $[\text{Pb}^{2+}] = 2.1 \times 10^{-11}$ (molar solubility)

99. 7.66

101. (a) $K_{sp} = [\text{Mg}^{2+}][\text{F}^-]^2 = (1.21 \times 10^{-3})(2 \times 1.21 \times 10^{-3})^2 = 7.09 \times 10^{-9}$ (b) $7.09 \times 10^{-7} M$

(c) Determine the concentration of Mg^{2+} and F^- that will be present in the final volume. Compare the value of the ion product $[\text{Mg}^{2+}][\text{F}^-]^2$ with K_{sp} . If this value is larger than K_{sp} , precipitation will occur.

$$0.1000 \text{ L} \times 3.00 \times 10^{-3} \text{ M Mg(NO}_3)_2 = 0.3000 \text{ L} \times M \text{ Mg(NO}_3)_2$$

$$M \text{ Mg(NO}_3)_2 = 1.00 \times 10^{-3} \text{ M}$$

$$0.2000 \text{ L} \times 2.00 \times 10^{-3} \text{ M NaF} = 0.3000 \text{ L} \times M \text{ NaF}$$

$$M \text{ NaF} = 1.33 \times 10^{-3} \text{ M}$$

ion product = $(1.00 \times 10^{-3})(1.33 \times 10^{-3})^2 = 1.77 \times 10^{-9}$ This value is smaller than K_{sp} , so no precipitation will occur.

(d) MgF_2 is less soluble at 27 °C than at 18 °C. Because added heat acts like an added reagent, when it appears on the product side, the Le Châtelier's principle states that the equilibrium will shift to the reactants' side to counter the stress. Consequently, less reagent will dissolve. This situation is found in our case. Therefore, the reaction is exothermic.

- 103.** BaF_2 , $\text{Ca}_3(\text{PO}_4)_2$, ZnS ; each is a salt of a weak acid, and the $[\text{H}_3\text{O}^+]$ from perchloric acid reduces the equilibrium concentration of the anion, thereby increasing the concentration of the cations
- 105.** Effect on amount of solid CaHPO_4 , $[\text{Ca}^{2+}]$, $[\text{OH}^-]$: (a) increase, increase, decrease; (b) decrease, increase, decrease; (c) no effect, no effect, no effect; (d) decrease, increase, decrease; (e) increase, no effect, no effect

Chapter 16

- A reaction has a natural tendency to occur and takes place without the continual input of energy from an external source.
- (a) spontaneous; (b) nonspontaneous; (c) spontaneous; (d) nonspontaneous; (e) spontaneous; (f) spontaneous
- Although the oxidation of plastics is spontaneous, the rate of oxidation is very slow. Plastics are therefore kinetically stable and do not decompose appreciably even over relatively long periods of time.
- There are four initial microstates and four final microstates.

$$\Delta S = k \ln \frac{W_f}{W_i} = 1.38 \times 10^{-23} \text{ J/K} \times \ln \frac{4}{4} = 0$$
- The probability for all the particles to be on one side is $\frac{1}{32}$. This probability is noticeably lower than the $\frac{1}{8}$ result for the four-particle system. The conclusion we can make is that the probability for all the particles to stay in only one part of the system will decrease rapidly as the number of particles increases, and, for instance, the probability for all molecules of gas to gather in only one side of a room at room temperature and pressure is negligible since the number of gas molecules in the room is very large.
- There is only one initial state. For the final state, the energy can be contained in pairs A-C, A-D, B-C, or B-D. Thus, there are four final possible states.

$$\Delta S = k \ln \left(\frac{W_f}{W_i} \right) = 1.38 \times 10^{-23} \text{ J/K} \times \ln \left(\frac{4}{1} \right) = 1.91 \times 10^{-23} \text{ J/K}$$
- The masses of these molecules would suggest the opposite trend in their entropies. The observed trend is a result of the more significant variation of entropy with a physical state. At room temperature, I_2 is a solid, Br_2 is a liquid, and Cl_2 is a gas.
- (a) $\text{C}_3\text{H}_7\text{OH}(l)$ as it is a larger molecule (more complex and more massive), and so more microstates describing its motions are available at any given temperature. (b) $\text{C}_2\text{H}_5\text{OH}(g)$ as it is in the gaseous state. (c) $2\text{H}(g)$, since entropy is an extensive property, and so two H atoms (or two moles of H atoms) possess twice as much entropy as one atom (or one mole of atoms).
- (a) Negative. The relatively ordered solid precipitating decreases the number of mobile ions in solution. (b) Negative. There is a net loss of three moles of gas from reactants to products. (c) Positive. There is a net increase of seven moles of gas from reactants to products.
- $\text{C}_6\text{H}_6(l) + 7.5\text{O}_2(g) \rightarrow 3\text{H}_2\text{O}(g) + 6\text{CO}_2(g)$
 There are 7.5 moles of gas initially, and $3 + 6 = 9$ moles of gas in the end. Therefore, it is likely that the entropy increases as a result of this reaction, and ΔS is positive.
- (a) 107 J/K; (b) -86.7 J/K; (c) 133.2 J/K; (d) 118.8 J/K; (e) -326.6 J/K; (f) -171.9 J/K; (g) -7.2 J/K
- 100.3 J/K
- (a) -198.1 J/K; (b) -348.9 J/K
- As $\Delta S_{\text{univ}} < 0$ at each of these temperatures, melting is not spontaneous at either of them. The given values

for entropy and enthalpy are for NaCl at 298 K. It is assumed that these do not change significantly at the higher temperatures used in the problem.

29. (a) 2.86 J/K; (b) 24.8 J/K; (c) -113.2 J/K; (d) -24.7 J/K; (e) 15.5 J/K; (f) 290.0 J/K
31. The reaction is nonspontaneous at room temperature.
Above 400 K, ΔG will become negative, and the reaction will become spontaneous.
33. (a) 465.1 kJ nonspontaneous; (b) -106.86 kJ spontaneous; (c) -291.9 kJ spontaneous; (d) -83.4 kJ spontaneous; (e) -406.7 kJ spontaneous; (f) -154.3 kJ spontaneous
35. (a) The standard free energy of formation is -1124.3 kJ/mol. (b) The calculation agrees with the value in [Appendix G](#) because free energy is a state function (just like the enthalpy and entropy), so its change depends only on the initial and final states, not the path between them.
37. (a) The reaction is nonspontaneous; (b) Above 566 °C the process is spontaneous.
39. (a) 1.5×10^2 kJ; (b) -21.9 kJ; (c) -5.34 kJ; (d) -0.383 kJ; (e) 18 kJ; (f) 71 kJ
41. (a) $K = 41$; (b) $K = 0.053$; (c) $K = 6.9 \times 10^{13}$; (d) $K = 1.9$; (e) $K = 0.04$
43. In each of the following, the value of ΔG is not given at the temperature of the reaction. Therefore, we must calculate ΔG from the values ΔH° and ΔS and then calculate ΔG from the relation $\Delta G = \Delta H^\circ - T\Delta S^\circ$. (a) $K = 1.07 \times 10^{-13}$; (b) $K = 2.51 \times 10^{-3}$; (c) $K = 2.73 \times 10^4$; (d) $K = 0.219$; (e) $K = 16.1$
45. The standard free energy change is $\Delta G^\circ = -RT \ln K = 4.84$ kJ/mol. When reactants and products are in their standard states (1 bar or 1 atm), $Q = 1$. As the reaction proceeds toward equilibrium, the reaction shifts left (the amount of products drops while the amount of reactants increases): $Q < 1$, and ΔG becomes less positive as it approaches zero. At equilibrium, $Q = K$, and $\Delta G = 0$.
47. The reaction will be spontaneous at temperatures greater than 287 K.
49. $K = 5.35 \times 10^{15}$; The process is exothermic.
51. 1.0×10^{-8} atm. This is the maximum pressure of the gases under the stated conditions.
53. $x = 1.29 \times 10^{-5}$ atm = P_{O_2}
55. -0.16 kJ
56. (a) 22.1 kJ; (b) 98.9 kJ/mol
58. 90 kJ/mol
60. (a) Under standard thermodynamic conditions, the evaporation is nonspontaneous; (b) $K_p = 0.031$; (c) The evaporation of water is spontaneous; (d) P_{H_2O} must always be less than K_p or less than 0.031 atm. 0.031 atm represents air saturated with water vapor at 25 °C, or 100% humidity.
62. (a) Nonspontaneous as $\Delta G^\circ > 0$; (b) $\Delta G = \Delta G^\circ + RT \ln Q$,
 $\Delta G = 1.7 \times 10^3 + (8.314 \times 310 \times \ln \frac{28}{120}) = -2.1$ kJ. The forward reaction to produce F6P is spontaneous under these conditions.
64. ΔG is negative as the process is spontaneous. ΔH is positive as with the solution becoming cold, the dissolving must be endothermic. ΔS must be positive as this drives the process, and it is expected for the dissolution of any soluble ionic compound.
66. (a) Increasing the oxygen partial pressure will yield a decrease in Q and ΔG thus becomes more negative. (b) Increasing the oxygen partial pressure will yield a decrease in Q and ΔG thus becomes more negative. (c) Increasing the oxygen partial pressure will yield an increase in Q and ΔG thus becomes more positive.

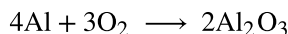
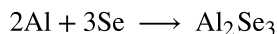
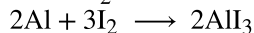
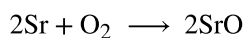
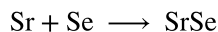
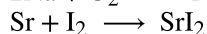
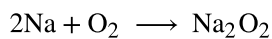
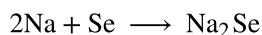
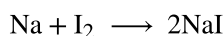
Chapter 17

1. (a) reduction; (b) oxidation; (c) oxidation; (d) reduction
3. (a) $F_2 + Ca \longrightarrow 2F^- + Ca^{2+}$; (b) $Cl_2 + 2Li \longrightarrow 2Li^+ + 2Cl^-$; (c) $3Br_2 + 2Fe \longrightarrow 2Fe^{3+} + 6Br^-$; (d) $MnO_4^- + 4H^+ + 3Ag \longrightarrow 3Ag^+ + MnO_2 + 2H_2O$
5. Oxidized: (a) Sn^{2+} ; (b) Hg; (c) Al; reduced: (a) H_2O_2 ; (b) PbO_2 ; (c) $Cr_2O_7^{2-}$; oxidizing agent: (a) H_2O_2 ; (b) PbO_2 ; (c) $Cr_2O_7^{2-}$; reducing agent: (a) Sn^{2+} ; (b) Hg; (c) Al
7. Oxidized = reducing agent: (a) SO_3^{2-} ; (b) $Mn(OH)_2$; (c) H_2 ; (d) Al; reduced = oxidizing agent: (a) $Cu(OH)_2$; (b) O_2 ; (c) NO_3^- ; (d) CrO_4^{2-}
9. In basic solution, $[OH^-] > 1 \times 10^{-7} M > [H^+]$. Hydrogen ion cannot appear as a reactant because its concentration is essentially zero. If it were produced, it would instantly react with the excess hydroxide ion to produce water. Thus, hydrogen ion should *not* appear as a reactant or product in basic solution.
11. (a) $Mg(s) \mid Mg^{2+}(aq) \parallel Ni^{2+}(aq) \mid Ni(s)$; (b) $Cu(s) \mid Cu^{2+}(aq) \parallel Ag^+(aq) \mid Ag(s)$; (c)

- $\text{Mn}(s) \mid \text{Mn}^{2+}(aq) \parallel \text{Sn}^{2+}(aq) \mid \text{Sn}(s)$; (d) $\text{Pt}(s) \mid \text{Cu}^+(aq), \text{Cu}^{2+}(aq) \parallel \text{Au}^{3+}(aq) \mid \text{Au}(s)$
13. (a) $\text{Mg}(s) + \text{Cu}^{2+}(aq) \rightarrow \text{Mg}^{2+}(aq) + \text{Cu}(s)$; (b) $2\text{Ag}^+(aq) + \text{Ni}(s) \rightarrow \text{Ni}^{2+}(aq) + 2\text{Ag}(s)$
15. Species oxidized = reducing agent: (a) $\text{Al}(s)$; (b) $\text{NO}(g)$; (c) $\text{Mg}(s)$; and (d) $\text{MnO}_2(s)$; Species reduced = oxidizing agent: (a) $\text{Zr}^{4+}(aq)$; (b) $\text{Ag}^+(aq)$; (c) $\text{SiO}_3^{2-}(aq)$; and (d) $\text{ClO}_3^-(aq)$
17. Without the salt bridge, the circuit would be open (or broken) and no current could flow. With a salt bridge, each half-cell remains electrically neutral and current can flow through the circuit.
19. Active electrodes participate in the oxidation-reduction reaction. Since metals form cations, the electrode would lose mass if metal atoms in the electrode were to oxidize and go into solution. Oxidation occurs at the anode.
21. (a) +2.115 V (spontaneous); (b) +0.4626 V (spontaneous); (c) +1.048 V (spontaneous); (d) +0.727 V (spontaneous)
23. $3\text{Cu}(s) + 2\text{Au}^{3+}(aq) \rightarrow 3\text{Cu}^{2+}(aq) + 2\text{Au}(s)$; +1.16 V; spontaneous
25. $3\text{Cd}(s) + 2\text{Al}^{3+}(aq) \rightarrow 3\text{Cd}^{2+}(aq) + 2\text{Al}(s)$; -1.259 V; nonspontaneous
27. (a) 0 kJ/mol; (b) -83.7 kJ/mol; (c) +235.3 kJ/mol
29. (a) standard cell potential: 1.50 V, spontaneous; cell potential under stated conditions: 1.43 V, spontaneous; (b) standard cell potential: 1.405 V, spontaneous; cell potential under stated conditions: 1.423 V, spontaneous; (c) standard cell potential: -2.749 V, nonspontaneous; cell potential under stated conditions: -2.733 V, nonspontaneous
31. (a) 1.7×10^{-10} ; (b) 2.6×10^{-21} ; (c) 4.693×10^{21} ; (d) 1.0×10^{-14}
33. (a) anode: $\text{Cu}(s) \rightarrow \text{Cu}^{2+}(aq) + 2e^-$ $E^\circ_{\text{anode}} = 0.34 \text{ V}$; (b) 3.5×10^{15} ; (c) $5.6 \times 10^{-9} \text{ M}$
 cathode: $2 \times (\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s))$ $E^\circ_{\text{cathode}} = 0.7996 \text{ V}$
34. Batteries are self-contained and have a limited supply of reagents to expend before going dead. Alternatively, battery reaction byproducts accumulate and interfere with the reaction. Because a fuel cell is constantly resupplied with reactants and products are expelled, it can continue to function as long as reagents are supplied.
36. According to kinetic theory, chemical reactions occur more slowly at lower temperatures due to the decreased rate of collisions between reactant molecules. The rates of the redox reactions taking place within batteries decrease with temperature, resulting in a decreased production of electrical current.
38. Mg and Zn
40. Both examples involve cathodic protection. The (sacrificial) anode is the metal that corrodes (oxidizes or reacts). In the case of iron (-0.447 V) and zinc (-0.7618 V), zinc has a more negative standard reduction potential and so serves as the anode. In the case of iron and copper (0.34 V), iron has the smaller standard reduction potential and so corrodes (serves as the anode).
42. While the reduction potential of lithium would make it capable of protecting the other metals, this high potential is also indicative of how reactive lithium is; it would have a spontaneous reaction with most substances. This means that the lithium would react quickly with other substances, even those that would not oxidize the metal it is attempting to protect. Reactivity like this means the sacrificial anode would be depleted rapidly and need to be replaced frequently. (Optional additional reason: fire hazard in the presence of water.)
46. (a) mass Ca= 69.1 g ; (b) $\frac{\text{mass Li}= 23.9 \text{ g}}{\text{mass H}_2 = 3.48 \text{ g}}$; (c) $\frac{\text{mass Al}= 31.0 \text{ g}}{\text{mass Cl}_2 = 122 \text{ g}}$; (d) $\frac{\text{mass Cr}= 59.8 \text{ g}}{\text{mass Br}_2 = 276 \text{ g}}$
48. 0.79 L

Chapter 18

- The alkali metals all have a single *s* electron in their outermost shell. In contrast, the alkaline earth metals have a completed *s* subshell in their outermost shell. In general, the alkali metals react faster and are more reactive than the corresponding alkaline earth metals in the same period.
-
-



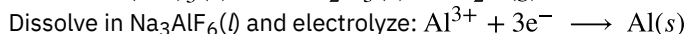
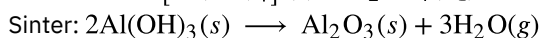
5. The possible ways of distinguishing between the two include infrared spectroscopy by comparison of known compounds, a flame test that gives the characteristic yellow color for sodium (strontium has a red flame), or comparison of their solubilities in water. At 20 °C, NaCl dissolves to the extent of $\frac{35.7 \text{ g}}{100 \text{ mL}}$ compared with $\frac{53.8 \text{ g}}{100 \text{ mL}}$ for SrCl_2 . Heating to 100 °C provides an easy test, since the solubility of NaCl is $\frac{39.12 \text{ g}}{100 \text{ mL}}$, but that of SrCl_2 is $\frac{100.8 \text{ g}}{100 \text{ mL}}$. Density determination on a solid is sometimes difficult, but there is enough difference (2.165 g/mL NaCl and 3.052 g/mL SrCl_2) that this method would be viable and perhaps the easiest and least expensive test to perform.
7. (a) $2\text{Sr}(s) + \text{O}_2(g) \longrightarrow 2\text{SrO}(s)$; (b) $\text{Sr}(s) + 2\text{HBr}(g) \longrightarrow \text{SrBr}_2(s) + \text{H}_2(g)$; (c) $\text{Sr}(s) + \text{H}_2(g) \longrightarrow \text{SrH}_2(s)$; (d) $6\text{Sr}(s) + \text{P}_4(s) \longrightarrow 2\text{Sr}_3\text{P}_2(s)$; (e) $\text{Sr}(s) + 2\text{H}_2\text{O}(l) \longrightarrow \text{Sr}(\text{OH})_2(aq) + \text{H}_2(g)$
9. 11 lb
11. Yes, tin reacts with hydrochloric acid to produce hydrogen gas.
13. In PbCl_2 , the bonding is ionic, as indicated by its melting point of 501 °C. In PbCl_4 , the bonding is covalent, as evidenced by it being an unstable liquid at room temperature.

- countercurrent
fractionating
tower
15. $2\text{CsCl}(l) + \text{Ca}(g) \xrightarrow{\text{tower}} 2\text{Cs}(g) + \text{CaCl}_2(l)$
17. Cathode (reduction): $2\text{Li}^+ + 2\text{e}^- \longrightarrow 2\text{Li}(l)$; Anode (oxidation): $2\text{Cl}^- \longrightarrow \text{Cl}_2(g) + 2\text{e}^-$; Overall reaction: $2\text{Li}^+ + 2\text{Cl}^- \longrightarrow 2\text{Li}(l) + \text{Cl}_2(g)$

19. 0.5035 g H_2

21. Despite its reactivity, magnesium can be used in construction even when the magnesium is going to come in contact with a flame because a protective oxide coating is formed, preventing gross oxidation. Only if the metal is finely subdivided or present in a thin sheet will a high-intensity flame cause its rapid burning.

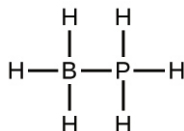
23. Extract from ore: $\text{AlO}(\text{OH})(s) + \text{NaOH}(aq) + \text{H}_2\text{O}(l) \longrightarrow \text{Na}[\text{Al}(\text{OH})_4](aq)$
Recover: $2\text{Na}[\text{Al}(\text{OH})_4](s) + \text{H}_2\text{SO}_4(aq) \longrightarrow 2\text{Al}(\text{OH})_3(s) + \text{Na}_2\text{SO}_4(aq) + 2\text{H}_2\text{O}(l)$



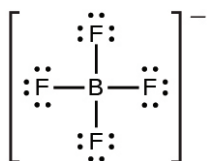
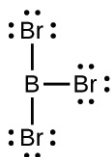
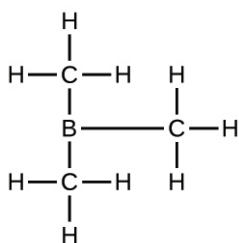
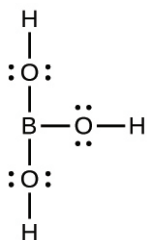
25. 25.83%

27. 39 kg

29. (a) H_3BPH_3 :



- (b) BF_4^- :

(c) BBr_3 :(d) $\text{B}(\text{CH}_3)_3$:(e) $\text{B}(\text{OH})_3$:31. $1s^2 2s^2 2p^6 3s^2 3p^2 3d^0$.

33. (a) $(\text{CH}_3)_3\text{SiH}$: sp^3 bonding about Si; the structure is tetrahedral; (b) SiO_4^{4-} : sp^3 bonding about Si; the structure is tetrahedral; (c) Si_2H_6 : sp^3 bonding about each Si; the structure is linear along the Si-Si bond; (d) $\text{Si}(\text{OH})_4$: sp^3 bonding about Si; the structure is tetrahedral; (e) SiF_6^{2-} : $sp^3 d^2$ bonding about Si; the structure is octahedral

35. (a) nonpolar; (b) nonpolar; (c) polar; (d) nonpolar; (e) polar

37. (a) tellurium dioxide or tellurium(IV) oxide; (b) antimony(III) sulfide; (c) germanium(IV) fluoride; (d) silane or silicon(IV) hydride; (e) germanium(IV) hydride

39. Boron has only s and p orbitals available, which can accommodate a maximum of four electron pairs. Unlike silicon, no d orbitals are available in boron.

41. (a) $\Delta H^\circ = 87 \text{ kJ}$; $\Delta G^\circ = 44 \text{ kJ}$; (b) $\Delta H^\circ = -109.9 \text{ kJ}$; $\Delta G^\circ = -154.7 \text{ kJ}$; (c) $\Delta H^\circ = -510 \text{ kJ}$; $\Delta G^\circ = -601.5 \text{ kJ}$

43. A mild solution of hydrofluoric acid would dissolve the silicate and would not harm the diamond.

45. In the N_2 molecule, the nitrogen atoms have an σ bond and two π bonds holding the two atoms together. The presence of three strong bonds makes N_2 a very stable molecule. Phosphorus is a third-period element, and as such, does not form π bonds efficiently; therefore, it must fulfill its bonding requirement by forming three σ bonds.

47. (a) $\text{H} = 1+$, $\text{C} = 2+$, and $\text{N} = 3-$; (b) $\text{O} = 2+$ and $\text{F} = 1-$; (c) $\text{As} = 3+$ and $\text{Cl} = 1-$

49. $\text{S} < \text{Cl} < \text{O} < \text{F}$

51. The electronegativity of the nonmetals is greater than that of hydrogen. Thus, the negative charge is better

represented on the nonmetal, which has the greater tendency to attract electrons in the bond to itself.

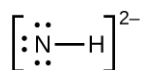
53. Hydrogen has only one orbital with which to bond to other atoms. Consequently, only one two-electron bond can form.

55. 0.43 g H₂

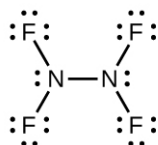
57. (a) $\text{Ca}(\text{OH})_2(aq) + \text{CO}_2(g) \longrightarrow \text{CaCO}_3(s) + \text{H}_2\text{O}(l)$; (b) $\text{CaO}(s) + \text{SO}_2(g) \longrightarrow \text{CaSO}_3(s)$;

(c) $2\text{NaHCO}_3(s) + \text{NaH}_2\text{PO}_4(aq) \longrightarrow \text{Na}_3\text{PO}_4(aq) + 2\text{CO}_2(g) + 2\text{H}_2\text{O}(l)$

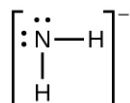
59. (a) NH_2^{2-} :



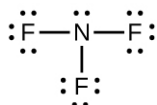
(b) N_2F_4 :



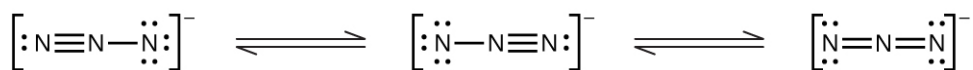
(c) NH_2^- :



(d) NF_3 :



(e) N_3^- :

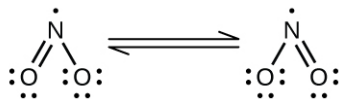


61. Ammonia acts as a Brønsted base because it readily accepts protons and as a Lewis base in that it has an electron pair to donate.

Brønsted base: $\text{NH}_3 + \text{H}_3\text{O}^+ \longrightarrow \text{NH}_4^+ + \text{H}_2\text{O}$

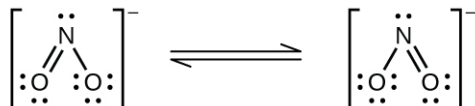
Lewis base: $2\text{NH}_3 + \text{Ag}^+ \longrightarrow [\text{H}_3\text{N} - \text{Ag} - \text{NH}_3]^+$

63. (a) NO_2 :



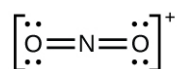
Nitrogen is sp^2 hybridized. The molecule has a bent geometry with an ONO bond angle of approximately 120° .

(b) NO_2^- :



Nitrogen is sp^2 hybridized. The molecule has a bent geometry with an ONO bond angle slightly less than 120° .

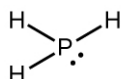
(c) NO_2^+ :



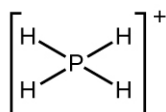
Nitrogen is sp hybridized. The molecule has a linear geometry with an ONO bond angle of 180° .

65. Nitrogen cannot form a NF_5 molecule because it does not have d orbitals to bond with the additional two fluorine atoms.

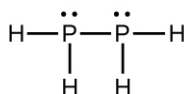
67. (a)



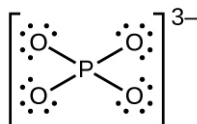
(b)



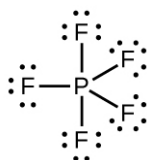
(c)



(d)



(e)

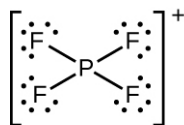


69. (a) $\text{P}_4(s) + 4\text{Al}(s) \longrightarrow 4\text{AlP}(s)$; (b) $\text{P}_4(s) + 12\text{Na}(s) \longrightarrow 4\text{Na}_3\text{P}(s)$; (c) $\text{P}_4(s) + 10\text{F}_2(g) \longrightarrow 4\text{PF}_5(l)$; (d) $\text{P}_4(s) + 6\text{Cl}_2(g) \longrightarrow 4\text{PCl}_3(l)$ or $\text{P}_4(s) + 10\text{Cl}_2(g) \longrightarrow 4\text{PCl}_5(l)$; (e) $\text{P}_4(s) + 3\text{O}_2(g) \longrightarrow \text{P}_4\text{O}_6(s)$ or $\text{P}_4(s) + 5\text{O}_2(g) \longrightarrow \text{P}_4\text{O}_{10}(s)$; (f) $\text{P}_4\text{O}_6(s) + 2\text{O}_2(g) \longrightarrow \text{P}_4\text{O}_{10}(s)$

71. 291 mL

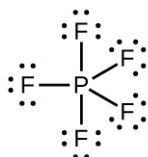
73. 28 tons

75. (a)



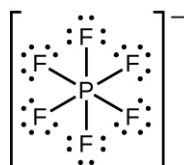
Tetrahedral

(b)



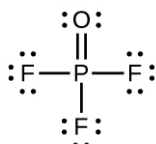
Trigonal bipyramid

(c)



Octahedral

(d)



Tetrahedral

77. (a) $P = 3+$; (b) $P = 5+$; (c) $P = 3+$; (d) $P = 5+$; (e) $P = 3-$; (f) $P = 5+$

79. FrO_2

81. (a) $2\text{Zn}(s) + \text{O}_2(g) \longrightarrow 2\text{ZnO}(s)$; (b) $\text{ZnCO}_3(s) \longrightarrow \text{ZnO}(s) + \text{CO}_2(g)$; (c) $\text{ZnCO}_3(s) + 2\text{CH}_3\text{COOH}(aq) \longrightarrow \text{Zn}(\text{CH}_3\text{COO})_2(aq) + \text{CO}_2(g) + \text{H}_2\text{O}(l)$; (d) $\text{Zn}(s) + 2\text{HBr}(aq) \longrightarrow \text{ZnBr}_2(aq) + \text{H}_2(g)$

83. $\text{Al}(\text{OH})_3(s) + 3\text{H}^+(aq) \longrightarrow \text{Al}^{3+} + 3\text{H}_2\text{O}(l)$; $\text{Al}(\text{OH})_3(s) + \text{OH}^- \longrightarrow [\text{Al}(\text{OH})_4]^- (aq)$

85. (a) $\text{Na}_2\text{O}(s) + \text{H}_2\text{O}(l) \longrightarrow 2\text{NaOH}(aq)$; (b) $\text{Cs}_2\text{CO}_3(s) + 2\text{HF}(aq) \longrightarrow 2\text{CsF}(aq) + \text{CO}_2(g) + \text{H}_2\text{O}(l)$; (c) $\text{Al}_2\text{O}_3(s) + 6\text{HClO}_4(aq) \longrightarrow 2\text{Al}(\text{ClO}_4)_3(aq) + 3\text{H}_2\text{O}(l)$; (d) $\text{Na}_2\text{CO}_3(aq) + \text{Ba}(\text{NO}_3)_2(aq) \longrightarrow 2\text{NaNO}_3(aq) + \text{BaCO}_3(s)$; (e) $\text{TiCl}_4(l) + 4\text{Na}(s) \longrightarrow \text{Ti}(s) + 4\text{NaCl}(s)$

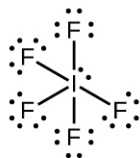
87. HClO_4 is the stronger acid because, in a series of oxyacids with similar formulas, the higher the electronegativity of the central atom, the stronger is the attraction of the central atom for the electrons of the oxygen(s). The stronger attraction of the oxygen electron results in a stronger attraction of oxygen for the electrons in the O-H bond, making the hydrogen more easily released. The weaker this bond, the stronger the acid.

89. As H_2SO_4 and H_2SeO_4 are both oxyacids and their central atoms both have the same oxidation number, the acid strength depends on the relative electronegativity of the central atom. As sulfur is more electronegative than selenium, H_2SO_4 is the stronger acid.

91. SO_2 , sp^2 4+; SO_3 , sp^2 , 6+; H_2SO_4 , sp^3 , 6+

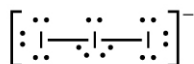
93. SF_6 : S = 6+; SO_2F_2 : S = 6+; KHS : S = 2-

95. Sulfur is able to form double bonds only at high temperatures (substantially endothermic conditions), which is not the case for oxygen.
97. There are many possible answers including: $\text{Cu}(s) + 2\text{H}_2\text{SO}_4(l) \longrightarrow \text{CuSO}_4(aq) + \text{SO}_2(g) + 2\text{H}_2\text{O}(l)$ and $\text{C}(s) + 2\text{H}_2\text{SO}_4(l) \longrightarrow \text{CO}_2(g) + 2\text{SO}_2(g) + 2\text{H}_2\text{O}(l)$
99. $5.1 \times 10^4 \text{ g}$
101. SnCl_4 is not a salt because it is covalently bonded. A salt must have ionic bonds.
103. In oxyacids with similar formulas, the acid strength increases as the electronegativity of the central atom increases. HClO_3 is stronger than HBrO_3 ; Cl is more electronegative than Br.
105. (a)



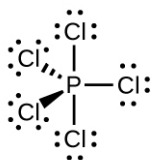
Square pyramidal

(b)



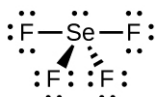
Linear

(c)



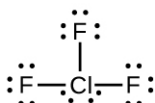
Trigonal bipyramidal

(d)



Seesaw

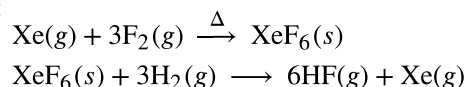
(e)



T-shaped

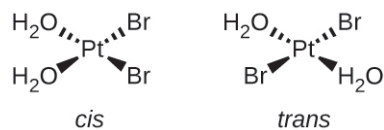
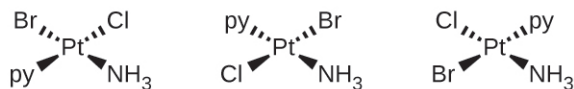
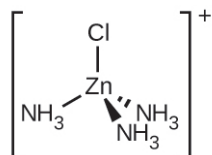
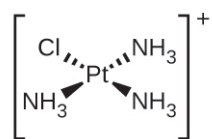
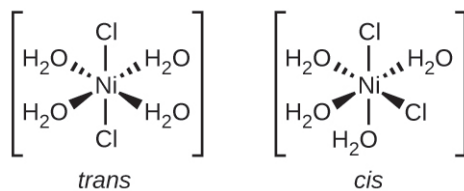
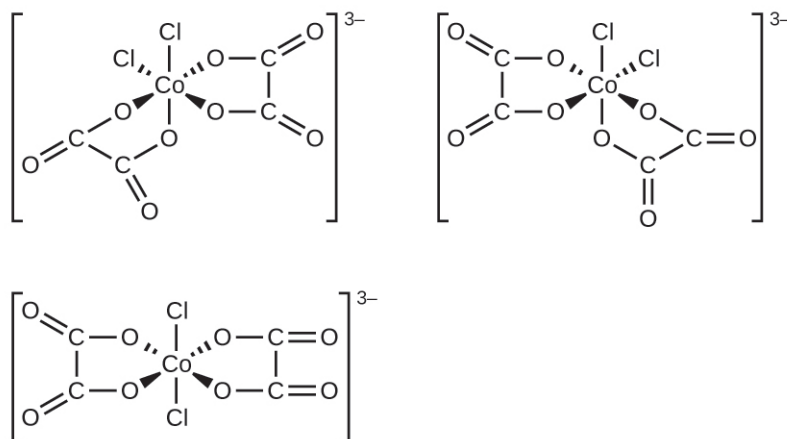
107. (a) bromine trifluoride; (b) sodium bromate; (c) phosphorus pentabromide; (d) sodium perchlorate; (e) potassium hypochlorite
109. (a) I: 7+; (b) I: 7+; (c) Cl: 4+; (d) I: 3+; Cl: 1-; (e) F: 0
111. (a) sp^3d hybridized; (b) sp^3d^2 hybridized; (c) sp^3 hybridized; (d) sp^3 hybridized; (e) sp^3d^2 hybridized;
113. (a) nonpolar; (b) nonpolar; (c) polar; (d) nonpolar; (e) polar

115. The empirical formula is XeF_6 , and the balanced reactions are:



Chapter 19

1. (a) Sc: $[\text{Ar}]4s^23d^1$; (b) Ti: $[\text{Ar}]4s^23d^2$; (c) Cr: $[\text{Ar}]4s^13d^5$; (d) Fe: $[\text{Ar}]4s^23d^6$; (e) Ru: $[\text{Kr}]5s^24d^6$
3. (a) La: $[\text{Xe}]6s^25d^1$, La^{3+} : $[\text{Xe}]$; (b) Sm: $[\text{Xe}]6s^24f^6$, Sm^{3+} : $[\text{Xe}]4f^5$; (c) Lu: $[\text{Xe}]6s^24f^{14}5d^1$, Lu^{3+} : $[\text{Xe}]4f^{14}$
5. Al is used because it is the strongest reducing agent and the only option listed that can provide sufficient driving force to convert La(III) into La.
7. Mo
9. The CaSiO_3 slag is less dense than the molten iron, so it can easily be separated. Also, the floating slag layer creates a barrier that prevents the molten iron from exposure to O_2 , which would oxidize the Fe back to Fe_2O_3 .
11. 2.57%
13. 0.167 V
15. $E^\circ = -0.6 \text{ V}$, E° is negative so this reduction is not spontaneous. $E^\circ = +1.1 \text{ V}$
17. (a) $\text{Fe(s)} + 2\text{H}_3\text{O}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \longrightarrow \text{Fe}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) + \text{H}_2(\text{g}) + 2\text{H}_2\text{O(l)}$; (b) $\text{FeCl}_3(\text{aq}) + 3\text{Na}^+(\text{aq}) + 3\text{OH}^-(\text{aq}) \longrightarrow \text{Fe(OH)}_3(\text{s}) + 3\text{Na}^+(\text{aq}) + 3\text{Cl}^-(\text{aq})$; (c) $\text{Mn(OH)}_2(\text{s}) + 2\text{H}_3\text{O}^+(\text{aq}) + 2\text{Br}^-(\text{aq}) \longrightarrow \text{Mn}^{2+}(\text{aq}) + 2\text{Br}^-(\text{aq}) + 4\text{H}_2\text{O(l)}$; (d) $4\text{Cr(s)} + 3\text{O}_2(\text{g}) \longrightarrow 2\text{Cr}_2\text{O}_3(\text{s})$; (e) $\text{Mn}_2\text{O}_3(\text{s}) + 6\text{H}_3\text{O}^+(\text{aq}) + 6\text{Cl}^-(\text{aq}) \longrightarrow 2\text{MnCl}_3(\text{s}) + 9\text{H}_2\text{O(l)}$; (f) $\text{Ti(s)} + x\text{F}_2(\text{g}) \longrightarrow \text{TiF}_x(\text{g})$
19. (a) $\text{Cr}_2(\text{SO}_4)_3(\text{aq}) + 2\text{Zn(s)} + 2\text{H}_3\text{O}^+(\text{aq}) \longrightarrow 2\text{Zn}^{2+}(\text{aq}) + \text{H}_2(\text{g}) + 2\text{H}_2\text{O(l)} + 2\text{Cr}^{2+}(\text{aq}) + 3\text{SO}_4^{2-}(\text{aq})$; (b) $4\text{TiCl}_3(\text{s}) + \text{CrO}_4^{2-}(\text{aq}) + 8\text{H}^+(\text{aq}) \longrightarrow 4\text{Ti}^{4+}(\text{aq}) + \text{Cr(s)} + 4\text{H}_2\text{O(l)} + 12\text{Cl}^-(\text{aq})$; (c) In acid solution between pH 2 and pH 6, CrO_4^{2-} forms HCrO_4^- , which is in equilibrium with dichromate ion. The reaction is $2\text{HCrO}_4^-(\text{aq}) \longrightarrow \text{Cr}_2\text{O}_7^{2-}(\text{aq}) + \text{H}_2\text{O(l)}$. At other acidic pHs, the reaction is $3\text{Cr}^{2+}(\text{aq}) + \text{CrO}_4^{2-}(\text{aq}) + 8\text{H}_3\text{O}^+(\text{aq}) \longrightarrow 4\text{Cr}^{3+}(\text{aq}) + 12\text{H}_2\text{O(l)}$; (d) $8\text{CrO}_3(\text{s}) + 9\text{Mn(s)} \xrightarrow{\Delta} 4\text{Cr}_2\text{O}_3(\text{s}) + 3\text{Mn}_3\text{O}_4(\text{s})$; (e) $\text{CrO(s)} + 2\text{H}_3\text{O}^+(\text{aq}) + 2\text{NO}_3^-(\text{aq}) \longrightarrow \text{Cr}^{2+}(\text{aq}) + 2\text{NO}_3^-(\text{aq}) + 3\text{H}_2\text{O(l)}$; (f) $\text{CrCl}_3(\text{s}) + 3\text{NaOH(aq)} \longrightarrow \text{Cr(OH)}_3(\text{s}) + 3\text{Na}^+(\text{aq}) + 3\text{Cl}^-(\text{aq})$
21. (a) $3\text{Fe(s)} + 4\text{H}_2\text{O(g)} \longrightarrow \text{Fe}_3\text{O}_4(\text{s}) + 4\text{H}_2(\text{g})$; (b) $3\text{NaOH(aq)} + \text{Fe(NO}_3)_3(\text{aq}) \xrightarrow{\text{H}_2\text{O}} \text{Fe(OH)}_3(\text{s}) + 3\text{Na}^+(\text{aq}) + 3\text{NO}_3^-(\text{aq})$; (c) $\text{MnO}_4^- + 5\text{Fe}^{2+} + 8\text{H}^+ \longrightarrow \text{Mn}^{2+} + 5\text{Fe}^{3+} + 4\text{H}_2\text{O}$; (d) $\text{Fe(s)} + 2\text{H}_3\text{O}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \longrightarrow \text{Fe}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) + \text{H}_2(\text{g}) + 2\text{H}_2\text{O(l)}$; (e) $4\text{Fe}^{2+}(\text{aq}) + \text{O}_2(\text{g}) + 4\text{HNO}_3(\text{aq}) \longrightarrow 4\text{Fe}^{3+}(\text{aq}) + 2\text{H}_2\text{O(l)} + 4\text{NO}_3^-(\text{aq})$; (f) $\text{FeCO}_3(\text{s}) + 2\text{HClO}_4(\text{aq}) \longrightarrow \text{Fe(ClO}_4)_2(\text{aq}) + \text{H}_2\text{O(l)} + \text{CO}_2(\text{g})$; (g) $3\text{Fe(s)} + 2\text{O}_2(\text{g}) \xrightarrow{\Delta} \text{Fe}_3\text{O}_4(\text{s})$
23. As CN^- is added,
 $\text{Ag}^+(\text{aq}) + \text{CN}^-(\text{aq}) \longrightarrow \text{AgCN(s)}$
 As more CN^- is added,
 $\text{Ag}^+(\text{aq}) + 2\text{CN}^-(\text{aq}) \longrightarrow [\text{Ag(CN)}_2]^- (\text{aq})$
 $\text{AgCN(s)} + \text{CN}^-(\text{aq}) \longrightarrow [\text{Ag(CN)}_2]^- (\text{aq})$
25. (a) Sc^{3+} ; (b) Ti^{4+} ; (c) V^{5+} ; (d) Cr^{6+} ; (e) Mn^{4+} ; (f) Fe^{2+} and Fe^{3+} ; (g) Co^{2+} and Co^{3+} ; (h) Ni^{2+} ; (i) Cu^+
27. (a) 4, $[\text{Zn(OH)}_4]^{2-}$; (b) 6, $[\text{Pd(CN)}_6]^{2-}$; (c) 2, $[\text{AuCl}_2]^-$; (d) 4, $[\text{Pt(NH}_3)_2\text{Cl}_2]$; (e) 6, $\text{K}[\text{Cr(NH}_3)_2\text{Cl}_4]$; (f) 6, $[\text{Co(NH}_3)_6][\text{Cr(CN)}_6]$; (g) 6, $[\text{Co(en)}_2\text{Br}_2]\text{NO}_3$
29. (a) $[\text{Pt(H}_2\text{O)}_2\text{Br}_2]$:

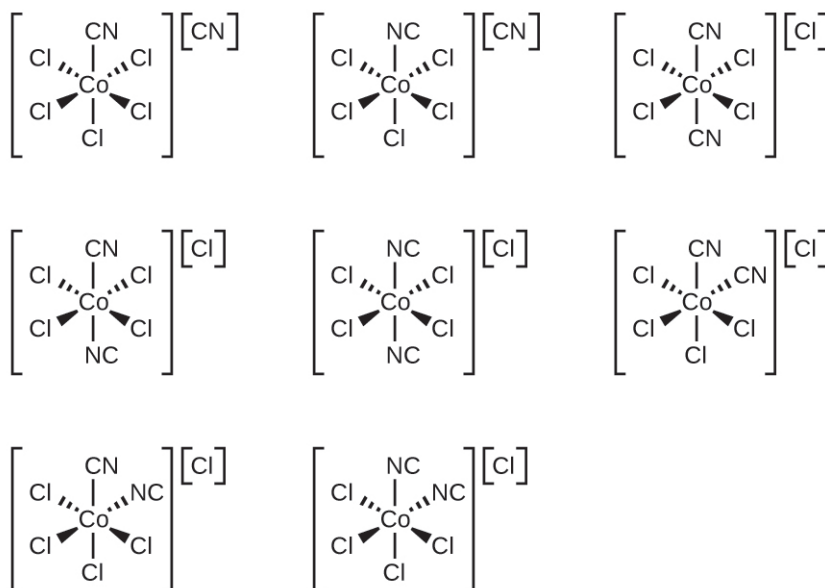
(b) $[\text{Pt}(\text{NH}_3)(\text{py})(\text{Cl})(\text{Br})]$:(c) $[\text{Zn}(\text{NH}_3)_3\text{Cl}]^+$:(d) $[\text{Pt}(\text{NH}_3)_3\text{Cl}]^+$:(e) $[\text{Ni}(\text{H}_2\text{O})_4\text{Cl}_2]$:(f) $[\text{Co}(\text{C}_2\text{O}_4)_2\text{Cl}_2]^{3-}$:

31. (a) tricarbonatocobaltate(III) ion; (b) tetraaminecopper(II) ion; (c) tetraaminedibromocobalt(III) sulfate; (d) tetraamineplatinum(II) tetrachloroplatinate(II); (e) *tris*-(ethylenediamine)chromium(III) nitrate; (f)

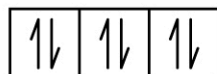
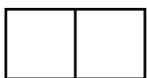
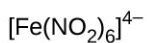
diaminedibromopalladium(II); (g) potassium pentachlorocuprate(II); (h) diaminedichlorozinc(II)

33. (a) none; (b) none; (c) The two Cl ligands can be *cis* or *trans*. When they are *cis*, there will also be an optical isomer.

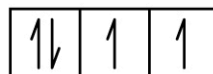
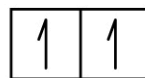
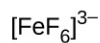
35.



37.



Low spin, diamagnetic, $P < \Delta_{\text{oct}}$



High spin, paramagnetic, $P > \Delta_{\text{oct}}$

39. $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_2$ with three unpaired electrons.

41. (a) 4; (b) 2; (c) 1; (d) 5; (e) 0

43. (a) $[\text{Fe}(\text{CN})_6]^{4-}$; (b) $[\text{Co}(\text{NH}_3)_6]^{3+}$; (c) $[\text{Mn}(\text{CN})_6]^{4-}$

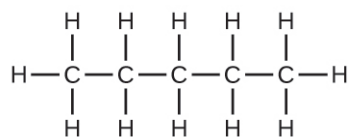
45. The complex does not have any unpaired electrons. The complex does not have any geometric isomers, but the mirror image is nonsuperimposable, so it has an optical isomer.

47. No. Au^+ has a complete 5*d* sublevel.

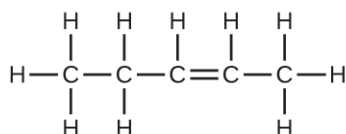
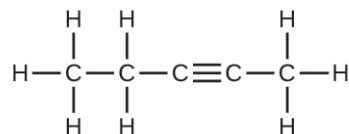
Chapter 20

1. There are several sets of answers; one is:

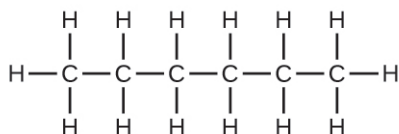
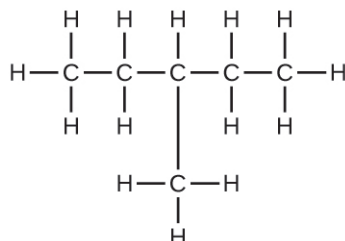
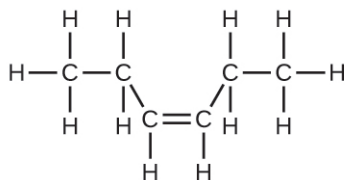
(a) C_5H_{12}

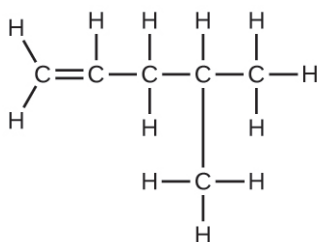
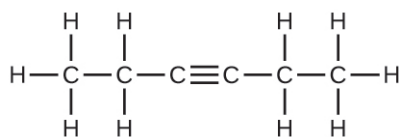
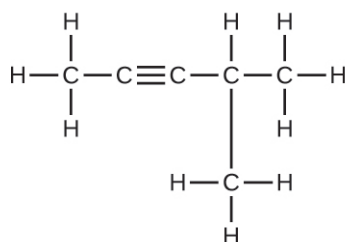


(b) C_5H_{10}

(c) C_5H_8 

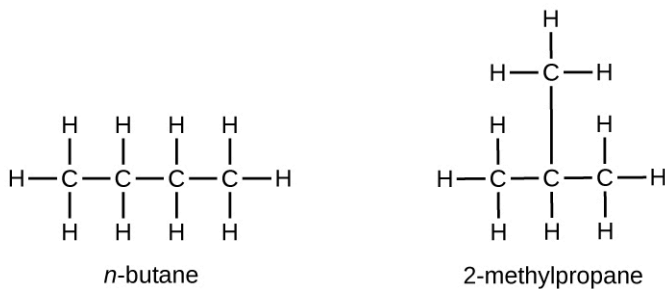
3. Both reactions result in bromine being incorporated into the structure of the product. The difference is the way in which that incorporation takes place. In the saturated hydrocarbon, an existing C–H bond is broken, and a bond between the C and the Br can then be formed. In the unsaturated hydrocarbon, the only bond broken in the hydrocarbon is the π bond whose electrons can be used to form a bond to one of the bromine atoms in Br_2 (the electrons from the Br–Br bond form the other C–Br bond on the other carbon that was part of the π bond in the starting unsaturated hydrocarbon).
5. Unbranched alkanes have free rotation about the C–C bonds, yielding all orientations of the substituents about these bonds equivalent, interchangeable by rotation. In the unbranched alkenes, the inability to rotate about the C = C bond results in fixed (unchanging) substituent orientations, thus permitting different isomers. Since these concepts pertain to phenomena at the molecular level, this explanation involves the microscopic domain.
7. They are the same compound because each is a saturated hydrocarbon containing an unbranched chain of six carbon atoms.
9. (a) C_6H_{14}

(b) C_6H_{14} (c) C_6H_{12} (d) C_6H_{12}

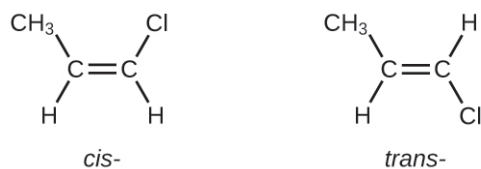
(e) C_6H_{10} (f) C_6H_{10} 

11. (a) 2,2-dibromobutane; (b) 2-chloro-2-methylpropane; (c) 2-methylbutane; (d) 1-butyne; (e) 4-fluoro-4-methyl-1-octyne; (f) *trans*-1-chloropropene; (g) 4-methyl-1-pentene

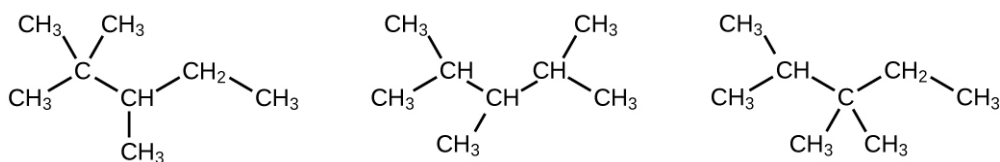
13.



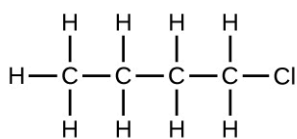
15.



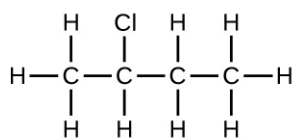
17. (a) 2,2,4-trimethylpentane; (b) 2,2,3-trimethylpentane, 2,3,4-trimethylpentane, and 2,3,3-trimethylpentane:



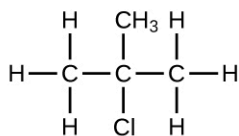
19.



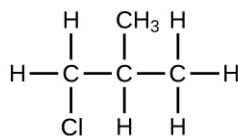
1-chlorobutane



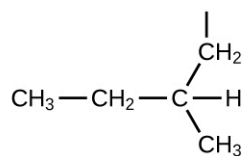
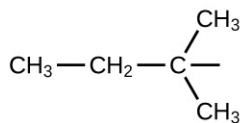
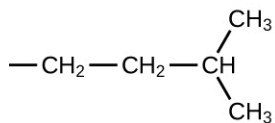
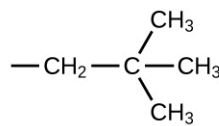
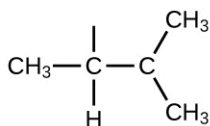
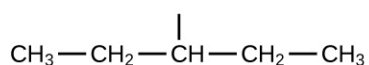
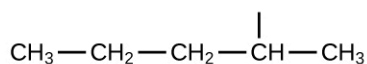
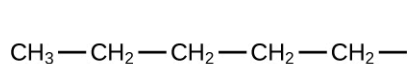
2-chlorobutane



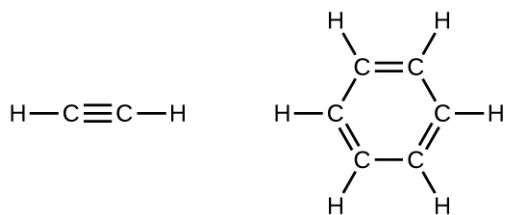
2-chloro-2-methylpropane

1-chloro-2-methylpropane
(1-chloro-2-methylpropane)

21. In the following, the carbon backbone and the appropriate number of hydrogen atoms are shown in condensed form:

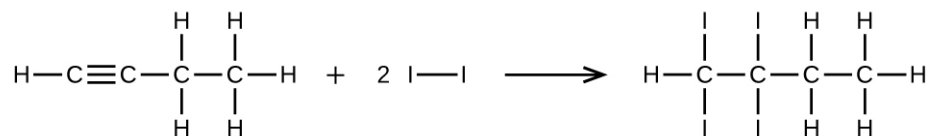


23.

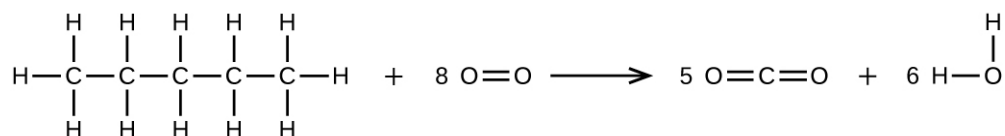


In acetylene, the bonding uses sp hybrids on carbon atoms and s orbitals on hydrogen atoms. In benzene, the carbon atoms are sp^2 hybridized.

25. (a) $\text{CH} \equiv \text{CCH}_2\text{CH}_3 + 2\text{I}_2 \longrightarrow \text{CHI}_2\text{CI}_2\text{CH}_2\text{CH}_3$



(b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3 + 8\text{O}_2 \longrightarrow 5\text{CO}_2 + 6\text{H}_2\text{O}$



27. 74.5 g

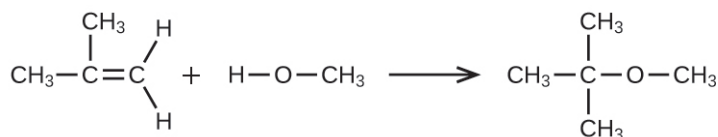
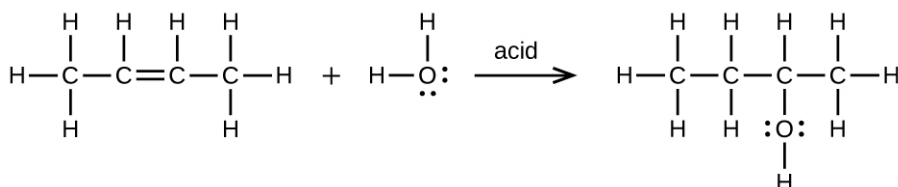
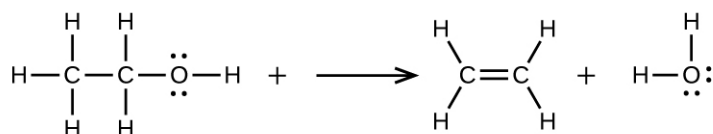
29. 9.328×10^2 kg

31. (a) ethyl alcohol, ethanol: $\text{CH}_3\text{CH}_2\text{OH}$; (b) methyl alcohol, methanol: CH_3OH ; (c) ethylene glycol, ethanediol: $\text{HOCH}_2\text{CH}_2\text{OH}$; (d) isopropyl alcohol, 2-propanol: $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$; (e) glycerine, 1,2,3-trihydroxypropane: $\text{HOCH}_2\text{CH}(\text{OH})\text{CH}_2\text{OH}$

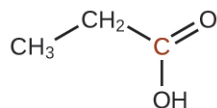
33. (a) 1-ethoxybutane, butyl ethyl ether; (b) 1-ethoxypropane, ethyl propyl ether; (c) 1-methoxypropane, methyl propyl ether

35. $\text{HOCH}_2\text{CH}_2\text{OH}$, two alcohol groups; $\text{CH}_3\text{OCH}_2\text{OH}$, ether and alcohol groups

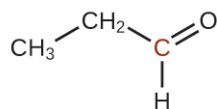
37. (a)

(b) 4.593×10^2 L39. (a) $\text{CH}_3\text{CH}=\text{CHCH}_3 + \text{H}_2\text{O} \longrightarrow \text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$ (b) $\text{CH}_3\text{CH}_2\text{OH} \longrightarrow \text{CH}_2=\text{CH}_2 + \text{H}_2\text{O}$ 

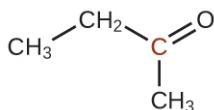
41. (a)



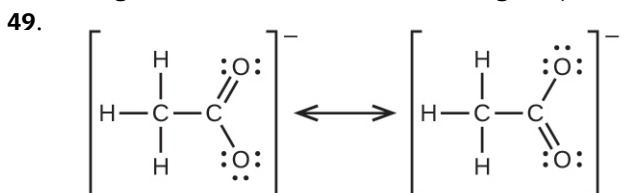
(b)



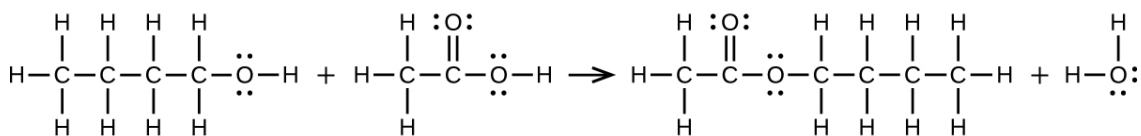
(c)



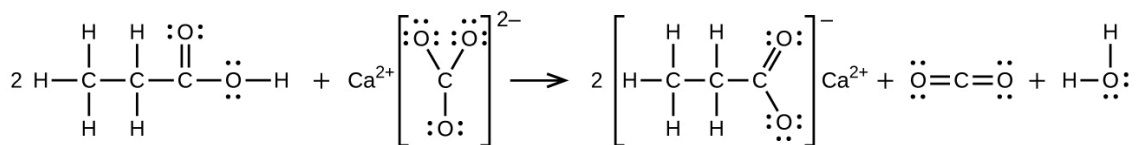
43. A ketone contains a group bonded to two additional carbon atoms; thus, a minimum of three carbon atoms are needed.
45. Since they are both carboxylic acids, they each contain the -COOH functional group and its characteristics. The difference is the hydrocarbon chain in a saturated fatty acid contains no double or triple bonds, whereas the hydrocarbon chain in an unsaturated fatty acid contains one or more multiple bonds.
47. (a) $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$: all carbons are tetrahedral; (b) CH_3COCH_3 : the end carbons are tetrahedral and the central carbon is trigonal planar; (c) CH_3OCH_3 : all are tetrahedral; (d) CH_3COOH : the methyl carbon is tetrahedral and the acid carbon is trigonal planar; (e) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{CH}_3)\text{CHCH}_2$: all are tetrahedral except the right-most two carbons, which are trigonal planar



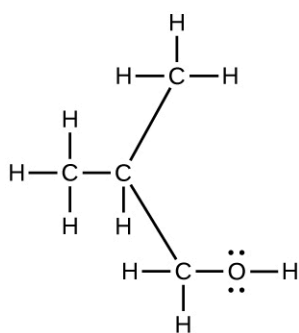
51. (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} + \text{CH}_3\text{C}(\text{O})\text{OH} \rightarrow \text{CH}_3\text{C}(\text{O})\text{OCH}_2\text{CH}_2\text{CH}_2\text{CH}_3 + \text{H}_2\text{O}$:



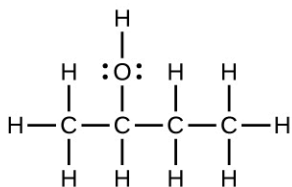
- (b) $2\text{CH}_3\text{CH}_2\text{COOH} + \text{CaCO}_3 \rightarrow (\text{CH}_3\text{CH}_2\text{COO})_2\text{Ca} + \text{CO}_2 + \text{H}_2\text{O}$:



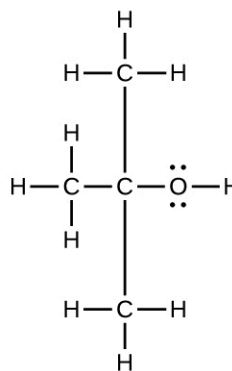
53.



Compound A



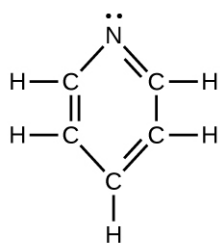
Compound B



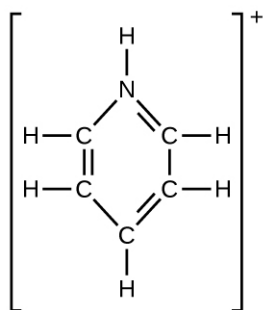
Compound C

55. Trimethyl amine: trigonal pyramidal, sp^3 ; trimethyl ammonium ion: tetrahedral, sp^3

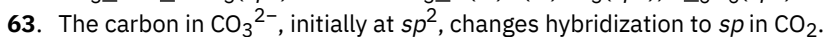
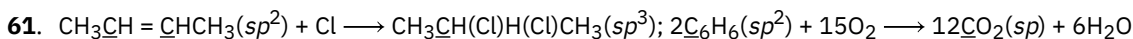
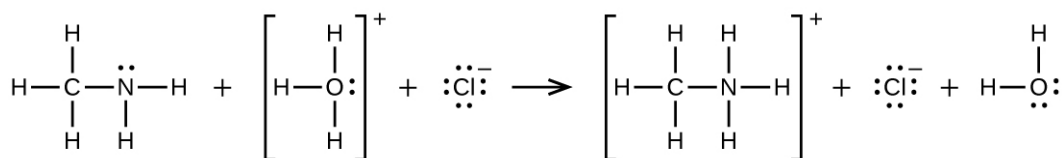
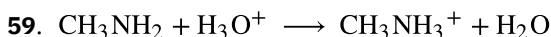
57.



pyridine,
trigonal planar, sp^2
(nonlinear or bent)



pyridinium ion,
trigonal planar, sp^2
(trigonal planar)



Chapter 21

- (a) sodium-24; (b) aluminum-29; (c) krypton-73; (d) iridium-194
- (a) ${}^{34}_{14}\text{Si}$; (b) ${}^{36}_{15}\text{P}$; (c) ${}^{57}_{25}\text{Mn}$; (d) ${}^{121}_{56}\text{Ba}$
- (a) ${}^{45}_{25}\text{Mn}^{+1}$; (b) ${}^{69}_{45}\text{Rh}^{+2}$; (c) ${}^{142}_{53}\text{I}^{-1}$; (d) ${}^{243}_{97}\text{Bk}$
- Nuclear reactions usually change one type of nucleus into another; chemical changes rearrange atoms. Nuclear reactions involve much larger energies than chemical reactions and have measureable mass changes.
- (a), (b), (c), (d), and (e)
- (a) A nucleon is any particle contained in the nucleus of the atom, so it can refer to protons and neutrons. (b) An α particle is one product of natural radioactivity and is the nucleus of a helium atom. (c) A β particle is a product of natural radioactivity and is a high-speed electron. (d) A positron is a particle with the same mass as an electron but with a positive charge. (e) Gamma rays compose electromagnetic radiation of high energy and short wavelength. (f) Nuclide is a term used when referring to a single type of nucleus. (g) The mass number is the sum of the number of protons and the number of neutrons in an element. (h) The atomic number is the number of protons in the nucleus of an element.
- (a) ${}^{27}_{13}\text{Al} + {}^4_2\text{He} \longrightarrow {}^{30}_{15}\text{P} + {}^1_0\text{n}$; (b) ${}^{239}_{94}\text{Pu} + {}^4_2\text{He} \longrightarrow {}^{242}_{96}\text{Cm} + {}^1_0\text{n}$; (c) ${}^{14}_7\text{N} + {}^4_2\text{He} \longrightarrow {}^{17}_8\text{O} + {}^1_1\text{H}$; (d) ${}^{235}_{92}\text{U} \longrightarrow {}^{96}_{37}\text{Rb} + {}^{135}_{55}\text{Cs} + 4{}^1_0\text{n}$
- (a) ${}^{14}_7\text{N} + {}^4_2\text{He} \longrightarrow {}^{17}_8\text{O} + {}^1_1\text{H}$; (b) ${}^{14}_7\text{C} + {}^1_0\text{n} \longrightarrow {}^{14}_6\text{C} + {}^1_1\text{H}$; (c) ${}^{232}_{90}\text{Th} + {}^1_0\text{n} \longrightarrow {}^{233}_{90}\text{Th}$; (d) ${}^{238}_{92}\text{U} + {}^2_1\text{H} \longrightarrow {}^{239}_{92}\text{U} + {}^1_1\text{H}$
- (a) 148.8 MeV per atom; (b) 7.808 MeV/nucleon
- α (helium nuclei), β (electrons), β^+ (positrons), and η (neutrons) may be emitted from a radioactive element, all of which are particles; γ rays also may be emitted.
- (a) conversion of a neutron to a proton: ${}^1_0\text{n} \longrightarrow {}^1_1\text{p} + {}^0_{+1}\text{e}$; (b) conversion of a proton to a neutron; the positron has the same mass as an electron and the same magnitude of positive charge as the electron has negative charge; when the n:p ratio of a nucleus is too low, a proton is converted into a neutron with the emission of a positron: ${}^1_1\text{p} \longrightarrow {}^1_0\text{n} + {}^0_{+1}\text{e}$; (c) In a proton-rich nucleus, an inner atomic electron can be absorbed. In simplest form, this changes a proton into a neutron: ${}^1_1\text{p} + {}^0_{-1}\text{e} \longrightarrow {}^1_0\text{n}$

23. The electron pulled into the nucleus was most likely found in the 1s orbital. As an electron falls from a higher energy level to replace it, the difference in the energy of the replacement electron in its two energy levels is given off as an X-ray.
25. Manganese-51 is most likely to decay by positron emission. The n:p ratio for Cr-53 is $\frac{29}{24} = 1.21$; for Mn-51, it is $\frac{26}{25} = 1.04$; for Fe-59, it is $\frac{33}{26} = 1.27$. Positron decay occurs when the n:p ratio is low. Mn-51 has the lowest n:p ratio and therefore is most likely to decay by positron emission. Besides, $^{53}_{24}\text{Cr}$ is a stable isotope, and $^{59}_{26}\text{Fe}$ decays by beta emission.
27. (a) positron emission; (b) α decay; (c) positron emission; (d) β decay; (e) α decay
29. $^{238}_{92}\text{U} \rightarrow ^{234}_{90}\text{Th} + ^4_2\text{He}$; $^{234}_{90}\text{Th} \rightarrow ^{234}_{91}\text{Pa} + ^0_{-1}\text{e}$; $^{234}_{91}\text{Pa} \rightarrow ^{234}_{92}\text{U} + ^0_{-1}\text{e}$; $^{234}_{92}\text{U} \rightarrow ^{230}_{90}\text{Th} + ^4_2\text{He}$
 $^{230}_{90}\text{Th} \rightarrow ^{226}_{88}\text{Ra} + ^4_2\text{He}$; $^{226}_{88}\text{Ra} \rightarrow ^{222}_{86}\text{Rn} + ^4_2\text{He}$; $^{222}_{86}\text{Rn} \rightarrow ^{218}_{84}\text{Po} + ^4_2\text{He}$
31. Half-life is the time required for half the atoms in a sample to decay. Example (answers may vary): For C-14, the half-life is 5770 years. A 10-g sample of C-14 would contain 5 g of C-14 after 5770 years; a 0.20-g sample of C-14 would contain 0.10 g after 5770 years.
33. $\left(\frac{1}{2}\right)^{0.04} = 0.973$ or 97.3%
35. $2 \times 10^3 \text{ y}$
37. 0.12 h^{-1}
39. (a) 3.8 billion years; (b) The rock would be younger than the age calculated in part (a). If Sr was originally in the rock, the amount produced by radioactive decay would equal the present amount minus the initial amount. As this amount would be smaller than the amount used to calculate the age of the rock and the age is proportional to the amount of Sr, the rock would be younger.
41. $c = 0$; This shows that no Pu-239 could remain since the formation of the earth. Consequently, the plutonium now present could not have been formed with the uranium.
43. 17.5 MeV
45. (a) $^{212}_{83}\text{Bi} \rightarrow ^{212}_{84}\text{Po} + ^0_{-1}\text{e}$; (b) $^8_5\text{B} \rightarrow ^8_4\text{Be} + ^0_{-1}\text{e}$; (c) $^{238}_{92}\text{U} + ^1_0\text{n} \rightarrow ^{239}_{93}\text{Np} + ^0_{-1}\text{e}$, $^{239}_{93}\text{Np} \rightarrow ^{239}_{94}\text{Pu} + ^0_{-1}\text{e}$; (d) $^{90}_{38}\text{Sr} \rightarrow ^{90}_{39}\text{Y} + ^0_{-1}\text{e}$
47. (a) $^{241}_{95}\text{Am} + ^4_2\text{He} \rightarrow ^{244}_{97}\text{Bk} + ^1_0\text{n}$; (b) $^{239}_{94}\text{Pu} + 15^1_0\text{n} \rightarrow ^{254}_{100}\text{Fm} + 6^0_{-1}\text{e}$; (c) $^{250}_{98}\text{Cf} + ^{11}_5\text{B} \rightarrow ^{257}_{103}\text{Lr} + 4^1_0\text{n}$; (d) $^{249}_{98}\text{Cf} + ^{15}_7\text{N} \rightarrow ^{260}_{105}\text{Db} + 4^1_0\text{n}$
49. Two nuclei must collide for fusion to occur. High temperatures are required to give the nuclei enough kinetic energy to overcome the very strong repulsion resulting from their positive charges.
51. A nuclear reactor consists of the following:
1. A nuclear fuel. A fissionable isotope must be present in large enough quantities to sustain a controlled chain reaction. The radioactive isotope is contained in tubes called fuel rods.
 2. A moderator. A moderator slows neutrons produced by nuclear reactions so that they can be absorbed by the fuel and cause additional nuclear reactions.
 3. A coolant. The coolant carries heat from the fission reaction to an external boiler and turbine where it is transformed into electricity.
 4. A control system. The control system consists of control rods placed between fuel rods to absorb neutrons and is used to adjust the number of neutrons and keep the rate of the chain reaction at a safe level.
 5. A shield and containment system. The function of this component is to protect workers from radiation produced by the nuclear reactions and to withstand the high pressures resulting from high-temperature reactions.
53. The fission of uranium generates heat, which is carried to an external steam generator (boiler). The resulting steam turns a turbine that powers an electrical generator.
55. Introduction of either radioactive Ag^+ or radioactive Cl^- into the solution containing the stated reaction, with subsequent time given for equilibration, will produce a radioactive precipitate that was originally devoid of radiation.
57. (a) $^{133}_{53}\text{I} \rightarrow ^{133}_{54}\text{Xe} + ^0_{-1}\text{e}$; (b) 37.6 days
59. Alpha particles can be stopped by very thin shielding but have much stronger ionizing potential than beta particles, X-rays, and γ -rays. When inhaled, there is no protective skin covering the cells of the lungs, making

it possible to damage the DNA in those cells and cause cancer.

- 61.** (a) 7.64×10^9 Bq; (b) 2.06×10^{-2} Ci

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